
The Associate CET Study Guide

Sixth Edition

Published by ETA International

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re The Associate CET Study Guide - 6" Edition This book has been developed by the members of ETA and is published by: The Electronics Technicians Association, Int'l., Inc. (ETA® International) § Depot St. Greencastle, IN 46135 (800) 288-3824 www.cta-i.org eta@eta-i.org Cover Design by: BryanAllen, CSM, CSS Edited by: ETA® International Staff Page Layout by: BryanAllen, CSM, CSS Printing by:

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Preface

academics. Sample wall certificate This book is intended as an official study guide for students and electronics technicians Who are seeking the status of CERTIFIED ELECTRONICS TECHNICIAN, ASSOCIATE LEVEL (CETa). It is also intended for technicians who are planning to become Journeyman (CET), Senior (CETsr) or Master (CETma) CETs. The latter three levels of certification require passage of both the Associate exam and one or more Specialty Option exams. The book is not planned as an engineering or scientific text. While it contains many in-depth discussions of electronics technology basics, it is intended as a practical guide for working technicians without the mathematics, chemistry or physics requirements of rigorous That isn't to say ETA isn't interested in electronics technicians advancing to studies of galactic or quantum physics. Advanced study is encouraged. In fact, students who become electronics technicians, understanding the world of electricity, photonics, communications, electronic components and methods of measurement are already a step ahead of much of the population. Competent electronics technicians often find their interest attracted to hobbies or vocations which involve intellectual leading-edge technological comprehension of the Cosmos, as well as the atomic world we live in. Possibly the most important technical worker today is the electronics technician. He or she may be involved in any one of hundreds of vitally important jobs. To-day's technicians do not know it all. They must specialize in only a few segments of the electronics universe. That is because the technology has not only spread out vertically, but horizontally too. To become knowledgeable and experienced in every phase

of electronics would require hundreds, if not thousands, of times the quantity of knowledge a technician needed half a century ago. The fact still remains that every technician is a better technician if he or she has a solid core, or basic, electronics knowledge. In a book that you can read in a day, ETA presents that core knowledge that can help you become a qualified electronics technician. Here is your pervanat cenificaion cord ETA' Internationa! Pepe,

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new Sample wallet card With an Associate CET status, you can qualify for hundreds of types of electronics jobs. You can stop there and become comfortable in a routine position, or you can use your knowledge to step up the ladder to Journeyman or higher recognition. With the basics firmly implanted in your mind, you will have a confidence in your work which can make your job gratifying, not drudgery. It can make many aspects of your life more fulfilling as you now have an understanding of how most things in life work. Good luck in your career as an ELECTRONICS TECHNICIAN. fit (A Dick Glass, CETsr, EHF CEO Emeritus

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Acknowledgements

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Authors:

Sohail Anwar, Ph.D., CHTsr, Altoona, PA John W, Baldwin, CUT#r, Mankato,

MN James Bruening, CUT, Bremerton, WA David Caldwell, Wr, Brownville,

PA Dick Glass, CHTsr, WF, Greencastle, IN Thomas F, Glass, CET, Indianapolis,

IN Robert Ing, CUSma, Toronto, Ontario Tom Janca, CL Tsr, Denver, CO Richard Kirkland, r, Las Vegas, NV Rollin Okerberg, CE'Tsr, Mountain Home AFB, ID Hdward Parady, CUT, Pittsburgh, PA

Fitz Herbert Reid, Fyzabad, Trinidad

David Samsell, CL Tsr, Scranton, PA Daniel Siddall, CETsr, Boise, ID John Ufer, CET sr, Williamsburg, MI Tom Schulte, CET, Sioux Falls, SD

Chief Editor:

Dick Glass, CET sr Cont: incl, jous editions): atti mettoraiin peor John Baldwin, CETsr Richard Booth, FOT Carolyn Carson, CSS

Karl Kilers

Dale Firmin

Mmily Hatfield, CSS Carole Ann Lentz, CS: 'Teresa Maher, CSS Jeannette Stroud, CSS Ray Stroud, CET Arlene G, Tinchier, CST

Graphics Assistance:

Bryan Allen, CSM, CSS Rie Thayer, Erie, PA

Printing:

Greencastle Offset Printing and ©jraphics, Greencastle, IN

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ETA' International Certifications

Basic Electronics:

Apprentice (APP) Associate Electronics Technician (CETa) Electronics Modules (EM1-5)

DC (EMI)** AC (EM2)**

+ Analog (EM3)

+ Digital (EM4) Comprehensive (EMS) Student Electronics Technician (SET)

Biomedical:

Biomedical Electronics Technician (BMD) Biomedical Imaging Equipment Technician (BIET)

Communications:

Broadband Voice over Internet Protocol (B-VoIP) Certified Satellite Installer (CSI) Distributed Antenna Systems (DAS) General Communications Technician-Level 1 (GCT1) General Communications Technician-Level 2 (GCT2) Line and Antenna Sweep (LAS) Mobile Communications and Electronics Installer (MCEI) Passive Intermodulation Testing (PIM) RF Interference Mitigation (RFIM)

RADAR (RAD) 7

TRN Wireless Communications Technician (TRNI) Telecommunications (TCM) Wireless Communications (WCM) Fiber Optics and Data Cabling: ARINC Installer, Technician (AFV/AFT) Data Cabling Installer (DCI) Fiber Optics Designer (FOD) Fiber Optics Installer (FOI) Fiber Optics Technician (FOT) Fiber Optics Technician-Outside Plant (FOT-OSP) SAE Fabricator (SFF) Termination and Testing Technician (TTT) ETA is also one of 13 COLEMs for FCC Licenses: Photonics and Precision Optics: Photonics Technician Operator (PTO) Photonics Technician Specialist (PTS) Specialist in Precision Optics (SPO) Technician in Precision Optics (TPO)

Renewable Energy:

Electric Vehicle Technician (EVT) Photovoltaic Installer-Level 1 (PVI-LVL1) 'Small Wind Installer-Level 1 (SWI-LVL1)

'Smart

Home:, Certified Alarm Security Technician (CAST) Residential Electronics Systems Integrator (RES!) + Audio/Video (AV) + Computer Networking (CN) * Closed-Circuit TV (CCTV) + Security/Surveillance (SS) + Environmental Control (EC) Master Residential Electronics Systems Integrator (RESIma)

'Information Technology:

Computer Service Technician (CST) Electronic Security Networking Technician (ESNT) Network Computer Technician (NCT) Network Systems Technician (NST) Wireless Network Technician (WNT)

'Additional Certifications:

Avionics (AVN) Certified Service Manager (CSM) Commercial Audio Technician (CAT) Customer Service Specialist (CSS) Gaming and Vending Technician (GVT) Industrial Electronics (IND) Radio Frequency Identification Technical Specialist (RFID) ETA International certifications are accredited by:

ICAC

Marine Radio Operator Permit (MP) General Radiotelephone Operator License (PG) Global Maritime Distress and Safety System Operator (DO) H Global Maritime Distress and Safety System-Restricted (RG) | Global Maritime Distress and Safety System Maintainer (DM) International certification f Radiotelegraph Operator (T) { Sodas comes

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Addendum to This Book

we whe hay we or -_- ' _ was never intended to be an ex h thins i ' © 10C hnology. It was intended to ho wansthve terthook about plain English. By involving multiple authors, some of the mater:

Ca practical haok that i erin of the plain language can help those studying to become elec i, tal is repeated oceavianally %

concepts and speed their understanding. Because everything Ween technicians overcome i referenced, here are some additional truths that may help aati

ti mre: Practice exams hag nat heey"

1. A video signal (such as TV or satellite) contains the video information plus the synchroni. : s onization si Signals ne display the video plus other information content or coding and messaging. It does not contai

s ain audio infor mation

te~ Audio or an audio signal includes frequencies between 20 Hz and 15,000 Hz. It is co

as the actual limits of the audio frequency band. mmon to see it defined as 20 to 20,006 4

3. Troubleshooting: When faced with a product that is showing signs of a short, perhaps blow a fuse that you can insert a low ohms, high wattage resistor in series with the AC power which are fuses or circuit breakers,

it may, much danger of further damage or fuse blowing. A variac also can allow you to reduce the input, please see without a power in a similar manner,

4. When identifying power supplies, if a filter type capacitor is found to be electrolytic in the

leads and the rectifiers, it indicates the circuit is likely a voltage-doubler or the power transformer secondary,

5. DMM's (digital multimeters) are invariably more accurate than analog multimeters. after 7 discuss

6.

In the

Chapter 7 discussion of motors, know that motor speed controllers may use microprocessors including op-amp comparator circuits. Processors or several other methods

7. A fuse should be replaced with the exact size and type. If a substitute must be made the

original. The voltage rating can be less but refers to the arc-over potential after the fuse blows. (it should not be greater than the

than 250 V, a blown 250 V fuse could be replaced with a 125 V). + (e.g.: in a circuit with no more

8. A power source (battery or generator) for electronics power supplies might compare

in a municipal water system. compared with the water tower/tank in a city:

9. Transformers often use soft-metal iron core to increase lines of force concentration.

10. Because a small portion of the current in a transistor flows from emitter through the base, the most current flow in a transistor is in the emitter, not the base or collector.

11. 'Q' is used in Chapter 3 to denote 'charge' in a capacitor (in Coulombs). 'Q' is also used to denote the 'selectivity' of a tuned circuit.

12. Electrolytic capacitors: Applying a 'backward' (reverse) voltage across this type will destroy it unless it is a special non-polar capacitor.

13. (See Ch 5) The 4th band on a resistor indicates tolerance (how far from its indicated value it may vary), Or reliability (failure rate/1000 hrs.). Check the specs and measure to be sure.

A balun is an impedance matching transformer often used to connect coaxial cable to

le to 300 Ohm transmission line.

accompanying lines (as at the 14.

15. In performing an ohmmeter check of a transistor, checking the resistance between the emitter to base forward-to-reverse reading and may be much higher. The E-B and C-B legs should show at least 300 ohms.

-apter 1 by Dick Glass, CETsr

In The Beginning

The CET program was initiated in 1965 by a national association incorporated under the name of NEA®, National Electronic Associations. There were few credentialing organizations back in those days. Most certifications were aimed at personnel in the health care industries: RN, CNA, etc. There were licensing programs in some states (Indiana, Massachusetts, Louisiana, Florida) for TV technicians and the 'biggie' was the FCC Radiotelephone Operators License, administered by the U.S. Government. NEA and the USDL In the early 60's the Bureau of Apprenticeship and Training, U.S. Labor Dept., and NEA had initiated a retraining program which required on-the-job education, plus formal school classes. The course length was 8,000 hours, or four years. At the conclusion of the apprenticeship, if the student was successful in passing the final examination, he or she was given a certificate by the U.S. Government and NEA attesting to his/her schooling, abilities and knowledge. The title for the successful student now became "Journeyman Electronics Technician." It was a type of 'degree' - not an engineering degree - but more on the order of an Associate technical school degree. Jerry Barkoukis, CET As the program was implemented under the direction of NEA's contracted coordinator Jerry Barkoukis, CET, of Akron, Ohio, the association began to think about the many Radio-TV technicians already working all across the country. Many of these were well trained ex-military electronics technicians or those who had completed public school or commercial electronics training courses. DeVry, NRI, CIE, ITT, etc. had begun training technicians also. The question was: Should these well-trained technicians be able to prove their knowledge, abilities and training and receive a comparable 'degree' as those who completed the U.S. Labor Dept. 8000-hour program? The answer, of course, was "Yes." So, under the direction of Dick Glass, CETsr, who was the Executive Vice President of NEA at the time, the CET program was initiated. The CET Program Description Dick Glass is an author of previous ETA publications as well as more than a dozen books: *Deal Logy and Bersines!* management, published by Howard W. Sams Co, TAB Books, and McGraw-Hill. He has also written *Electronics Technology and Maintenance Publications*. Dick is a former Navy Avionics Technician, Industrial Electronics Commercial License examination modernization committee Chairman in 1995 and 2005, inductee into the ETA Hall of Fame in 2004, currently CEO Emeritus of ETA. He has a broad background in all phases of work performed by motor technicians also written hundreds of articles in trade journals and is a Computer Technician, Technical Writer. FCC Com A Program for Radio and TV Techs The first examination was aimed at Radio-TV technicians. Questions pertaining to antennas, transmission lines, basic components used in electronics, mathematics and formulas, test equipment, radio and TV and basic electronic circuits, electrical safety, audio and video, RF electromagnetic signal propagation, amplification, block diagrams and so forth were considered proper basic materials for the first certification exams. There had been some earlier attempts to establish certification for professional technicians in the Seattle area and in California. Once a national association (NEA) took on the task, these programs joined in the national effort. In the beginning there was only one examination. It included material related to radio and TV, simply because there were few other jobs in electronics. There were not many biomedical or industrial electronics products, satellites or fiber optics, and the avionics technician could still fall under the 'radio' portion of the certification. One exam for all. Certification is Good Certification proved to be a good program. It allowed proper recognition of those who had gained a level of education and had acquired sufficient on-the-job or experience credits. Technicians who were proud of their high tech job, and who recognized that the service they were performing was extremely important, could show their credentials to potential employers. TV-radio customers could affirm that the person working on their expensive new entertainment equipment was qualified, schools could determine just what needed to be taught to bring a student up to qualified worker status, and some states even used certification as the requirement for state licensing where laws were passed to regulate the repair and installation business. Not many years after the CET program had begun certifying professional electronics technicians, most of whom were TV repair people, it was found that everyone wasn't involved in TV repair. Some technicians were exclusively wireless communications or radio workers. Many of these were amateur radio buffs and some were auto radio specialists. What to do?

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Chapter 1

A Core Exam ~ the Associate

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SIV on other alhed fields, the association leaders serommnat that ate y Te should he requiredto pass An CX- svnaron MISH an the core elements of electronics of which S/T RNAMCAME Shou have knowledge. The idea was thata tevhown Working wath (for instance) computers (which by now had hvome Important electronics products) should not he rquired to pass a test pertaining to the operation of a color TV. The sxeiahized technician should know about solder- ing and basic electronics components and how to read block shagrams and schematics and to not get killed by electrocu- fon. He peed not know about the shut-down protection circuit mache TV even though this might be useful to him now or m the future. So,a core CET Exam was put in place and the

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ET"as expected to pass it, PLUS another writ- fen exam Petting to the field of electronics work in which he or she speialized.

Overthe

years,the Specialty Exams were developedby technicians. instructors, employers and others, with each spe- cialty CET program having its own committee chairman and committee members. These groups of subject matter experts averagedabout a dozen members each. Here are the Specialty areas as of 2016: Avionics (AVN)

Biomedical Electronics Technician

(BMD)

Biomedical Imaging Equipment Technician**(BIET)****Certified Alarm Security Technician****(CAST)****Certified Satellite Installer****(CSI)****Commercial Audio Technician****(CAT)****Computer Service Technician****(CST)****Distributed Antenna Systems****(DAS)****Electronic Security Networking Technician****(ESNT)****Fiber Optics Technician****(FOT)****Fiber Optics Technician - Outside Plant****(FOT-OSP)**

General Communications Technician - Level 2 (GCT2) Industrial Electronics (IND) Line and Antenna Sweep (LAS)

Network Computer Technician (NCT) Network Systems Technician (NST) Radar (RAD) Radio Frequency Identification

Technical Specasst (RFID) TRN Wireless Communications Technician (TRN) Telecommunications (TOG

Wireless Communications Wan**(WNT)****Wireless Network Technician****The Cer**

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Associate Status

So we now have a beginners or apprentice-level certifica- tion program, as Ww cll as a program to certify the knowledge of one who is the mainstay of the work force. In older times that was called a journcyman or skilled craftsman. Today the term journcyman is less favored and rarely used except in formal Jabor union advancement programs. Customer Service Specialist (CSS) It was later decided that a certification devoted to human relations, people skills, problem resolution and job values was needed in the electronics industry. In fact, it was needed in all industries. It became apparent, however, that

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personnel, delivery and sales people, supervisors and non- electronics workers should not be testes for ol my a specialty electronics knowledge (even though some of this night ephthem better do their jobs). So,ETA, withthe push of Don Howell, CETsr, of Lake Washington Technical

College

in Seattle, in 1990 designed an examination which could set a people skills level and examine workers

to see if they know the practical thinking processes which can raise the standard for co-worker and customer skills. It was called the Customer Service Specialist or CSS certification program. Because it does not require an experience prerequisite or technical skills, it is different than the Associate or the Jour- neyman levels of the CET program. It does, however, require the examinee to also sign and accept a code of conduct which commits the CSS to just plain, professional, decent behavior. Have it Both Ways Other certifications, technical in nature, also seemed to fit the stand-alone format. For instance, fiber optics installers would do well to have basic electronics knowledge. They would benefit from a thorough understanding of data com- munications. This isn't required in order to work with a crew of fiber installers. In the same manner, a keyboard technician (CST or A+) would be a more versatile worker if he or she knew basic electronics or cabling fundamentals, but can, in many jobs, serve well without those nice-but-not-required skills. So CST can also be a stand-alone certification by ETA. The successful CST cannot use the term CET after his name, as he isn't a Certified Electronics Technician. He can display CST after his name and can show his ETA certification prov- ing that he has the required knowledge to perform as a com- puter keyboard worker. Thus, ETA has allowed the computer service technician to be certified and to also be a CET if he feels that certification will allow him to advance, or if a new job requires a CET. So, now a worker can obtain an Associate, beginner or ap- Prentice level, certification. He can become a journeyman or professional certified electronics technician, or he can obtain specialty certifications that fill special needs. TheCETProgramDescription Chapter 1 Senior CET Today, ETA requires re-certification for most CETs and ape cialty certification holders. Prior to 1996 when re- certification was officially instituted, once you obtained an NFA, ISC ET or ETA certification, it was for life. With the high- tech world demanding even more qualified workers, ETA instituted a four-year re-certification program. Before 1996 the Senior CET was offered to journeyman technicians who could show at least eight years of schooling/experience in electronics. The Senior also must retake the CET exam and this time. pass with a higher score: 85% rather than the ordinary 75% The Senior program allowed conscientious technicians to Prove they were keeping up with advances in electronics. The motive was to not copy the policy of the Federal Communica- tions Commission which allows a technician to be licensed once, then never have to prove he or she ever learned the new technologies or gained any further skills. So, ETA solved another problem by instituting re-certification and thus requiring all CETs to keep up with technology. With re-certification the Senior certification was not dropped. In fact, it is still popular. The time requirement was reduced to six years, the passing score retained at 85%, and a beautiful wall plaque is offered. A majority of the electronics instructors who certify do attain the Senior status, as do many working technicians. So now, a technician has several choices: He or she can be a beginner and

attain the Associate CET status; within two years advance to the professional level of Journeyman (by far the most popular certification ETA issues), add a specialty certification such as the Customer Service Specialist, Fiber Optics Installer, etc., or he or she can attain the Senior CET which shows he/she has a very high level of knowledge, plus plenty of on-the-job experience. An employer can credit the applicant with the exact level of abilities for the job requirements. Master CET One almost revered level of certification is the Master CET. Over 350 technicians had attained this highest level of certification in ETA as of 2016. Why would you want it? The Master CET must be able to prove himself or herself in not just one electronics category, but he or she must master the Associate CET exam, plus pass at least six of the Specialty exams at the 75% level. Because it is difficult to be proficient in several categories (i.e. Avionics and Biomedical, for instance), Master is not for everybody. It does qualify the professional technician as being proficient in several areas,

Chapter 1. The CET Program Description. For instance, a technician may have served in the military for 8 years. While in the service he or she may have become heavily involved in audio and video. Upon detaching from the service he or she may have gone on to study computers, thus gaining the CST and CNCT certifications. While working on the job he or she also might have received an FCC Radar endorsement and decided to get an FCC Radio endorsement for his FCC license. At this point he or she might pass five of the specialty exams. He or she might need to choose an additional Specialty exam and he or she has enough credits passed to become qualified as a MASTER CET. This too is a unique honor and is recognized by many employers for the specially-trained and experienced electronics technician. It attests to, so now you know about all five levels of the CET program:

. Student

(high school students)

+ Associate

+ Journeyman

> Senior

+ Master

(all categories are listed in the front of this book) What are the costs to become certified by ETA?

They

are much less than most other similar type technical certifications. Here are the established fees* (current as of 2016):

Electronics Technician (CET). . . . \$75-\$150 ~Each additional Journeyman certification is \$30, if taken with the Associate at the same time

see ARINC Installer, Technician (AFI/AFT) -

Apprentice

Electronics Technician (APP). . . . Associate Electronics Technician (CETa). . . -Associate exam FREE if taken with a Journeyman exam at Biomedical Electronics Technician (BMD). Biomedical Imaging Equipment Technician (BIET). . . . \$75 Broadband Voice over Internet Protocol (B-VoIP). . . \$150 Certified Alarm Security Technician (CAST). Commercial Audio Technician (CAT) . . - Certified Satellite Installer (CSI). «eee

-Satellite Endorsement . :

Certified Service Manager (CSM). Computer Service Technician (CST). Customer Service Specialist (CSS) Data Cabling Installer (DCI). Distributed Antenna Systems (DAS)

Electronics Modules. . . 2 \$2!

Electronic Security Networking Technician (ESNT), Electric Vehicle Technician (EV T) Fiber Optics Installer (FOTN). . . .
 >> Fiber Optics Technician (FOT). - Fiber Optics Technician - Outside Plant (FOT-OSp, Fiber Optics Designer (FOD). . . .
 - ' Gaming and Vending Technician (GVT). .

General Communications Technician

~ Level 1

(GCT))

General Communications Technician

- Level 2 (GCT2) Industrial Electronics (IND) SER © BIRTEy Line and Antenna Sweep (LAS). - - Network Computer Technician (NCT). - + * Network Systems Technician (NST). - Master RESI (RESIma). Mobile Communications and E!

Passive Intermodulation Testing (PIM). - = °° > ' .

Photonics Technician Operator (PTO). - - °° °° 7* ape Photonics Technician Specialist (PTS). © - "7° 7 °>. Photovoltaic Installer - LVL] (PVI-LVL). - > > - - RF Interference Mitigation (RFIM). --- °° ° °>. Radar (RAD) 2% swe

siceSys « mma ee Fe * =

Mie Radio Frequency ID Technical Specialist (RFID). .. Residential Electronic Systems Integrator (RESI). .. -Basic RESI and one endorsement is \$ 75, if taken, same sitting -RESI Endorsement. eee SAE Fabricator (SFF).---- +007" BERS Small Wind Energy Installer - LVLi (SWI-LVL}). .. Specialist in Precision Optics (SPO). - - - ----+ Student Electronics Technician (SET). ---- ----+ -- Technician in Precision Optics (TPO). - - ----+ Telecommunications (TCM)... ----+ --- ste Termination and Testing Technician... -.-.-.-.- Wireless Communications (WCM). ..-- ----+: TRN Wireless Communications Technician (TRN) Wireless Network Technician (WNT)... -- ----+ Master Exam.--oeoeoeereersrrr sce Master Upgrade with plaque. .----- ++++++55550> Senior Exam with plaque. .----- +----+ +525

e> Senior Upgrade. -Senior upgrades must be done within the first certification *Fees and certification availability, and procedur change. Please check the ETA website (www.eta-i most up-to-date information.

The CET

Program

Description

Formal Schooling is Not Required One might think 4 well-thought-out program hike the ETA Certification would require mandatory classroom contact hours, That isn't the case. Since 1966, the association's atti- tude has been: If you know your stuff, can prove it by showing your work and or training experience, and you can pass the CET exams, you then qualify,

Certification Maintenance

ETA initiated a certification maintenance program for the majority of its certification programs in 1996. CETs can retest or prove that they have attended 40 total contact hours of training during the previous four years. In 2003, the associa- tion implemented an even easier continuous certification main- tenance process. This requires submitting upgrade education information, address changes, etc. each year, modeled upon the ISO 17024 standard.

The Certification Administrators

The success of the CET program in NEA, ISCET and ETA has been due to the individuals who promote and proctor the actual examinations. A typical CA is an electronics or computer instructor at vocational, technical, engineering public or commercial educational institutes. A list of the current test sites is included in the appendix of this publication and on our website (www.eta-i.org). The CAs maintain security of the examinations and, with another school official, plan and proctor the examinations. Sometimes the exams are given at the conclusion of a technical course and are used as the final exam. Some schools will not graduate a student who cannot pass the appropriate exam, CAs also are a major part of each of the certification examination development committees of ETA. Every year several CAs are elected to major ETA Board of Director and Officer positions. Additionally, the CAs also keep a close watch on the content

of each of the program examinations. The interaction of actual teachers between ETA and the school, with real knowledge of how well the student did during the course, helps both the school and ETA validate the certification. For instance, if a class of twelve students did very well in a tech school course, passing the textbook quizzes, the written and oral exams composed by the instructor and exhibiting a thorough understanding of the subject, but then failed the ETA exam, it would be a red flag to the instructor, the school and to ETA that something is not right, either with the training or the examination,

Free Retakes

Because everyone doesn't acquire ETA study guides, or may not be totally familiar with the content of the ETA exams, a free retake policy has always been in effect. One free retake is allowed after waiting at least thirty days, and before one year has elapsed. The association has always felt that the period between

a failing of

an exam and then retaking it and passing, 's a time of maximum learning as the examinee goes back to the books, or engages in discussion of exam topics with other technicians or his instructor. What's the Point? The point of the entire CET Program is to establish a level of technical ability and knowledge that assures the world that when it comes to electronics, cabling and computer installation, wireless communications, maintenance and service, that CETs are the best qualified workers. That the cost of gaining

an outstanding technical education pays off in higher compensation and personal satisfaction in knowing your work is an important part of the human experience on this planet. Each new technician who becomes certified raises the overall level of the entire profession. Each individual counts. CETs are proud of their profession and their part in it. They respect other CETs and the educators who help make it all happen. CETs belong to a great fraternity!

NOTES

Chapter 1

The CET

Program

Description

Chapter Quiz:

1. When did the CET certification program begin?

- a. 1965
- b. 1976
- c. 1986

d. 1996

2. When did ETA initiate Certification Maintenance?

- a. 1966
- b. 1976
- c. 1986
- d. 1996

3. All ETA Certification categories require the passage of the Associate CET Examination.

- a. True
- b. False

4. To qualify for MASTER CET, the examinee must attain a score of 85% on each separate exam.

- a. True
- b. False

5. An Associate CET cannot attempt one of the specialty examinations.

- a. True
- b. False

6. Master CET status is lifetime and does not require certification maintenance.

- a. True
- b. False

7. The Journeyman level:

- a. denotes competency in one specialized area.
- b. requires 75% passing.
- c. requires 2 years of work or school.
- d. all of the above,

8. Certification maintenance requires payment of a fee and retesting or proving at least 40 hours of upgrade retraining within a four-year period.

- a. True
- b. False

9. A worker may become certified by ETA as a

Cy

(Computer Service

Technician), without also being certified as a CET

(Cert

- a. True
- b. False

- 10. The Associate CET exa Journeyman exam,
 - a. True
 - b. False

ANSWERS on page 223

ified Electronics

Technician

im may be taken after the

ooo ter 2 by Dick Glass, CETsr Hib Chie an mithor of previous A publications ae well ae more than adh Hommapeniienh grstabioueddfey Blea mnddWh events Hatton poilalte wtioone Hie btm | eit fer tal Eteonee exannination ie CPO Pit AEE TA He tie a hind tae

Shock

Techotcinna work with cleetiely, Therefore, they are in the vteinity of eleetie Chater during much of their workday Thur, theyfe potentially in a mote hazindons position than the avernpe worker On the other hand, beedtime eleetronion techies dadorstind the nature and the danger of electric shook. nome think they dre in lowe danger han thane who May Hot always Hotice a hirkng dangerous situation, Either way, HEpvaye to Know ae much ax possible about the dangers of working with eleetronion, Thheinelides the danger to your poron, Ay Well as other danger to component or other in the workplace, or the workplace tsell! Vloctricnt shock ie the number one concern we have when we apoak of cloctiical safety, Hveryone haw experienced cloctrical ahovk, even IP ithe only the attic dinecharge we get in dry weather when we foueh # door knob after walking across a carpet, Chonge of Character Dozens of years ago the U.S, Navy ndvined personnel who worked in and around electricity that even though contuet with electricity might not be lethal, that any amount of clectile current panning through the body could caune a change in character, For some of unthin might be good! On the other hand, since the profound change might make iN Unknown and unpredigtable, we should be concerned, No one knows whether a slight change in DNA, ora shifting of brain cella taken place, We do know thateven small doses of electri current tond to hurt, So, while you can't completely climinate electric shock, you can respect elvetricily and keep shocks to a minimum, A lethal doe of elevtrivity ix a good place to start in this

discussion, Some xay that the 115 VAC trom the wall socket

isn't lethal, Most ofthe time it ixn't, But ifyou make a really good contact with it, say while you are standing ina pool of Water in your bare feet, you may be killed, CO, TAN Hobe, and MiGraw Hill He ta Safety Precautions for Electronics wen be Healing with elec tromiea technology ane he Hees written hen ada af artic hes im trade yenurrate ane fee Navy Astonia Lech ian, Industrial Hlectronice Computer fechnician, fechmeal Weiter HOC Com HHIZAHON COnUnttEe Chairman in 1998 and 2004 Inductee intes the Hlectromics Hall aff vene and currently byreund in all phases of work performed by electronics tev hntetune i8+ or supply voltages for the tubes of transistors in radios or other products used to be much higher, from \$o volts to 500 Volts, Today, transistor products may utilize low voltages hike 5V and +12 V, which are very unlikely to be lethal, AC pow-

ered products, however, still have 116 VAC coming in on the

line cord, so you should be concerned even with these much friendlier electronics products Reaction to Shock While many of the products technicians work on are not danger Thanks to solid) oun, SOMetiMeES a shock stat circuitry, most q from a lower voltage is

x id home: such a surprise that the reaction to the shock causes some damage. If you react to the shock by dropping your screw- driver into the circuitry, you may destroy the product. If you jerk back from a small shock and fall off your chair or ran into someone, you may feel pain, even if it is not caused by electrons flowing through your conductive body, So the reactions to the trauma that even small electric currents can cause are dangerous to life and limb, electronics products operate from much safer! voltages

Where Are The Volts?

Just where are dangerous voltages if we are mostly dealing with transistor circuitry, chips and such? Keep in mind that there will still be legacy equipment to service such as TV and computer monitor kinescope displays. The high voltages which operate these high-vacuum picture tubes is as much as 26,000 volts, While the pigtail which connects this voltage to the button on the side of the picture tube is insulated With an insulating suction cup protector and the pigtail itself has an insulation value of well over 30 KV, it is possible to work your way into the circuitry, Today's plasma screen typically use 300-400 volts and LCDs use 1,500 volts, Which you need to be aware of and cautious around,

Chapter 2 .

To check the high voltage levels you use a high AT ast probe which contains a small voltmeter and a large resistor thus making it a safe way to check the presence and value of the second anode supply to the picture tube. It can become a shock hazard if you accidentally remove the ground wire and clip from the chassis, but even then, because of the internal resistance in the probe, the current is less than that which would burn your skin or produce trauma.

High Voltage Probe

The Electric Chair

How much voltage does it take to kill you? You might consider that the famous electric chairs in prisons use only 2,000 volts. Compare that with the 30 KV on the picture tube. As little as 10 milliamps of current can be felt, with 100 mA being lethal. Wet skin can lower your normal skin resistance of about 100k ohms so that small currents can be felt and may be dangerous. Other common items which can zap you are stun guns and even bug zappers. But far more people are injured by power company service than these. Working with cable, you may find you have occasion to work on power poles. You might find in your career that you deal with antennas, towers and satellites, on roofs or in locations where you are in the vicinity of power lines. The insulated power service cables which deliver 230 volts to your electric meter base are not much of a hazard. The insulation on them protects against accidental touching by hand or with antenna elements, poles, etc.. Still, they are to be respected and not touched.

Lineman

It is the bare primary power wires which most often kill. While they are usually high enough that a person really has to try to touch them, there are accidents which happen. You could be enjoying cutting a field with your tractor when the tractor tire catches the guy wire for the pole, breaking the pole and bringing the bare wires down on you. Or you could be working on an antenna tower or tall pole which breaks and falls across the power primary wires. During construction of One day, on a street in Greencastle, spread a small bridge, one of the tri-axle dump trucks was spreading gravel, with the dump bed in the up position. It caught some drooping cable TV hard line which angled across the street near where the work was taking place. The cable wires stretched, then broke, then snapped back and coiled around the wheel. While no one got killed, many of the men had 4500 volts or more attached to them. No one was hurt but thousands of dollars in damage was done. Anyone holding onto the wrong line in the have been electrocuted as could workers, at least in the way of the snapped off aluminum cables might be.

Hot Chassis

Today's products often have only a small amount of current, their mounting chassis. Thus, even if the chassis is connected to the AC line, and the house wiring, the AC outlet is reversed, there is no observed hazard,

but sets and other electronic products are now that any exposed metal even the signal input connector case, is isolated from the chassis. The technician will need to expose the chassis in order to service the unit, in which case a "hot chassis" may become hazardous. Often, "Hot chassis" is a name given to equipment which is not isolated from the AC power source by a transformer. Instead: to reduce costs, the AC power hot side of the AC line is connected directly to the power supply diodes. The low side of the line is connected to the chassis. Under normal circumstances: there

is

no hazard in servicing the unit. For instance, you could be holding the chassis with one hand and be touching the workbench ground with the other and no harm occurs. What happens to make the "hot

“hot chassis” is the AC power outlet is wired incorrectly and the receptacle has the black hot side of the outlet back to ground and the white wire to the

One way to solve this problem at the shop is to always try to use an isolation transformer to operate your bench equipment. Another cause of “hot chassis” type shocks is that lightning may travel down the power line into the house and are across the ‘hash filter’ capacitors which are connected between each side of the AC power input to the product and the chassis. With these capacitors shorted, even a transformer type product becomes a possible shock hazard. Most often you will notice the problem before a shock occurs. You will notice when you connect the cable TV coaxial cable to a DVD player or TV, that a small arc occurs. Static? Possibly, but probably not. When this occurs, you may even notice a tingle of a shock. It is a red flag, and should be checked out. The causes for the wrong voltage on the AC line are many. It isn’t always easy even to check to see that they are wrong. You can usually use the AC power receptacle 3rd wire as a ground for your AC meter. You might use the cable TV or antenna coax shield as a ground. But you need to reference your voltage readings to ground. I once tracked such a problem back to a reverse wiring of an AC power outlet, right by the sink in the bathroom! Black AC power wire to bronze colored screw (hot) on smaller prong. Check for proper wiring at the AC outlet. Most hot chassis problems go unnoticed as the product is in a house, isolated from good grounds. But when the TV or DVD player, etc. is now moved to the damp basement, it can become a lethal problem. Technicians should always seek the cause of a wiring problem, even if this means warning the homeowner of the problem and suggesting an electrician be called to inspect the wiring. More often the problem is another piece of equipment connected on the same circuit. One that comes to mind is the case of the antenna amplifier that burned out frequently. The customer, of course, thought we were just installing weak or defective products. But after a while we decided to check the trailer home wiring, especially since our techs seemed to get a ‘tingle’ sometimes when dealing with the pre-amp or antenna coax cable. It turned out the skin of the trailer was ‘hot’ at times. These ‘times’ happened to be when the home heating system was in a particular mode. Which furnace wire was backwards, we didn’t find out, but we did find out it was the

furnace that was the cause of the hazard. It wasn’t the homeowner who discovered the problem [the most often the technician who recognizes something is wrong and who should advise the owner that there is a problem that could show up

in other ways. Hot side of AC plug is the small

Prong

Product makers specify that technicians should check the power input cord for resistance every time the unit is serviced. Any leakage current above 500 microamps is too much. Search out the problem.

Capacitors

Large capacitors (electrically) are a lot like lead-acid storage batteries. If you short the terminals of a battery, you can do a nice welding job on your screwdriver. If you short the terminals of a large capacitor you can do a small welding job, or a large human shock job. The battery, at least, does have an internal resistance which can limit the short-circuit current somewhat. Capacitors have very little internal resistance when shorted and can supply all of their stored energy between your right hand and your left hand, right across your heart! Capacitors can be made to have virtually no leakage and thus can even be used in place of batteries in some types of equipment. Some are expected to retain a charge for ‘years’! With that in mind, the technician must respect power supply and other filter capacitors. The voltage retained in a capacitor has been called residual voltage. Caps hold charge. You assume there is no hazard, having unplugged the unit before attempting to service it, yet now, you get a severe shock after accidentally touching the capacitor terminals. It becomes second nature to technicians to discharge capacitors before using an ohm meter in the circuits or working with circuit components, so

Arcing

Touching a wire or part which has a hazardous voltage potential is one thing to avoid. But higher voltages can jump and shock you the same as if you physically touched the metal part. Most common are the high voltages in TV and monitor displays. But RADAR and many other products operate with more than 1,000 volts, inadvertently passing your hand or a meter lead by a high voltage potential can draw an arc. How far can voltage jump? Only three kilovolts can jump 1/4th inch. Could the

30,000 voltson picture tube jump over two inches? Think about it.

Capacitor

a color

Water

We mentioned the isolation transformer above as being a safety device. It allows your work bench or equipment to be ed while putting a barrier between your ground and a 'hot' groundsome other piece of equipment may havein your shop or home. If someone defeats a polarized extension cord, grinds off the ground prong of an adapter, or the basement wiring is backwards, the isolation transformer can partially remove the hazard. most never the solution. The solution is to find But peopleare more likely to defeat polarized and extension cords in garages and basements than

in aes rts of the home. Basements are notorious for being neler ing pools of water on the floor. AC power hazards dap Oe : to happen in these areas. Technicians can run waiting :

oe

oe further potentials when they work in wet crawl spaces padthemselves doing wiring in monsoon type conditions

or outside a building. That is a! the problem. cords should be replaced. Don't trust the insulation. Fi: try to keep AC potentialscovered. If they must be left uncoveredtemporarily,leave a danger tag attached.

Modern Equipment

Computers are much safer to service than TV's. ie ae ate the memory and program chips and transistors rom 12 volt DC sources. The only danger, other than the

me is the power supply section where only the AC

ee fuses, power transformer, filter capacitors and bri a vii en fier are waiting to zap you. These are usually covere insulating box or metal cage. Most electronic equipment is protected from with one or more fuses. Some equipment, such as

be i vi has circuit breakers rather than fuses. TV sets ivers, fuses. some

c have interlocks which disconnect the some other equipment

Safety Precautions

power if the cover or back is removed, Defeat is usually necessary in order to tr

he feat them with care and do not le; oubleshoot the 'ave them eXpos,

One Hand

Anule followed for nearly a hun- dred years is for the technician to work with one hand where possible. Keep the other hand in your pocket when taking a voltage reading on powered equipment. Don't hold on to the chassis rail and make voltage even hold onto a grounded workbench or rest yout cabinet of the test equipment. Second rule is use the buddy system, Don't work tronics products without having someone else neart you do get shocked.

Shock Victims

When a person gets shocked badly, they pass out or Some shock victims drop onto the hot voltage source. is to remove the power and attempt to save that persor There is a debate as to whether you should apply resuscitation procedures after removing the electrical hazard. If. not trained in CPR, don't start learning now. If you are t now is the time to do it. Mouth-to-mouth resuscitation is best method of getting a person breathing again. Ox-~"-v 4 oie £2; a? 4

Safety Symbols

Mouth-to-mouth resuscitation Of course, attempting to get medical emergency help first is common sense. M-M resuscitation requires about 12-18 breaths per minute after you have checked to see that the throat is clear and have held the nostrils closed. In addition to breathing problems, a shock victim may have severe burns, Treat these burns the same as other types of skin burns. Fortunately, most lesser electrical shocks cause the skin to bum and the charred bum contact point becomes an insula- tor of sorts, thus limiting the electron flow through your body.

Safety Precautions

W

Grards

The first rule about fire hazards is to always attempt to Prevent them. Why Use @ grinder to sharpen your screwdriver, letting the sparks shower down onto your open gas can? Many chemicals can catch fire with a spark or from your hot soldering iron, Keep your work area neat, don't smoke, and be on the constant lookout for potential fire hazards. Keep fire extinguishers handy and checked for operation and let everyone know how to make them work. Powdered chemicals such as CO₂ are used on electrical fires, rather than liquids. But whatever type of extinguisher you use, realize that the chemicals may put out the fire, but you should clear the area and ventilate it afterwards to protect your lungs. When soldering, use a proper soldering stand. Don't breathe solder fumes - use ventilation. Beware of paper and cloth as well as chemicals which might start a fire when accidentally touched by the iron. Don't leave soldering equipment on after work is over. Product makers have recognized that some parts, especially in TV receivers, are critical so far as their composition and current carrying capabilities. They can cause fire or other problems if operated wrongly or if replaced with non-exact parts. These are usually shaded on the schematic diagram. The rule is to always replace these parts with exact replacements.

Burns

Of course, you can get a burn by touching the muffler on a lawn mower, or on your truck when you are trying to get something that may have rolled under the vehicle. But technicians also get burned from soldering irons. They may find themselves working with heat shrink which covers cabling hardware and splices. Using a propane torch is a potential fire and burn hazard. The soldering iron may have a 600 degree temperature.

Burns

of the first degree, those minor burns which redden the skin and hurt, but don't bubble up with blisters, usually require no treatment. Running cold water on the bum for a few minutes will relieve the pain. Then dry them off and protect them but don't apply a bandage. Second-degree burns should not have water put on them or ice. Also don't break the blisters, if possible. Third-degree burns are cases for the doctor. No water or ice should be applied. Only a thick clean bandage should be put on the area prior to getting to a doctor or hospital.

Chemicals

We have already discussed the fire safety rules when working with chemicals. There are other hazards

Breathing

carbondetrachloride fumes is something that progressively damages your lungs. Before about 1980 it was commonly used as a solvent on technicians' workbenches.

Also

the

Bane

at alum capacitors. Don't play with

small oxide transistor cases are hazardous

to your lungs. Use ventilation and don't expose yourself to the chemicals or fumes.

Working

with shop tools, technicians may use the bench grinder or power hand tools. Safety glasses are required. If you have never tried to polish or grind an odd shaped piece

of metal and had it fly across the room like a bullet, you may not recognize the power of these tools used carelessly, or accidentally. Don't take a shortcut by doing a small job without glasses. An eye is hard to replace. Your skin, your eyes and lungs, and those of your co-workers are dependent on your work safety. Obey the rules your company has set up and practice safety as an example to others around you.

Body Injuries

Glasses for eye protection; ear plugs for hearing protection; taking special care when cutting or working around glass, wood and abrasives; taking extreme care when handling kinescope tubes - these are common sense practices that will help you and your company avoid unnecessary and lost-time accidents. Remember that people don't plan most accidents. They are just that: accidents. Recognize the potential for them. X-Rays and EMR Electromagnetic radiation is a subject about which very little is known. The genetic damage caused by small or large doses of radiation is not fully understood. What is known is that a microwave oven does work. X-rays do cause burns and possibly initiate cancer and other ailments. No one knows if cell phones, operated next to your brain, cause brain damage, but there are a lot of weird people walking around these days! Even low-powered satellite Internet transceivers are a concern for workers or users. Subjecting yourself to a direct beam of radiation is a potential hazard. Like smoking, everyone doesn't die of radiation sickness. But the overwhelming evidence is that it increases your susceptibility to life threatening sicknesses. That is why your x-ray technician always steps out of the room you are in and puts on a leaded apron before zapping you!

Chapter 2

Even the old TV sets (tube types) were very hazardous if technicians tried to operate them beyond their designed limits (or if parts started to fail causing abnormal current flow while using up to 30 KV potentials). Technicians must know the proper operation of the equipment they are working on and recognize that re-engineering the product can produce radiation or fire hazards.

NECS

The National Electrical Code is a bible for electricians and everyone else installing, designing, modifying and working with electricity. The NEC rules have been designed to prevent electricity from harming humans and starting fires. The rules are modified every three years to bring about the latest changes needed to protect the public and workers. All of the NEC rules need not be studied or memorized by technicians. Those rules that apply to your area of work should

he studied and abided by: For instance, if you install fiber Optics, you should know the TIA and EIA communications wiring standards, but you should also know the electrical code for wiring buildings. You might say: "But I only want to run a single fiber cable through the attic and down a wall, so why should I be concerned with NEC rules?" As an example, if you installed a cable which when burned releases toxic chemicals, that would be the reason to know about plenum cable NEC rules. There are dozens of NEC rules related to cabling and they must be followed. Technicians are in the installation, maintenance and repair business and thus in a position to either violate the rules, or conform to them.

OSHA -

Occupational Safety and Health Agency of the federal government also has established rules for workplace safety. It is each technician's duty to follow these rules. For instance, technicians often climb poles, towers and buildings, OSHA requires that workers who might fall as much as six feet must have body harnesses which protect them should they fall. Violating the rules may save a moment's time, but the penalties for ignoring them can be severe in two ways, #1: You have a better chance of hurting yourself or a co-worker if you ignore them. #2: OSHA penalties are severe in monetary costs to your company. NEMA, the National Electrical Mfrs. Assn., also publishes rules for installation and safe operation of electrical equipment. Depending on your job, you may need to become familiar with NEMA rules also, though that word may not be in the dictionary, it has come to mean signs that are posted to communicate to people, One need to communicate is when there is a human hazard potential and want to warn people of it to prevent an accident,

Safety

Precautions

'Open Manhole' might be a good sign to keep people from falling in,

just 'Hard Hat Area' reminds people that the area may have falling hammers or debris that could kill them, 'Radiation Hazard' or 'Ear Protection Required' might be other valuable signs to post, temporarily or permanently. Electronics technicians need to post signs if they have exposed electric hazards to the public or co-workers. If you are working on an industrial motor and must have the electricity off while doing this, you will need a sign on the on-off switch or breaker to warn others not to turn the power on while you are working. Someone might do this thinking the power was turned off by mistake. One of the most common warning devices is the street cone used to advise traffic that work is going on near or on the road and thus the drivers need to be cautious and on the lookout for the workers. In addition to signs, there is a system of lockouts. These are tags or actual padlocks and other devices which are placed on switches and equipment covers to prevent inadvertent energizing of a circuit while work is going on. ESD - Static and Component Damage ESD is known as Electro Static Discharge and does damage to small sensitive components. The technician is the person with the best opportunity to destroy circuit components via this static route or in other ways as he goes about his troubleshooting and repair tasks. Today's electronics technicians are aware of static discharge problems with CMOS technology. The problem is the thin substrate which makes these transistors so sensitive and valuable, but a very small static charge can blow a hole in the static sensitive substrate, destroying the sensitive component. The answer: discharge your body substrate before and while working on or near computer type circuitry. While working, wear a wrist strap that is grounded through a mat which drains off any static charge,

Safety Precautions See Deriving

- Unsoldering

You can do a lot of damage attempting to remove a transistor or IC chip. It takes a steady hand and knowledge of good soldering practices to do the job right. Even then, repeated solder-unsolder work will steadily deteriorate the PC boards. Troubleshooting with the idea that you must narrow the possible bad part down to only one has to be your creed.

Simply

replacing parts until you find the bad one is not the way to do it.

Desoldering station

When you do remove a bad part, it is sometimes best to clip the legs and remove the part before attempting to unsolder the remaining legs and then solder in a new part. Too much heat can harm adjacent transistors or other components. Use heat shields (heat sinks) on the legs of pn junctions and use the least heat possible from the soldering iron. While troubleshooting today's electronic circuits, it is increasingly difficult to measure voltages and signals without slipping off the lead you have aimed for and shorting between two of the legs. This often immediately destroys the transistor. Thus, taking your time and using thin test probes and taking every precaution is required. Break the glass picture tube base. Legacy equipment such as kinescopes (picture tubes and vacuum display devices) can be ruined also. The base and pins are smaller and lighter than in the early years of TV. They can be broken off. The high voltage anode connection in some tubes is coaxial and also contains a connection for the focus element. A special tool is required to remove the HV pigtail. Without the special tool, you can damage the connection, destroying the expensive tube. Being an unwieldy, heavy and bulky part anyway, removing, storing, shipping and handling are hazardous undertakings. If the tube is removed and headed for the trash, most technicians remove the implosion hazard by breaking the neck of the kinescope. They protect those nearby by putting the tube in a corrugated box to accomplish this procedure.

ES

Tools & Test Equipment

A dull screwdriver that falls off the table most of the times you try to use it wastes time and does bad work. A DMM (digital Multimeter) that needs new batteries and has an intermittent test lead may be worse than no meter at all. The best investment a shop-owner can make is taking the time to make your work bench as efficient as possible and to assure that you have all of the needed test equipment and tools. Digital multimeter. In checking hard-to-reach components, perhaps near dangerous exposed voltage sources, any meter will work, but a meter that includes a hold feature can allow you to concentrate on getting the leads on the right spot and not between two transistor legs or between the fuse and ground. An isolation transformer and a variable AC power source are both safer and very useful products to use daily on your bench. I have never seen a shop with too much test equipment. Some of the equipment will not accept unlimited voltages on the signal inputs. A good technician becomes familiar with the limits and specifications of the equipment he uses and thus avoids 'frying' his oscilloscope or signal generating equipment. Weller air filter

Clothing

What has your attire to do with safety? It depends on the type of electronics work you do. In all cases, however, a technician is at risk if he has dangling jewelry, trailing clothing, exposed skin and any of the varieties of skin piercings. Some jobs require steel toe boots. In some you need climbing boots, even gaffs if you climb power poles. Many tech jobs require hard hats some or all of the time. Goggles are necessary in every shop on occasion. Handling poles and working with heavy construction equipment requires full time gloves as do many cabling and communications jobs. Some jobs, for instance, working in auto-sound installation and repair, require the removal of all finger rings. If you have never gotten a ring between twelve volts and ground, you can't imagine the excitement it produces! Long locks may be in style, but in work where moving equipment, fans, shock hazards and plenty of wires are involved, short hair wins every time. Loose sleeves look nice.

Safety Precautions

get caught on everything and limit your movements. ER sec the shuttle astronauts with loose clothing. It is the same for cool, about to fall off, holey, sloppy trousers. Don't wear them to work in a technician job. When you work with chemicals, use a breathing mask or get good ventilation. If you work with heavy objects, wear foot protection. They also the a i items described in this chapter, then f out all about the special requirements of your special ieece You NEV

NOTES

First Aid

As stated in this chapter, there is concern when you attempt to help a person who is an accident victim. I wouldn't advise you to try to adminis- ter CPR if you aren't trained and experienced at it. You can do more damage if you do it wrong. That's probably why mouth-to- ite resuscitation method, but mouth is today's favor- CPR may be the right thing to do. If you aren't trained at it, it is better to seek professional help even if you might dearly love to help the person. First Aid kits are required in all work trucks and in all work ee places. Most aren't elaborate. They have some band aids and . alcohol or hydrogen peroxide; perhaps some small bandages eed

a and tweezers and scissors. They probably have some aspirins. Know where the kit is and help keep it stocked to the level required at your workplace. If you work outdoors, on or under buildings or in trenches or in the suburban areas you should carry hornet spray. You may find dog repellants are necessary if you do cable hookups or home repairs and are frequently in unfamiliar neighborhoods. I haven't mentioned truck safety before, but it is one place that has more than its share of accidents. Of course, prudent driving habits are required and technicians have a pretty good record as a group. Problems arise, however, when the truck is a LO S:*~CS~S~S not regularly serviced and racks aren't made secure. Trucks carrying poles and other equipment that can become flying a objects in the event of a crash must have fences behind the driver and passenger seats. Ladder racks are often added as an afterthought, which is better than trying to carry ladders around loose, but racks can be hard to either remove or replace the ladders on. The ladders also can be secured with bungee straps which can break and leave your ladder flying down the cc road at someone. Trying to get by on worn out tires and never giving the vehicles a close inspection for bad tie rods, loose wheel nuts, broken shocks, dangerous metal fender or door Se parts hanging out waiting to tear your flesh are all common sense things that need to be concerns. Always carry a fire cee remreereceeneee a a ee extinguisher in the van. Don't use the two-way or cell phone while negotiating city streets or crowded roadways. Pull off. Each individual job a technician performs will have addi- tional safety rules and needs. Your job is to understand all of

Safety Precautions Chapter 2

Chapter Quiz

1. Ba or supply Voltages in electronics products, are 8. If you have never received formal training and considered to be: practiced CPR ong person or dummy. in the event of finding a fellow worker not hreathing. you _

a. +12 V DC.

bd. 0 to 30 VDC. a. should use what knowledge you do have of the

©. +12 to 300V pc. Procedure and attempt it

d. the anode or plate voltage supply. b. should not attempt it, but call for help ¢. should attempt it after calling for help

2. Leakage current in an electronic product, between d. should, if handy, read the instructions and attempt it in either of the AC power cord conductors and the chassis order to save the life or any exposed metal parts should not exceed A

9. When working at heights above 6 feet, a is
 - a. 5 microamps required by OSHA rules.
 - b. 500 microamps ec. 5 milliamps a. lineman's belt
4. 50 milliamps b. lineman's belt attached to a lanyard
 - c. full body harness attached to a lanyard which prevents
3. Shaded components on electronics schematics mean: more than a 6° drop
 - d. good insurance policy
 - a. only factory service personnel may change the part.
 - b. only exact replacement parts may be used. 10. Residual voltage is associated with_____.
 - c. the shading makes for better readability.
 - d. high voltage is present on these parts. a. Northern Lights
 - b. corona arcing from electrical generators
4. Ingested solder fumes or particles 5 c. charge retained by inductors
 - d. charge retained by capacitors
 - a. are not harmful to the human body
 - b. are toxic
 - c. from lead-tin solder aren't harmful ANSWERS on page 223
 - d. from tin-alloys are most harmful
5. Second degree skin burns should :
 - a. not be treated with ice and bandage pack
 - b. be covered with thick bandages
 - c. be treated with cold water or ice
 - d. be treated by draining blisters
6. Rules describing the fire or toxicity retardant types of cabling which may be installed in buildings is contained in NEC Article 770 UL 568B TIA/EIA standard 578 NEMA code 1316 peop ba! ESD and static discharge is most destructive to, . CMOS semiconductors . TTL semiconductors Germanium PN junctions Silicon PN junctions

Rogp

apter 3 Math for Elect; by Fitz Herbert Reid, CET Fitz Herbert Reid, CET, is a former Director of Technical/Vocational Education in his homeland of Trinidad and

Tobago. Heh

42 years in education at Primary and Tertiary and Administrative Levels. Fitz is the holder of BSc. (Electronics Technolo: , ns special traming in Testing and Evaluation. He has written three booklets for the Caribbean Examinations Council, where 2 if ° a Examiner These texts are Supplemental Study Guides for the Associate CET examination, Powers of Ten Refer to Table 1: The powers of TEN are used extensively by engineers and technicians. Powers of Ten allow very large and very small numbers to be shown in a compact form. The name of the commonly used prefix is also listed in the table.

$$1,000,000,000 = 10^9 \text{ Giga}$$

$$1,000,000 = 10^6 \text{ Mega}$$

M

U

L

T

I

P

L

Y

$$1000 = 10^3 \text{ kilo}$$

Table 1 - Units & Powers of

$$1 = 10^0 \text{ Base Unit}$$

Example: 1000 pF

$$1000 = 10^3 \text{ milli}$$

Example: Convert pF to 1

a. 1 pF

$$.000001 = 10^{-6} \text{ micro}$$

b. 0.001 pF

c. 0.01 pF

d. 0.1 pF

$$.000000001 = 10^{-9} \text{ nano}$$

SOLUTION:

$$.000000000001 = 10^{-11}$$

i

For We are moving "down" so we divide by 10^6 . The number of zeros determines the index or power to decimal point to the left (6 places). which 10 is raised. By shifting the decimal point to the right and left the following equations are true:

$$1000 \text{ pF} = 0.001000 \text{ pF}$$

0.001 pF

125,000 = 1.25 × 10⁵

125 = 1.25 × 10²

Example: Convert 700 KHz to Hertz (Hz)

a. 700

07 = 7 × 10⁻² b. 70

c. 0.7

007 = 7 × 10⁻³

d. 700,000

SOLUTION:

Move the decimal point to the right for large numbers and to the left for small numbers. : The arrows on the Table tell us whether we should (shift the decimal point to the right) or divide (shift the decimal point to the left). How do we relate the Power of Ten to the units used in electronics?

ca Converting from kiloHertz to Hertz, we must multiply by 1000 or shift the decimal point three places to the right

Therefore: 700 KHz

700,000 Hz Choice (d)

Example: Convert 700 KHz to MegaHertz (MHz)

.70 . 0.7

ae

ao

SOLUTION:

The arrows on the table tell us whether we should multiply (shift the decimal point to the right) or divide (shift the decimal point to the left). Converting from kilohertz to MegaHertz, we must divide by 1000 or shift the decimal point three places to the left.

Therefore: 700 KHz = .7 MHz

Choice (c) Example: Convert 2000 pF to pF

a. 20 pF

b. 0.002 pF

c. 0.02 pF

d. 0.2 pF

SOLUTION:

We are moving down so we divide by 10^6 or shift the decimal point to the left (6 places).

$$2000\text{pF} =$$

$$0.002000\text{ pF}$$

$$= 0.002\text{ }\mu\text{F}$$

Choice (b)

Equations

An Equation is a mathematical statement. It tells us that two quantities or numbers are equal. By using letters to represent numerical quantities, any number can be used to form an equality,

$$\text{Let } a=6, \text{ } b=3 \text{ and } c=2$$

$$\text{If } a=b \times c \quad 6=3 \times 2$$

On rearranging the equation to maintain equality:

$$b=a/c, \text{ then } 3=6/2 \text{ which is also true}$$

Math

for Electronics

Chapter 3

One -acbiten 2 -# 4 Which also forme an equality In the equation ty ©2." y" 4 termed an unknown quantity because we do not know the value of °y However. we can colve for "y" this

$$4y=2; y=24: y-05$$

Ohm's Law

In an alternating current (AC) or a direct current (DC) circuit the relationship between Voltage, resistance and current is

called Ohm's Law.

This relationship is expressed mathematically as

$$V=IR \text{ - where } V \text{ is the applied voltage}$$

I is the current R is the resistance

$$I=V/R \text{ and } R=V/I$$

With any two

of the quantities of a circuit known, the third unknown quantity can be found. Fig. 1 (It's common for E (electromotive force) to be used instead of V in electronics math) A triangle (Fig. 1) or circle is sometimes used to show the relationships between V, I and R. V is always on top or to the

right in the Ohm's Law equations. Covering up the unknown

with your finger in the triangle will help you remember the arrangement of the form of the equation to use. To find I, cover up I and you see V over R To find R, cover up R and you see V over I To find V, cover up V and you see I times R

Watt's Law

Power is measured in units of Watts. The Power

consumed in the circuit is calculated this way: $P=VI$

Chapter 3

$$\text{Power (P) = Current (I) x Voltage (V)}$$

$P=IV$

rearrange to find $V =$

P/I and $T = P/V$

Fig. 2

As with Ohm's Law you can also use a triangle to help

you find the unknown: To find I , cover up I and you see P over V . To find V , cover up V and you see P over I . To find P , cover up P and you see I times V .

With substitution from quantities in Ohm's law, the others

can be found: F

$P = IR$ or $P = V/R$

These are handy to find power when current and resistance are known or with voltage and resistance known:

$R = \sqrt{P/I}$ and $V = \sqrt{P \times R}$

With these combinations, 12 formulas are available for

problem solving with Ohm's and Watt's Laws combined.

A graphic illustration in Figure 3 is provided in the shape of a wheel to help remember and use each. See

Fig. 3 Formula combinations (12) of Ohm's & Watt's laws

Math for Electronics

Internal Resistance

All electrical devices which produce electric

Energy

affected by their own internal resistance. These devices include generators, batteries, etc. Why a battery is two or more cells connected in series, also, a cell is sometimes referred to as a battery. Let us take, for example, the inside of a battery.

Why Fig. 4 Battery connected to a load The total voltage developed by the chemical action is called the electromotive force (EMF). This EMF however, is not available at the terminals because a part of this voltage "sits" across the internal resistance whenever a current flows in the circuit. The size of this voltage drop is determined by the value of the current drawn by the load.

Terminal voltage = EMF -

VR Example: A cell with an EMF of 1.8 V delivers a current of

2.5A through an external resistance of 0.62. Calculate the internal resistance. - 0.122 . 1.22 12Q . 0.1252

op

ap

SOLUTION:

Vigg

$$= \text{EMF} - I r_{\text{int}}$$

$$I (R_{\text{ext}}) = \text{EMF} - I r_{\text{int}}$$

I(Ryeg)

$$2.5(R_{\text{ext}}) = \text{EMF} - 2.5(0.62) \quad 18-15$$

$$R_{\text{ext}} =$$

$$0.37 \Omega$$

$$R_{\text{int}} =$$

$$R_{\text{int}} = 0.120 \Omega$$

Pytatt

Example: The EMF of a cell is 1.8V and its internal resistance is 0.250. Calculate the current when the external resistance is 62. 24mA

$$0.24 \text{mA}$$

$$2.4 \text{mA} \mid 240 \text{mA}$$

Pp

ap

SOLUTION:

TR, +

$$V_{\text{ext}} =$$

EMF

$$y =$$

-EMF. $R_{\text{ext}} +$

Vind

p=

SV

6.25Q

I= 240mA

Choice (d) Resistors in Series and Parallel Resistors in series and parallel will be covered in Chapter 4 and 5 in detail, but the formulas are listed here for reference.

Resistors in Series $R_t = R_1 + R_2 + R_3 + \dots + R_n$

Resistors in Parallel R_t

$=$ or $R_t =$

where n is the number of resistors in parallel.

n = number of equal

value resistors With a calculator, the parallel formula can be entered: $R_t = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}}$

$R_t = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}}$

$\frac{1}{R_t}$ where $\frac{1}{R_t}$ is the reciprocal key.

The Sine Wave

on

+V Fig. 5 AC Sine Wave Fig. 5 represents one complete cycle of an AC sine wave which can represent voltage or current. The following formulas are important in order to calculate various values pertaining to the AC sine wave:

Amplitude Conversion:

V Peak to Peak (V_{p-p}) = $2.828 \times V_{rms}$

V Peak

(V_p) = $1.414 \times V_{rms}$

Root-mean-square (V_{rms}) = $0.707 V_{peak}$

Peak-to-peak = Peak value $\times 2$

Peak

peak to peak

Time and Frequency Conversion:

Ds L/T

= seconds

Hertz (Hz)

Period (T)

Frequency (f) nou i]

Example: The RMS value of a 60Hz sine wave voltage is

120V. What is its peak-to-peak value?

339.4V . 42.4V . 84.8V

169.7

V esp

e

SOLUTION:

i]

Peak value= RMS = 120 =

169.7 volts

0.707 707

Peak value x 2 = 339.4V

Peak-to-peak Choice (a)

Chapter

Frample; 4 sine wave has a firgirenc) of 60 2, Find the period,

T:

a) 1668 ©) Ole

ms

b) 16.6ms

d) 16 sec Choice(b) opposes & change of volt the Farad, whereas the in Coulombs. A capacitor

in
an electrical circuit. The basic unit of capacitance
is charge stored in a capacitor is measured

$$Q = C \times V$$

where:

Q is the charge C is the value of the capacitor in Farads V is the voltage in Volts Example: A 12V DC supply is connected across a capacitor of 500 pF. Calculate the charge:

- a. 60 Coulombs
- b. 0.6 Coulombs
- c. 600 Coulombs
- d. 60 pCoulombs

SOLUTION:

$$Q = C \times V$$

$$= (500 \times 10^{-12}) \times 12$$

$$= 6000 \times 10^{-12}$$

$$= 6000 \text{ pCoulombs}$$

Choice (d) The basic unit of energy is the Joule. The energy stored in the electric field between the plates of the capacitor is:

$$E = 0.5 C V^2$$

Where: E is energy in Joules C is the capacitance in Farad V is the voltage in Volts

Math for Electronics Example: Calculate the energy stored in the capacitor from the previous

$$\text{example where } V = 12 \text{ V and } C = 500 \text{ pF}$$

Sup

- a. 360000 Joules
- b. 0.360000 Joules
- c. 360000 pJoules
- d. 3.600000 pJoules

SOLUTION:

$$\text{Energy stored (E)} = 0.5 C V^2$$

$$= 0.5 \times (5 \times 10^4) \times (12 \times 12)$$

jou (0.5 x 5 x 12 x 12) x 10⁴ joules, 360 x 10⁴ joules 360 x 10⁴ joules 360 joules Choi,

Capacitors in Series

Capacitors

in series are like Resistors in Parallel.

Any capacitance can be found by the sum of the reciprocal

For 2 capacitors in series (a simpler formula):

$$C_1, C_2,$$

$C_1 + C_2$, For equal value capacitors in series:

$C_1 = C/n$ where n = number of capacitors For capacitors in parallel: $C_{\text{total}} = C_1 + C_2 + C_3$, Example; If three equal value capacitors of 10 nF each are connected in series, calculate total capacitance. a. 33 nF

$$\text{b. } 0.33 \mu\text{F}, 330 \mu\text{F}$$

4. 33 μF SOLUTION;

$$C = C/n$$

$$10/3 = 3.3 \mu\text{F}$$

Choi

Example: Calculate total capacitance of two 10 nF capacitors in parallel,

$$\text{a. } 20 \text{ nF}$$

$$\text{b. } 10 \text{ nF or } 3 \mu\text{F}$$

4. 0.33 μF

SOLUTION:

$$C_1 = C_2, C_1 = 10 + 10 = 20 \text{ nF} \text{ a2 12 Choice (a)}$$

Time Constants

A capacitor takes 5 time constants to be fully charged and 'one time constant equals the value of the resistance multiplied by the capacitance.

$$1 \text{ Time Constant (TC) in seconds} = R \times C$$

The charge takes place exponentially. The discharge action follows a reverse pattern (see Figure 6). Fig. 6 Charge/Discharge Curve of Capacitor Example: A series RC circuit is connected across a 100V

DC supply. If $C =$

SF and $R = 10 \text{ k}\Omega$, calculate one time

constant.

- a. Ssec
- b. 0.05 sec
- c. 5 psec
- d. 0.05 psec

SOLUTION:

One time constant = $R \times C$

$10^4 \times 5 \times 10^{-6}$

= 0.05 sec

Choice (b)

Math

for Electronics

Example: If 100 v is the applied voltage, after one time constant, what approximate voltage can be expected across the capacitor?

- a. 0.63V
- b. 63V
- c. 63V
- d. 0.063 V

SOLUTION:

63.2% OF 100V = 63.2V Choice (b)

Capacitive Reactance

Capacitive Reactance, X_C , measures the opposition offered by a capacitor to the flow of an alternating current. X_C is mea-

sured in Ohms and its formula is:

$X_C =$

$\frac{1}{2\pi}$

fC 2 π is 6.28

f is the frequency in Hertz (Hz)

C is the capacitance (farads)

where: A memory aid that can help you remember that in a capacitive circuit current "I" leads the voltage "E" (E is voltage, not energy in this representation), hence the word "ICE". Refer to Fig. 7 (A), circuit diagram; (B) waveform; and (C) vector diagram. In Fig. 7B, the peak voltage is represented by V , and not E . In Fig. 7C, the phasor pointing straight down is voltage (E) which is 90 degrees out of phase with the current (I), with its phasor pointing to the right.

Fig. 7 Capacitive Reactance Phase Representations

Fig. 8 RC circuit can be in series with resistance and also in parallel as shown in Fig. 8. When resistance is present, the phase angle between voltage and current is less than 90 degrees, and can be determined. See chapter 6 for details.

Capacitors

Capacitors with resistors

When a capacitor is connected in series with a resistor, the current in the circuit is found with vector or phasor representation details.

Inductance

Inductance is that property of a coil or inductor which opposes a change of current in a circuit. Its unit is the henry (H).

Inductors in Series Inductors in series are calculated like resistors in series, with mutual inductance added (if magnetic fields add) or subtracted (if the magnetic fields oppose) when the coils are close to each other. $L_T = L_1 + L_2 + \dots + L_n \pm 2M$ for any mutual inductance, where $2M$ equals $2 \times$ the Mutual Inductance) Inductors in Parallel Inductors are treated as resistors in parallel, with any mutual inductance either adding or subtracting from the total.

Inductors in Series Inductors in series are calculated like resistors in series, with mutual inductance added (if magnetic fields add) or subtracted (if the magnetic fields oppose) when the coils are close to each other. $L_T = L_1 + L_2 + \dots + L_n \pm 2M$ for any mutual inductance, where $2M$ equals $2 \times$ the Mutual Inductance) Inductors in Parallel Inductors are treated as resistors in parallel, with any mutual inductance either adding or subtracting from the total.

$L_T = L_1 + L_2 + \dots + L_n \pm 2M$

Example:

Three inductors, each 4mH, are connected in series. Calculate the total or effective inductance.

- a. 12
- b. 12
- c. 12mH
- d. 12mH
- a. 12mH

SOLUTION:

$$L_T = L_1 + L_2 + L_3 =$$

$$4\text{mH} + 4\text{mH} + 4\text{mH}$$

$$= 12\text{mH}$$

Choice (a)

Math for Electronics Example: Three inductors parallel. Calculate the total inductance

- a. 13mH

- b. 0.13mH
- c. 0.013mH
- d. 1.3mH

SOLUTION:

If equal in value, $L = \frac{L_{total}}{n} = \frac{4}{3} = 1.3\text{mH}$

Where $n =$

number of equal value inductors

Ch

Inductive Reactance

Inductive Reactance, X_L , measures the opposition offered by an inductor to the flow

of AC current. X_L is measured in Ohms, and its formula is

$$X_L = 2\pi fL$$

A. AC & Q

Peak

peak

B. C.

E

Fig. 9 Inductive Circuit Phase Representations

It is important to note that in an inductive circuit the voltage lags the current, hence the word "ELI," where E is shown as V_{peak} on the horizontal axis. In Fig. 9(A), (B) and (C), the waveforms for inductive reactance can be found

MathforElectronics

Impedance

When both capacitors and inductors are present in a circuit in either series or parallel, along with resistance, special conditions take over and make the math calculations more difficult. The total opposition to current flow in an RLC circuit is called Impedance and its formula symbol is Z . Z is made up of (R) resistance, (X_C) capacitance reactance and (X_L) inductive reactance values, which are inherently dependent on frequency of operation. The formula for impedance in a series RCL circuit is:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

R x

3a

20Q Fig. 11 Series RLC Circuit Example: In the circuit in Fig. 11, calculate impedance, Z.

a) 39Ω

c) 332

b) 5Ω

d) 7Ω

SOLUTION:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

a

Z

Zz

Zz

Z Choice (b) In examining the phase relationships between the voltage and current

of the circuit in Fig. 11, it can be plotted on a graph to represent the 90 degree difference between resistance and the reactance. There is 180 phase difference between the values in the reactances of the capacitor and inductor, therefore they will subtract from each other. Action through the resistance is in phase with the supply voltage, but there is "leading" and "lagging" with regards to the inductor and the capacitor. Because the X_C capacitive reactance dominates this circuit (it has the higher value), the circuit is said to be capacitive. $- \frac{1}{j\omega C} - \frac{1}{j\omega L}$ $ee-----$

The circuit is more capacitive than inductive

x P94

$$R = 3\Omega$$

Example: (for R and X in parallel calculations, use the formula): $\frac{1}{Z} = \frac{1}{R} + \frac{1}{X}$

$$Z =$$

Resonance

Resonance is a special condition when the capacitive reactance and the inductive reactance are equal in a circuit (and exactly opposite in value, as noted on phasor representations). The circuit is said to be "Resonant."

$$X_L = X_C$$

Substituting the formula equations that contain the frequency and basic units of capacitance and inductance, we can find the resonant frequency, where this equality occurs: Where f_r is the resonant frequency and

$$L = \text{units of inductance in henries of the coil}$$

$$C = \text{units of capacitance in Farads of the capacitor}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f = \frac{1}{2\pi\sqrt{LC}} \quad (22VIC)$$

Also, the current in the capacitor and inductor are at maximum and the phase angle between the voltage and current is zero. A circuit containing resistance and reactance is said to be in resonance when the supply voltage and the current are in

phase. This makes the phase angle zero, and $X_L = X_C$.

Example: A 200 μ H coil whose resistance is 5 Ω is connected in series with a 200 pF capacitor. The resonant frequency of the combination is:

- a. 795 KHz
- b. 795 Hz
- c. 7 MHz
- d. 7.95 KHz

SOLUTION:

Given:

$$L = 200 \mu\text{H} = 200 \times 10^{-6} \text{ H}$$

$$C = 200 \text{ pF} = 200 \times 10^{-12} \text{ F}$$

$$f = ? \text{ KHz}$$

° Choice (a) Units for Electronics Formulas:

Unit

Abbreviation

Formula

Unit

Quantity

$$f = \frac{1}{2\pi\sqrt{LC}}$$

[2 |e] Math for Electronics

Greek Symbols

Many electronic abbreviations and Greek symbols for identification of and The Greek name along with upper and lower case summarized in the table

Capital Lower Case Greek Nante

after 4 by Rollin R. Okerberg, CETSr e for lenmming new ideas from fellow members

Introduction

A student starts with a basic point as with any subject of study, a student has a basic understanding before pursuing more advanced topics. We will start with the atom. A technician needs to understand the structure of the atom to see how electrical circuits behave the way they do. Troubleshooting is a final part of this chapter.

The Atom

The hydrogen atom, the most basic element known in the universe, has one electron orbiting one proton. Its atomic number is one; this number tells us how many electrons and protons are in any particular element. They are the basic building block of all the known elements in the universe. Yes, of course, these particles can still be further broken down into more fundamental subparticles, but to do so is really beyond the scope of this chapter.

~ Electron

+ Proton

Fig. 1a Hydrogen Atom Electrons and protons have equal but opposite charges. The electron has a negative charge, and we will see later on that it will play an important role in this chapter's discussion. The proton, of course, has a positive charge and is located in the nucleus of the atom. Another part of the nucleus of the atom is called the neutron and, as the name implies, has a neutral charge. This particle also contributes to the atomic mass of an element. Having an equal number of electrons and protons, an element is electrically neutral. In a few words, some elements present themselves as good conductors while others are not conductors at all! Atoms have up to 7 shells around the nucleus. These shells represent energy levels that electrons are allowed to occupy, and each one is designated with a different letter as shown in Figure 1b. Rollin Okerberg, CTTS, has been employed by the Department of Defense for 11 years. He started his technical training at Lowry AFB, Colorado. He also worked for and testing test programs for circuit cards and in a replaceable unit. He is currently living in Irvine, California. He earned his Associate Degree at American River College, Sacramento, California. He possesses a Genet® Radar Endorsement. He is currently an active member of FTA International and feels he DC Circuits is a part of his Army at Sacramento's Army Depot as a clerk in the maintenance section. He owes ten

Ad the telephone

A secretary has given him new and interesting information

\ Nucleus

Conduction band

Fig. 1b

In electronics, the electrons we are concerned with are in the outermost shell; this is where conduction takes place when the electron is subjected to an external electrical source such as a DC power supply. In the conduction band, free electrons are allowed to move in a material that is subjected to a voltage source. Three categories of materials in electronics work are:

1. Conductors
2. Semi-conductors

3. Insulators If electrons move easily, the material is a good conductor. However, a material not having free electrons available for its conduction band is said to be an insulator. In an insulator, a large amount of energy is required to move electrons to this conduction band. Examples of good conductors are: silver, gold, copper, aluminum, and iron. Examples of insulators are rubber, ceramics, glass, and wood. Some materials are also poor conductors, but not as poor as insulators. These are nichrome, carbon and silicon. Now that we have an understanding of conduction and the atom, let us go on to the next discussion.

Chapter 4

Basic Units in Electronics A table at the end of chapter 3 summarizes many of the commonly used electronic units and their assigned symbols and abbreviations. Formula symbols are used in math equations as unknown quantities. Abbreviations are used so that you don't have to write down the entire name of the unit when describing it. The Coulomb is a unit of charge that is equal to

6.28×10^{18} electrons. The formula symbol for the unit of charge is Q . The Ampere (A) is the unit of current flow in electronics. Current flow is measured in Amperes and is abbreviated A. More specifically, it is a rate of one Coulomb moving past a given point in 1 second as defined in:

$$1 = Q \text{ (in Coulombs)} / t \text{ (in seconds)}$$

Voltage is defined as energy per unit of charge. The unit of voltage is the volt with a formula symbol of V , and the abbreviation is also V . This unit expresses the amount of force needed to move a given charge from one point in a circuit to another. The joule (J) is a unit of work or energy. The volt is a unit of electromotive force (EMF) that causes a current to flow in a circuit. Voltage is expressed as Joules per Coulomb in the formula:

$$V = E$$

(in Joules) / C (in Coulombs) It should be noted that in some circuits and references, "E" is used as the formula symbol for voltage. However, in this study guide, "V" will represent voltage, as it is in most new textbooks, so as not to confuse the student with the symbol "E" being taught in science classes as the symbol for energy. Energy efficiency is important today in product design and operation. In any circuit:

Power

$\times 100$

$$\% \text{ of Efficiency} =$$

Pin Power (P) is defined as an amount of work per unit of time. It

is measured in Watts and is the amount of heat that is dissipated

from a component such as a transistor or a resistor. Wasted energy ends up as heat and not as useful output.

$$P_{in} - P_{out} = P_{dissipated} \text{ (heat)}$$

The difference between applied power (input) and the consumed power (output) is the power dissipated as heat. Component sizes are rated with power ratings to express how much heat a device can dissipate. Technicians need to pay close attention when replacing components with an equivalent part, otherwise it may burn up if it has too small of a power dissipation,

Ohm's Law

Ohm's law is one of the most important tools for technicians

and it is repeated several times in this book. You will use Ohm's

Law frequently. Voltage is known as "potential difference" DC Circuits

Ma between two points, Voltage is also the product of current and Resistance as shown in

$$V = I \times R$$

When resistance is held constant, as in the case of a fixed resistor, and current increases, the voltage increases. If, on the other hand, current through it is allowed to decrease, voltage across will also decrease. So one can say that voltage is directly proportional to a change in current. If we rearrange this basic formula to calculate current, we can see another relationship develop here. To calculate the amount of current flowing through a component, the current (I) is equal to voltage divided by resistance:

$$I = V / R$$

If voltage is held constant and resistance decreases, Current increases. Or if resistance increases, current decreases. What this is really saying is that current is indirectly proportional to resistance. Resistance is defined as the opposition to current flow. Its unit

is the Ohm (Ω), named in honor of George Simon Ohm. One

Ohm lets one Ampere of current flow through a circuit having one

volt applied. Rearranging terms in the Ohm's law equation above,

resistance can be calculated from knowing the voltage and current. The relationship is shown below:

$$R = V / I$$

Again, notice that resistance is dependent on either voltage

or current as discussed above. The three formulas above are important tools in the electronics trade. Please take the time to memorize them to be able to use them with ease.

Power (P) in Watts can also be shown as a product of current

and voltage and is known as Watt's Law:

$$P = I \times V$$

If either voltage and/or current increase/decrease, power will also increase or decrease in the same manner. If resistance and current are known, square the current and multiply by the resistance to find power. The basic formula for this is:

$$P = I^2 R$$

After examining this formula, one can see a new relationship between resistance, current, and power. When the resistance doubles in a circuit, the power also doubles. If the resistance is cut in half, the power is cut in half. _____ x DC

Circuits

But what about current? If we double current, what happens to power? The answer to this question is that the power goes up by 4 times its previous value! Now, what happens when current is halved? The power also would be one-fourth of its previous value. Another formula that is often used in the electronics world is the voltage squared divided by resistance:

$$P = V^2 / R$$

If an increase in resistance occurs in a circuit, the power decreases. To put this concept in another way, let's use an example. Suppose the resistance is cut in half. What happens to power? Power is doubled. Suppose resistance is doubled from a previous value? Power is now cut in half. Understanding the relationships that go on in circuits makes circuit analysis quite simple. The whole idea is to simplify the complex circuitry to something more manageable to see what really is going on. The same relationship goes on with voltage, as does current in the previous equation. If voltage were to be doubled from a previous value, the power is increased 4 times its previous value. Another important unit of measure that is not used quite as often is known as conductance. This unit is opposite of resistance and expresses least opposition to current flow, or how well devices conduct current. This unit has been known as the "mho" which is ohm spelled backwards. Conductance (G) is defined as the reciprocal of resistance as shown:

$$G = 1 / R$$

More recently, the conductance unit mho has been changed to Siemens. It is expressed in the same way and its symbol is the S.

Series Circuits

A series circuit is shown below. Fig. 2 As with any circuit, a complete path is necessary. Electrons need to have a destination to reach. Without this path, current flow would not be possible and electricity would be nonexistent. Electrons will flow from the negative terminal of the battery in Chapter 4 shown around the circuit and to the Positive terminal. This is known as electron flow and will be discussed from here on. There is another explanation of current flow, and this is "conventional flow." This explanation says that current flows from positive to negative and Ben Franklin originated this idea before electron theory was understood. As a practicing electronics technician, you interpret schematic diagrams. In examining the series circuit in the schematic diagram in Fig. 2, you note that there is only one path for current to travel. That current is the same at any point because there is only one path in the single loop. In this arrangement, only one current flows, therefore the current is said to be constant, and remains the same through any component in that path. In series circuits, it is

easy to find unknown values using Ohm's and Kirchhoff's laws.

The total resistance of a series circuit can be found by adding all of the resistance values, as shown in:

$$R_T = R_1 + R_2 + R_3 + \dots + R_n$$

Another important fact about this circuit is that the total voltage drops around a series circuit can be also found in the same way. All of the voltage drops across each resistor can be found using

Ohm's Law. The voltage drop across each resistor is added to

arrive at a total. This total should be equal to the source voltage as shown in: $V_T = V_1 + V_2 + V_3 + \dots + V_n$ With this, another law known as Kirchhoff's voltage law developed. This is stated as:

$$\sum V = 0$$

This says that the sum of all the voltages in a closed loop is zero. You notice that a ground is used as a reference point for the sake of measuring, and in this case is a negative point with respect to any other point in the circuit. Notice the polarity of each voltage as we go around the circuit. If we add each voltage, while noting each voltage polarity, we will come to the same conclusion about this circuit as Kirchhoff's law states: The algebraic sum of all the voltage drops in a closed loop is equal to zero. Voltages add up in a series circuit, and so does the total power. Each resistor, or load, is working to contribute to the total amount of work that a circuit is doing as a whole. Everyone likes to be recognized for his/her work and to receive credit, and so does each load in a circuit. The power expended from each resistor can be added to arrive at a total power. This is expressed as follows:

$$P_T = P_1 + P_2 + P_3 + \dots + P_n$$

$$P_T = P_1 + P_2 + P_3 + \dots + P_n$$

$$P_T = P_1 + P_2 + P_3 + \dots + P_n$$

$$P_T = P_1 + P_2 + P_3 + \dots + P_n$$

Chapter 4

If the total voltage is known, total power can also be found by using Ohm's Law. This is done by multiplying the current by the voltage applied to get the total power:

$$P = VI$$

Parallel

Circuits

What is interesting about parallel circuits is that current flow is divided through several branches or paths of the circuit. Depending on an individual branch resistance, one branch can carry more current than another.

This

can

be shown below where you can see that a 1000 ohm resistor will allow more current to flow through it than a 3000 ohm resistor, because the smaller resistance allows

more current to flow. Ohm's Law shows that higher

current takes the easiest path. which has less resistance. A B c

wv

Fig.3

Current

through branch

$$A = 10/1000$$

$$= 10 \text{ mA}$$

Current

through branch

$$B =$$

$$10/2000 = 5 \text{ mA}$$

Current

through branch

$$C =$$

$$10/3000$$

$$3.3$$

mA

As

result

of this current division, the total current in a parallel circuit is the sum of each branch current,

Another

important law that has to be mentioned here is Kirchhoff's

Current

Law. It states that the sum

of current flowing into

a junction (or node)

is equal to the current flowing out of that same junction.

This

is shown

in Fig. 4.

Node**Point**

by Fig. 4 mathematical form,

$$I_T =$$

I_T is shown in Fig.

5, which explains how current behaves in series and parallel circuits. Before the mnemonic was discovered,

Gustav

Kirchoff, a mid-nineteenth century scientist, discovered these two important laws (Voltage and) which make it possible to analyze different circuit configurations.

The total current in a parallel circuit is expressed

as follows: The total current in a parallel circuit is

Since a parallel circuit consists of more than one branch, the total resistance of this circuit is less than the least resistance

is this fact possible? Well, let us take a look at the above circuit. The total resistance can be found with the following equation:

$$1/R_T =$$

$1/R_T = 1/R_1 + 1/R_2 + \dots + 1/R_n$. Notice that the total parallel resistance is the reciprocal of the sum of the branch resistances. The total resistance of a circuit can also be calculated as:

$$R_T = V_T / I_T$$

V_T is the source voltage, and with this voltage, along

with the total current, the total parallel resistance can be found. Notice that each current shown below (V_T/R) has the same source

Voltage

(V_T) applied to each resistance and is divided by that branch resistance.

$$V_T/R_T = V_T/R_1 + V_T/R_2 + \dots + V_T/R_n$$

V_T/R_T . This is why current divides: there is more than one path for current to flow. Each branch resistance determines how much current will flow through it. A little algebra is used to show that V_T or V_{source} is divided into both sides of the equation to get the resultant equation for total resistance:

$$1/R_T = 1/R_1 + 1/R_2 + \dots + 1/R_n$$

If each branch resistance is equal to one another, then the total resistance is equal to the branch resistance divided by the number of branches. With this in mind, total resistance of parallel circuits with equal branch resistance values can be found quite easily:

$$R_T = R/n$$

Series-Parallel Circuits

Two simple forms of a series-parallel circuit can help you understand many of the possible combinations of circuits. Circuit can be in all different shapes and sizes. Doing a little circuit analysis is really all that is required to simplify a complicated circuit. Do not let them confuse you, as they are not what they seem to be—a different guise to confuse the sharpest technician who needs to get a piece of equipment working. Shown in Figure 6 is one form of a series parallel circuit.

DC Circuits

Find the total resistance of Fig. 5, add the two resistors in series on the right side. Then calculate the resultant resistance the single resistor in parallel on the left. If all three of these are 2 kΩ, then the total will be:

$$2\text{ k}\Omega + 1\text{ k}\Omega = 3\text{ k}\Omega \text{ equivalent for the right hand resistors.}$$

Then the 3 kΩ equivalent value is in parallel with 2 kΩ. This can be solved with the combination formula: $R_T = \frac{R_1 R_2}{R_1 + R_2}$. Sometimes this circuit in Fig. 5 is referred to as the "2/3 rule" circuit arrangement when equal value resistors are used, as the total is 2/3rds of the value of an individual resistor. $1/2 \times 2/3 = 1/3$ of the value of an individual resistor. $1/3 \times 3\text{ k}\Omega = 1\text{ k}\Omega$

The other simple form of a series-parallel arrangement is shown in Figure 6. First find the equivalent value of the two resistors in parallel on the lower part of the circuit. Use the combination formula again: $R_T = \frac{R_1 R_2}{R_1 + R_2}$

$$R_T = \frac{5\text{ k}\Omega \times 5\text{ k}\Omega}{5\text{ k}\Omega + 5\text{ k}\Omega} = 2.5\text{ k}\Omega$$

$$R_T = 2\text{ k}\Omega + 2.5\text{ k}\Omega = 4.5\text{ k}\Omega$$

Or

$$R_T = R/n$$

where n is the number

of equal value resistors in parallel.

$$1\text{ k}\Omega / 2 = 0.5\text{ k}\Omega$$

Then add 5 kΩ to the upper resistor of 1 kΩ (1) because it is in series. $R_T = 1\text{ k}\Omega + 0.5\text{ k}\Omega = 1.5\text{ k}\Omega$ Figure 6 is sometimes referred to as the "3/2" circuit because the total is one and a half times the value of one of the equal value resistors.

R₁

1 k

R₂ 2 k

Ka ® R

R, Fig.7 A more complex circuit is shown in Figure 7. To approach this circuit, simplify the most interior circuit combination first. It is the series circuit of the two 2K resistors, the right hand path of the parallel branch at the far right side made up of R₁ and R₂. The resistor values of R₁ and R₂ are added to equal 4k or 4000 ohms. Now our circuit reduces to a circuit shown in Figure 8, a 4000 ohm resistor in parallel with R₃. R₃, 1000 R₃, 2000 R₃, 1000

Fig. 8 A further reduction of the circuit with two resistors in parallel will equal the equivalent resistance that is in series with R₄ and R₅. To find this value, use the following: $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$. Substituting each value in the above equation results in: $R_{eq} = \frac{2000(4000)}{2000 + 4000} R_{eq} = 1333.3$ Ohms

$$1333.3 \text{ Ohms}$$

The parallel combination now becomes a resistor in series with R₄ and R₅, as shown in Figure 9, eee eee

we proceed Fig 9 combination to finally arrive at the total resistance: 3333.3 ohms. final answer of 8 total resistance © RoR ARR

$$= 1k + 133k + 1k$$

$$= 3.33k$$

As you can see from this it reduced to something more

Now that we have found the total resistance of the series parallel circuit, suppose we need to find the voltage drops and currents through each branch of this circuit. In addition, a complex circuit can be manageable.

1. First find R_T, the total resistance of the series parallel circuit. A practical technician may want to simply measure the complex circuit with a digital VOM (volt-ohm-meter) to find the total resistance.

Next

to find the total current.

This is found by dividing the applied voltage, 12, by the total resistance, 3.33kΩ.

$$I = V / R_T$$

$$= 12V / 3.33k =$$

$$3.6$$

mA The result is 3.6 mA. With this current, we can now proceed to find each voltage drop around this circuit.

3. To find the voltage drops across each resistor, the current is multiplied by each resistance.

V_y

$$I \times R = 3.6 \text{ mA}$$

3.6kΩ

3.6 volts The voltage across R_1 is 3.6 volts. The voltage across the parallel (equivalent) portion of the circuit shown in Figure 9 is found by multiplying the total current times the total parallel resistance 1333.3 ohms. Volek,

$$= 3.6 \text{ mA} \times 1,333.3 = 4.8 \text{ volts}$$

The voltage across this portion of the circuit is 4.8 volts. The voltage across R_2 is found in the same way as R_1 , just mentioned. The voltage across this resistor is also 3.6 volts because the resistor is the same value as R_1 , and the total current (3.6 mA) also flows thru it.

DCCircuits IE Kircha's Voltage Law says that the sum of the voltages all of the voltages around the loop is zero. The voltage drops across each resistor follow Voltage across R_3 ,

$$= 3.6 \text{ volts}$$

$$\text{Voltage across } R_2 = 4.8 \text{ volts}$$

$$\text{Voltage across } R_1 =$$

3.6 volts Adding all of the above voltages shown saw:

Ohm's Law

drop equals 12 volts across

$$3.6 \text{ volts} + 4.8 \text{ volts} + 3.6 \text{ volts} = 12 \text{ volts}$$

$$12 \text{ volts} \sim 12 \text{ volts} =$$

volts Since we know the voltage drop across the parallel network can find the branch currents in the original circuit in Figure 7. The branch current for resistor R_3

Find

by using the 4.8 volts across this resistor by the resistor

a value 2000 ohms. This equals 2.4 mA as shown: inane?

$$I_g =$$

Voltage

$$= 4.8 \text{ V} /$$

$$2 \text{ K}\Omega = 2.4$$

mA To get the branch current for R_3 and R_4 subtract 2.4 mA from the total current 3.6 mA to get

1- Tgs.ravrance ip-

$$T_q = 3.6 \text{ mA}$$

$$- 2.4 \text{ mA} = 1.2 \text{ mA}$$

Another way to do this is to divide the parallel network by the total resistance. None is Pd RA:

Vaseancr/

Resreveren

$$= 4.8 \text{ V} / 4 \text{ k}\Omega = 1.2 \text{ mA}$$

Then, to find the voltage drop across R_3 and R_4 , take the current of 1.2 mA times the resistance of each: V_{as}

$$= I_{\text{current}} \times R$$

$$R_y =$$

$$1.2 \text{ mA} \times 2 \text{ k}\Omega = 2.4$$

$$\times R_4 = 1.2 \text{ mA} \times 2 \text{ k}\Omega = 2.4 \text{ V}$$

View

Tas

ra branch

Voltage Dividers

Now that we have discussed the series, parallel, and series-parallel circuits in great length, our discussion takes us to an application of the above circuits. Voltage dividers are used in extensive power supplies to distribute voltages to various parts of a circuit for biasing amplifiers and such. An example of this is shown in Figure 10.

DC

Circuits

WO VIC

$$180 \text{ VDC}$$

Fig. 10 In Figure 10, this voltage divider could be used for distributing the B+ plate voltage for tubes. You might think that tube circuits are a thing of the past, but do not be surprised if during your everyday encounters with circuits you find a few around. Voltage dividers can also be used in setting up reference voltage for one of the inputs to a comparator IC circuit such as the one shown below in Figure 11. Fig. 11 In Figure 11, an LM139 voltage comparator is shown. The voltage divider sets the threshold voltage for the input to cause a change in the output when 1.67 V is reached. This voltage is also known as the

trip point. The output will change from one maximum voltage to another, depending upon whether the (-) input exceeds the reference threshold voltage. This is a good example of where a voltage divider circuit is found in electronics.

Voltage dividers can be utilized in voltage comparator circuits to set the threshold voltage with respect to common ground. Use the following formula to determine the output across a lower resistor (R_2) in a voltage divider:

$$V_{out} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right)$$

Voltage drop across

R_2 ,

To find the threshold voltage find the aim of and from the values shown in Figure 11

$$2k + 1k = 3k \text{ or } 3000$$

ohms. Now from The next step is to divide $R_2(1k)$ by the sum of R_1 and R_2

$$\left(\frac{1k}{3k} \right) = .333$$

Multiply this ratio with the V_{source} which is +5 vdc

$333.058 \text{ v} = 1.67 \text{ volts}$ As you can see from this discussion, the voltage divider rela-

tion is a powerful tool in finding a voltage in many electronic circuits, 77: The next part of this chapter takes us to another very important concept.

Maximum Power Transfer

A load is said to be matched to the source if the load impedance is equal to the source impedance.

With Resistance $R_L = R_s$, for DC circuits

With Impedance $Z_L = Z_s$, for AC circuits

Examples of impedance matching conditions when you will find maximum transfer of power are: 8 ohm amplifier connected to an 8 ohm speaker 50 ohm transmitter connected to a 50 ohm cable & antenna 75 ohm TV antenna connected to 75 ohm cable to TV input 600 ohm telephone connected to a 600 ohm line transformer Common industry standard impedances are established to ensure maximum power transfer between devices from that same industry. For example, the RF industry uses 50 ohms, Video uses 75 ohms, Audio uses 8 ohms, and telecommunications uses 600 ohms devices.

Wier'

At this point this is shown in Figure 12

As a result of this, an amplifier has an internal impedance of 16 ohms. The maximum amount of power is also dissipated in the load. Notice that the load impedance. The amplifier and the speaker is the

maximum power is dissipated in the internal impedance of the amplifier. About the condition of maximum power the circuit is matched, with equal impedances. The power delivered to the load is transferred and an interesting phenomenon is that when the impedances are matched, only 50% of the total power is found. This is the best that can be realized with a passive network. The efficiency of a matched condition is 50%. There are, however, certain instances where matching impedances is not necessary simply because it is not very practical, but the load resistance has to be equal to or higher than the source internal impedance.

Efficiency

will go up with lighter loads that are higher in resistance. Take an example of a car battery and a small dome lamp in the vehicle. It is easy for the large capacity automotive battery with low internal resistance to power the dome light with moderate resistance. Another example, a battery having an internal resistance of, let's say, 1 ohms at 12 volts. What would be the maximum current capacity

of this battery?

The answer, using Ohm's Law, is

120 amps. From this information, we can understand that a load being connected to this battery (like a car starter) cannot be less than the internal resistance of this battery or power source. Otherwise, more power is being expended internally to the power source than it can handle. This is why you would not place a load having a resistance lower than what a power supply can handle.

Troubleshooting

As a technician, you will find that troubleshooting involves doing a lot of circuit analysis as well as doing the divide and conquer method. You will also find that you will develop your own method of isolating a fault causing a problem in a circuit. There is no certain way to troubleshoot a faulty circuit. Just as a writer has his own writing style, a technician also has his own unique approach to troubleshooting a faulty circuit. You develop your way of doing this. Troubleshooting is a form of logic. For starters, take a look at equipment service manuals for a piece of equipment. You will find a section that is devoted to troubleshooting. In this section there should be a flow chart that will direct you by using listed possible symptoms. In addition, these flow charts help the technician be in the right section in order to decrease the amount of man-hours required for troubleshooting. Use them if such charts are available. They are good troubleshooting tools. They can get you in the right side of the ballpark.

DC Circuits 4 Shown below is a simple series circuit to illustrate an

my

ing problem that is often overlooked, 1000

Ohms

ae «IZ Vo 2000

Open 3000

ohare Fig. 13 After examining the above illustration, one can see there is an open that makes the series circuit incomplete, so electrons flowing from the power source have no way back home. You see, electrons seem to have a sixth sense about complete circuits. If one is not complete, current cannot flow. What voltage reading will be indicated on the voltmeter if one were connected to this circuit? The answer is that approximately 12 vdc will be indicated on the voltmeter, provided it is on a high enough scale. Why is this so? The voltmeter circuitry has high internal impedance, so most of the voltage applied to this circuit is felt across this measuring device. Not only is this true, but an open is essentially a high impedance anyway, so most of the voltage is felt across it. This is one symptom that should be considered carefully when encountered in a complex circuit. Let us take a look at another circuit that has a similar problem as discussed above. open 12V 500 1000

DC Ohms Ohms R,

2000

Ohms

Fig. 14 A parallel circuit having resistance values as shown has an open in one of its branch circuits. What are the symptoms? First of all, the total resistance of this circuit has increased from its 285.7 ohm original value. The total current has decreased from 42 mA to now 36mA. The total voltage is really the same, even though an open exists in one of the branch circuits. What about current through each branch circuit?

Will

they change

as a result of an open somewhere in the circuit?

Well, let's take a look. The branch with 40 ohms is drawing the usual 24 mA, and the branch with 1000 ohms is drawing 12mA current. Now looking at the branch with 2000 ohms, there

is

an open. Yes, this does explain the total current decrease

No current is flowing through this portion of the circuit

Keep

in mind that these examples are simple in comparison to more complex circuitry, but with a good understanding of basic circuit theory, the most complex circuitry can

be reduced

to

many

Chapter Quiz

1. The atomic number of an element is the same as the element's atomic weight. **False**
2. An increase in voltage with resistance held constant causes current to
 - a. decrease
 - b. increase
 - c. double
 - d. none of the above
3. The ampere,
 - a. is volts/sec
 - b. is a unit of charge movement per second past a given point
 - c. is coulombs/charge
 - d. is an increase in resistance
4. The total resistance of a parallel circuit containing equal branch resistances is found by the following:
 - a. by adding all of the branch resistance values,
 - b. by taking the product of all the branch resistance values. . by dividing the resistance value of a branch by the number of branches.
 - c. by dividing the resistance value of a branch by the number of branches.
 - d. the value cannot be found. **Correct**

4K

2k + open DC Circuits

5. An open is found in the series connected load of the transistor amplifier circuit as shown, voltmeter were connected from the open side of the resistor (tn, point of open as shown) to ground, what will be the reading?

- a. 0volts
- b. -12 vdc

c. Half of V_{CC}

- d. of the above

6. What is the voltage drop across the 2000 ohm resistor as shown in the amplifier?

- a. 3.3 vdc
- b. 8.0 vdc
- c. 9.0 vdc
- d. none of the above

7. If there is NO open in the circuit, what is the voltage across the 3000 ohm emitter resistor as shown above?

- a. 5.1 vdc
- b. 6.0 vdc & 8.0-vdc
- d. of the above

8. A 12 vdc power supply has a 15 ohm internal resis What load resistance can be connected to transfer maximum power?

- a. 15 ohms
- b. 20 ohms & 10 ohms
- d. 50 ohms % Maximum power transfer occurs when the load impedance is equal to the source impedance.
- a. True
- b. False

ANSWERS on page 223

Review Questions « DC Theory, Ohm's Law, Series/Parallel Circuits

Chapters 3 and 4

1. What is the current through the resistor in the schematic below?
 - a. amps
 - b. 2 amps
 - c. 5 amps
 - d. 50 amps
2. In the circuit above, what is the power dissipated in the resistor? 6 watts 12 watts 2 watts 24 watts
3. What will happen to the current through a resistor if the resistor value increases? Current increases Current remains unchanged Current decreases Current and the voltage across the resistor increases

ae

op

4. Which type of circuit has the same voltage drop across each component?

Parallel

Series-parallel

Series

Combination

people

5. Total power in a series circuit is determined by the:
 - a. source voltage times the current.
 - b. total voltage applied to the circuit.
 - c. current flowing through the switch.
 - d. average wattage rating of each resistor.
6. Which is the equivalent to 100 microamps?
 - a. .1 milliamps
 - b. 1 milliamps
 - c. 10 milliamps
 - d. 10 nanoamps

Unit

7. If the current drain from the battery is 4 mA, what resistance is the rheostat in the drawing below?

1.0k ohm

2.5k ohms

5.0k ohms 10k ohms aese -_y}--

II

8. In the circuit below how much current is flowing in R2? 1 amp 2 amps 10 amps 100 amps

RE

TP

lov =

109 109

ANSWERS on page 223

nance and Fs itance, Inductance, Reso ity,

Chapter 5

Director of Electrical Engineering Technology, Northeast State University system. He is also serving & Electronics Technology Faculty & coordinator.

Universe

point

Assisted by Dr. Soheil Anwar, Ph.D, CET senior associate professor of Retasall, the first precision geology & FAD fahisinal Vocative! education With more on

geophysical Smd

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Science

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t= member

of ASEE, IEEE. multiplier Review of Resistance In chapter 4, resistance was covered with DC circuits and formulas discussed. The resistor symbol was used to represent loads for the circuits. Keep in mind that resistance and resistor have the same effect in both DC and AC circuits. Resistance is defined as opposition to the flow

of electricity. It is designated as R and its schematic symbol is shown in Figure 1. Fig. 1 Resistor schematic symbol Resistance R is measured in ohms symbolized by the Greek letter omega (Ω). The resistance of a material is one ohm (1Ω) if a voltage of one volt (1 V) applied across the material results in a flow of one ampere (1A) of current through the material. Fig. 3 Resistor color code

Meanings

The resistor color code is interpreted in the following way:

1. The first band is the first digit of the resistance, the second band is the second digit of the resistance, and the third band is the multiplier.

2. The second band is the second digit of the resistance.

Common multiples of the unit ohm are kilo-ohm (K Ω) and megaohm (M Ω). 1K Ω stands for 1000 Ω and 1M Ω means

1,000,000 Ω or 10⁶ Ω .

For example, a resistor is an electrical component which is specifically valued to have a certain amount of resistance. Two

Principal : 5

oo.

pe a resistors are: fixed resistors and variable resis- 3. The third band is the multiplier (number of Zeroes), tors. A fixed resistor has a specific ohmic value which is set

during manufacturing and pi be altered 4. The fourth band gives the tolerance of the

Resistor and

easily. Fixed resistors are available in a large variety of is usually gold or silver. For example, a 10% tolerance resistance (ohmic) values. Various methods and materials (silver band) indicates that the acceptable

Resistance

are employed for their construction. A common type of fixed value can be within 10% of the Color-coded value. So resistor is the carbon-composition type. Other types of fixed a 200 ohm resistor with a tolerance of 10% can have

p sesistor, carbon film metal oxide, metal acceptable resistance values in the range of 180 ohms are conduc. to 220 ohms. ig how to obtain resistance values us A few examples shown are presented below: film, metal blaze, and wis ing the resistor color code

1. The color bands for a certain resistor are: first band is is violet, third band is yellow

Figure 3

ein te color q 8reen, second band and the fourth band is silver. Using the resistor color code shown in Figure 3, the resistance value with

tolerance is $570,000\Omega + 10\% =$

$570\text{K}\Omega + 10\%$. 'in the 5%, 10% and 20% tolerance ds to indicate the resis.

Mi

A}

rr Capacitance, Inductance, Resonance and Filters Therefirre the acceptable resistance value for this resistor ranges from $513\text{K}\Omega$ to $627\text{K}\Omega$ in measured value.

2. The color bands for a certain resistor are: first band is orange, second band is brown, third band is red and the fourth band is gold. Using the color code shown in Figure 3, the resistance value with tolerance is

$3100\Omega + 5\% = 3.1\text{K}\Omega + 5\%$, Therefore the acceptable

Resistance values for this resistor range from 2945Ω (or $2.945\text{K}\Omega$) to 3255Ω ($3.255\text{K}\Omega$).

3. The color bands for a certain resistor are: first band is blue, second band is red, third band is green, and the fourth band is silver. The resistance value with toler-

ANCE is $6,200,000 + 10\% = 6.2\text{M}\Omega + 10\%$. Therefore

the acceptable resistance values for this resistor range from $5.58\text{M}\Omega$ to $6.82\text{M}\Omega$. Some color-coded resistors have a fifth color band also. This color band is the indicator of the resistor's reliability

in Percent of failures per 1000 hours of use. The reliability color Code is shown in Figure 4. As an example, a red fifth band indicates that if a group of similar resistors are operated under standard conditions for 1000 hours, 0.1% of the resistors in that group will fail.

Brown 1%

Red 0.1%

Orange 0.01%

Yellow 0.001%

Fig. 4 Resistor Reliability Band

between terminals | and Yop hetecen Dand Vean be y aried by moving the contact up or down A hee: device shown in Fipure (Ry fat 14a O8 terminal The power rating of a resistor is the maximum amount of Power that a resistor can dissipate without being damaged by a buildup of an excessive amount of heat. Power rating is expressed in watts (w). The power rating of a resistor is not Related to its resistance value but rather it is determined by the Physical size and the shape of a resistor. The larger the surface area of a resistor, the greater amount of power it can dissipate Carbon-composition resistors are usually available in standard power ratings ranging from $1/8$ watt to 2 watts. The carbon- film and metal-film resistors have ratings up to 10 watts and wire wound resistors have power ratings up to 225 watts or greater.

Series Connected

When connected in series, resistors form a single path for the flow of electrical current, Figure 6 shows some examples of resistors connected in series, In Figure 6(A) resistors R_1 , R_2 , and R_3 are connected in series forming a single path for the flow of electrical current The second category of resistors comprises variable resistors that are designed so that their resistance values can be changed with a manual or an automatic adjustment. The potentiometer and the rheostat shown in Figure 5 constitute two basic types of manually adjusted resistors. from point A to point B. In Figure 6(B) resistors R_1 , R_2 , R_3 , and R_4 are connected in series forming a single path for the flow of electrical current from point A to point B. In Figure 6(C) resistors R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 are connected in series forming a single path for the flow of electrical current from point A to point B. When the resistors are connected in series, the total resistance is the arithmetic sum of the ohmic values of the resistors. For example, the total

resistance between the points A and B in Figure 6(A) is $R =$

$R_1 + R_2 + R_3$, Similarly, the total resistance between the points

A and

B is

Figure

6(C)

is

$R_1 + R_2 + R_3 + R_4 + R_5 + R_6$

+R,+R,+R,+R,+

As illustrated in Figure 5(A), a potentiometer is a three-terminal electrical device. Terminals 1 and 2 have a fixed resistance between them which is the total resistance. Terminal 3 is connected to a moving contact. The electrical resistance Fig. 5 Potentiometer (A) and Rheostat (B) In general, for any number of individual resistors connected in series, the total resistance is the sum of the individual resistances.

Therefore, the total resistance $R_t = R_1 + R_2 + R_3 + \dots + R_n$

The combination of adding resistors in series is given by $R_t = R_1 + R_2 + R_3 + \dots + R_n$. In Figure 7, resistors R_1 , R_2 , R_3 , and R_4 are connected in series between the points A and B.

Figure

combination

The total

resistance $R_t = R_1 + R_2 + R_3 + R_4$

$128 \Omega = 380 \Omega$

Fig. 7 Series resistance

Parallel

Connected

When two or more resistors are individually connected between two points they are said to be in a parallel combination. A parallel circuit allows more than one path for the flow of electrical current. Figure 8 shows some examples of the parallel combinations of resistors. Figure 8(A) shows resistors R_1 , R_2 , and R_3 connected in parallel between the points A and B. Figure 8(B) shows resistors R_1 , R_2 , R_3 , and R_4 connected in parallel between the points A and B. The equivalent (total) resistance R_t for any number of resistors connected in parallel between two points is given by the formula:

where R_t is the total resistance and $R_1, R_2, R_3, \dots, R_n$ are resistors connected in parallel ("n" is an undetermined number). In order to illustrate the formula given above, an example involving parallel combinations of resistors is presented here: Figure 9 Parallel resistance Capacitance, Inductance, Resonance and Filters

In Figure 9 resistors R_1 , R_2 , and R_3 are connected in parallel between the points A and B. The equivalent resistance (or total resistance) between the points A and B is determined, follows:

$\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$R_1 = 10 \Omega$, $R_2 = 100 \Omega$, $R_3 = 100 \Omega$ So $R_t =$

$= 10 \Omega + 13 \Omega = 23 \Omega$

An important point should be noted here: the equivalent resistance of a parallel combination of resistors is always less than the smallest resistor connected in parallel. The example which was presented above attests to the accuracy of this statement.

When

all the resistors connected in parallel between any two points are equal, the equivalent resistance or the total resistance of the parallel combination is simply the

value of an individual resistor divided by the number of resistors connected in parallel. Thus if twenty 1.5kΩ resistors are connected in parallel, the total equivalent resistance of the parallel combination (R_T) is found by:

$$R_T = R / n$$

where

where n = number of resistors

$$= 752$$

Series-Parallel Resistors

Finally, an example is presented below to show how to determine total resistance in series-parallel resistive circuits. Fig. 10 Series-Parallel resistance In order to determine the total resistance between the points A and B, we start from the extreme right of the given series-parallel circuit. It is noted that R_1 and R_2 are in series. So their series combination gives 2kΩ which is in parallel with R_3 , and also in parallel with R_4 . This is shown in Figure 11. Fig. 11 Series Parallel resistance

Capacitance, Inductance, Resonance and

Filters

The equivalent resistance R_T of the parallel combination of R_1 and R_2 is determined as follows:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_T} = \frac{1}{1.5k\Omega} + \frac{1}{2k\Omega}$$

$$\frac{1}{R_T} = 11.1$$

Now R_T is in series with R_3 as shown in Figure 12. Fig. 12 Series resistance Thus R_{AB} or the total resistance between the points A and B

$$R_{AB} = R_T + R_3$$

to

$$= 2111\Omega$$

Capacitance

A capacitor is an electric device that can store electrical charge. A capacitor consists of two parallel conductive plates separated by an insulating material called the dielectric. Figure 13 shows the schematic symbol for a capacitor.

to

Fig.13 Capacitor schematic symbol The ratio of electric charge Q of a capacitor to the voltage V across its plates is called capacitance C . Thus the capacitance C of a capacitor is: $C = Q/V$ where Q is the electric charge stored by a capacitor and measured in coulombs. V is the voltage across the parallel plates of a capacitor and measured in volts. The unit of capacitance is called a Farad (F). However, if

is quite a large unit, so its fractional units, called microfarad

(μF), nanofarad (nF), and picofarad (pF) are used instead. One

μF equals 10^{-6}F , nF is equal to 10^{-9}F , and pF equals 10^{-12}F . Several examples are presented below to show how various calculations related to capacitors are done:

1. A capacitor stores 100 microcoulombs (100 μC) with a voltage of 40 volts across its plates. Find its capacitance.

Is 2 Spr

2. Find the voltage across the plates of a capacitor having a capacitance value of 500 pF and storing 40 μC of electric charge at some 194 μs

C 500 pF 500 nTOU

3. Convert 1000 pF to μF . 1000

$$1000 \times 10^{-12} \text{ F} = 1000 \times 10^{-6} \text{ pF} = 10^{-6} \text{ pF} = 0.001 \text{ pF}$$

10* The amount

of electrical energy that can be stored by a ca-

pacitor is $W = \frac{1}{2}CV^2$ where W is the electrical energy in joules

(J), C the capacitance in farads, and V the voltage in volts.

Thus the energy stored by a .01 μF capacitor having 10 volts

$$\text{across its plates is } W = \frac{1}{2}(.01 \mu\text{F})(10 \text{ volts})^2 =$$

SpJ. The breakdown voltage or the working voltage, for a capacitor is the maximum DC voltage that can be applied to a capacitor without damage to the device. If this voltage rating is exceeded, permanent damage to the capacitor can result. The permittivity of a medium is the measure of ease with which an electric field can be established in the material. Permittivity is represented by the Greek letter epsilon (ϵ). The permittivity of vacuum is (ϵ_0) and

$$= 8.85 \times 10^{-12} \text{ F/m. The}$$

relative permittivity (ϵ_r) of a dielectric material is the ratio of the permittivity of the dielectric to the permittivity values for selected dielectric materials as shown in Fig. 14,

Dielectric Material

Fig. 14 The capacitance C of a capacitor is given by:

ϵA

d

ϵ_r

where C is the capacitance measured in farads (F)

ϵ_r is the relative permittivity of the dielectric material used in the capacitor

$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$

A is the area of cross section of the parallel plates of the capacitor in square meters (m^2) d is the separation between the plates measured in meters (m)

Chapter

As the distance between plates increases, capacitance decreases. As the area of the plates increases, the capacitance increases, and greatly affects the value. Both the distance between the plates and the area of the plates are used in variable capacitors to change value, capacitors are generally categorized according to the type of dielectric. The most commonly used dielectrics are ceramic, plastic film, and electrolytic.

Ceramic capacitors

have capacitance values in the range of pF to 0.1 μF and voltage ratings up to 50V. Ceramic capacitors are usually about the size of a dime, and are frequently marked with values such as 47pF may be marked as 47k, Plastic film capacitors have several different types.

Examples

Some of the dielectrics used for these capacitors include mylar, tantalum, and polystyrene.

Their capacitance

values are up

to 100,

F.

Electrolytic

capacitors have high capacitance values up to and even over 200,000 μF , however, they have relatively low breakdown voltage.

The capacitors

described above are fixed capacitors.

There

is another class of capacitors known as variable capacitors,

They are

used

in radio frequency circuits when changing frequency or tuning is required.

Athe

plates separate

or open, capacitance decreases.

The schematic

symbol for a variable capacitor is illustrated in Figure

15. The most common types of variable capacitors are air capacitors, trimmers, and varactor diodes, which are voltage controlled.

'i Fig.

15 Variable

capacitor schematic symbol

When

the capacitors are connected in series, the total capacitance

C_t is determined

by the formula: $\frac{1}{C_t} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$ where

$C_1, C_2, C_3, \dots$ are capacitance values

of the individual capacitors in series. When the capacitors are connected in parallel, the total capacitance value is the sum of all the individual capacitance values.

$$C_t = C_1 + C_2 + C_3 + \dots$$

$+C_4 + C_5 + \dots$

C_t where

C_t is the total capacitance and $C_1, C_2, C_3, \dots$ are the individual capacitance values.

'Two examples

illustrated below show how to perform calculations

Culations

related to the series and the parallel combinations

of capacitors, or T @

Fig. 17 Parallel

Capacitance

Fig. 16 Series capacitance

Capacitance, Inductance, Resonance and Filters

1. What is the total capacitance C_t of the series combination,

of capacitors shown in Figure 16?

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

So,

$$\frac{1}{0.026 \text{ F}}$$

2. Find the total capacitance

C_{total} of the parallel combination of capacitors shown in Figure 17.

$$C_{\text{total}} = C_1 + C_2 + C_3 =$$

$$0.02 \text{ nF} + 0.01 \text{ pF} + 0.05 \text{ HF}$$

$$0.084 \text{ F}$$

Inductance

Inductance is defined as the property of a coil of wire (called an inductor or a choke) that opposes a change in electric current. When there is a flow of electric current through an inductor, a magnetic field is established. When the current changes, the magnetic field also changes. An increase in electric current expands the magnetic field and a decrease in electric current reduces the magnetic field. Inductance or self-inductance is a measure of a coil's ability to set up an induced voltage across it as a result of a change in the electric current that flows through it. The schematic symbol for an inductor (or a coil) is shown in Figure 18.

Inductor schematic symbol Inductance L is measured in a unit called Henry (H). Inductances are usually measured in a unit called milliHenry (mH).

$$1 \text{ mH} = 10^{-3} \text{ H}$$

An inductor stores energy in the magnetic field established by a flow of electric current through it. The inductor energy is given

$$\text{by the formula } W = \frac{1}{2} Li^2 \text{ where } W \text{ is the energy mea-}$$

sured in Joules,

$$L \text{ is the inductance measured in henrys,}$$

and i is the electric current.

The physical

properties that determine the inductance of a single layer air-core coil are shown in the following formula: $L = \frac{\mu_0 \mu_r N^2 A}{l}$

$$\text{Where } N = \text{number}$$

of turns

$$T =$$

diameter

or radius of the coil spacing of turns or length of the coil L (inches)

As the length or spacing between, P , of the coil turns increases, the inductance

increases. The material around which a coil is formed is known as the core. A core made of ferromagnetic material permits the establishment of a stronger magnetic field. Examples of ferromagnetic materials include iron, nickel, steel, cobalt, or various alloys. Inductors are categorized according to the type of core material. Figure 19 shows three common types of inductors classified according to the core materials used. Fig. 19 Inductor schematic symbols Iron and ferrite cores increase the inductance of a coil. Iron core types have large inductance measured in base units of Henries and are frequently called "chokes." Ferrite cores are in shapes of round torroid coils or in radio frequency (RF) types that have adjustable moveable cores. Some RF coils also have aluminum or brass cores, which decrease the inductance, rather than increase like iron cores. These are typically used where a very small amount of inductance is required in VHF and UHF circuitry. The inductors described above are fixed inductors. Adjustable (variable) inductors have an adjustment that moves a sliding core in and out, causing a change in the value of inductance. Figure 20 shows the schematic symbol for a variable inductor. Fig. 20 Adjustable inductor schematic symbol When inductors are connected in series, the total inductance of the series combination is given as the arithmetic sum of

$$L_T = L_1 + L_2 + L_3 + \dots + L_n$$

where L_T is the total inductance of the series combination and $L_1, L_2, L_3, \dots, L_n$ are the inductances of the individual inductors connected in series. When the inductors are connected in parallel, the total inductance value L_T of the inductors in parallel is given by the formula

1/Capacitance, Inductance, Resonance and Filters

where $1/L_T$ is the sum of the reciprocals of the inductance values of individual inductors connected in parallel

Transformers

A transformer is a device that transfers energy from one circuit to another by electromagnetic induction. The energy is transferred without a change in frequency. The basic construction of a transformer is shown in Figure 21

Fig. 21 Transformer

The transformer core is a closed loop of magnetically permeable material. Two coils of wire are wound around the core. The primary winding is connected to an AC power source and the secondary winding is connected to the load. The primary winding produces a changing magnetic flux through the secondary winding which induces an AC voltage in the secondary winding. If the number

of turns of wire in the secondary winding are more than the number of turns in the primary winding, the voltage induced in the secondary winding will be more than the voltage in the primary winding. This is called a step-up transformer. If the number of turns in the secondary winding is less than the number of turns in the primary winding, the voltage induced in the secondary winding will be less than the voltage in the primary winding. This is called the step-down transformer. This relationship is expressed mathematically as:

$$V_p = V_s \cdot \frac{N_p}{N_s}$$

Vs Ns

where:

N_1 = number of turns in the primary winding

N_2 = number of turns in the secondary winding

V_1 , voltage in the primary winding

V_2 =

voltage in the secondary winding

a = turns ratio

The currents in the primary and secondary circuits are related as follows: $I_1 = I_2 / a$, where: I_2 , current in the secondary circuit

I_1 =

current in the primary circuit The apparent power at the input to the primary circuit is equal to the apparent power rating at the secondary. This is expressed as:

$V_1 I_1 = V_2 I_2$

The efficiency of a transformer is given as:

$\eta = \frac{P_{out}}{P_{in}} \times 100$

where, η = percent efficiency

P_{in} > input power

P_{out} =

output power A commonly used type of transformer is called autotransformer.

Figure

Figure 22 shows two autotransformers.

Fig. 22 Autotransformer

An autotransformer consists of a single winding that serves both as primary and secondary. Autotransformers can be used whenever both the input and output have a grounded line. Autotransformers are smaller, lighter in weight, and use less copper. The disadvantage of an autotransformer is that it does not provide the isolation of load from the power line.

Filters

Filters are frequency-selective electric circuits that permit signals of certain frequencies to pass from the input to the output while blocking all the other frequencies. A low-pass filter allows signals with lower frequencies to pass from input to output while rejecting higher frequencies. The frequency response of a low-pass filter is shown in Figure 23.

Fig. 23 Frequency response of a low-pass filter A high-pass filter allows signals with higher frequencies to pass from input to output while rejecting lower frequencies. The frequency response of a high-pass filter is shown in Figure 24. Capacitance, Inductance, Resonance and Filters V_v out Fig. 24 Frequency response of a high-pass filter

A combination of low-pass and high-pass filters can

be to obtain a band-pass filter. A band-pass filter allows

band of frequencies to pass and rejects all frequencies below and above the pass band.

The frequency response of a band-pass filter is shown in Figure 25.

V_{out}

frequency Fig. 25 Frequency response of a band-pass filter Series RC

(Resistance-Capacitance)

networks exhibit a frequency-selective characteristic and therefore function as basic filters. A low-pass filter is implemented by taking the output across the capacitor shown in Figure 26. A high-pass filter is implemented by taking the output across the resistor shown in Figure 27. A band-pass filter is realized by taking the output across the resistor in a series resonant RLC circuit shown in Figure 28.

R

To

Load

I

Fig. 26 A low-pass filter

Capacitance, Inductance, Resonance and Filters

C

ac

ss

Fig. 27 A high-pass filter L C

V_{Out}

Fig. 28 A band-pass filter The opposite of a band-pass filter is a band-reject filter, sometimes called a 'trap.' It is used to eliminate specific frequencies that may cause interference with desired signals. Fig. 29 Frequency response of band-reject filter

NOTES

Chapter Quiz

stores an charge measured in coulombs in a circuit causes a voltage drop from one end of the resistor to the other end dependent on lines of force or same as

a. 1 thousand ohms 1 thousandth of an ohm ϕ . 10^0 ohms 10000 ohms

3. Resistor has color bands orange, red, brown. It is 232 ohms . 3.2 megohms

2.3 kohms . 320 ohms pose

4, A circuit has one 500 ohm resistor in series with three 500 ohm resistors that are connected in parallel with each other. The total resistance would be

- a. 666 ohms
- b. 167.7 ohms
- c. 750 ohms
- d. 1000 ohms

5, The voltage rating of a capacitor. .

- a. must be factored into the reactance formula
- b. relates to the maximum voltage that should be applied to it ϕ . indicates the maximum current the capacitor can allow
- d. is the same as the power factor

6. Calculating the total capacitance of several capacitors connected in parallel

- a. is not necessary as they are not used this way
- b. causes a need to consider inductance factors
- c. is done by adding each of the capacitance values
- d. is the same as the power factor

7. Ferrite :

- a. is a capacitor dielectric
- b. is a common inductor core material ϕ . is the most common resistor coating
- d. causes faulty saturation of inductors j Capacitance, Inductance, Resonance and Filters

8. The term Henry refers to

- a. the inductive Property of a coil
- b. the inventor of inductors
- c. of LR circuits
- d. resonant frequency of tank circuits

9. A certain transformer has 250 turns in the primary winding and 1250 turns in the secondary winding, turns ratio is

10. When the turns ratio of a transformer is 10 and primary AC voltage is 8V, the secondary voltage,

- a. 80 Volts
- b. 0.8 Volts
- c. 8 Volts
- d. 64 Volts

11. Which of these color code patterns is found on a

270.5% resistor?

- a. Red, violet, brown, silver

- b. Red, violet, brown, gold
- c. Red, violet, black, silver
- d. Red, brown, black, silver

12. One microfarad is equivalent to

- a. 10
- b. 1000
- c. 100,000
- d. 1,000,000

13. The capacitance of a capacitor with 100C of electric charge and 100V across its plates is

0.1pF 100F

ANSWERS on page 223

a Pa) Sah ads <apter 6 by John Ufer, CETsr John Ufer. CT Ter, has taught electromice ae am adnumet metructor at Northwestern Michigan College fir AC Circuit Operation?

development for Sears, Syntec Inc, American Natural Gas, and Cone Drive Corporation His background cal service manuals for the gaming industry. He is currently teaching a Business Computer Technology program ata prrann facility seated om Northern Michigan.

Series

RLC Circuit AC circuits can contain four different types of basic a resistive component, electromagnetic component, RI Seid component, and a semi-conductive component.

aN perv' AC circuit, any one or all of these elements can be

CI found.

i 'lement of

a circuit, i When trea how allthe conework. LI may require the understanding © Se cledi reat

iH ish a speedy and efficient repair. togetherto accompl ° Voltage Addition within In this chapter we will examine AC circuit devices and a Series RLC Circuit - uit actionsfrom a technician's viewpoint. circuit action idwster v,) Figure 1 shows a series AC circuit.

Inthis

circuit acommon current is shared by all components, with a division of the source voltage by the various circuit devices. Individually, the voltage drops of each device may exceed the supply voltage put, added together, all voltage drops combined will equal the supply voltage. The total impedance Z , is a result of the ap- plied voltage divided by the circuit's total resultant current. You can also find total impedance in a series circuit from the formula:

$$Z = \sqrt{R^2 + X_{net}^2} \quad \text{Characteristics of a RLC Series Circuit}$$

$$X_{net} = \text{total series reactance}$$

Division of Voltages

One Common Current

Sharing of Energy between the C and L components In the series circuit shown in Figure 1, if in this circuit there is a single frequency applied, the resultant voltage phaseangle and component voltage drops around the circuit would not change. In circuits that have various frequencies applied, the resultant addition of voltage drops and circuit current would changebased on the applied frequency value. This is the main difference in looking at circuit actions with AC, rather than an applied DC voltage.

Resistive (V)

Resultant V ;

Capacitor (V,)

If X_L is Larger than X_C , then X_L Subtracts from X_C ,

Figure |

Reactance will vary within the capacitor and inductor as the frequencies applied to the devices change. The change in reactance is directly proportional to frequency changes in the inductor and inversely proportional to changes in frequencies applied to the capacitor.

Inductive Reactance is $X_L = 6.28 \times \text{Frequency} \times L$

Capacitive Reactance is $X_C =$

$\frac{1}{6.28 \times \text{Frequency} \times C}$ Frequency goes up, then X_C changes up. Frequency goes down, then X_C changes up.

AC Circuit

Operations

Chapter 6

The effect of changes in reactance of the two components will react in changes in total circuit current. The reactive component voltage (V_{OT} V,) each subtract from each other, the reactive components would add at 90-degree phase shift with the net reactive component voltage. Inductive voltages, capacitive voltages add to their capacitive voltages. The inductive and capacitive voltages subtract directly from each other. The net reactive and total resistive components add at a phase angle of 0 degrees from each other. A maximum circuit current would occur when X_L is equal to

Maximum

is called Series Resonance. X_L In a series circuit this

Frequency

of Resonance

$f_r = \frac{1}{2\pi\sqrt{LC}}$

VLC In a series resonance circuit the frequency at which both reactances are equal is based primarily on the circuit component values of the inductors and capacitors within the circuit. The bandwidth frequency response of the Figure | circuit on the previous page (below: or above the resonance frequency) is a factor of the resistive components of the circuit and the Q factor of the inductor. The higher the non-reactive resistance seen in the circuit, the broader the bandwidth response of the circuit. Changes in circuit frequency from the resonance frequency will make the circuit voltages more inductive as frequency changes above the frequency of resonance and the circuit will have higher capacitive voltages as the applied frequency drops below the frequency of resonance.

L = Henries of Inductance

C = Farads of Capacitance

In a series AC circuit containing capacitors, inductors and some level of resistance, circuit current is based on the sharing of potentials developed across each device. The capacitor electrostatic field contains a phase difference from the inductive component by 180 degrees. As the voltage across the capacitor goes positive, the voltage developed across the inductor will be negative. The two components can share energy as loading conditions reduce each field's energy level. Current demands on inductive and capacitive devices will reduce the back-EMF developed within each device, causing additional current to flow through the circuit. Most series circuits in electronic devices drive output devices. Output devices in series with reactive components increase peak current demands and may have short time periods in which to use the stored energy within each device. Power needs to be available to the output device sometimes within a fraction of the source wave time period. Capacitors and inductors working together can supply energy during low voltage wave times of the source. The current limiting factor at resonance is the circuit impedance in series with the filter components and the

internal resistance of the inductive device. From a troubleshooting standpoint, an open in the inductor or capacitor results in an inability for either device to share energy from the electrostatic field of the capacitor

or electromagnetic field of the inductor. A short in either device will result in additional circuit current and could result in loss of current to an output device if the output device uses the circuit energy in short duration peak demand cycles. The source voltage should appear across the open with no circuit current in or through the device. Because the devices are using signal current (in many cases) to drive the capacitor or inductor, a lack of output signal may indicate an open in this circuit. If you measure across the open you may "see" the signal voltage. The best measurement is at the output to ground rather than across the device. Comparing the input signal to the output signal level from ground may show you where the open exists. Parallel RLC Circuits In the RLC parallel circuit, the common factor that is applied to each device in the circuit is the total applied circuit voltage. Each device, or series group of devices, has its own current path from and back to the source voltage. The reactance or resistance of each circuit component will limit the current through the device. The true total inductive or capacitive current is based on the subtraction or addition of each reactive device's current. Capacitive currents add to other capacitive currents, and inductive currents add to other inductive currents. Capacitive currents subtract from inductive currents. When you add or subtract currents in a parallel circuit & voltages in a series circuit, you must do so in a uniform way. Negative or positive results do matter, especially when dealing with more complex circuits. For the technician, use of schematic diagrams can be invaluable in "seeing" circuit actions and providing you with critical voltage measurement levels. In a parallel circuit, current is shared between the capacitor and inductor. The phase difference between the inductor and capacitor are 180 degrees apart. As a result, a capacitor is developing a positive voltage across itself while the inductor develops a negative voltage. Current flows from a difference in potential. So currents will flow from the electrostatic field of the capacitor to the electromagnetic field of the inductor if a difference in potential exists. When the inductor and capacitor circuit devices have the same level of energy in their magnetic and electrostatic fields, the circuit is at parallel resonance. Total circuit current is at a minimum; circuit impedance is at a maximum.

AC

Circuit

Operations

When the fields have the same level of potential, there is a difference in potential, and current stops flowing between the source and the devices. A voltage is present, just no additional current flows out.

$$I_T =$$

Voltage + Inductance only the out-of-phase currents

Inductance + Capacitance represent

4 Parallel

RLC circuit has:

+ Separate

conductive paths for current flow through the circuit. ; : /

+ Supply

voltage potential that is supplied to all parallel circuit devices. + Current that is shared by the inductive and capacitive components. , « The electrical potentials of the electromagnetic and electrostatic fields in the circuit that are 180 electrical degrees apart. + Minimum source circuit current flows at the parallel resonance frequency. Q and Resistance effects in Parallel RLC

Circuits

In a parallel circuit, internal resistance in the coil can make a difference on the selectivity of the bandwidth

Response

of the circuit. This is why it is critical when tuning a coil or troubleshooting defects in frequency responses that you do not think that, because an inductor is not open, it must be working correctly. As the reactance of the inductor or the reactance of the capacitor changes, the circuit allows current to flow out of the capacitor circuit and to the load as a result of these changes. x Q

BW=

For The level of this output current is based on the interactions at specific frequencies of both devices. Shorted turns on inductors or leakage through a capacitor's dielectric can cause major changes in the way these devices share energy, and how they react to energy demands at specific frequencies. For the technician, response of the circuit, or distortions in output signals, can be a result of a malfunction of a circuit component rated value. These types of problems are not always caused by just a short or open in a device. Formulas for RLC Circuits

Vector Additions

In a series circuit the variable quantity is voltage. To find the unknowns of a series circuit you would first find the total

Chapter 6

impedance, then the circuit current, and from there. the voltage drops around the circuit. As a technician, this should help you have a better understanding of how a malfunction would cause changes in circuit readings. See eeeeeeeeeeeeeeneneeee

a Reactive net total difference between the addition

of all inductive reactance minus the addition of all capacitive reactances

« Remember -

$$X_L = 6.28 \times \text{Frequency} \times L$$

$$= X_C =$$

$1/6.28 \times \text{Frequency} \times C$ L is in Henries and C is in Farads In a series circuit: (Current in a series circuit is the same through each component.)

$$R_{\text{net}} =$$

the addition of all series resistive components

$$\text{Source } 120V - 30KHz$$

$$R_{100} 8K\Omega C_{200} 40pF I_{300} 20mA$$

$$X_C = \frac{1}{6.28 \times \text{Frequency} \times C}$$

$$6.28 \times 30 \times 10^3 \times 40 \times 10^{-12}$$

$$\times 10^3 = 132.6K\Omega$$

X_L

$$= 6.28 \times$$

$$30 \times 10^3 \times 20 \times 10^{-3} = 3.7K\Omega$$

$$Z_T = \sqrt{R^2 + \text{Reactance}_{\text{net}}^2}$$

$$Z = 124K\Omega$$

$$= V_Y Z,$$

si

$$I_{me} = 9.7 \text{ mA}$$

$$\tan \theta = B/A = 9.5K / 2/8KQ = -55^\circ$$

CIRCUIT PHASE

ANGLE

$$\sin \text{Angle} = B/C$$

Cos

$$\text{Angle} = A/C$$

$$B \tan \text{Angle} = B/A$$

The analysis is based on a common frequency being applied to all components. If the frequency changes, the total circuit current and each reactance would change with it.

'laa . . « te Geant ta the Het (vires ee ree \ . aoe yee § % AT Vo ¥h yee Povenel

Crew

ay cater the EWTCE voltage is applied to each

oy hee components aC current and imped- hared (subtracted) between aTets RC jenny And CAPACHOT components

Inductor

ts add a currents. capacitor current arent Resistive elements are in sync with supply currents. All resistor branch cur- branch currents.

t nage

ecurstve

at

NO aw ae * verefeen thet

Souree

Voltas a ego

PTAC

ioH Cre ioopt

Ry - ON

x vane gan 17210 10° SRD ae 6.28850 x 10) (1005107)" 31.8K2 aA Ri 400mA

$$T = I_{ad}$$

$I_{1,2} V_X$ 3TH LP represents only the
 I, V_X ,

$$I = 38.2 \text{ mA out-of-phase currents}$$

$$I = [I_{re} + I_{ear}] = 401 \text{ mA}$$

sex

$$I = \text{Difference between the Inductive and Capacitive}$$

as

Currents

impedance of an AC Circuit Impedance is the total opposition to current flow in an AC circuit; it has two components- magnitude and phase.

$$Z, =$$

fe,

I, AG Circuit Operations Parallel RLC Circuit

Figure 2

$$V = V, , 7$$

487, T a 12 R? In Figure 2 the current through the inductor the current through the capacitor, so the net cu circuit is a vector addition of the resistive curr resultant inductive current - (I, I,) - ve branch was ec If the current in the inducti a the circuit woul rent in the capacitive branch, parallel resonance and the circuit current wou current through the resistor branch section anc leakage current through the reactive compone resonance, circuit current is at a minimum. C circuit is a) Another name for a parallel Li lectrostatic fir tank energy from the capacitor & the energy from the inductor's electromagneti the circuit to share current between the induct The bandwidth response of the tank at resona the DC resistive characteristics of the inducto tive devices in the circuit.

Parallel Circuit Phase Angle F«

$$\sin @ = \frac{\text{Net Reactance}}{\text{Current}}$$

Total Current in the circuit cn cnEE EETEEIEEDEIEn

SESS

Cosy Curent in the Cieu

_ Net Reactive Current

Resistive Current

of Series and Parallel Ci combinations el Circuits

Or

Output

Anant

Figure3

combination of a series and parallel configura-

f

This

isan example of the inputchoke (inductor) ton is SHOW work usedto helpreduce lightningand power and capacitor
"icircuits.This configuration is very common surge ann This circuit has the characteristics

of inae resonant andparallel resonant circuit. potha seri inputfrequencyincreases (suchas with a electrical noise),the inductor's
reactance in- itor reactance decreases. Energystored harge through the capacitor with isola- by the resistor to ground. in
Figure

htning strikeoF

we whilethe capac on ss inductor Will discharge or andcurrent limiting tion ACPassive

Filters

use for the series-parallel configuration

Soak

The combination ofa series and parallel ac js usedto pass signals

of specific frequencies

ee acetitd is provided for those frequencies outside ped resonant frequencies. L1o and C1o provide asseries

= bandpass

filter while L2o and C2o provide a parallel inti bandstop circuit. L1o C1o TTI} C2o I2o

Figure

At specific frequency ranges the series circuit will pr «the Fesonance frequencysignals, while the parallelcircuit will block the
flow of these frequenciesto ground The 1o and L1o series components will attenuate those frequenciesthelow or above the
resonance frequency. 2o and C2o will aid in conduction of frequencies ahove and helaw the frequency of Tesonance to
ground. Each circuit component will have an effect on frequen- Cies passedthroughthe filter outside the bandwidth of the
frequency of resonance. As frequency goes up. capacitive Teactance will dropwithin C1o and C2o. L1o and L2o induc- tive
reactance will increase as frequenciesrise above reso- nance frequency, The inductive reactance will increase, while the
capacitive reactance decreases. At frequenciesbelow the frequency of resonance, the capacitivereactance will increase in C1o
and C2o, while the inductive reactance of L1o and L2o will decrease. At resonance the series components L1o and C1o will
have their smallest impedance, and L2o and C2o will have their maximum impedance to the signal frequency.This type of

combination allows set bandwidths of frequencies to pass or be conducted to ground. Signal processing is a very common use for these devices within highly integrated circuit applications. Use of a scope and voltmeter can easily locate faults in these filter circuits if you know what signal characteristics you need to find, and what levels of signal should be present. Basics of AC Devices with R, L and C Let's look at how AC circuits function from a technician's viewpoint. See Figures. When approaching an AC circuit troubleshooting problem you need to have a good understanding of how devices function under various frequency characteristics, as well as various circuit configurations. Some devices can defy logical thinking (such as capacitors). How does a capacitor really work? One mystery to many is how can you couple (pass) an AC signal through a capacitor. The dielectric of the capacitor is an insulator that will block current flow, so how does the input signal reach the output side of the capacitor if its path in the circuit is blocked by the capacitor? - Vv

= Circuit Current will rise and

fall C1

Rt

Figure 5

C1 will charge to 24 Volts - The AC source will increase and decrease the voltage drop across R1.

How can a capacitor stop the flow of DC current through the device, yet allow the flow of an AC current in the circuit? The answer is simply to understand that current flows from a difference in electrical potential. When the voltage of the signal is applied to the dielectric of the capacitor it creates a difference in potential in the dielectric material inside the capacitor. The stored charge induces a reflected voltage across the capacitor through its plates to the circuit it lives in. The capacitor dielectric stores charge, which is why some capacitors can hold a charge even when power

is removed. The capacitor dielectric's difference in potential causes the circuit to have a reflected difference in potential

in the

ri

the voltage level within the capacitor, current follows paths that are felt by the circuits connected to the capacitor's leads. So how can this help the technician?

Suppose

a capacitor develops

an open in the lead of the device. The capacitor will never develop a charge in the dielectric, so no charge - no potential difference in the conductive paths connected to the circuit. You would measure an input signal to the capacitor and may even see a small signal at the output of the capacitor. With a smaller potential difference into conductive paths from the capacitor, the circuit driven from the output side of the capacitor may reduce or stop its operation.

Power Use - True Power & Apparent Power

Capacitor devices and inductive devices have two distinct characteristics not shared by most purely resistive devices. Inductor components will develop opposition to the flow of current by the creation of a magnetic field that will oppose the source current. Capacitor devices develop electrostatic charge, which will oppose the source, as the voltages are developed directly opposing the source voltage. Resistive device applies a direct relational resistance to an applied source. Secondly, capacitors and inductors store energy and then return it to the circuit. The electrostatic field held in the dielectric of a capacitor has a potential voltage in an opposite polarity to the inductive device magnetic field. When two sources of stored energy are connected in a conductive path, the voltages will either aid or reduce the effect of the energy stored in each device. The inductor may be a motor, with big current demands when it first starts to turn. In the motor circuit could be a capacitor with

stored energy in its electrostatic field to aid in applying energy to the motor. The source voltage has a lead "90 degrees before it can supply a maximum potential into + motor. So when the motor inductor demands are at their k, the motor's supply voltage is at 0V from the source. The AC Circuit

Operations

capacitor voltage is available at this time to aid the inductor, its energy demands. True power is the actual power used in the circuit to

Re work out

of the circuit devices. This comes from the source, and is always equal to current squared times the resistance seen by the source. Apparent power is the supply Voltage times the supply current times the power factor. Transformers, motors, and other inductive devices are rated by the VolAmps, (VA rating) they use. They not only consume power but supply power that is then used to create work by motors, lighting circuits, and other devices driven by inductive circuits, Power factors can be adjusted by adding capacitor banks to inductive circuits to bring the power factor closer to unity. Unity power factor of 1 allows maximum power to be used. The higher the power factor, the more in-phase the voltage ; current are with respect to each other. To efficiently use power you would want the source to be able to directly supply energy as needed to each device. When a perfect balance of sharing power takes place, the circuit has a power factor of 1.0. A power factor below 1 indicates an imbalance between the apparent and real power of the circuit, The technician must be able to look at circuit devices and "see" how the devices function together to produce the real power needed, not just the apparent power of each device, independently.

AC Circuit - Real Power = True Power

$P_r =$

Apparent power

$= I^2 R =$

(or $V \times A$)

$P_r, = \text{True Power} = IR =$

$(VA) \times \cos \theta$

P_r (True Power)

$PF =$

Power factor = $\cos \theta =$

P_a (Apparent Power)

R (Resistance)

PF also equals $PF =$

Z (Impedance)

In the Figure 5 circuit, the capacitor gives the motor additional starting current. This can be as high as four times its Tun current, Sensors detect a slow motor start and shut the mo- tor off to protect it from overheating. You need to "see" how the capacitor helps the motor at Starting times to repair this circuit. Capacitors can supply energy to an inductor field, Analyzing an AC Circuit If you find changes in voltages in series sections - remember

ye capacitor stores signal voltages in an electrostatic e 5 Reld n the capacitor has leakage within the device's dielec re. its phility to store signal energy is reduced. An open in the ot reduce, signal energy from being shared thot a ee capacitor call stop with another component In parallel circuits. selectivity is the key factor. Distortion of the FM envelope. of passage of unwanted signals would be an indicaton of problems.

This

may result in largespikes that can destroy sensitive electronic devices, as occurs in TV horizontal output or high power amplifier circuits.

The Whole Picture

an RF (Radio Frequency) Signal Circuit: A an AC signal out into the air. It travels across asize to allow the wavelength of the signal

an a ily resonate across the antenna andinduce signal wer

m which travels into the receiver. The signal contains a ier signal and a modulated signal with the encoded trans- mitted information. The signal must pass throughcircuits and/ ateblocked, or shunted to ground, by other

AC circuits. Due to the extreme low level of voltage of the signal, the signal must also be amplified. Lets start with transmitter sends The signal will travel into a series ACcircuit tuned to spe- cific frequenciesby the values of the inductor and capacitor. Resistors are used to set levels of current and biasingofother circuit devices such as transistors or IC chips. The series com- ponents allow specific bands and bandwidths of frequencies to pass through to the next stage. Connected across the signal pathmay be parallel tank circuits designed to take to ground signals outside of the selected bands and block the flow

to groundof specific frequencies passing through the series com- ponents. As the signals are amplified and selectively pass through the circuits, any unwanted signals are shunted to ground, and the final clean signal is sent to the final amplifiers, such as video or audio amplifiers. When troubleshooting, you need to know what is passed in each section of the circuit and what level these passed signals should be. Substitute where needed to isolate faults, and watch closely for any changes in voltage measurements from those supplied on your schematics. For example, you measure a positive voltage where you should have a negative voltage. This may be an indication of an open or short in a capacitor in the circuit. Why a capacitor? They are the most likely culprits. AC Circuit Configurations AC circuits are used to amplify an AC waveform, or create

AC CircuitOperations

Chapter 4

A MALE ape Oth ae ay nie Maring Coe gneit en torment onfrequency phase adynis ene

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te pe ailians feel MPplies AC creme ae cee for many additional cirenit chara Dee ee

ae on erate ve atechinics view, understand when Working in an 7 oma Crete crineal ency. shape, magnitude and phase of y
troubleshoot. Avoid th S iz Avoid the "Got Some' syndrome - understand that frequency 19 critical im onder to ros energies
through an AC circuit as well as magnitude A a Penvoltage may just be a signal being supplied around a defective circuit
through other components and conductive paths.

NOTES

-00000000000000000000

Chapter 6 i

Chapter Quiz

4, Total impedance In series circuit is found by: adding up all the series resistance and reactance all the inductive reactances from all the finding the net reactance each separately. add them to require Total of the same the applied voltage components AC circuit with a resistor, inductor and same resistance or reactance, & voltage drop by the addition of all the currents, 2 In a capacitor, each with the voltmeter each individual would find:

- a. the voltages equal to each other.
- b. the reactance voltage of the capacitor or the inductor be greater than the resistance voltage measurement. the resistance voltage would be higher than the voltage across the or the capacitor. 4 the supply would be measured across the resistor and the voltage across the capacitor and across the inductor would measure OV.

3. Which of the following is not a characteristic of a series circuit?

- a. components have a division of the supply voltage.
- b. All components share one current. Energy is shared by reactive devices.
- d. There is a different level of current in the reactive components than in the resistive components.

4, In an AC circuit, as the applied voltage frequency changes upward. the inductive reactance in the circuit will: a decrease.

- b. increase. c. stay the same.
- d. either increase or decrease. In an AC circuit, as the applied voltage frequency increases the capacitive reactance in the circuit will: a decrease.
- b. increase. c. stay the same,
- d. either increase or decrease.

ANSWERS on page 223

Impedances

6. A 50 KHZ AC voltage is Applied to a Reactance at this frequency is

- a. 3140 Ohms.
- b. 300 Ohms.
- c. 3100 Ohms.
- d. 300 Ohms.

7, capacitor in a circuit with a frequency,

50 KHz applied has a reactance of: X_C

- a. 1.59 Ohms.
 - b. 159 Ohms.
 - c. 159 Ohms.
 - d. 15.9 Ohms
- 8 In a series circuit when the inductive reactance, the capacitor reactances the circuit has an impedance.
- a. maximum
 - b. double the inductors reactance, minimum
 - c. double the inductors reactance, minimum
 - d. no

9. In a parallel circuit when the inductive reactance, the capacitor reactances the circuit has impedance.

- a. maximum
- b, double the capacitor's reactance c. minimum d.no

10, Resonance is when

- a. all resistances in an AC circuit are equal to each other,
- b. all reactances are equal on each reactive component c. the total of the inductive reactances in the circuit equal the total of all capacitive reactances
- d. is flowing through an AC circuit when a voltage is applied

Generators, Alternators and Motors

Captain James Bruening, CET Pe 3 by chief the US Navy as an Aviation Electronics Technician Chief

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as a Senior Serre link: aobtery comipbien, and electronic Im his 20 yeury Of Naval service MAU #3408 ey ters

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James

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da ciel I, he has a BS in Electronics Fechnology(Magna (aim Uauate) from

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Figure 1 -

The "Leg Hand Rute for Gen .

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Alternators

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C Generat

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i Ore, the reversin current flow ind eo ae flow and "euddle finger of the left in the rotating conductor Yields our AC output,
Ve econ forefingeran other to determine ee tu licular to ne cuarieate THE thes The basic componentsof an alternatorae the
Feld,the een between motion, rie of Figure ! should eam, the slip rings, and brushes.Alt,

HOTS

typically use bose 'andacompanyiNg

el

Sctromagnes

"o Set up the magnetic field through which the ats is concept. Conducting loops are fotated. The wi

h ; ing this Con fi - RS Which

q

is visualizing

is primarily to illustrate this magnetic field are called the field Coils), of simply, the Left Hand Rule is applied. The armature is rotated within the field. The armature contains the windings in which the AC current is induced if a conducting loop is rotated. The output is taken. In most alternators, the induced voltage can be directly

H

the generated reverse electromotive current is known as Controlled by adjusting the field Coils' excitation voltage,

ing continuously

ns This is an (AC). The slip RINGS provide electrical contact from the

Totating

oo Current loop to an external circuit via the brushes. The slip rings may

F Provide current to the rotor

Figure 2 - Simplified AC Generator part of the alternator from The armature in an alternating-current stationary windings. field, are known as stators are known as the stator. Remember, the armature

is the which the output voltage is taken. rotor may be the rotating windings of the The rotating windings, be they armature or of the rotor. The stationary windings; Three-Phase Power - Older automobile generators (pre 1960) are single phase machines. The output taken from the armature was a single sine wave. However, the most common AC power is produced in three-phases. Three-phase power consists of three distinct sine waves, each lagging the phase before it by 120 degrees when viewed on an oscilloscope.

AB 2000

8-C Ovs

CA 2000 AM Te A-N 12000s aN Tes BN

woke CN 2 cans C-N (2000)

Figure 3 -

Three-phase power connections Three-phase power can be manipulated to obtain the desired voltage level and the desired number of phases by altering the electrical connection of the stator windings. For three-phase power, the stator is used as the armature. The two basic methods of connecting AC power are Wye or Delta connection of Generators, Alternators and Motors and

the stator See Figure 3 for an

Mustration

connections

of three 4 DC Generators - Recall that an AC current is taken from slip rings. In a DC generator, the

i

is placed with a commutator The commutator consists of a ring, cut into sections that are insulated from each other and that encircle the shaft of the rotor Each section

any commutator is connected to one end of the field winding

of the "loop" such that a complete current path is provided by each section of the commutator to another, via the

Conducy

om

mg The effect of commutation is that the altemating

or the rotating loop is picked off the commutator as Volta, 8 singlepolanty, i.e., Direct Current (DC) Figure 4 ile a basic commutator and its effect on gencrator output 'ty

a

Figure 4 -

Simplified DC Generator DC Generator Output Ripple - The single rotor loop De generator in Figure 4 is for illustration only. A single rotor loop would produce an output that looks like the top signal in Figure 5. In practice, the rotor will consist of many windings to produce a cleaner DC signalwith greater voltage potentia} and less ripple. The bottom signalof Figure 5 illustrates the effect of addingarmature coils. Generator Self Excitation - Generators, both AC and Dc, are often constructed such that the field excitation voltage is provided by the generator itself. This is accomplished by wiring the field coil in series with the armature, shunting the armature (in parallel), or wiring the field as a compound com- bination of both series and shunt.

The effect of multiple

ee oils ina DC generator ature © are basically wired as ilhustrated

d ert eldcoil generators are not

Practica

en oo 'vanes with the load curren!

output

VO

4a

- Compound

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Reid

ited generator 6- Self-excit/

Pe dd configurations

i hunt the armature with rs basically s| ot ¥ ane figure 6b. Shunt wound fielda sh4 coil(s)- Re! e of the generator to vary inverse yot "a output onewound generators are also not pi caused current. th load be s reason.

i enerators are a practical method ofgen- - The field coils are

a

in

E anna ifexct

ing the undesired ow - aren and og mo for an illustration of acompound dj cancel. See tions

aoand genet - Most el' lectric machines experience losses re Losses tcl hi wanted conversion of usable energy to heat. the un dyelo. nerators, Alternators and Motors

-- Ge Chapter 7

These losses are known as Copper loss, eddy current loss, and hysteresis. Copper loss in the windings of a generator or a motor occurs because heat is produced whenever an electric current flows in

a copper wire. Copper loss occurs when some of the energy of the current escapes in the form of heat.

Motors

A motor is an electric machine that converts electrical energy to mechanical energy. Recall that magnetic Poles of Opposite polarity attract each other, while poles of the same Polarity repel each other. Motors use Magnetic attraction and repulsion to do work. The magnetized by the poles of the field. DC Motors - A DC motor is constructed exactly like a generator, except that current is supplied instead of taken from it.

Current

the armature windings with each revolution, the current is passed through the DC motor commutator. This reversing action of the current in the armature coils causes the magnetized armature to reverse polarity as it rotates through the field. It is constructed exactly like a generator, but the direction is reversed.

in the rotation of the armature

as

The commutator elements are placed such that the

"a polarity attracts and repels the field polarity at the moment needed to promote rotation, on configurations - DC motors, like a generator, have the field windings in series and/or in shunt with the armature. This allows the same

Also

current to magnetize the armature and

to set

up the field the field strength of a motor can be adjusted by adjusting field voltage. The series wound

DC motor's speed varies greatly with the mechanical load placed on the motor. With

no load applied, the series wound

DC motor will speed dangerously out

of control. For this reason, unloaded operation of a series wound DC motor is not recommended.

Series

wound

DC motors have very good starting torque, making them ideal for use

in small battery operated appliances and portable power tools. The field strength

of the shunt wound

DC motor is not dependent upon the physical

Load

on the motor

as it is in the series wound motor. This is because the field current is in parallel with, and thus independent of, the armature current.

Shunt

wound motors have less starting torque than series wound motors, making them less attractive for applications where large torque from

a dead stop

is desired. Compound wound DC motors enjoy the characteristics of both the series and shunt wound fields.

A compound wound motor has strong starting torque (from its series wound field windings) and some speed regulation regardless of physical load (from its shunt wound field windings). See Figure 6 for

summary of series, shunt, and compound connections (note that instead of an output voltage, a motor will have an input Voltage). DC Motor Starting Current - The starting current of a DC motor can

be

as high as

percent of its rated current. For this reason, resistors are added

to the armature circuit during start-up, limiting armature current to rated values. These resistors are removed from the armature circuit either manually

or automatically as motor speed increases. Some external resistance may be left in the armature circuit to control final motor speed. Figure 7 illustrates motor current versus speed. RDN HM ON mm oo

Meter

Speed

Figure 7 - Motor current vs speed

Generators, Alternators and Motors AC Motors - AC motor speed depends

on the voltage supplied to it. This makes AC speed constant for many applications. In a single phase AC motor, the speed is directly dependent upon the frequency of the supply voltage. The alternating current causes the armature to alternate, promoting rotation as discussed previously.

In the case of DC motors, multiphase AC motor operation requires an external rotating magnetic field to be set up by the stator. Remember that the stator windings are stationary and the rotor will rotate. The primary power to the motor is a multiple

phase AC signal that is applied to the stator and creates a field that rotates around the rotor. The rotor can be provided with external power to turn it into an electromagnet. The rotor follows the rotating stator field similar to a

Compass

allowing a magnet rotated around it.

The most common types of multiphase AC motors use three-phase power. Each phase of the supply voltage is applied to one of three sets of windings in the stator, producing the desired rotating field. Types of AC Motors - Three types of AC motors are the series AC motor, the synchronous motor, and the induction motor. Series AC motors are wired similar to series wound DC motors. These motors are single-phase machines and their speed is directly dependent upon the frequency of the supply voltage. The series wound AC motor has strong starting torque just like the series DC motor. Having strong torque from a dead stop, series AC motors are ideal for fans, small appliances, and power tools.

Synchronous

motors are most commonly three-phase machines. As such, they rely on a rotating magnetic field from the stator to operate. Synchronous motor speed is synchronized to the speed of the rotating magnetic field and is thus constant regardless of physical load. Without assistance, a synchronous motor has no starting torque. This is overcome by wrapping the rotor windings in a conducting cage known as a squirrel cage. Enough current is induced in the squirrel cage to nudge the rotor up to a speed at which it can catch hold of the rotating magnetic field. See Figure 8 for an illustration of a rotor squirrel cage.

squirrel-Cage

Winding

Over

Salient Pole Windings

starting synchronous AC motor see picture

These are the most common type of AC motor. The rotor windings of an induction motor are in a squirrel cage. The current in the cage provides all of the rotor polarity to run the motor. The induction motor rotor gets its supply from the stator. The rotor in an induction motor, not

electrically

connected electrically to any external voltage, needs no end or brushes: unlike an induction motor is first powered up, the rotating magnetic field crosses the conducting loops of the squirrel

cage motor, This induces a current in the rotor that

causes the rotor to rotate in the direction of the field rotation and it soon catches up to the rotating field. The rotor eventually catches up to the rotating field because to do so it polarizes the rotor (remember that a conductor must cut off flux for a current to be induced). The difference between the speed of the rotating magnetic field and the speed in an induction motor is called slip. An induction

motor is proportional to the physical load placed on the motor. of AC induction motor is a single phase machine

It uses a capacitor and a starting coil in series to create a starting circuit. This starting circuit is wired in parallel with the main field winding. The combination of resistance and capacitive reactance in the starting circuit produces starting-winding currents 90 degrees out of phase with the current in the main field winding.

The two-phase rotating field which drives the

motor. Additional Info: The result of Lenz's Law: Lenz's law explains the need for greater effort from the prime mover when a generator is heavily loaded. The law of electromagnetic induction states that a conductor in motion in a magnetic field will produce a voltage in the conductor. The left hand rule applies to generators. In a generator's output is coupled via slip rings. In alternators, the output voltage can be controlled

by varying the field winding voltage. Induction motors have slip in order to operate. The amount of slip is

100% Generators,

Alternators and Motors

A commutator

of AC In a motor, Counter emf The field Strength of a

Subtracts

from the field vol Motor can be

Y

Voltage

ad) Husted by adjusting pecvoltage.

An induction motor

m A capacitor start AC

lust have sp in order to =

Magnetic field.

NOTES

a ee

Chapter7 |

Chapter Quiz

r cells are prone to battery memory: a Zim b NiCad

Alkaline

Lithum

motion through magnetic lines juctil in duced in it. The physical law that of flux has a current in explains this is what?

- a. Lenz's Law The law of inertia
- c. The Compton shift equation

4. Karchoff's current law

3, Am AC generator is also known as a

- a. regulator
- b, altemator
- c. phasor do shpmng ed from an AC generator

or

4, Three-phase power is obtained fro by wiring the armature windings in a configuration. a alpha, beta

- b. treble, in-phase c. polymorph, one-to-three d delta, wye

5, What element of a DC generator distinguishes it from an AC generator? 2 Slip rings

- b. Commutator c. Series wound field coil
- d. Armature a Generators, Motors

- 7. AC motor speed dependson
 - a, the frequency of the voltage applied to it
 - b. the voltage level applied to it
 - c. the phase of the voltage applied to it
 - d, the power left in the dilithium crystals
- 8. Series wound AC motors have strong torque from dead stop. This makes them ideally suited for use in small appliances.
 - a, True
 - b. False
- 9. What element of a synchronous AC motor provides startup torque for the motor?
 - a. Slip rings
 - b. poles
 - c. Squirrel cage
 - d. Armature
- 10. An AC induction motor requires what to operate?
 - a. Input power
 - b. Slip rings
 - c. Shp
 - d. Rotor battery

ANSWERS on page 223

6. What type of motor will speed dangerously out of control if not physically loaded?

- a. Induction motor
- b. Shunt wound DC motor
- c. Series wound DC motor
- d. Series wound AC motor

-- - AC Electronics U ° t wo gi eee Chapters 5, 6 and 7 i ni if the spacing 4. The circuit below wilt require haw many R 7 to the value of a capacitor Constants (o charge the capacitor «ot sen decreases? voltage? 1 al eases b 2 ons thes 63

d. 100 ¢ es ver e? R p waveform below Is a sine wave wie! A IN c our b B

Cc

ze Eat (ey (D) 5. A variable capacitor's Plates can open o · (B) Plates open, capacitance: form is: voltage of the below wave a. remains the same, a b. capacitors aren't made this way, ©. decreases.

d. increases, A 4 BY 6. An inductor with the same length of wire but whose rf 107: turns are spaced further apart will have:

- a. less inductance.
- b. more inductance. equal inductance,

= 4. more mutual inductance.

Ay · 7. What frequency will the Parallel LC circuit below be T resonant to? ia a. 100 Miz

: b. 159 Hz

c. 6.28 MHz

| d. 1590 KHz

| 20 mH

Il 50 pF

i

Review Questions ~ AL Electronics & Which formula is used to calcula following schematic: te reactance in the znFC

- d. c. Xoo
- a. $X_c = 2\pi LFC$
- a. X_e

100 Hz

40V

9. An inductor in an electrical circuit: uses current to lead voltage. f AC current. AC current. st to AC current.

- a. cal
 - b. opposes the flow of
 - c. appears capacitive to
 - d. appears as a short circuit
- he combination restricts the flow of alternating current is:
- a. volt-amperes.
 - b. impedance.
 - c. susceptance.
 - d. power factor.

Is the value of a 500 picofarad capacitor and a .5

iM.

nanofarad capacitor identical?

12. In a generator, a DC voltage applied to the field windings controls the output level.
- a. True
 - b. False
13. Which of these is within tolerance for a resistor color coded red, black, red, gold,
- a. 1850 ohms
 - b. 2150 ohms
 - c. 1950 ohms
 - d. 2250 ohms

Chapters 5, 6 and 7

14. Which one of the below is a common filter shown below? Multiple choice

la

Bandpass 16

Low pass L Low pass T High pass T

Lt

15. What is the combined reactance of the two capacitors in series in the drawing below? Multiple choice, Approximately 1,198 ohms
- a. b. Approximately 265 ohms
 - c. Approximately 24 ohms
 - d. Approximately .00377 ohms

0 Hz 2 uF

40V Cc, 4 uF

16. Which one of the below will cause a generator to produce DC voltage rather than AC voltage?
- a. Brushes
 - b. Armature
 - c. Commutator
 - d. Field windings

17. Total capacitance of capacitors connected in parallel is like: resistors connected in series, capacitors connected in series, resistors connected in parallel. inductors connected in parallel,

AC Electronics

Review Questions -

A Chapters 5, 6 and 7 Unit factor determines the value of a capacitor? and d 8

3. The sine wave's plates are shown below. Calculate the dielectric constant frequency in case of the battery covering the area.

d. 500 Hz inch diameter by ceramic dielectric #47" is most likely what value? 10 V at 7 mF. Overens to current flow in a capacitor when the voltage across it has increased to the applied voltage flow is maximum in capacitor. Load & File current flows. 4 polar reverses. drop maximum. ANSWERS on page 223

Inductance in a series tuned circuit at the resonant frequency will cause

- shown to increase-
- decrease:
- there will be no change. become unstable. Answer

IN

= Series RLC Circuit

3. The filter configuration shown below is commonly called: an LC filter, a filter, or an LCL filter.

- active filter. is OUT

WH

Introduction

Commonly In this chapter, we will discuss the components used in electronic devices today. Batteries by James Bruening

Types

Batteries are devices in which chemical energy is converted into electrical energy. Batteries deliver direct voltage power to a device independent of an external circuit.

im

Primary cells cannot be recharged and are disposable

Primary cells are rechargeable. Types of primary cells include zinc-carbon, alkaline, and lithium. Lead-acid, nickel-cadmium (NiCad), Nickel Metal Hydride (NiMH) and lithium ion. Primary carbon-zinc cells have powered flashlights for years. A new carbon-zinc dry cell provides 1.5 vdc. Alkaline batteries have greater capacity than carbon-zinc and will provide a 1.5 vdc output three to eight times longer. Lithium batteries are the tiny button cells used in wrist watches and on PC motherboards to retain BIOS settings. Lithium cells provide 1.5 v, and some are 3 v. A lithium cell will provide a minute amount of current for 5 to 10 years. And it is ideally suited to power portable micro-circuitry with low current demand such as volatile memory. Rechargeable lead-acid batteries are used in automobiles, motorcycles, etc., to provide starting power and a reliable source of DC. Each cell of a lead-acid battery delivers 2.1 vdc when fully charged and exhibits very low internal resistance

with very high current output demands. A rechargeable type of alkaline cell will provide 1.5 vdc and withstand being recharged about twenty-five times before performance degradation occurs, nickel-cadmium (NiCad) cells deliver 1.2 vdc when fully charged. In practice, a NiCad cell can be recharged 1000 and will often develop a "memory effect" that degrades

FIECronic C

Ryhy

there for 35 years servicing computer systems A

4.yea, I,

ng an 1998 after ser izing in radio, audio and small applians At : _ specializing PPLANCES HH 2 aay Agi olMarian Distress and Safety System) FCC examination se catty of the FCC

GMDS'

Pou DON

yy charging capacity. To avoid the "memory effect

ea should be completely discharged before beingre, cha ay Newer NiMH and Lithtum Ton battertes do not have memory effect and have greater energy-to-weigh, ratios

Battery Ratings

The power delivering ability of a battery is given in "y hours (Ah)." Abattery rated at 5-Ah will deliver the . of 5 amps for one hour. To arrive at the practical POWER

4,

My ering ability of a battery from the rated Ah, divide theta

Ih amps by the number of hours of desired current. For ex, 5-Ah is Just a convenient way of saying that a batterywill' deliver one tenth of 5 amps (500 milliamps) for ten hours

Ny

Table |

Interny

Primary

Cells

vee Ah Resistanc,,

Lithium

| 15-36

Zinc-Air

Pas fil - | setae] | | Piensa [ar [ow] a | J ce, fa Ds [metim]tow|

Nickel Metal

Lithium-Ion

rechargeablebatteries are currently used in a majority of small consumer products. Other power Producing components

Some

devices like thermocouples produce a small amount of energy that is measured in the millivolt range. Other

eh ait

3a

x VET

a. ce Fee cewson

ma

Electronic

Components

{produce 5 volts plying @ voltage across a resistor a very low output will solar cells

Fent

(amperes) to flow through resistor will cause a cell's Is have

f cell stall

MOTE

eric CY

nN amenint of cur. hit, deapating 9 cert equenc f power (watts) if th 410 amonnt ble frequencies o i Power eating ere

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Will fail (overheat or open) ded. the reator vt ' yore F appre pen Ee nd" okers f) ox fe - vit Bre s d cl d signed to fail or interrupt

on vices

ied, protecting # circuit sç jon de exceeded,

i

fv ecu nt level is current flow usually sive

co ent flow ee accident and can result in on! nents and circuits:

Potentiometer

po! A resistor of 5 ohm would be sm: Slo- Blow fuses £500 h id be al all im value, whereas one often have their ol meg-ohms wou a lar; wire looped around Be value. They May be similar Resistance value is

Much

8. SiZç vanes, where V4 w n physical size

ms in physical size even thought th fail adilenoad different. But with power ratin: small, while 20 watts is large i Tolerance is how much the resis tance value from the rated value. Actual Tesistance may i error

Circuit

Breaker rated value by as much as 20% and is indicated on

when the rated current is exceeded. They are tors. Common tolerance values are 5%, 1% and 20% alias two neJ or thin wire. When they fail, they newer precision and resistor packs as low as 1%

+ With

ete parallel "slo-blow," are made so that

f W nayaced:

Some

vdor so, thus allowing short spikes Most resistors are fixed in resistance. But there are special (be ye after

on come in a great number of types such as the rheostat, which is adjustable: the potentiometer only 1 to pass: . mounting configurations. A fuse of cter, which has a moveable 'ap, the varistor, which will iom-

vf

oe \$,el small, 30A would be large. resistance depending on the Voltage applied, Vary in sn const s) ee we | fuses that fail when their environ- 1A a ais? are exceeds a design level. These protect a eure ear the fuse. | oa erating OCCUS mer when y ice jrcuits in the same manner as fuses dev rotect circuits Fixed Resistor Mponents will also y; after failing. They usually are made of Other special co mal switch when the design current crs tie "ese thermist

Cire

gy be "Tes ermistor will vai

N aty resistance. The

te gost ill open

a in Ty resistance dependin, de in a wide variety of shapes, Negative temperature coefficient (NTC) ti temperature. In a that y are Mal t eTmistor, when the acoil

a

figurations. 'Cmperature increases as with the addition of mote heat, the eli8 ing ©O" Tesistance decreases, ev"oad moun! re: joted wi : ' ' ns «sculated wire is commonly found in electronic ailoo8° insulal think of wire as a component, but it is bt You may not een many devices. Two systems are

Resistor Pack

ci pect tink ae the American Wire Gauge (AWG)

a thee sentify wire SIZE- umber the smaller the con- Night depen "ststor (LDR), which is a resistor soto The larger the gauge ni 'ons list wire tables that that varies depending on light being, applied to it also dectea

od

BE ameter.

Electrical

rm ah" a s resistance with more light. Resistors have the same effec

di apacity of ca - AC and

DC ductor eters and currentcapa on ACani :

go dae?

Capacitors

resistors i Ita:

e voltage -caors are usedto control current flow a fio A capacitor, often called a "condenser," is made by pl

R ircuits.They are made of ϵ 'ii fo (ueot two plates of metal in close proximity to each other, ser Cee materials.They are designed for a va by a dielectric material. As a voltage is applied toit, a: ote resistive Jin "ohms." With a given resistance, ap-

Chapter 8 |

ys developed across 1 They are used to couple signals between circuits or to filter signals in a circuit. They also are used in timing and wave-shaping circuits. High voltages | 500"

33,0004) do not use electrolytic filter capacitors. Paper, ceramic or mica are used

Capacitor

The size of a capacitor (capacitance) is defined in "farads"

and it will have a maximum voltage limit. The size is determined by the area of the plates, the distance separating them and the dielectric constant of the dielectric material. The maximum voltage limit is determined by the dielectric material and thickness. This material might be paper, mica, oil, air, ceramic, plastic, electrolytic or some other material that allows electromagnetic signals (AC) to pass through, but not electrical current (DC). Mica has a dielectric constant of 1.00059 (a vacuum). A capacitor works by having a voltage applied across it. Initially, there is a high rate of current in the charge, but as it charges, this rate decreases until the full charge is achieved. The larger the capacitance, the longer a full charge takes. Thus a smaller capacitor will react better to higher frequencies while larger ones react better to lower frequencies. A capacitor will charge to 63.2% of full charge in one RC time constant. In SRC circuits, it will be virtually fully charged. Capacitors are used in circuits involving AC signals and

pulses. A one farad capacitor will store one coulomb when one

volt is applied to it. The discharging rate is the same as the charging rate. Many capacitors are fixed and constructed so that the plates are separated by the dielectric, then that combination is encased in some solid material. Variable capacitors usually have a plastic or air dielectric and are made up of one or more pairs of plates that can be adjusted for more or less overlap (interleave). The more they overlap, the larger the capacitance is. These are often found in radio tuning circuits and other circuits that must be tuned to certain frequencies. A capacitor will be used in DC circuits to smooth out or hold constant the DC voltage. These are called "filter condensers."

That

capacitor will charge to the basic DC voltage of the circuit and then either charge more or discharge as variations (ripple) occur, resulting in a more pure DC voltage. Filter capacitors may be thought of as reservoirs of electrons.

Electronic Components

Treatment

was

re

WA

a Filter capacitor circuit 'a

on Capacitors are used to couple signals from one

ra another (coupling capacitor) These will be selected reactance close to the frequency of the desired signal and those may pass and unwanted signals do not pass. Capacitors also are very often used in conjunction

with resistors or coils to either block or pass signals of certain frequencies. These are called bandpass, bandstop, or bandpass filter circuits. Some filters are designed to change the signals to meet special needs.

Another important variation of these filters is to be used as timing circuit for many types of oscillators. Oscillator with capacitor tuning. Connecting capacitors in parallel results in a capacitance equal to the sum of all of them, Connecting them in series results in a capacitance less than that of the smallest one,

Actually, capacitors rated in "farads" would be very large

and rarely used, so the most common references would be in

"microfarad" (mf), "micro-microfarad" (mmf) or "picofarad"

(pf). A 1 mf capacitor would be very large, while a 1 pf would be very small,

u

Inductor |

Inductors, sometimes called "chokes," are coils of wire encircling a core that might be metal or air. While they do have a

Resistance

in the wire itself, they are used for their inductive ability that creates a magnetic field. This inductive quality causes an opposition to a change in current. They are rated in this amount of inductance in "Henrys." A change in current of

one amp will induce one volt across a one henry inductance. |

ii

Inductors

Electronic Components

Chapter 8

called an "autotransformer" determined by the size of wire etc. as

Other

windings may be added to the number of turns and the depending on the winding needs of the circuit, the tone of the we need across the built windings. Size and length of wire and number of windings set a voltage for magnetism (force per unit). In each coil affects stepping up or down of signal frequency by the ratio of the lines of magnetic flux.

Of course, in

A. (When voltage is constant, current will flow in proportion to resistance)

When resistance is constant, current will be

proportional to the voltage. When voltage is constant, the magnetic field of force will begin to build up in the coils, which then results in a signal.

im eA NE Pte ttage nce wth oul ndvern rough the inducta

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od py having & voltage ie sit

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ws ate es at the maximunt The larger the kes. nee a full charge t@ wee ine ae ne The transtormer is des) I| react better to higher frequen- uctor

Wt ind frequencies An Iie react better (o lower ba nn '! a sie 63 2% of full charg ale rn yar e io tants, it will be fully c rh- wil ye tie Ore 'oulomb when one vo! oF o8 fy tore One

ne orypmne as the charge Fale harge rate : here they will DC eircunts

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aa hs current, tne ction with aresistor, capaci col nan

ie ysedin conyun wets, These circuits are designed ath application transformers are known as "IF."

a 150 stqnk" circu! ds of frequencies or 'ower, yback," "Constant Voltage." "Isolation," "Power iH" al or "ta or bani o:44 a

io)puters Or frequencies Output." "Balun," "Wariac" or "RF input." oF tlt rec uency: ber ore) freq

of 5OT aegiven most inductors will

Vacuum Tubes

eh' eto size and usaBes

I

Fs ary of micronenty Fang These are co apse llibenty Mponents that conduct Current from a cathode wi yn the toa plate and, under vanous designs, are able to Perform a ms aril number of special Jobs. In order to do this, the cathode must

be orm es of coils (inductors) made of a maternal that will emit electrons easily and will be ns) the properti 16 e use

of ually, they will have two wind- heated to promote this Process. The electrons given up by the { 3rom great

Seely?

secondary.Applying an AC cathode will be drawn to and collected by a positiv ely charged come Mepam Em[I result in an induced voltage plate. They require a vacuum to do this effectively.Beyond

a8 called the pnmary WI"windings were exactly alike, this, the makeup of the tube wall y ee recondst if the aadas an isolation transformer, yo the Id simply

1058 er WOU vany Purposes. 9 circuits. ged between two circul i nn une Many vacuum tubes are known by the Purposes they serve.

<<I

Some of these are: rectifier, diode, triode, tetrode, Pentode, re will be alter f the transformer deter- beam-poweramplifier, en, the! construction o se . t. Listed here are the more common the ill do | sd gas diode, thyatron, klystron, magne- tron and cathode ray.

In some diodes the heating filament and cathode are one and

F

Conduction in the tube depends on the plate being

Laminated

This might be achieved

by cote PEscant effect on the coup! :fide applying negative voltage to the cathode or positive voltage to | ta es concetra the lines of Force. the plate. Grids are added to vary what the tube can do,

I

One oF more, depending on applica- of windings: eae ith uni inding and selective taps is coil with unique WH | sg Asingle

gt

Hs a

Chapter

Legere ai to the controlgndcal tl ti

the tube and thus an mnvert

Often. emalways, this results in an wad : in oteyarations theees ees the tube current is referredio as

d, which too will be second "screen" gm wi seal aan Ike "reductng ees a ~ This makes late and control end ; influence pias . ~~

Li ven reduce the secondary emrssions that can occur wi aaa siostwons srk the plate

'Atube

wath three ends

ca pentode

Vacuum Tube

Gas diodes effectively become & short when the ml , threshold of the gas (about 65v) occurs between the plate cathode.

They

are diodes because the cathode can only ent and the plate can only receive electrons. The behavior o 'oe gas is similar to neon and fluorescent tubes. Thyratrons s! the gas property, but become a short when a gnd raises to a potenuat that ignites the gas. It may be extinguished, or cut off, by breaking the plate or cathode curcutt. Kystrons are tubes that will oscillate by themselves for a short period when the cathode is pulsed.This occurs because of it having a set of "buncher" grids. The spacing of these two gnds controls the frequency of oscillation, and ths is adjust- able from the outside of the tube. Magnetrons also oscillate when the cathode is pulsed, because of its contaming reso- nant cavities inside. These cavities are designedto a certain . The magnet acts as a guide for the electron stream, necessary in the operation of the tube. Magnetrons operate in a frequencyrange much higher than klystrons and are capableof much more power. RADAR and microwave ovens use magne- trons. Cathode Ray tubes (CRT) use an "electron gun" to emit electrons in the direction of a phosphorcoated surface such as a

TV screen, The gun has elements that affect the speed and focus of the electrons. Turning the gun off and on modulates the electron stream producing light or dark from the phosphor coating. An "aquadag coating" on the tube's inner surface is made positive to draw the stream. Elements outside the tube

wesc will act to cause deflection from, "ie The phosphor coating might heat

Color images are possible by: threat Cle Ny

VS arranged in very small "dots and th

is St he outside deflection coil (or internal ae of ihe ns rd the stream of each gun to the colo; don Plane j re

mt

May Wy

a Vacuum tubes vary in size depend)n, Sf from very small glass tubes with fine weat ' for hey lar tra

MV

ty industry has developed, more and more vay " tions have been taken over by semiconiictor gas '4, tubes have the inherent problems of

Power

co Hgy MP power tubes used in radio, TV and ra,

yy ASum, ure,

is

No and noise and limited life due to filament fay) " undesired internal gas buildup and cat) on mand cathode detent a Semiconductors oo

rs There are many devices referred to as "semicondy, os and modern technology continually creates new ones

e

LU share several fundamental properties. They usually are of silicon or germanium. As long as they are not abuse, can have a very long lifetime, maybe infinite (they don' ogy or burn out). They are composed of at least two types aba nal, referred to as "P" or "N," which means it has

Immun, :

that make it relatively either positive or negative. They

as be made very large or very small and, as a rule, are much expensive than the components they replace ley, Generally, these devices might be broken down into thy, groups: diodes, transistors and integrated circuits, (IC) s; ach may be modified to serve many different purposes Diodes: Composed of two materials, N as cathode ang p anode. Current will flow when positive voltage is applied tg the anode or negative to the cathode. The materials, usually, silicon or germanium, are impregnated with impurities to make them slightly negative or positive relative to the pure material. This allows current to flow from the N to the Pq only in that direction. One of the characteristics of the diody would be the current handling, or power dissipation limit, Exceeding this, either in a forward or a reversed direction, wil result in catastrophic failure of the diode. " A

Diode

Bipolar Transistors: are made up of three sections called the emitter, base and collector in either the NPN or PNP format. A positive voltage will be applied to the collector of an NPN while a negative voltage will be applied to the collector of a PNP. The base serves like the control grid does in a triode (o vary transistor current. Conduction occurs when the base is properly biased in conjunction with the a ropri icati of voltage to the emitter and collector. lillie ae

Electronic Components

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PY ae holes travel from on while i i vonds that occur back-

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Transistor

ly, mod- ing rather simply, its (ICS)! icin ricgrated circutt types. 'ree myriads oft red

c \$ created functions. The inventory 6 ogy

Da ple complex Integrated circuits get galt no lOBjumcral nential rate ce sean

at arn CXPO strate, these components aa iss tarting with & -

st net ues cated

an Is and msula- » resistive material "a oe by spelyink

ye Sact as resistors, diodes and

bY 10 'and connec! C is designed to do 'gee what the | Pd ay gocording

ro miconductor fYPCS

sc tion, a voltage is strikes a PN junc ode? ei adv ato - ohotdiodes form of p oto odes els arc a t. Solar pan! tes when current
JED): Illumina ting diode iuoan and materials, these may different

Haye

different colors. These colors hein a --s in that infra-red is near

it f the lig! ile ultra-violet is near rules Nhe visible spectrum

wh lications using, ery useful in app!

rules ght eensitive devices. The hght conductors

OO om and incoherent.

is very Fiber optics cables tihze photons generated by LEDs to transmit intelli- gent signals . That light is . t when tured on.
de: cepa and may be me,see small beams niesubstitute for atrue (gas, solid-state Se an en oem used in many printers.) laset
orby te lagi ially to conduct t is designed especially ode: A ior'thtcertain voltage levelis exceeded, seoem a voltage regulator or
limiter. While usu- pad eae it does have functions in AC.

in alyuse? ide diode. It performs . tal/metal oxide dios attky diode? Yee conducts high current and wellat higher enerdiode
types. These are found in

Josses

pie converter) power supplics. racing" (OC agile!

Tunnel dode: \ application) there rates at switching. OM Decal are ve evele thy Oscillator ane

Metal Oxide

Varts

that is designed to co Ferverse bias level tor (MIC WPA « Pecal ty Matuet in cither Lad of diode direc tien
wy They are nit Pol are the basis of the hest SUES Suppressors Capacitance can Vary ast changes

Diac: Two

zeners used in an AC eny connected

Teverse

design maximum

iD

Paralle}

ronment to tum On, or valanch, ic, bias is exceeded, ete and Photo.

Operational

Amplifier

(Op-amp):A

Single

circuit made of two transistor Circuits that Samplesignals at two inputs.

Depending

on the design,those two Signalscause the Op- amp to output a third signal level, Integrated Circuit (IC): Basically, a small 14-Pin component that is designedfor particular ap- plications. They might contain some simplegates, amplifiers, oscillators or combinations of all,

They

have evolved from the original 14 pin dual in line

24, 28, 40 and KR pn LSIs (Large scale mtegranon) and vancus therconfiguranans. They also arepackaged
w "flat packs' and cylindrical cases. Read Oah Memory (ROM): Comes i many sizes and comhgurstors The common
characterise is thattheyare matrafactured so as to retain fixed information (data) when then be sampled by aed tothem Thes
data may (oetr ® are used toprovide the invbal data

Jered

on,

as well asmnumerable package devices

8, 16. Random Access Memory (RAM): Integrated anathat

E ily as 8K, gre deugned especuallyasdata Usual s JOA, 32K ofOAK. they wall stare data im an 8-bit format Each data location
addressable, and thus available at any time for there is power writing or reading Data is retained as long 85 applied
totheRAM This ys usually provided for

by the power supply of an on-board battery Programmable Read Only Memory (PROM): A program-mad ic ROM, but only with special equipment. Once pro-grammed,rt may BOt be altered. Read Only Memory (EPROM): has @ window that allows it to afier which it may be

Erasable Programmable

Lake the PROM except that tt be erased by use of ultraviolet energy programmedagain circutt, probably rather

Microprocessor:

Anmiegrated

large (32,

ormore pins) It is designed and made tobe the controlling component for an electnc ofelectronic device. It actually contains all the circuits, memory and programming that will allow i to perform the job of controlling device under all predictable condinons They currently are used to control nearly all electrome devices including automobile igni- thon and electncal system, compulers, beating systems, rockets and coffee makers.

SUEEEETERST

ETD

E

AA gtaspanapsytt

ty

4 bepestaps are very small components, maybeno larger pared

ee or nickel, that might be permanentlywired or jon Hed

Lsoehets

They might be simply a semiconductor or '¥ mightalso include such thingsas capacitors, coils, transformers, crystals and thelike. perform special functions, a TS oe

Electronic Components

The hist of integrated Circuit ty includes many other types such we sali Tate Encoders, Decoders, DRAM SDRAM. Mux 2%, growing and packing more and more me am

UA Packages A person must research ace ve wie she encounters if he wants to stay up ide compe? nents that are introduced into newerWee Nhe ¥ Thermocouple: A thermocouple is a junc, different metals that produces a voltage relay

a ture difference Thermocouples are a Widely c, te ators temperature sensor for measurement and we saiSieg" be used to convert heat into electric power The, Mocouples vanes according to type. but is most Output

ot 100 microvolts per degree of temperature change< Balun: This type of transformer couples an unbay transmission line (coax) to a balanced line (t¥In leag etc. Thus the name for it. bal (balanced) un (unbalan.,P, Piezo-clectne crystals create fairly high but very me when flexed or struck They also work in reverse when

bi age is applied across them, they flex. They are often vk, receiver-speakers in cell phones. A crystal can be ne ta aericonets at a very specific frequency, making siya

v4 one means of generating very preeise frequencyTefey

hy

NOTES

SS

a

se avlt "one Mo icrowave oven

= vision

ator ty of a jun ly sperheterodyne receiver device would most likely be

Vv er SUP! psc" ig most likely be used as a signal wou! ould be used ina · pate resistor in Is at a set level aks in AC sive certain ae
als above pare hile blocking the AC gna ve le om getting into an amplifier. control the functions of an

to w rred to as: : ost likely ould be refe! ice · rd most likely be rated: you! rue of a PNP transistor? toff is reached. when the cu
ard. gat well garde CaN ill be appli

ni b c heated to operate. ¢ ode must be 7 tube transistor has over & vacuum vantage # advan! i rode ist nduct from the
cathode to the anode. jade cannot require an anode load. bias ; mally has 2 longer life. no}

Electronic Components

The most likely use of a transformer would be: & to from one circuit to another b to Alter out unwanted signals · a8 a
regulator in power supplics

to protect @ from excessive currents Which of these would @pply mostly to a

Semiconductor

Impedance matching device Integrated circuit Over voltage protection

a

b

c

10. DIP refers to: double mtegrated Protection.

a b dual ion positron

ec discrete icon Positioner. intine package

ANSWERS on page 223

'Chapter 9. Semicondy,

Ct

oy by Dick Gloss, CETsr

Fick

Glam

wth

of eva CTApotion aswell mre han

a dven fe dealing th clr py wna management. rushed

by

Horwand

WSamm Co, TABBooks, and McGrew

-Hill

Hehas

also written hundreds of Ales

in trade My, ecccwnon publication, Dak wa former

Navy

Avvowucs

Techancan,

Industnal

Electronics

Computer

Tevhnicisa, Teche

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and 208, mixno the I lextrnics Wilt

Fareeaagg

i) CEO

Linentas

ofETA,

He has 1 bevad back ground

m all phases

of work performed

by eelectronics tevhnic

as In Fig2, the 100hm resistance allows the 10 volts

Of Semiconductors tncal 'pressure' to send| amp of current around the tygyy

Jostabouteveryone realizesthat, without electronicsemt=

Aresistor composed of a less-conducting material (sucha, conductors,our workd would still appear

to

be nthe dark carbon) measuring 200hms(Fig.3) now allows onlyhal,

Yeracho

around with many electrons to move around the circuit.

Electrenty

«yy ages Imagine trying to carry your poral tts five electronund and large buttenes Yourwatchwould

be moves at nearly the speed of light(186,000 MPS) but on, spemgdriven. Onlya few businesseswouldhave computes halfamp
is flowing(Fig.2,3,4 and Sshow clstton fly

Typewnters

would form the

Jeter

wntng chores.

There

seul a eae CS cx cell phones

or CDs. Wecould controlthese electrons better if we also put g Our world today is affectedin nearlyeverything

Wedoby

switch in the senies circuitt (Fig.4 andFig 5) semiconductors.

Early

Semiconductors

Since

the early days of the 20th century, electrical equipment has used semiconductors.

Here

is

a picture of one in Fig.

So, power, instead

of blowing up the battery by shorting the + and - terminals with a screwdriver,

or

by substituting a 10 ohm

= resistor

for the screwdriver and thereby reducing the electroning it even further with a 20

ohm (current) flow to amp,

of reduced ohms resistor

to where the current is only 1/2 amp (or 500 mA), we can turn the current

on

or off with the switch. The resistor, unlike a straight wire made of copper, aluminum

or

steel, resists the flow of electrons. It doesn't slow the circuit and the buzzer

on. In fact, we could turn it down, but it puts some logjams or roadblocks in its way for a short time or for a long

time and communicates its way. It sort of narrows down the pathways in the partially any one close enough

to hear the buzzer by using Morse Code.

One shot

is a

one shot 638,

Hey! That

means the letter "A." dot-dash or "A"

Fig. 6 automobile starter circuit and of a solenoid in the circuit instead

of an automobile starter to crank the engine. We may use

semiconductors before reusing it. It was discovered that for some reason, if metal will

be able to flow

in only one direction through, the gated side of a metal-selenium plate was connected to the negative terminal and the metal side connected to the positive terminal. When the connections were made, current flowed. Reverse the connection and no current flows.

Semiconductor

Non-Rectifying

connect the rectifier stack so that the terminal is connected to the negative potential on the non-rectifying side, causing current to flow. Connect it in an opposite way and no current will flow.

pm

{ Current

pen in plate and

Electrons

flow in the metal plate by (electrons) flows against the arrow and

Note

the diode symbol in Fig. 8 conductors symbol.

a way we have biased it. No current flows.

Look

Fig. 10. Fig. 10 shows the most basic AC power supply and how the diode converts AC current into pulsating DC current. It is as follows:

At Flow & reverse biased drawing in Fig. 8 as

we are creating a barrier between the arrow and the perpendicular line part (the cathode). The difference between the negative voltage and the greater the barrier becomes, the barrier becomes as a thin slice between the semi-

Ue

n

Chapter

Conductor

Material

on the metal plate, and this slice gets

Electrically

wider with higher forward voltage, you start to see how semiconductors work. With reverse bias, no current flows, but importantly (as you will see later on) this barrier becomes less of

More

intense depending
on the reverse

Voltage

levels applied.

With a

simple rectifier circuit 4 shown in Fig. 10, that stress barrier doesn't matter much, other than that it doesn't allow current to flow in the reverse direction. In Fig.

9, we reverse the bias:

Forward

bias, Current now flows. The barrier - Net positive electrons are enticed away from the barrier and the electrons can

Forward

base or 12
of
be AC or

Reverse

on the other

Figure

10 shows the most common power supply circuit. On positive across the diode junction in a C. Thus, across the resistor load bottom to top of the resistor, leaving a positive voltage with reference to ground. That is exactly what you wanted to create in order to supply a voltage to

Properly

operate the transistors in your little radio (or TV or computer), common transformer half-wave
V_e half-cycles, current flows

Counterclockwise

direction, there is a

Voltage

drop from On the negative half-cycle of the AC waveform which the transformer supplies to the diode, the top of the transformer

Secondary

is now negative, So is the anode of the diode, thus Current can't flow, So we just wasted half of our power from the transformer!

There are ways to not waste the half-cycle

of power from the transformer, but

we need

to

do some other

material That leaves a no-man's land between the P and N materials or a thin layer of full-width material

Send

how this Material can be used

Why it works. Why it fails

Fade

the SCT area

the trouble with works You take an inductor

is the

we have one kind of Semiconductor on the Junction

a wong pai Gel

[Eero

(Actual

Vow) HOLES! (search 'transistor depletion zone' online for more) P type material (on the left side in Fig 12) current flow is considered to be by holes rather than electrons. This is not easy to understand. In the N material current flow is normal, and negative electrons move through the crystal material, But in the P material it is the holes that move from the positive terminal of the P material to the negative terminal. It becomes more confusing when you join P and N materials. The junction of the two is called the 'depletion' area, or barrier (as we discussed above). It is called a depletion area since there is a lack of free electrons, or holes that electrons could move into. (Fig. 12) One way to think about this mysterious barrier area is that the barrier slice between the P and N material has positive and negative ions which are attracted to the edges of the depletion area. The negative ions cozy up to the P material and the positive ions snuggle up alongside the N (and negative to the N material) reduces and

Md

in the thin barrier,

Applying

positive (forward) bias

Voltage

So let's see, the

clog May,

that exists between the positive and negative terminal. By reducing the electrostatic field, the electric field in the junction (and hole by the opposite direction) is reduced.

JUNCTION

LEGEND

10) Hole

Free))

© Negrverg

© reverse, . For 8 ton, Reversing the bias connecting the negative Voltage

to P material and positive to the N material, the barrier is even wider (electrically). The electrostatic field is strong, thus it is much more difficult to move electrons or holes

At the barrier.

Fig. 13 Diode symbol (K is used here as C is already used for capacitance.)

Forward Bias

Plus to P material. Minus to N material. Silicon diodes require .7 volts forward bias to overcome the barrier potential and allow conduction to occur. Germanium diodes require about .3 volts. The natural barrier, or depletion slice, is narrower in germanium. Once the barrier field has been overcome, an increase in bias voltage of only a few hundred millivolts-up to several volts-will cause an increase in forward current. There are limits specified for inverse voltage that can be tolerated before the junction breaks down and shorts out the diode for good. See Fig. 14

=A

axe

an that can be read current U forward in the PN junction or with an ammeter

y My,

Techni

are fast Tenants of the semiconductor

W z if further S and yourself

of

N Junction of the P and N semiconductor - " truths which ARE as Fig 14 symbol (Fig. 14) the diode symbol

Recognize

A). y are the + voltage on the

ay for

e that hode. on a diode is the cat! mer tthe denfyng yen at a higherpositive poten- naer®
i the anodeneeds 3 ct, haps the positive excur- . or il on eee electrons from the cath- cave, W tive

He 60 Hz sine

saan ina power supply

Cae

Jeaving

want a diodeto see itu ~e oot al\$0519 eet is placed on the

os msi recog

Oe ns You notice the ea eae y apowersupply. istfor poste ratty aks likeat eoGenerally ve aynbol tio of 10 to 118 OK.

at aig (or any other PN vari aeyeeYOU wi dae voltageor allowing too hig]

ones too large - Comoed antec a 3 oe flow, such lena mi the voltage is being

is alloft (which is al or i near the AC input " sr nner oe re the most likely pecause dioee ered product, theya

sceh} lightning or power surges.

be posto

io

15 Simple

transistor configuration Fig.

Transistors

Ifyou Understand

py;

JUNK

thons then you will ha r

Problem

seeing how transist ve little lors: work NPN We will talk Mostly about NPN transistors pry ally opposite in ey CTYFespect,so

If You under Vi Understand PNPs tog tand NPNs yoy that the hase should the emitter. That mal Thereis a Very smal base
current flowing tn an NPN transistor if the base'emitter (BE) junction is

fo That means when YOU expect a transistor to 4circuit and you find ze and base Mward biased,

be doing its job in TO volts difference between the emitter something

is Wrong If you find 2 OF 3

Voltson

Fig. 16 - Common emitter NPN

Chapter 9 fo

Amplification

Drades ontinarity are used in an or 8 SS ns uae: fan conduct leas than the fill PORTAL TT bon if they are harchs biased ca
amt firfly Conduct to the penential unguet (roe lHe 1 Of Jor 7 volt) hutne more thar anampast

the The tranaretor has the ability f seni\$ee A COM Cor the emitter hut Fer's tah ant te eager appearing

on pheity) The amp' Te nave. but veut fimuration tar sum gn r hase signal ¥ the collector is not the signa! would think so
cinco Hrs 8o ext 16 | eversthing 1 braved corres Check core the NPN hase through ; Notice the smaticr sine 8 8E Se ie saltector
\$o the

j 3 much larger a the ae Q? through the PN base- iy 1 flowing from the EMM A based-on base allows poi bla sere: a net to the
collector The eer Ato flow: from Me } voltage across the load larger current produces # IATEST signa reustance RL. That as
amplincation Bipolar Transistor (NPN & PNP) Configurations Transistors are very useful They can be made to perform
hundreds of electronic funcuons You can study books which deta! all of the ways they are used. You will find that IC
(integrated Circuit) chips may have many transistors, all doing similar yobs. Or you may find iC's with dozens of transistors
E in them, each one-or groups of them- -performing different ; tasks They may be used in an oscillator circuit, clock.
regula- { tor, gate, Memory or power amplifier circuit for examples. Scrne are high power and some are used in small signal
cur- cuits. All will fall into one of three configurations:

1. Common emitter - like the circuit in Fig 16

2. Common base - where the input signal is applied

H to the emitter rather than the base | (highest voltage gain)

3. Common collector - has high input impedance (and low output impedance) Common emitter is the most common
hookup for a signal transistor. The emitter may be grounded or connected to a DC voltage supply. The emitter is
common, or the reference for both the base and collector signals. As the base voltage goes higher, more current flows
through the transistor to the collec- tor. More current lowers the collector voltage and the collector voltage then is the
opposite of that on the base, but it is still a replica.

a could work about the The commen Onathe signal may be jae ih \

in ee erent grounded) SO, Whether v., the

ok

Saal

ao the hase oF the emitter, the difference bey

aD

iw hase and emitter may be the same Thus, the cone'een : ase Meh ene you put the signal i on. and yy!" gS Juces the amplified
signal on the collector ih ah, common base configuration usually does not pi Matty, an amplification, but rfcan produce as
large a (yy, reana One plus to putting the signal in on the emitter

Tent

lector signal is not inverted That might be nice tha only wanting to amplify a single pulse and you Seus remain a negative or a positive pulse after amplifica _ i.

Aton

The common collector might be useful in

Sttuat

which you wanted the collector to be grounded yy, ne ty your auto radio where the transistor is mounted, meoi) might get shorted out by a screwdriver TV set horelly s put transistors and many other power transistors, are antal og! with the collector (which ts connected to the shell) py. Meg In thts case, the output is off ofthe emitter with the mpd

Ng the base Part number Identification Why are transistors numbered like 2N123A? The 2 for the number of junctions N means a semicondu, r wands 123 as the manufacturer's ID number An "A" or "Bmee lowing the last number identifies an improved or | _

ve

fo of the 123. It is best to consult product catalogs or ra Tn

ta to assure that a replacement transistor ts the ca

Identifying Transistor Leads

A single semiconductor maker ma:

or pin placement for the E - B - C lees But mim'avout other maker to follow suit. You can check uisleione

oa

wi ohm meter. Checking between the E - B - C legs, the rz

iy forward to reverse resistance should be at least 30to Nels

ef 100 to 1 is not uncommon.

aa PNP We have previously discussed transistors abov ; ve, usin| theNPN type. The PNP

ts just as popular. Since the ae inival (N) shows the normal collector voltage potennal quick check in-circuit of aPNP is to see if the collectae i negative. The base also should be negative within a volt tie emitter and the emitter will be most Positive. The polz reverse of theNPN type.

es The biggest problem je isi you have is in determini -

is

ig nce base or emitter. There js aS EN ha leg \ tion. Sometimes you can determine the C,E, or B by pete | to] ing

zemMiconductors today's world of two-sided

Pc but im it, hoards and that too r8 something only to be

Attempted

cunts. Ih the NPN, you can short the emitter amon

S

Bite

changing the transistor. This can show whether it is drawing current and thus if the collector

= means conducting.

your stor

we JFET's and MOSFETs

Transistors

effect is to increase the input impedance in 2 different ways: by using a Metal Oxide

a

Pitot

adsorption of water vapor

of the anode-cathode junction in NPN and PNP in bipolar devices and

BT

OPERATION. voltage across the gate- drain on a semiconductor

There is no material, whether it is a depletion or an enhancement type. Depletion mode MOSFETs are available and voltage applied to the control gate opens

channel conduction.

Enhancement

in source and drain metal contact path between the gate and the drain. A

reverse bias is applied to the gate (means below) and when voltage is

applied, it begins to conduct. The symbols for these devices

are shown below.

a

of the

G

Symbol for P-Channel JFET

N-Channel

Depletion Mode

Depletion mode MOSFET P-Channel MOSFET

N-Channel

mode

Enhancement Mode

P-Channel MOSFET enhancement mode MOSFET

rer”

Cher

we thET muhuctors, like MOSEE P¢ 6 stall

we suscepble te damage (rum

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ne accumulates in cold weather IN (

Work

ng with CMOS ICs and other Mean must wear aw mos pe vemice (Insulated Cate FET) slectrc ity that everyry STVironments

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components, the tech, TST gremind 'trap and the hench SURface must he Bounded I yeny are working m the fielg and are noe Certain of the Potential. yeny shouk! have Sople get chargey Up to hun- dreds of volts: This discharge rough

« (ANGMtOF can

Puncture

the thin

Metal

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Composed

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wode These dhivtes use the came symbol as an ordinary

a But m chown thes alan have a tiny capacttersymbol the cathode fect of the When the capacitive PA qunchan was first understood, vos created a revolution r” the ways < and TVs were tuned to the different statrons. The first varactors were ail tanststors A techmewn wou! Figits think he had discovered& transistor where someone had forgotten to solder aleg Actually

the product makers used the inexpensive transistor only as a diode-a varactor diode. If you page back to Fig. 12, you will note the use of a PN junction. As different voltages are applied to the materials, this depletion zone electrically expands or contracts. The electrostatic field then is, in effect, changing. If you used a variable capacitor in place of the varactor, you could vary the electrostatic field by varying the separation of the plates of the capacitor. The varactor accomplishes the same thing. But the varactor is a very inexpensive semiconductor and takes up

Intitle

space. It works great, so that is what you use for tuning and voltage control circuits today. You won't find any transistors being used in place of the varactors anymore. Increasing the voltage across the varactor PN junction decreases the effective capacitance. SCRs (or Thyristors) Since most diodes and transistors are made from silicon you may wonder why one of them is named silicon-controlled rectifier or SCR. It is also like a transistor and a little like a diode. Physically the SCR has four layers of N and P material.

With

no voltage

on the gate or control element the SCR can't pass current. With the gate positive, the SCR is turned on and current flow takes place. The current is limited only by a limiting resistor externally. By removing the gate voltage, pulse, signal, etc., the SCR does not turn off. Once it is turned on it can only be turned off by removing the anode potential. SCRs are used as high speed switches. They can rectify controlled amounts of AC power. They can be turned on and off in microcircuits and do not have contact problems such as a solenoid or switch will develop. Fig. 19 SCR One of the beauties of the SCR

Isthat

a very small signal

Voltage

on the gate can turn on large current through the SCR. It is like the auto starter where you energize a solenoid with a tiny current to apply the full battery power to your automobile starter.

Semiconductors

Triacs

Triacs are three terminal semiconductor devices. Your TV remote control turns on the TV power by actuating a triac. A triac can be constructed by connecting two SCR's back-to-back with a common gate and common terminals.

Optical Semiconductors

Important devices in this class are those with optical abilities. These are generally:

1. Photodiode
2. Phototransistor
3. Photocell

4. LED and LCD Light Emitting Diodes (LEDs) need only about 1.6 V forward bias to produce light. They draw around 10 mA of current and may last for over 100,000 hours. LEDs are interconnected to form alpha-numeric characters. The seven-segment LED display is a common example. The color of the LED produced depends on the material used in

Manufacture

SUPER BRIGHT LEDs are very energy efficient and

produce more light output (Lumens per Watt)

LCDs (Liquid Crystal Displays) are used in the same manner as LEDs but the liquid crystal display is energized to polarize the segments of the display to form alpha-numeric characters. LCDs use much less power than LEDs. The disadvantage of LCDs is that they do not operate well in very cold temperatures and need external illumination to be read in dim lighting.

Photodiode

The photodiode, phototransistor and photocell are semiconductors which are made to accept light through a protective lens which covers the photosensitive material. This PN junction has a certain resistance in darkness.

Fig. 22 Photodiode

When exposed to light the resistance decreases, allowing current to flow in direct proportion to the amount of light beamed onto it. A characteristic of photodiodes is that they respond instantly to light, making them very useful in card readers, light meters, optical scanning equipment and so forth.

x
of
ay
oe

Semiconductors

Chopper?

stor

NOTES

Much more sensitive to light than the photodiode is the photo transistor. These may be either of the PNP or NPN type. They may

be used after a photodiode in their base circuit to utilize the speed of the diode and the larger current of the transistor.

are met

are light-controlled variable resistors but with a wide

range. They are often used to control street lighting

in these circuits.

i

od

Fig. 24 Photocell

oo sovoltale

Cell

as they are called, convert light energy into ; The space shuttle and space station use many

ae cells as do satellite transponders. When light strikes the

ae

ce inctions of the photovoltaic cells, it produces movement

a pny etn Each cell produces about a half-volt.

of el =

Light has been proven to be both a

a wave and particles (or photons). Photons impinge onto the light sensitive PN material and cause a change in the barrier or depletion region of the junction and current flows. A photon

a has an energy, so a photon knocks - iconductor material. -_-_-__ eoo8 out of the semicon cells. optical

Coupler

pnicians frequently encounter opto-isolators, or optical me can be used to couple AC signals or pulses

rt tile completely frblock any nent. Capacitors can js also, but may not entirely gminate leakage current. The isolator is a low-current an be used to trigger rompers such as | Fig. 26 Optical Coupler gcRs and triacs.

Chapter? i.

Chapter Quiz

1. Diodes are marked with a band at the end. a cathode

b. anode & cemer have clement 2 a germanium diode, a forward-to-reverse ratio of 10-1 is indicative of good diode. a True

b. False

3. Toproperh bias a NPN silicon signal amplifier transistor, which voltages appear correct for a signal amplifier

a E=0. B=+.7, c=10

b B=2; c=+20

c C=93

ad B= c=10

4. To properly bias a PNP transistor which voltages appear proper (SI)?

a. $E=0$; $C=+10$

b. $E=5$; $C=+20$

c. $E=-1$ $B=-17$; $C=-10$

d. $E=0$, $B=0$, $C=-10$

5. In a transistor, which section is made very thin compared with the other two? a. Emitter b. Base

c. Collector

c Collector

@ Each are exactly the same

6. Which part of a transistor carries the most current? a. Emitter b. Base

c. Collector

d. Base and emitter carry the same amount

7. The primary difference between a NPN and a PNP transistor is:

a. shape.

b. DC voltages are reversed. c. one is silicon and the other germanium,

d. the polarity of the power supply voltage.

8. Which transistor circuit

9. Name one safety precaution taken before

Semiconductors

largest voltage gain? a. Common emitter b. Common base c. Common collector Transistor circuits produce current gain a. Bipolar transistor. a

a. Remove power

b. Remove transistor c. Short emitter to base to remove discharge

d. Put on rubber gloves for isolation

10. Zener and varactor diodes are usually

a. reverse, meaning the positive voltage connects N material to P material

b. reverse, meaning the negative voltage connects N material to P material

c. forward, meaning the negative voltage connects N material to P material

d. forward, meaning the positive voltage connects N material to P material

ANSWERS on page 223

by John W. Baldwin, CETsr va anal a:retces commune atioers

at

Scuth

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s core! ed circuits have not taken over the mas pr integral ie

yo why heh volume circu Jhere individual or "discrete"

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aaa

aM frequency (RF) circuits

Latte

mplified with a pare high frequencies to be amp!

Pant

ahat require Oe ts to reduce shunt capacitances watenumber of components y d circuit con- could be high-spee nu chs other areas Td from input to signal delay time 5Prd pat need minimum

id

5"

wt NPN transistor nents, such as the } 4 solidstate Sonn the device as shown in the pst ve circles

a. transistor were to be fabricated into eat symbol.

Me i:drawn around the device as shown no cl sees

ic

a eto

' tance adds up, as in paralle! use jection OPC PC,+C,+C,, high density semi- a circa perausing multiple transistor circuits ee"
- have a greater signal loss

=ae ;

aes -] filtereffect ofall ofthe shun casted the low-pass es due to Ace adding Up. i. ae

J nics for 8? years a8 6 pro! mnvolved in elect cae aM nidere has bees poe Ba 'od aed averufcstion adkrnvetratie foe probs

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Sa tes twa)

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chmenn Co en a techetical uppcat euperveecw anal erty euplycd

OY fF ty Ne eeaerevpeald the GNC Wireless U dha stir

id the Yous eward hee ha fe2006 Juha w weatnomee end teleccenenany

Discrete Solid State Circuits

'echetry mut hoe mit eary Whe -aeght rlee ner neut VIN the wes eninenutie Sevrgrate rad Yee arwat freon tee trite Me oy tg Categories of

Circuits

Solid state devices can be cate; ways - by the number Of junctions, by the Matenal or by theur function Diodes, transistors and iMegrated circurts make up the vast Majority of discrete dev ces, with special input-output sensors, displays and Teadout devices making up most of the rest One way to class HY circurts and electronic products is by their function m 4 troller, cell phone, copy a class 2 product Class 3 products are usually tho ught of as in the medical or aviation field, and if they fail, they Could cause loss of life. Another way to classi humber of semiconducto; holes and electrons, W; thin the 2 junction devices are bipolar Junction transistors (BJTs) and field effect transistors (FETS). With more layers and junctions, some dev Ices are variants of the basic types, and can control DC or AC circuits with gate inputs. Examples of these devices are silicon Controlled recti- fiers (SCRs) and tnacs, which are called thyristors, Multi-layer Semiconductor construction techniquesare used extensively in integrated circuits, with low, mediu: im and high numbers of transistors, or high gate density in digital applications.

Chapter

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Freqacmcnes

BCS dost SET . poet norm ofeyrected performance

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De sounded poor 3 bosons world

Types

of Components mito 4 groups 3c-

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of components can

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a

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are those designed coramg

to physical ritutes ic soldered more dicct i mount

3s toe coe and makng 2kpost ampossible

omnespans! means.

Four

groups

of CompOOeR

Types

aE:

L THROU**GHHOLE**

ences DIP. PGA, TO-. DO-, Axl and

Radial**Lead****(Components**

2 SMD

Chips

and

Discretes

secindes SOT, SOD. Resstors,

Cepacttors

Inductors, awe

Crysals

3.SMD Ks(lead frame types) incodes OFF. PLCC, TSOP, SOIC. SOJ. SSOP,

TSSOP

pps

4. BGA and

FLIP**CHIP**

includes

Ball

Gnd Arrays.

Bumped

Devices, CSP and

Land Gnd**Amays****Packages**

are the physical containers that house microelectronic devices.

Devices

can also

be classified

by package type into the above 4 types.

Examples

of these types of devices shown below,

Axial

lead component

Radial

lead component

Discrete**Solid State Circuits**

SIP

- Single Inline**Package**

DIP

-Dual Inline**Package**

DS

Chip

component. discrete semiconductor, SMT - discrete Surface Mount Technology QFP - Quad Flat Pack PLCC - Plastic (black) Leaded Chip Carrier or CLCC - Ceramic (white) Leaded Chip Carrier LCC - Leadless Ceramic Chip Carrier

Flat Pack

a Discrete Solid State Circuits

a or:

Thum Small Outline Package

known as Small Outline Package BGA - Ball Grid Array is a large scale integrated circuit (LSI) package. The Ball Grid Array (BGA) was first produced today, and

is used extensively

verage Oakes products. It increases reliability

Peso

and other chip sets rarely fail, except creepers not conditions due to over voltage. excessive soaking in water. In most cases today, trouble-

average counts reduced to only a few simple so-called service shop. Advanced semiconductor testing requires a specialist with the same test equipment as what is used in the manufacturing. A really store controlled equipment is needed for

weight sensor of complex circuit functions. A widget per month, today produces

1,000,000 widgets about one widget produced every 2.5 seconds

every second month). There is little time for parallel production lines, or micro-structured testing, where circuit tests are performed in

Parallel

Chapter 10

CICUIts protect mont 5 chock glass fuses for any sien of excee een BE 8o hard biran. 4 soft blo

1. Observe symptoms 2 Compare to normal

Operanon

3. Study flow charts and diagrams

Measure crrcut

Parameters

5. Analyze results

Troubleshooting

highdensity microelectronic boardsadds another dimension of how to

Sueccessfully

remove the faulty component. Proper tools and specialized de-soldening equip. ment ts required. An electro-stane discharge (ESD)compliant Workstation and soldenngCQuIPMent is recommended for repairing and replacing all new types of components, Also, with today's software contro! of hardware compo- nents, the appropnate test software is often Tequired. Auto- Matic diagnosis routines may be a tool that is available from the manufacturer of the product

OF diagnostic self-tests may be built into the equipment.

Discreers verre Tore

on

Chapter

10 | ee aren fre i aka for most software troudleshowing today That makes car hfe as

an electrons fecHS jase nec higher

OF

se cilevel ar bat ale

Infe

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to have come par troubleshooting AV!

NOTES

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of we pins petiole the sides

ms & ee

he ting discrete circuits involves a knowledge oo! yes tion ae t was made [ung temperature ren technology | ts tough metal thickness

an cn 7 we ded soldering gun unl static grounded soldering pencil or we pene temperature possible (o get the job the | © gone and 250 W gun | wet i wa au still find discrete

I

ae ipment ff video games sity boards ble handheld devices · per wot with a high of components at super high frequencies? | resistance lel inductance igh impedance

= capacitance

circuit parameter makes it difficult to build

rc

Discrete Solid State Circuits

7. When troubleshooting equipment that is software controlled, use of a computer may be required b AC signals are not used © DC voltages are always 5 volts

d. diagnostics will be easy BA category 2 type of electronics equipment could be a consumer home TV heart monitor fax machine FM walkman aogce

9. An example of a solid state device is Cermet resistor Tantalum capacitor stheon diode vacuum tube

10. A four layer PN dey ice is classified as a

ty

- a. diode junction
- b. transistor
- c. thynstor
- d. integrated circuit

ANSWERS on Page 223

Review Questions = Qnq

Chapters 8, 9 and 10 Sign.

Un 15 8. Which of the silicon diodes below will require proper voltages applied? Consider the voltages applied

1. Rewind

a

b. B

6.

4D will pose, ground potential will 12V 1.8V given voltage 7 isistor will: temperature coefficient thermal! a B) (c) temperature. (a) @) a with heat or cold. ϕ have no change in resistance with voltage. 9. If the gate voltage drops below the trigger level anode and cathode voltages are maintained, what will occur? cell, when charged, has a terminal

3. ANIMH battery voltage of:

a. The SCR turns off.

b. The SCR stays on. are c. The SCR turns on. : Prasad. An output pulse is generated. or 10. The transistor configuration shown here is;

4.4 thermocouple has an output in the range of: PNP.

b. NPN, ; 10 c. MOSFET.

d. FET, diodes: B IV forward voltage drop. often used in power supply circuits, A of heat than silicon diodes. forward voltage drop.

Capacitance, sag 11. What is the effect on a varactor (varicap) when the negative reverse bias voltage is increased?

4. Capacitance decreases and the tuned frequency increases,

6. Capacitance increases and the tuned frequency decreases, ϕ . The diode conducts more. @. The signal waveform becomes distorted,

ANSWERS on Page 223

by Ed Parady, CET Technician on the US Navy in 1971, works as a Sonar Cable Installer Instructor at the

Pinshargh

Job

Corps

operation as Unit develops systems using Magnetism and frequency (current technology to perform; fuses Inc. repair/renew heat exchangers. piping, pressure vessels, & erage each devrute testi suc AWW ca porntient plant equipment tion yao? jnician needs to understand power supplies. In perme you will find details on the: paper

po

i | net Pimple

DC (auct current) power supply.

y of each section in a DC power supply. fa DC power «functoOn» that make up each section o! 'components seas that indicate a defective DC power supply. vsof powersupplies. ekectro! i eof found in the typical piec

a ent Ssdee AC (alternating current) reinto DC voltage levels required for the vol pot tectonic circuiis. ion Y ly are the | jons of a simple DC power supp them tor filter and regulatorcircuit. Many apse ectie are available today, including complex ws of power that change the input AC to a higher frequency brainthe output to

on This chapter will cover the riot s,so letsget started. poten | e of a Power Supply ified in four categories: es are often classifi per opel | ico AC frequency converters «acioDC rectifiers || pctACinverters pppCeamerters George Westinghouseand Nicola Tesla,

ET nt A decid power distribution, over- ee of Thomas Edison using DC power as the wie of electricpower distribution. tegan, er became the standard utility. How- dectonic ae made up of amplifiers,oscillators,

a andother circuits that requirea steady,set | called DC. AC uses a changing voltage waveform and, nemust be changed to DC for systems to operate.

i

i

Sections of a DC Power supply The order of the sections within a maintained, that is, you cannot filter The sequence is diagrammed as: Power supply must be before you Teeny, etc,

Ac

oc (A regulator is not considered a basic Section of a power Supplybut is most often included.)

Transformer

The transformer is used to Step-up or step-down the Voltage.120 Volts AC can be changedto a smaller or larger amount by passingit through a transformer. Figure1 isa Picture of a small transformer and Figures2 and 3 show the schematic symbols. The AC receptacleor outlet will connect to the Primarywhile the rectifier will connect to the Secondary,

Fig. 1

+ To Power Sope

Verano) Igeconeen?

Fig. 2 Fig. 3 ce that it is noth Ifyou examine Figure }

Sonata

mon ork, Each loop iS than wire looped ae

me me and you have mary ras and econ ined by the ratio of the secondary is determin bar amo in the pnmry to the number of tums In cola and the voltage appliedto the primary. mary and tum inthe sec

Ifyou

have

turns inthe pri 5 would

be s 4 down ondary a pnmry vollage of 120 voll to 12 volts

on the secondary.

Rectifier Circuits

To review. AC is a waveform generated when current flows inone direction, stops and then flows in the opposite polarity. DC is

a current flowing the same polanty or direction. A graph of AC voltage versus time is shown in Fig. 4. 01 02 03 Of 05 06 Time (mSec) Fig. 4

A rectifier changes AC as sho: "Oho, ' Ore tly

HL Oy fn,

wn sating DC as in Fig 5 shown as halteMe 4 full wave ia

+4 & *2

01 02 03 04 96

g5 Time (mSec)

Fig. 5 Half Wave

Pulses

+4

so

01 02 03 04 05 06 Time (mSec)

Fig. 6 Full Wave Pulses

The heart of the rectifier is the diode. The diode allows the electrons to move in only one direction, changing from a flow in two directions (AC) to a flow in one direction (DC) Fig.7 The schematic symbol of adiode shows the anode (arrow)

NY on the left and the cathode (bar) on the right. Electron flow js againstthe direction of the arrow much like water flow check valve.

na The most common connection of diodes is shown below:

i:

Fig. 8 Half Wave Rectifier

The circuit in Fig. 8 creates half wave (as shown in Fig. 5) and is called a half wave rectifier. It eliminates the

ai

em by blocking the

al wavefe miconductor material) ae wave to be

g ve Rectifier

e

Wee

Fr crane former FAB er-tapped fier ridge

Rect!

Fig.

create full wave g and Fig. 10 oneas it will affect the si? peduodesr a Note that Fig. 8 - sibs

eA ng DC

OU 'and that Fig. 10 will also

f

Saami

because of the direction o yave! ii o

Bm oridge circuit. ulsating rectifier ac ag represent Ee aeae AC to pulsating DC. her mar is a steady level. A

I ee on

he oe DC" by filtering and regulating eve be steady for the required load. 10 as to \|" an values,

5° ulse per cycle fer Fig. 8 produces one p gave tifier hari rectifier Fig. 10ae wre Fig. 9

wr EAC. Asa result, the ripp! ; Ac oe sien will be ie pea oer hall f the full-wave or bri

pe hat

i the rectifier p -wave or bridge opens,

t garde se rteaenle frequency will be the

i shalf-w : aeACinput frequency.

<"apter 44

Fectifier

ten Ponts. This is nog theconstant De

Pict

we need Suppose Fig 5 represented water in a

Blass Ifthe

high points Fe present a full 21448, then the

Ponts:

would he

an empty glass Suppose we Fefilled the B)a89 after JUS a

Little

water was removed What would that do? Roghtr The gy would never be empty Viens Fig V1 and Fig 12 demonstrate this,

Full Fut

1 Empty

4 Anpple

Counter can XPose the

Tpple

frequen, Fig 11 ttle

Full Loss

ri ~ Circuit, ig. 13 The capacitor r educe the tippie or change i

Pulse

amplitude in the DC Output. snd reduces the How does the

Parallel}

connected Capacitor do that? Like the glass holdin: 18 Capacitor holds electric ity in the form of a charge. The Capacitor is charged of filled, and only loses a Portion of the charge before is filled up once again. The result is a fairly smooth DC voltage, How smooth is it?

Electronic

Power Supplies

Chapter

17. the lowest discharge The peak vast.» ienietqualto the percentage vahe. wea value that

is ager y of rple AC nppie change

Sof Ringe©

DC peak out The percent of nipples difficult to calculate _ o . the wave shane of the drecharging capacitor That es at* coetralie' ty the value of the load connected

supply, anni rtrs related we $TC = R \times C$ which is the hme

con: tami for the capacttos filter and kaad resistance

When the peak voltage is 10 V and

Discharge

is 1 V. The

This

would equal

$$U_5 = 26.2\%$$

Typical

percentages are

1%

or smaller being better? The larger the capacitor, the less ripple the DC output.

Notes:

« Filters consist of capacitors, resistors and inductors.

More

capacitance. that is, replacing a 10 mF with

a

mF, would cause a lower ripple voltage and smoother

DC output

BE CAREFUL! The capacitors

in a power supply hold

a charge even after the system is turned off.

Before

working on a power supply always:

Turn

off the power

Discharge

all capacitors

To calculate the

DC voltage output of a power supply, and the voltage ratings

of the capacitor required across the rectifier, the following

formulas

can

be used, when the AC voltage is known
at the transformer secondary.

$$V_{p-p} = 2.828 \times V_{is}$$

$$V_p =$$

$$1.414 \times V_{rms}$$

$$V_{om} =$$

$$.707 \times V_p$$

$$V_{ave} =$$

$\frac{1}{\pi} \times V_p$ For example if the secondary of the transformer
is rated

at

V_{rms} , then the peak

DC output of a half wave rectifier

is

$$V_p = 1.414 \times V_{rms} = 14.14$$

volts peak

$$= \frac{V_p}{\pi} \text{ to account for the voltage}$$

drop across the diode which may

be 3 V for Germanium and

Shockley

diodes and .7 volts for Silicon diodes, r_s is the resistance of the actual

DC output will

be

Trouble Symptom:

If the ripple voltage is higher than the frequency is correct, the filter is the, Mea 7 J roby 'my, lem \

Here Is

an example of how ripple shown (in the shaded gray area) specifications of « power supply,

+33VQ1IA

+5VSB@ 2A

You can research more specifications

by visiting www.pcpowercooling.com

Regulators

The regulator is used to keep the output voltage of the supply constant in spite of changes in the power demands of the system load connected to it. Did you know? Your radio demands more power when you turn up the volume than it does at a low listening level. Think of this the next time your favorite song is played. Actually, changing the volume level would be just another event to a regulator that must do its job with every little change in the music. How does it do that? The secret is in a very specialized part called a Zener Diode.

As voltage and maintain remain within a specified the circuit so that it is reverse-biased, 'vert range, so that it regulates the parts such as resistors and transistors and will also be found in will be at "et al Fig 14

Zener Diode

is a key part to regulation but other components are found. See Fig 15

ECON POWER SUPPLY

are being produced and for 07788"

Better

and found in Power tools and

Automobiles

computer,

Cell**Occasions****Technique**

a

Change

from

DC becomes Diode becomes defective,

a higher than NZ More popular back page could result. This higher

Voltage

is

to use 110 VAC

LaresPowerin a -

im je seco oF 5M i become def. ble power inverters maybe exp

he rat (gif ogWhenthe other sections are replaced Such as www volageconverter oe? "Ng web

or

D oot ly for a time but eventually

th - a a

ic y osne geschangingthe disk drive,hard ri hele

More

popular DC ers oy ce Yaa Wile comands; ound 19 our PC. m8

a switching poo PS 8 the one) AC line frequ; YPC that 3 quency to a much hy her anges the "oe Sigh Creates a power supply
ire Frequency Thisde. smaller components. For a nore efficient and a regulation power supplyvisit dcatled discussion On the a
www.howstuffworks, a vst) filload)

a SUS SOM /POW

Er supply

a, a Summary full losd)

we Take a look at the section Z yest of npple,

a lower the percentage of regu- 15. It consists of a elheae

Power

supply

in

Figure

y peer

POET

SUPP type filter and a Zenerdiode regulary scape lode regulator. all ofthe bee * ered in this chapter. basies coy. (ol : Rectifier ;

Fitter

Zener Reg

Step-up oF : AC to Pulsating :

Pulsating

DC to Mater ie 8 DC : RippleDC : Ripple

DC tosmooth, © pana

+ tegulated

DC

a

i

Chapter 11 EB

Chapter Quiz:

1. An electronic power supply may be designed to | a. change DC to DC change AC to AC & change the frequency | a change AC to DC above answers are correct
2. In the normal layout (block diagram) of a DC power supply the order of the sections would be:
 - a. Filter, Rectifier, Regulator. | b. Rectifier, Filter, Regulator. | c. Rectifier, Regulator, Filter. of the above. } 3. This section of the DC power supply changes AC into pulsating DC, | a. Transformer
 - b. Rectifier
 - c. Filter d. Regulator
4. This section of the DC power supply would step-up or step-down the voltage.
 - a. Transformer
 - b. Rectifier
 - c. Filter

Regulator

5. This section of the DC power supply would provide a steady DC voltage to a system.
 - a. Transformer \$ b. Rectifier { c. Filter ! d. Regulator
6. The zener diode is the most important component in this section.
 - a. Filter
 - b. Regulator c Both a & b
7. A ripple frequency of twice the AC frequency f_s is normally associated with:
 - a. a half-wave rectifier. a full wave rectifier.

Electronic Power

8. Changing the filter capacitor from 10 mF to 100 mF would result in:
 - a. ripple voltage.
 - b. a lower ripple frequency:
 - c. both a and b. y of the output of a full wave measures the same as the ripple this problem is the C
9. The ripple frequency of a rectifier power supply input. The section causes!
 - a. transformer
 - b. rectifier
 - c. filter
 - d. regulator in the specification,

10. The ripple voltage is higher than that causing this frequency Is correct. The real problem is the a by
- a. transformer
 - c. filter
 - d. regulator
- ea POST disk drive the A new disk drive was instantly displayed. The hard disk the cables were replaced and still the problem persists The motherboard was replaced and the system operated properly. After a day of operation returned. The most likely cause of the trouble is «(No reference in text, use logic.)
11. A new computer gave time it was powered up- and the same error was
- a. the power supply
 - b. the disk drive & the hard drive cables
 - d. the motherboard
12. The switching power supply used in a PC takes input from the AC line and changes it to °
- a. a lower frequency
 - b. a higher frequency

ANSWERS on page 223

Chapter 12 Amplifier Circuits, Types and dBs by David E. Caldwell, CET. David E. Caldwell is a senior certified electronics technician in avionics for the U.S. Air Force and state-of-the-art weapons systems for 20 years, before developing and teaching his current new certification.

Amplifier

Amplification is the process of increasing the magnitude of a quantity (especially voltage, current or power), of multiplying any other quantity. In the study of electronics, without increase the magnitude of a signal's voltage, current or

power, an amplifier consists of an active device,

which converts DC power and an input signal device into a function called gain. Gain may be expressed as: $\text{Gain} = \frac{\text{output signal voltage}}{\text{input signal voltage}}$

Gain:

Current

$\text{Gain} = \frac{\text{output signal current}}{\text{input signal current}}$

Active devices used as discrete amplifiers are: + Vacuum tubes (VT) + Field effect transistors (FET) + Bipolar junction transistors (BJT) integrated circuits (ICs) that incorporate some of their biasing circuits internally are in wide use and of the less complex ICs is known as an Operational Amplifier (Op Amp). We will briefly examine each of these characteristic curves and bias them to operate within

specified parameters. vacuum

Tube Amplifiers

Today, these days, they have vacuum tubes are not widely

used. Notice in Figure 1 a graph representing characteristic curves on the right and the corresponding transfer curves for each load line on the left. The load lines on the right are calculated by first assigning a maximum value for V_p

by using the tube is shorted and calculate I_p . For 24k, this equals $(300) / (24k) = 12.5 \text{ mA}$. For 48k, we set $(300) / (48k) = 6.25 \text{ mA}$.

$V_g =$

grid voltage, $V_p =$

plate voltage, $I_p =$

plate current, and

$R_p =$

load resistance

With

V_p and I_p calculated, simply draw a line connecting the two points on the plate curve. To establish the dynamic curves on the left, assign our values of V_p and note the intersections of V_p and your load line. Plot these points on the left side of the graph and connect them. This will yield the dynamic transfer curves on the left side of the graph.

pe

50 Hz to 15k Hz input

the dynamic curve on the $0V_g$ that will let the signal ride on the most linear portion of the load line. For our 48k load line, this would be -7 volts. Our signal will drive V_g down to -12V and up to -2V. With

$a_v = 77$, we have $i_{p2} = 5\text{mA}$

and

$V_{g1} =$

V_{g1} . This will be our quiescent operating point (Q), To determine R_g , we need to have the cathode

resistor more negative than the grid and we know we will have $I_P =$

to 2.5mA so:

$R_{y1} =$

$(Q_{y1})/(Q_{y2}) +$

$(7) / (2.5\text{mA}) + 28\text{k}\Omega$ Let X of $C =$

$(R_{y1})/(I_P)$ at the lowest frequency to be bypassed, so:

$C = (I_P) / (Q_{y1} \times f \times R_{y1}) = (2.5\text{mA}) / (6.28 \times 80 \times 2.8\text{k}) = 11\text{ }\mu\text{F}$ So we choose to use a standard 10 μF capacitor. TV note positive - For a grid resistor we will use a typical value for R_g of 470k Ω , this will allow any space charge that accumulates around the grid to bleed to ground,

: Amplifier Circuits, Types and dBs

Chapter 12

Class of Operation nal applied. Whee the ators pon of he rede other alterna When cvipot curren flows for 340° of ty, wd nine he say the amphiicr 1 · avs A nnd also \ 1 HI Aw We causing fy (o

wl poe x the meer a) wn WR

na Whe DX bras or the amplitude ofthe inp arene 1 NTN yosted, the class can change

i ous Vi © Ore Ith, \ gin of WAY 10°

a) wah if res Output Tf output current flows for 280°, m15 Clayy mately 200° rts Class B and if less than J; ir, itis Class C Class Camps are basically reserved for the tank ci cuit associated With there » VT, the rest ofthe signal Class AB and B can be y, ann reproduction if connected as push pull amps FET Operation

a

ud For our ne stexample, let's examine aren trolled device sumular to the vacuum tube. Ref, and 4 and note the following: Vg iscqunaleat to VA ol, Is coquivalent V, from our vacuum tube ¢ ier cit Vo! equivalent

Figure

cn (td

F

ce an dra ·00 Lat's compare the p RL we use the QUES av 13 he an she tube by US ang SN (anu) or (arp ea, for cur cucult yal

2 Ourp (2683

yl 10) cunts: canal

on

af hx cues =

Input

resistance) by 1 win V, 10)F

Ifweuse

imAcst when | ona le by te oral Voluge deals

rR ash) (Ake (193 Figure 3 FET amplifier cirey establish Ve a

i Note how the same results, bo weer, RBS not also affect gut components ·Tis is because He cae ote nea POF that we as kept he at a conducttg forthe

oo ing ad ine ad we kept of the puts cycle

cttoug,

Amplifier

Circuits,

Types

ond dag san apn nt

Me (AV) AN) sma

ne drain resistance: $r_D = \frac{V_D}{I_D}$ (A) V_{DD} of the circuit may be determined by: $V_{DD} = I_D R_D + V_{DS}$ of our load line.

If first establish the two end points V_{DD} power supply and I_{DSS}

ve owing the V_{GS} , I_{DSS} the JFETs based onal

he cog apes a ibe way (sua tt V_{GS})! (R_D) establish $I_{D,max}$ to be I_{DSS} . If we wish

to am- e input signal in a linear fashion,

we must establish V_{GS} alloth V_{DS} swing a $-V_{GS}$ from out

whlrein wiin

be

oc

of pn for pis jer apifcaion $-V_{GS}$ our

V_{GS} ,

..This

establishes our quiescent V_{GS} Van 2 Aah pingsa R_D 67 V_{GS} (V_{GS}),

$$V_{GS} = \frac{V_{DD}}{2} + \frac{I_{DSS} R_D}{2}$$

$$V_{GS} = 12V$$

$$V_{GS} = 3V$$

BAX of shuld beap. pena noofthis valu. ve apply our input signal, the gate voltage will de-

$$I_{D1} = 5mA. \text{ This results}$$

in

$$V_{GS} = 12V,$$

the next

$$\text{alteration the } V_{GS} = 3V \text{ decreasing}$$

I_{D1} to $3mA$

- This

results in V_{GS} , $4V$.

We achieve

a voltage pino! QN)

8.

'compareA=4totheamplification

factorof theJFET

fia=i 20VandSVanddrawa linehorizontally

Chopter

Hl, 6mA AV, SVwehneV

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V« Meche,- dmx

This

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BPOSVY

UAV)+118) (1260) 0130

Seven

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o.\$°SmA

at SV. and O38 mAatV,, DV, therefore,

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= (18)

(Sm) =WOK

Computing

voltage gainAV with these valuw

ATER VIR

+r) (120.013 3k) /(15 3k + 300K)+

Allowing

for errors reading the graph, we get nearly the same results. This also further illustrates how much impact the biasing circuit has on I_{out} (amplification factor) was approximately 120 but our G_m was held to or $\leq$

Bringing

the JFET into Ohmic

region will result in a non-linear operation similar to the VT but necessary if you plan on operating class AB, B, or

C. Bipolar

Transistor Operation

Our third device, the BJT

(Bipolar

Junction

Transistor}

is a

Current

operated device

Study

Figures

and

and note that if the transistor is biased off, it will have $V_{CE} \approx 12V$, and if we bias it to the point of Saturation,

we will have 82

mA

of

Draw

a load

Line

between these two points. Next, we establish the quiescent operating point (Q) for our amp. For

Class

A operation, choose a point midway along the load line.

We choose

$$I_B = 40.4 \mu\text{A}$$

prong V_{BE}

$$V_{BE} = 0.7 \text{ V}$$

of $I_B = 40.4 \mu\text{A}$,

Note

how

$$I_B = 40.4 \mu\text{A} \text{ controls}$$

$$I_B = 40.4 \mu\text{A}$$

mA gring β (beta) of approximately 120, β varies from various types of transistors and is not

Part number

transistors, Transistors can also break down from thermal runaway.

To compensate

for this and

to regulate out Current gain, we use different forms of biasing. For our example, we will illustrate one of the most stable forms of biasing known

as voltage divider biasing

Figure

$$V_{CC} = 10 \text{ V}$$

$$R_1 = 10 \text{ k}\Omega$$

$$V_{BE} = 0.7 \text{ V}$$

$$I_B = 40.4 \mu\text{A} \text{ so we can find } I_E$$

$$V_{BE} = 0.7 \text{ V} \text{ and } V_{CE} = 10.82 \text{ V. The } V_{BE} = 0.7 \text{ V and } V_{CE} = 10.82 \text{ V so we will}$$

use this stability factor

of

to calculate the size of R_2 ,

by using the formula:

$$S = \left(\frac{R_1}{R_2} + 1 \right)$$

$R_y) (R, X_{R_y}, R_y)$

We also

know that R_{e1}

is

9.18 times larger than R_{e1} , because of the voltage drop ratios $(10.821/1.179)$.

We can substitute $(9.18 \times R_{e1})$ for R_{e1} , $(9.18 \times 444) \text{ k}\Omega$ for R_{e1} . To verify, we use the voltage divider

id $[R_y R_{e1} / (R_{e1} + R_{e1})] \times 12 \text{ V}$ | ay One last component to calculate is the cap. For most applications, set its X_{C1} at the lowest frequency to be amplified

16, Let's apply a signal to our circuit

Figure

GOA to 30 nA causing the I_E to vary from 6 nA, $I_E / I_{E1} = (3.5 \text{ mV}) / (40 \text{ mV}) \approx 87.5 \text{ mA}$

We also know that R_{e1} is 9.18 times J_a of the voltage drop ratios $(10.821/1.179)$

er than We can substitute $(9.18 R_{e1})$ for R_{e1} .

$r_{e1} = (100 \times (9.18 R_{e1} + R_{e1})) / (1 + 4440 \text{ }\Omega)$

th

Is R_y , $(9.18$

$\times 444) \text{ k}\Omega$. ah) To verify, we use the voltage divider $C_{quat} [R_y R_{e1} /$

$R_y @ V_{ee}^*$

$[(444) / (444 + 4 \text{ k})] \times 12 \text{ V}$ 1.2 One last component to calculate is the sag of cap. For most applications, set its X_{C1} at the lowest frequency to be amplified. Let's apply a signal to our circuit. Figure band 60 nA to 30 pA causing the I_E to vary from 6 nA, $I_E / I_{E1} > (3.5 \text{ mV}) / (40 \text{ mV}) = 87.5 \text{ mA}$. I_E can be determined by first determining

ac the emitter. r_e ,

$$= 25 \text{ mV} / 1, -$$

$(25 \text{ mV}) / (4.79 \text{ mV})$

on To allow for variations we also determine the using $f_{(50 \text{ mV})} / (I_E) (50 \text{ mV}) / (4.79 \text{ mV})$ lag Next,

$$A_v = (R_{e1} / R_{e1} + 1) + (14 \text{ k}) / (100 + 5$$

9), Since we added a bypass

C_{e1} , we can use

$$R_{e1} / t > (1.4 \text{ k}) / (5.2) = 269 \text{ or } (1.4 \text{ k}) / (104)$$

134.6, Finally, let's determine power gain,

$A_v = A_{v1} \times 134.6 = 11.8 \text{ k}$, This may be expressed

in dB's;

$$A_v = 10 \log 11.8k$$

40.7

dB.

to bias our various amplifier circuits may use

commonly to compensate for amplifier constraints like impedance matching, frequency losses, and input

resistor amplifier circuits compared with FET and vacuum tubes:

two types of config!

common emitter (CE)

if

at

a). rise [ah]

Common CE

[| $A_v = \text{Low}$

power

$f_{\text{max}} = \text{moderate}$

| $Z_{\text{in}} = \text{low}$

| $Z_{\text{out}} = \text{high}$

power

Common Source

Sut

Common Source

$A_v = \text{Low}$

Figure 7

was created so you could visually compare the various configurations. The common-cathode (Ck), common-drain (Cs), or common-emitter (Ce) amplifiers have been used. Voltage and current gain are most widely used. Common-source (Cs), or common-drain (Cd), common-gate (Cg) amplifiers are used primarily as RF amplifiers because of their high power and voltage gain but no current gain. The

si 'Cp), common-drain (Cd) and common-col- es are used mainly as isolation amplifiers in power and current gain but no voltage gain,

IC Operational

Amplifiers

vanced, the IC (integrated cir-

As

on et op amp is one such device. It consists of the amplifier circuits internally : has a way as to give it a high input impedance so it won't draw the signal source and a low output impedance so it can drive a low impedance load. Op-amps may have open loop -- Amplifier Circuits,

Types

onddp,

, Chapter 12

a Of 200 o) One added advantage of the Op amp is its common mode REJECTION, which allows it to reduce hum and (Output Iau | Input wae ee

Figure &

inverting (A) and Non-inverting (B)

Op-Amp

If a signal is applied

m

Figure

8B, the

A, $V_o = -V_i \frac{R_f}{R_i}$ (for the inverting MPUt as illustrated in Figure 8A,

the $A = -\frac{R_f}{R_i}$.

Notice how gain is set by the ratio of R_f to R_i Op-

y 7 » Op-amp tends to drop off rapidly at higher frequencies

mA

Types of

Coupling

Three types of coupling are used in cascading amplifier Circuits together-

1. direct
2. capacitor

3. transformer. Figure 8 above uses Capacitor coupling. Figure 9 below Uses direct coupling and Figure 12 uses transformer coupling

Negative Feedback

Negative

Feedback in any amplifier reduces gain and increases bandwidth, where Positive Feedback causes oscillation. Amplifier circuits

Darlington

A Darlington

Pair is an example of a directly coupled pair of transistors. The collectors are tied in parallel while the base emitter junctions are in series. It has high input impedance

to reduce circuit loading and low output impedance with good Power gain. Often it is housed in a single package.

soe - ter 12 ! Push Pull RF Amplifier Our focus so far has been on Class A operation, but at times we may need to use 2 ac

fy amplifier configuration Push-Pull Miller n t amplify one alternation of the input signal 'sh

a A vice amplifies the remaining alternation of the Miller La - realize any increase in efficiency we must, teas

Low

operate as Class AB or B and still amplify the in fashion. See Figure 12 * par

9 Darlington

amplifier input matching requirements! nner oT et of 8

Teen

input

dv them differ - they are are external network. The output voltage "phase

St

Is

Horn

a balanced center point and impedance of each transistor

as well as Figure 12. Push Pull FET RF amplifier (the collector

in Op-Amp circuit input need are balanced signals At radio frequencies, a class C amplifier with

a parallel LC resonant circuit can be used to reproduce the electron beam and unbalanced. See Figure 13. Class C conducts for less than one-half cycle of the input signal. The same frequency is applied to the input and output tank circuit. The amplifier pulses to make it ring at the resonant frequency to produce output. Vee

Input B

Input A

so Differential Amplifier

Figure

MOSFET

elementary amplifier power MOSFET is shown below. One MOSFET is used in a series

Figure 13

RF frequency conversion Output An application of non-linear amplification is frequency conversion or heterodyning as use - two radio frequencies are fed to an amplifier in linear fashion, they will combine to produce both sum and difference frequencies. If the tank circuit in Figure 13 has those four frequencies, for example the difference frequency will be amplified and passed on to the

Figure 11

QW 3 A

The amplifier with center frequency is tuned to difference frequency to the following.

b) of the tank circuit will determine how wide range of frequencies may be passed along with the carrier frequency

of the tank This allows conversion of a radio frequency containing either amplitude or phase modulated intelligence to an IF amplifier (intermediate frequency)

Frequency

in the receiver When the received RF is mixed with a LO (local oscillator) frequency, the intelligence is recovered from the RF input to the IF stage of the radio through tuning the first RF amp of the receiver with the

Image

reject any undesired image frequencies that may mix with our local oscillator and produce our given IF. This is due to the receiver's selectivity (ability to reject image or other frequencies). More RF discussion will follow in RF Detection Another use of non-linear active devices is receiver detection - See Figure 14 Detection is the process of recovering the intelligence from the transmitted radio frequency. Once the

intelligence has been transferred to our intermediate frequency - it is first amplified to a suitable level and then fed to the detection stage. Figure 15 FM detector For an FM (frequency modulated) signal, see

Figure

where the IF is fed to a PLL (phase locked loop) circuit or discriminator. In a PLL, as the input frequency varies, the DC error voltage will vary in an attempt to keep the VCO on track. The variations in DC error correction voltage level and

= "Amplifier Circuits, Types and dBs | Chapter 12

frequency of those variations of the

DC error correction voltage - age equates to the intelligence that originally modulated the transmitted radio signal

The Decibel (dB) by Fie Reid

The Decibel (dB) is a sub-unit of the bel, named after the inventor of the telephone, Alexander Graham Bell There are

10 decibels in one bel (1 dB = 1/10 Bel). Since the decibel is

a ratio, it may represent volts, amperes, watts, acoustic power, etc. The decibel can be regarded as the smallest difference in sound intensity that a normal ear can distinguish.

An

average conversation in a quiet room is about 50 dB

above perfect silence or zero loudness. Let us avoid confusion with loudness and sound intensity. Intensity of sound is determined by the amount of energy carried by a sound wave. Loudness is the actual sensation produced in the brain. Loudness depends on: The intensity of the sound waves which strike the diaphragm of the ear, The functioning capability of the ear and its related hearing mechanisms.

Notice that 0 dB indicates the Presence of sound.

It does not mean NO SOUND. It is the sound intensity to which other intensities are compared.

Some Facts To Remember:

1. 4B is a ratio; therefore it needs a

Reference

2 - dB is positive when

it represents an increase above the reference level and dB is negative when it represents a decrease below the reference level

4. Gains are positive and losses are negative
5. dB may be added or subtracted directly
6. We hear sound in a logarithmic fashion

Power Levels

$$4B = 10 \log P_o / P_i$$

P_i ,

Where: P_o =

input power

P_o =

output power

'r vd th

Frxomple:

The ovtpat ofan ammpliter

Oe

a

e Fan8.8

Watts

Cakelate

em oF in ») *dR

b) dR

co) 03dB

4) 0348

SOLUTION

Gain

$$= 10 \log P; F,$$

$$= \log 7/3 \$$$

$$= 10 \log 2$$

$$= 10x (3010)$$

$$= 3 \text{ dB}$$

Choice (a) Example: A power of 10 milliwatts is required to drive an audio amplifier. The output of the amplifier is 120 milliwatts. What

is the gain expressed

in dB?

a) 108dB

b) 12dB

c) -124B

d) -10.8 dB

SOLUTION:

$$\text{Gain(dB)} = 10 \log P,$$

P_p ,

$$\text{Gain(dB)} = 10 \log 120$$

$$\text{Gain(dB)} = 10 \log 12$$

$$\text{Gain(dB)} = 10 \times 1.08$$

$$\text{Gain(dB)} = 10.848$$

Choice (a)

Example: A network has a loss of 16 dB. What approxi

mate power ratio corresponds to this loss?

- a) 40:1
- b) -16:1
- c) 16:1
- d) 4.01

SOLUTION:

$$\text{dB} = 10 \log P_1/P_2,$$

$$16 = 10 \log P_1/P_2.$$

(Divide both sides by 10) 1.6 ~

$$\log P_1/P_2$$

$$(\text{Taking antilog of 1.6}) = 39.8 P_1/P_2$$

$$= 40:1$$

Choice (a)

Example: A power splitter was developed to split the output of a circuit of 9 is modified so that the power to the load is 4 from 450 mW to 68 W. Calculate this power attenuation in dB %

- a) 16 dB,
- b) -16 dB

d) -16

SOLUTION

$$UB = 10 \log 650 / 459$$

"Ig hore,

Voltage Ratio Levels

'y The true formula for finding voltage gain in decihe, Say

$$\text{Gain dB} = 20 \log (V_1/V_2)$$

Example: The signal measured on a simple dipole is 500uV, What dBm does this signal represent? Men, |

- a) -2
- b) 2 c) 6
- d) -6

SOLUTION:

The reference signal used n RF work is 1000 nV. or 9 dBmV. Therefore: Gain,,

$$= 20 \log (VV)$$

$$\text{dB} = 20 \log (500/1000)$$

$$= 20 \times (-0.3)$$

$$=-6$$

Choice

(d)

a8 RATIO RATIO

-H

FE i it lee

ET Le

Table 2

ook at way:

S

$$=6\text{dB}+6\text{dB}$$

aya phe (Voltage Ratio)

=4 times

amplification factor

4) ification by the factor, One'

id

= 60

dB-6dB +3dB

ae (2) 57

Bo 00-22 XL.4

| = 700

ation by the factor 700) amplification: increase

or | UB dBs expressing | types of dBs where simplifying decibels by using a table. 30 referenced to .0002 (SPL) levels re! and pressure for audio sounds' seconds per minute power or voltage) at a point vel (a .

dB is referenced to | mW)

level specified, as in dBm, (the "m" suffix

ce

31

Relative

difference between two points a dB unit by the

bandwidth relative dB Measurement between two

Transmission

Level

Power (TL Power)

1) Acoustic sound pressure Also known as loudness of Perceived sounds

Set Engine

2000dyn

em? =

14008

$$20 \times \log \frac{2000}{0002} =$$

140 UB sound level

2) Absolute: dBA Acoustic threshold of hearing referenced

to 0 dB

dBm Power level referenced to 1 mW (See 48m chart) dBV Voltage level referenced to 1 volt dBmv Voltage level referenced to 1 mV dBrn Noise or signal level referenced to 1 pw dBrac Noise or Signal as above with C weighting dBrnC As above, but also normalized to 0 TL dBw

Power

level

referenced to 1 Watt

dBu Voltage that corresponds

to dBm in a 600-ohm circuit - about 78 volts

3) Relative:

4B =

(+dB) Gain or (-4B) Loss of

acircuit, NO Refer-

ence specified. (See "dB

chart)

TL =

Net gain or loss of transmission path between two TLP points or dBm levels. Tables on the following Pages are provided for your use and reference.

NOTES

iof

Chapter

12 +.

Relative dB chart

Power ratio

1,000,000
100,000
10,000
1 Unity Gain
0.0001 (10%)

Based on

dB
= 10
log

P/P, Voltage ratio

1,000.000
100,000
10,000

1 Unity Gain

(10) 0.0001 (10*) - 0.0001

Based

20 log V, /V,
dB

60 dB

0 dB



20dB

6dB

0dB

-6 dB

-20

dB

40 dB

-60dB

-80dB

-100dB

-120

dB

Gan

Gain

Loss

Absolute dBm Power Level; is Signal Power dBm 1000

W (1K W) 60 dBm 100 W 50 dBm 10W 40 dBm 1W 30 dBm

TW (100mW) 20 dBm 10 mW 10 dBm 1 mW 0dBm 1 mW (100mW) -10 dBm 10 mW -20 dBm 1 mW -30 dBm .1 mW (100 nW)
 -40 dBm 10 nW -50 dBm 1 nW -60 dBm .1 nW (100 pW) -70 dBm 10 pW -80 dBm 1 pW -90 dBm 1 pW (100 fW) -100 dBm 10 fW
 -110 dBm 1 fW -120 dBm 1 fW (100 aW) -130 dBm 10 aW -140 dBm 1 aW -150 dBm any

daw

aBy

daw OdBy "10 day -20

By 330 dRW -40 dBW -50 dBW -60 dBW -70 dBW -80 dBW -90 dBW -100 dBW

Brownian Noise Floor

-173 dBm per Hz

-133 dBm / 10KHz

-113 dBm/ 1 MHz

- 93 dBm / 100 MHz

Units

T = Terra 10°?

G = Giga 10^9

M = Mega 10^6

k =kilo 10^3

m = milli 10^0

m =micro 10*

n = nano_ 10^9

p = pico 10"

f =femto 10¹

a =atto 10'%

Charts courtesy of John Baldwin, CETsr

Circul

the μ formula $\mu = \frac{g_m}{g_{p1}}$ is used to find the aeternm "rpter Quiz asconductance of the vacuum tube g_{p1} plate resistance of the vacuum tube factor of the vacuum tube current transfer ratio of the vacuum tube Juss AB amplifier will allow A_c utput current to flow roximately for $\frac{1}{\mu}$ of the input signal. 100° 180°

Pago"

4. 360° formula $[4 * T_s]$ is used to calculate:
3. forward transfer current ratio of the JFET voltage gain of the JFET amplifier. amplification factor of the JFET. of the JFET. age tions se three of the following amplifier configurations yield the highest power gains. common-gate common-cathode common-plate common-source common-gate common-drain common-emitter common-base common-collector me Be bout a common- owing information al
- 5, Given a .01Vp-p signal on se current to vary from Hee collector aden ri aay h our load. Dete A through! to 2.5m. dynamic current gain.
- nnn 7 Types and dBs
- 6 Which of the fattowing amplifier configuration(s) wo You likely find being used in a linear Push-pull arrangement?

Class A

Class Bor AB

Class ·

All of the above aoge

18a 1.02 MHz signal an.

a common-emitter tran: in a non-linear manner the following frequenci da 1.475 MH signal are fed to sistor amplifier biased to operate we would expect to see which of es in the output? a 455

b. 102 Miz

© 1475 MHz

d 2495 Mie © Allofthe above

Sensitivity

Selectivity

Bandwidth

Figure 14 on Page 97 illustrates de.

Modulator

an Rk amplifier being used as a mixer ter amp used as a limiter 'or

a

b

c

d. detect

10. Which of the fall lowing circuits c; audio from a; Fin a FM receiv.

Phase Locked Loop

Class A, common-gate JFET amplifier Push-pull, class A JFET amplifier

Mixer

can be used to recover er?

ANSWERS on Page 223

~ preview Questions = AN

t voltage by present

1. In the drawing, what is the voltage across the load resistor?

ay

WV

av 10V 12V ranges

be

2. The output (A) of the regulator circuit below should be about:

a

b. 9.2 VDC

c. 10VDC.

d. 43.8 VOT

e. 9.2V Zener

3. In the circuit below, to reproduce the square wave input more faithfully at the output:

a. decrease the value of the resistor.

b. lower the frequency of the square wave. c. decrease the value of the capacitor.

d. increase the value of the resistor. 4. 4

Log LITCUTS, TOWET Chapters 11 and 12

Qn

4. In the below, what voltage would be present at point A if 117 volt AC were applied? 7V

24 VDC

DC arse

5. In the circuit below, what voltage is expected across the secondary of the TI transformer if the circuit is

connected to 117 VAC?

a. b. 17VAC

ce H7VAC

d. 60

cS

d= ae Sle 4

6. How do you calculate the gain of a non-inverting operational amplifier?

a. input output

b, - output input

a

7. If the frequency of the broadband input signal to an Op-Amp increases, then the output signal level

a. increases

b. decreases c. remains the same

d, increases the frequency

ee the q | 7

alog Circuits, Power Sy

estions > AN Chapters Vand and a 'plifiers pewer foa speaker, a doubli

6 dB increase.

b op-amp below is a 5 V sine wave a about +2 V). What will the output be?

ea vane wave centered about +1 V | A gine WAVE centered about -1 V Al gine wave centered about -2 V A ysine wave centered about -4 V

2K wrong with this circuit below? wnat | ¥ are backwards. 72 filters reversed. « bode ne 4 i

OH mer» wy Ny ond

Commonly

Useq

i

Chop

(fhe ep ht outReshnedateca quay : srevien THF ge bxaskgraliann lak Tipe ey ent sgemee Pe" wig own (ATincezms RSL, cei gueiaseon Minh yy, Ny paoMr scar eae

Mn ana

Aondoein

wis daaryR hey y beicldn nivale ested DiTasco dk mae

wt

py sepia dante ; earner grain

Dstt

San ceniqor iebentseopearran ele Tera, scaly fakemecianbin - febainsletestinyy, reerecpasenimsnes aaa mai : wy los thnd

Sdeaaten

tes

Voweardeomeene

Rory hy pvstavabstinberi' in, tse tabloid fru! cham aes iaemagk wit eat secs abt Te egy al eomcemrvarenh

Soubns

encanine

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s thattheame igre ha

gy fn pong Medco scieenpoesensi (ayefay oer dyb nts ep fekptcssastgenamsonms

- 'Ntdoehaydeeeeapcetin

hag

Gridemeraiogeccconavmeriry

- Vatboehog

delenit, idashereaabvistiaden

-Melbepay

ayn!

Pi earths Bs

bu main

a cstelon eh,

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Reson

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Braga

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V ar

Oy

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Bien

testo eae Fue) ·kr ·Ay

Sy Lie hus ih

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He sent WF) ah tl rsa te opt ing ey Ths vast eat has phy

Uh Spa

Kg ie? ye ry ah gt we ata ine

yt

te si Fai jr Buti Tn ·hep ·ey

Satin

dati gery wa andor gy vat ess hal be meen rds gage slg ih ed erating gd Wouter ine SPL Sebo pal (AC) ce con

NI gx cb ine A) rae bee tals fen sn thy 2 al oy

in ene htt aa The foc ford fy they Fins i

ya · Five sins ia) } doo 1 yylef inby pues othe yom 2 athe ti PW setaHig Satins the ease

Antena

Gio Fy

i!

Chopter

Figure 4 RE Section

Mixer/Local

Oscillator

the On«e the signals are amplified. : ned (or mixed) with & signal produced 2Sa qacillator carcunt (Shown m Figure

8) The output yes. Now as a rule will contain the sum & current is constructed (ai nha op) amount to the IF (intermediate receive frequency) v favonte radio If you want to receive your

igen also to have an TF of 455 kHz. (This

ape M radios), what should our oscillator

$$= 1000 \text{ kHz} + 455 \text{ kHz} = 1455 \text{ kHz}$$

(desired freq) + (IF) Osc. Freq,

Local

Oscillator

Figure 5 Mixer and Local Oscillator Circuit Frm

Tots

ar Sect

IF Sed

All night, the oscillator signal is at 1455 kHz and the origi-

nal station that we wanted to hear is 1000 kHz. Both will be

mined to produce the sum and difference frequencies. The difference of the two signals will be our IF band pass frequency.

$$1455 \text{ kHz} - 1000 \text{ kHz} = 455 \text{ kHz (IF)}$$

We always subtract to find the IF. Our old frequency of 1000 kHz has been converted to 455 kHz. IF Section

Now that we have established an IF of 455 kHz, we can

Proceed and see how this section works. The IF sections use a set frequency throughout. Also it is really an amplification network comprised of transistors, resistors, and capacitors. This is one of the major differences with our superheterodyne iposic Radio, RF signal Propagation and Lommonty vs. Fr, fecerver compared to the basic receiver, we have by coming signal with a local oscillator T

IF

ey band frequency Now you a high tuned frequency through! . above for fe ly 'be ane Heian pepatie Ot be amplifiedop MMi" erative that the mixing process of both Mixer and local oscillator produces only this frequency

"Komp

It is much easier to amplify just one frequency

he

ut

Figure

below. Ptelh s ho pr YNCd op Ay. Ce ttn, ' ATC AMPHin,

Wahi"»

YB the . 4

From

Maer

Cote

yy Figure 6 IF Section In Figure 7 below, I have compiled a list of

Commonly Use

IF's and their uses

10.7 MHz
44 MHz
1.6, 3.35, 9 and 40 MHz Figure 7 Common IF Frequencies

Detector Circuit

Detection, as it is noted, is the rectification of the RF carrier. Note Figure 7a below: Figure 7a Simple Detector Diode If you look at the diode, what is it doing? Is it amplifying, attenuating, filtering, or distorting the waveform? No, it's really rectifying the incoming RF signal. As the desired signal enters the anode and goes through the PN junction, the negative portion of the RF wave is "removed." Then the signal will proceed through other circuitry to have the RF 'carrier' removed, thus leaving only the modulation. AGC/AVC Section Automatic gain control (AGC) or automatic volume control (AVC), as it is called, actually samples the signal at the detector and develops a control voltage that is fed back to the first IF Amp. Ideally the operation of the AGC will amplify weak incoming signals and attenuate strong ones, allowing a constant, desirable output volume, Please note Figure 7 b, aeeee taba.

tion and Commonly used Frequencies

h

From

Laat Detector

To iF sect

Figure

The AGC Circuitation helps the modulation to deal with This is the pre-amp you will find that depends on the carrier. Modulation rate

by static, electrical noise to voice, Morse code,

needs "ation, and other analog, as well as digital in-

When you took an oscilloscope and hooked up a microphone to its inputs and spoke, what would you see? You'd see voice. On the display. Depending on the voice, the waveform is different than anyone else's. As you

it's a wave: you'd see HEISE and fall

"Frequency" depending on the carrier. Also, it would expand and contract, depending on the characteristics and frequency. See on modulation. \ me common modulation types. \ see 38

for

in the modulation section. See Figure 8a Modulation Types AF Section

The basic part of the superheterodyne receiver is the audio

Frequency

(AF) Section This is an audio amplifier that drives your speakers. Remember what we talked about before regarding antenna impedance matching? The load (in this case our speakers) must match the radio's circuitry (AC resistance wise). Proper impedance matching has to be observed not only to deliver maximum power to antennas, but must be observed with audio applications as well. This gives the speakers the maximum, matched output power possible. See Figure 8b. \

i

Detector

i. Figure 8b AF Section RF Signal Propagation

Principles

Radio frequency emissions travel quickly. They move at a rate of over 186,000 miles per second, the speed of light! If you refer to Figure 9a and b below, they show a few simple examples of signal propagation and antennas. *(Electromagnetic field around dipoles) Figure 9a Signal Propagation

Radio, KFH =

Chapter

SE

vertical

element

of the

Dipole Antenna

Figure

8 Simple 12 WINE "OEY ay Sea. narrow beamwidth as dese Med hey degrees The more director etc Mente Ore, the gain rating and narrower by litt thay Yoru a est the left. are the directors Ayan neal The more directors that you have. the high is ity. The second element from the arm

and this is the part where the coax is, a tiny W element is the reflector I does just that

Chey 5

as

radiates onto the driven element Your beam is about, not a specific station, unlike our omni direction

is the element in Figure 10 which transmits and receives equally in all directions, Without getting into antennas too deep (full 16),

only hope this, give you a flavor of what some basic PCF type Fe, antenna type and their characteristics, YP es

Orwen

the EME the average SN is not Dipoles or Me are the pattern

of

det an shown in Figure 10a new "polar pattern

an antenna has a polar plot which

a vertical "omni direction pattern of @ full 360 degrees:

Overtors

son in mea- used as a reference and as

COMFA is about

A dipole antenna. Gain will be as 'as a wave dipole antenna,

is thereby having a reference for Figure 11 Yagi Style Antenna

Our Atmosphere

Ionosphere What is the ionosphere? It is part of our atmosphere that is some 40-400 miles above our planet. The sun SECTNS to be an endless supplier of the earth's radiant energy. As the sun warms up the ionosphere, bizarre things happen. If you

Tefer

and Beds even over the cat antenna electromagnetic (23 200°

Laas

Figure 10 Omni directional Antenna (Vertical) -wise, is the So what is antenna gain? Transmitting-wise, amount of effective radiated power from the antenna

Compared with that of a reference antenna. Given the same input power parameters, receiving also benefits, thus being able to receive weak and often very distant stations. The more gain the better. Why is gain important? It is important not only for long distance reception but also for distant transmission of RF, since it actually concentrates the signals on transmit and receive. Please note Figure 11. This type of antenna is called a yagi. Figure 12, we will discuss what happens to radio signals. This layer of our atmosphere is heated. First, let's look at the D Layer. It is approximately 30-59 miles above the earth. It is most prevalent during daylight hours, and practically disappears at nighttime. This is one layer that short wave radio listeners do not enjoy. This layer actually absorbs radio signals under 10 MHz, especially those

from 1-8 MHz during daylight hours.

The E Layer is less discriminating in a sense. It is located 60-90 miles above our planet. This layer will, at times, pick and choose both High Frequency (HF) and Very High Frequency (VHF) signals. Some of these signals will be reflected back to earth, depending on the frequency used and time of day. Now

Why

The F₁, F₂, and E₂ Layers of the Ionosphere. This type of "beam" antenna focuses all of its energy in a direction that reflects the furthest out

of all the layers, from 250 miles and OUT! Now these

points are situated at the

surface, since most of the wave is reflected

More

on earth. The reason for reflecting at these frequencies, with the length of the reflected back to us, unlike the waves on other layers, which pass right on

Chart

Ionosphere

7: Cable Chart or radiowoman will be required for use. In Figure 13 below, I have a set cable system which will generally be

used for the (O.D.), outside diameter

Semi

of the

Cable

effect of cable, at 100 and above the velocity factor, and the center conductor

is connected to the 100 MHz. We will discuss each

Be

with

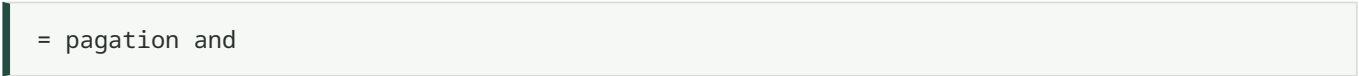
18 Gauge

Solid

RF Coaxial Cable Chart

Figure

13 Sample 'a



Commonly Used

Frequenc).

Chapter 13

Type

Installations free because the 1 lone would be too

Ohms

near 1

Determine

Figure 14 RG-58 Coax The impedance of the coaxial cable is determined by the spacing of the center conductor to the shield and the dielectric, not the diameter.

of

(Outside

Diameter)

which cable to use and testing of the line with an ohmmeter.

dB Loss

No matter what materials manufacturers will always have some signal loss, whether in receive or transmit. We use decibels (dB's)

In Measuring signal loss or gain--in this case, loss. If we refer back to Figure 13, the chart clearly indicates the loss at 100

and 400 MHz, both using 100 foot runs

of cable. The loss is

4.2 dB and 9.0 dB, respectively. Notice the difference in loss as we go higher in frequency. As another rule of thumb, the higher the frequency, the greater the RF loss will be and vice versa.

down to a minimum: transit results.

Chapter

Thos

part of the coax has multiple purposes [not only sets up characteristic impedance but also prevents RF from sagging the center conductor. This protects the entire cross-section of the insulating dielectric housing shielding characteristics of a cable are extremely important, not to mention the loss.

but also for reception

of

RF loss rated in per cent. We should always strive for maximum shielding and low loss, resulting in greater efficiency.

Velocity

Factor

Don't get overly caught up with this stem. For most applications, the manufacturer has already taken this factor into consideration when designing the cable. RF energy slows down a bit when it travels through coax. The velocity factor of a cable is dealing with the speed of the transmission of RF through the coax, as compared with the same signal propagated in free air. Our example RG-58 cable has a velocity factor of 71%. In other words, the transmitted signal would be traveling at 71% of the same signal speed propagated in free air.

As you already know, RF energy travels at the speed of light. Even if it slows down a bit, it is still traveling very quickly.

Center

Conductor

Thiss

is the heart of the cable that carries the signals to and from the antenna. It is commonly rated in a standard gauge form. The gauge we used in our cable was 18-gauge solid copper per

Some

center conductors can

be

a stranded wire

as well,

or

a combination of both. The thicker the center conductor of the cable, the lower the signal loss. RF Safety-SAR

Specific

Absorption

Rate (SAR), as it is termed, is your body's ability to absorb

RF energy

As a technician working with radio transmitters, antennas and cables, you should be aware of the potential dangers with RF. Generally, the higher the frequency you attain, the more it affects your body. It takes lower frequencies to do the same damage.

at some frequencies with higher power. Radio frequency emissions, at certain frequencies and power levels, resonate and produce a heating effect of human tissue. Be careful in the

of the main organs that

we need to be concerned with when. There is evidence that strong, near-field RF energy

Your

eyes over the long term
 from the formation of cataracts,

SPEEA

LTS

SOMO

First, Now that I have seen that it can be reduced. The first one, Never touch a transmission, can result. Never repair RF % TOs
 per Wel Wilke per, as with legs "s hint removals as a fiber optic cable or wave guide as well as all equipment, make low
 Power trans; and terminate all connections that could affect the : deal true with microwave emissions [fy "8 "i ite good sense,
 observe the situation, analyze the effects, and limit our problems, we should enjoy RF '

Poses

lions and pride ourselves in the fact that we are.

Electronics Technicians

do you BITE,
 gg HIM is tein, Oth
 nn
 TR

Bagg

Se antenna My th, mbly Wet lag MISSIONS with Min Con,

i The following chart will list some commonly used frequencies and their appropriate uses. It would be wise to commit them to
 memory. In closing, I truly hope that the information provided herein was informative, educate fun. Good luck with your career
 in electronics, may it be a rewarding one not only for yourself, but also for others YOU help along the way. Commonly Used
 Frequencies FCC Band Plans The Federal Communications Commission (FCC) establishes all of our frequency rules,
 regulations, and frequency allocations Here are some common frequencies that should be committed to memory Radio
 Frequency Spectrum

y

Medium Frequency**300 Kilohertz-3 Megahertz****Extremely****High Frequency****30 Gigahertz-300 Gigahertz**

aS

CE

power RFS caght sideband

ss modulation (SSB) is a refinement of am- rd at more efficiently uses electrical power othi ss closely related to vestigial sideband

F wo th tr to AM TV broadcasting) a (8

vd ral Jation produces a modulated output signal

a alihe bandwiath of the orginal baseband signal, a cae ulation avords this bandwidth doubling, | y ted on a camer, at the cost of somewhat

Seen

complexity:

e | AO ignal Propagation and Commonly Used Frequencies

Chapte

Chopter Quiz:

nee mean? a of tb Fanancaa! Commissaty

Federal

a Communications (Faraday Commission of C ne radio receiver? does

2. What spot a part a Frequency Cirvutt b Radio Frequency Circurt

ç Audho Frequency Circuit

4. Radiation Rewstance

Mixer Circuit

4. Match up the correct items:

Amphitude Modulation

Weather (no chapter reference)

Audio Frequency

Radio Frequency

Intermediate Frequency

Frequency Modulation

RF

AF

IF

AM

WX 4 Which part of a superheterodyne recelverrectifies the RF carrier?

- a. AF

b. Detector Circuit

Ç. Low Pass Filter

ç. RF \$. How long should a $\frac{1}{4}$ wavelength vertical antenna be

that is designed for operation on 146 MHz?

a. 1.1 Feet

b. 1.6 Feet

Ç 19 Feet

d. 2.3 Feet

32 Feet

6, What is the common impedance of radio communication and television antennas respectively? $50\ \Omega$ and $75\ \Omega$ ohms

6, $50\ \Omega$ and $75\ \Omega$ ohms $75\ \Omega$ and $50\ \Omega$ ohms $75\ \Omega$ and $100\ \Omega$ ohms $50\ \Omega$ and $100\ \Omega$ ohms

RF Signal Propagation and a Basic Radio, RF Signals and Commonly Used

1. Which layer of the ionosphere absorbs high radio waves during daylight hours? a. D Layer

b. E Layer

c. F Layer

d. F₂ Layer

Ç. F₂ Layer

1. The

8, Which style of antenna radiates energy with a pattern resembling a figure

a. Omni directional

b. Yagi

c. Dipole

d. Cubical Quad

Log-Periodic

9, What frequency band is an average broadcast radio capable of receiving:

a. 535 to 1700 kHz

b. 1810 kHz to 8100 kHz

c. 3.8 to 4.2

10, Why is RF safety important?

1. To insure maximum power output at all times

b. Always assuring variable tuning capability Capability to receive distant radio To limit human exposure to RF for health reasons bandwidth spacing Δf , Maintain adequate FCC

ANSWERS on page 223

a7" by Chief Petty Officer

Hames

currently serves in the

1S Navy

ag Jom has fixed radar, radios, sonar, data link, military thom

ao

an

Industrial

Uelectromics

CET

he has

a iss, ; o.

5 (Monor)

from San

Diego

Mesa

College

rs cally convert intelligence into a

Signal that

oso ipsadistance to a receiver The receiver idcanes transmitted signal back into the original

o cote intelligencecan be MorseCode, voice, video, gem tion. The method of

Converting

intelligence ee sre signal is most often by modulating

a signal ns the intelligence o a receiver. Modulation

Is ye of umposing intelligence on a carry signal

ae jency is usually in the

Radio-Frequency

gi les ao enaae Spectrum. RF signals lie

af rey range of 3

KHz to 300 GHz,

RF signals can travel at the speed of light (3.0×10^8 Meters per second)

Power of the radiated signal depends on the power - COMMUNICATIONS transmitters typically

30 Watts to MHz frequency

range Satellite com.

operate in the GHz frequency

range. RADAR also uses a

the GHz. RADAR is used to determine

the direction of an object, uses Wave (CW) Transmitters ~ The output of a CW is a constant amplitude carrier wave that is
4 The CW signal is keyed on or off to transmit Morse code that can be interpreted at the receiver

usually

Morse is an example

of Master-oscillator-power amplifier with keying in the buffer stage, Basic CW Transmitter

Figure 1

Morse Transmitters ~ AM transmitter is the intelligence to be transmitted

is the amplitude of the carrier wave of the envelope

the

q Transmitters,

power

EE

Transmission

Lines and

Antennas

James

Bruening, Center for Aviation

Electronics

Teaching (PME) and electronics as a Chef

im

Fectrowics**Tech****Peay****"PRON**

Other It hig 2

Ne

Colle

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Phyvg

The RF intelligence ro Will exactly

Tesemble

the Modulat,

Signal In

lular Pion, the ecewver de lates

be

RF hal hon, ligencers

Again

AVailable SS removed nd the intel. Figure245 the block be diagram c The Mtelligence eof 'ypreal AM

S VOICEaudio from a micr hehe Fent driver and a m latingamplifier stren, ams audio A Continuous,

Ww ave ts ed by applied to a

Power

amplifier Via a buffer

Stag

SlageIsolates the oscillator ern: TOM ext the oscillator frequency

Stable The

po' Ullatorsignal The amount Of am;

Plificat

determined bythe

Modulator**Output This 15**

accomplished biasing the Power amplifier with the modu

an envelope of RE epresentative of the the voice

an

Oscillator

ang The buffer lalcircuitry, keeping

on

Figure

2BasicAM

Transmitter

Frequency

Modulated

Mitters, the frequency

of Sent the intelligence, (FM) Transmitters InFM trans. the carrier wave ig Shifted

to

Tepre-

Figure

3BasicFM

Transmitter

Referring

to Figure

3, a modulating signal is applied to a varicap. The changing amplitude of the modulating signal Varies the reactance of the Varicap. The varicap is Part of the frequency determining

Circuitry

of the oscillator, The oscil. lator frequency is thus varied about its center frequency

as the reactance of the

Varicap

is varied. The varying Output of the oscillator is frequency multiplied to radio frequencies and

FM modulation The hand o' een oll: { d audio is © vNAM Lat TV video anc wd aN? eon herween a TV video transmitter erence pacarts the video SYRC signal generator camera sweep in anon allows the recetver to display the preture syte crrcudtry sync with the cameta sweep crow Ja for calculating t

C Transmitters

- The formu a . weinera signal as teow, where 'T' is the eer mncters, "v' as the sped of light (3 o be. MTs the frequency of the inal

an bert. Fre -- ae a a meeeavelengths. This is yyeldvety 8r ene mackerel

I.O

GHz and 100 0 GHz are defined

why as microwave frequencies jderations becaust ts require specialcons! sien accsoweve signal. If pobre much of the microwave signalwould unwanted radiation. signalalso requires special cit Generating a microwave deacon as

Jectron

sed

to pom lta tubes act as oscillator and amplifier for the microwave signal.

Figure 4 - A Klystron Tube

A klystron tube is constructed such that the electron stream emitted from the cathode transits the tube in bunches. The physical construction of the klystron determines the fre- quency at which the electron stream bunches up. Some of the energy of the pulsatingelectron stream is picked off by the tube itself to sustain oscillations. Most of the pulsatingelec- ton stream, however, is removed from the tube as a micto- wave signal (sine wave). Klystrons are used in low to medium Power microwave applications,

"ug

Figure

5 - Electron Behavior in a Magn, tron, A magnetron tube produces a microwave Signal f high power applications The magnetron tube ue

Or to resemble a ning of pockets, or cavities, suttooundte electron-emitting cathode as can be seen from f'ences magnet mounted in contact with the tube causes th

"s stream inside the tube to bunch up ina circular pay the cathode. Some of the electron energy is transfe, the cavities as the bunched electrons pass the cavity 9 The energy transferred into the cavities sets up cure met tions within the skin of each of the cavities The frequen oscillations is determined by the physical size of the at of Asmall probe mounted in one of the cavities picks of ie microwave signal to be uscd. elects tem an RADAR Transmitters ~ RADAR transmitters are mer wave devices. A typical RADAR transceiver wil] transmy 4 high frequencypulse of RF and wait for the pulse to retum af ter having been reflected from some object. The time the Pulse takes for the round trip will be used to calculate the distance the object. The direction from which the pulse returns is used to determine the direction to the object. ARADAR is designed to use the DOPPLER Effect if the speedof an object is desired. For Doppler RADAR, the amount of frequency shift of the returning pulse is used to determine the speedof an object.

Satellite Transmitters -

High power microwave transmit. ters are required for satellite communications. The power rating for asatellite up-link transmitter is in kilowatts. Satellite

up-link typically occurs with a 6 GHz carrier and down-link at

about 4 GHz. The transmitter signal is beamed at the satel-

lite via a highly directional parabolic antenna, which will be discussed later.

Transmission Lines

A transmitter is often separated from the transmitting antenna by some distance. When this is the case, it is necessary to route the signal to be transmitted to the antenna via a transmission line. Ideally, a transmission line will get the RF from the transmitter to the antenna with a minimum amount of power loss. The two

lines you will likely encounter are coaxial, and waveguide.

OT ad. fe®

Lines

- The two leads of the twin lead are uniformly spaced by embed-

We

are used. The simple construction of lossless waveguide, however, excessive losses are reduced and excessive noise pickup are avoided.

type of line. Twin lead line is most of this antenna to the television antenna and a television con-

ad

Transmission Line

qin

Lead

Transmission Lines - A major advantage of shielded transmission lines is that unwanted radiation is minimized. Losses due to magnetic fields of the line are confined to the conductors. Higher frequency

lines experience unacceptable losses

are, however

As a result

coaxial (is com

get eA.

OUTER CONDUCTOR

Coaxial Flexible Transmission Line

Flexible coax in use are designated RG6 and RG59, which are capable of carrying

RG6 and RG59 with its 20-gauge center

is designed for carrying a signal at 20% of the maximum, is capable of handling less attenuation than RG59. RG59 has a higher center or inner conductor of 22-gauge (smaller diameter).

ws

a

Use of Microwave Circuit using Waveguides and Coaxial

Ex

by technicians

F using

Waveguides

Standing wave of a circuit that

AVES ~

Recall that it takes time for the electric fields induced in the receiving antenna, to transmit and receive, depends upon the RF frequencies it will be transmitting or receiving. range of frequencies, with the frequency for which the antenna

chopter

maximum is that the resonance

If the length of the antenna is equal to one-half of the wavelength, the antenna is at

at

he dissipated

as well as the maximum energy is lost

Because of this, ground waves are composed

of a surface

in wave. The surface wave hugs the surface of the earth.

at frequencies below

2 MHz

can

be propagated via surface over great distances, Frequencies above 2 MHz are attenuated because of absorption by the ground, making long range ground wave communications impractical for higher frequencies.

Space

waves travel directly from the transmitting antenna to the receiving antenna. VHF frequencies

or above, i.e., above 30 MHz, are often propagated

as space waves. Space waves are known as line-of-sight when used for radio communications in the VHF and UHF bands. Tropospheric ducting often allows VHF and UHF signals to travel hundreds of miles. Skywaves bounce off the ionosphere and return to the earth at a distance of wave and a space

FARE

FITTS

SES

So, at the resonance, much of the heat. Resonance

Occurs over the horizon from the

Transmitter

band is limited to a narrow

Tangeof

Meg)

My too high frequency waves) SIMPLY sheeey hare and pass into space They occur become the the high frequency signals XceEMS that energy

i) wation Low frequency Propagation, as given. try! not achieved with surface waves Mac say the HF frequencies (3 to 30 MHz) are ideal, Propagation. A HF signal does not have enough

Wave

simply pass through the Ionosphere and radii

Chery

but bounces back towards the earth as does KC Pace, on the other hand, would simply pass through the,

H

en because of its high energy level This is why AM is all over the place,

My radio can be received far from the transmitter and by radio, Mating, are local to the transmitter Antenna

Directivity and Gain - The directivity of an antenna is the property that causes radiation or reception, in a particular direction.

Omnidirectional antennas radiate, receive equally in all directions. Uni-directional antennas

radiate strongest in one direction, and bi-directional say that an antenna is highly directional is to say that it is directional of bi-directional. No,

Litas

Un).

Darectivaty

is directly proportional to antenna But a unidirectional antenna has high gain in one direction because

i

of the radiated or received energy is focused to or from one direction. An omni-directional antenna, by comparison, has poor gain because the transmitter or receiver

COCEBY

radiate, or received via all directions, i Antenna Polarization - Polarization is an important consideration in positioning an antenna. A horizontal antenna transmits a horizontally polarized wave and a vertical antenna transmits a vertical wave. A receiver's antenna must be polarized the same as the transmitter antenna for ideal reception

Signal

polarization is defined by the orientation of the electric lines of force of the electromagnetic wave

Passing

through the antenna. An electromagnetic field is composed of both electric (E-Field) and magnetic (H-Field) lines of force at right angles

to one another as can be seen in Figure 11. When the E-field is vertical to the horizon a wave is considered to be vertically polarized. A horizontally polarized wave has its E-field horizontal to the horizon. Figure 12 - E-Field and H-Field in an

Electromagnetic

Wave

Electromagnetic waves are usually omnidirectional. Frequencies of the transmitted signal range from the high end of the radio spectrum to the microwave region. Radio signals propagate as ground waves and are less prone to be reflected by the ground than horizontal ground waves. A vertical antenna is more directional than a horizontal antenna. Recall that a vertically polarized antenna receives only vertically polarized signals. With a horizontal antenna, reception can be controlled to reduce the amount of interference from unwanted signals.

As the frequency of the signal increases, the wavelength decreases. A half-wave antenna, which is isolated from ground by an insulator, has an impedance of 70-75 ohms. The impedance of a dipole antenna is

POLE ANTENNA

A pole antenna

is a

TRANSMISSION

LINE TO

TRANSMITTER,

Figure 12 Dipole Antenna - A simple folded dipole is a parallel combination of two conductors mounted parallel to each other at a distance of several inches. The purpose of the antenna is that the antenna

operates over a wide range of frequencies. This makes the antenna useful for Transmission Lines and

Antennas

Chapter 14

fed wave antenna is frequently used in mobile receiver test

Mutual

applications These are normally vertically polarized Figure 13) Quarter-Wave Antenna used for

Mobile Communications

Parasitic Arrays Array antennas are constructed of more than one element. The parasitic array consists of a driven element, usually a dipole, and one or more parasitic elements. The driven element is the element of directors on

Yagi

Figure 14 -

A

Parasitic

Array Antenna

. Parasitic elements concentrate the signal feedable in the desired direction. For instance, the impedance of the folded dipole is greater than that of a single dipole. The impedance of the folded dipole is greater than that of a single dipole.

as well as other types of antennas are normally used

"4 le antenna.

1, a half-wave dipole antenna - A grounded quarter-wave antenna has the same frequency as an ungrounded half-wave antenna. In Figure 13, the grounded quarter-wave antenna is shown.

: speed of light (3.0×10^8 m/s), and f is the frequency

7 Micromet

an microwave Antennas. The antenna is a half-wave antenna. The antenna is a half-wave antenna.

ested =

om Te betus a flashlight has one that focuses the light of the bulb into a beam. A microwave antenna, all the energy is

into the parabolic reflector and

is concentrated. All the received energy is concentrated at the focal point for reception

unidirectional

in this reason, parabolic reflector antennas are used. For example, one direction antenna is used for satellite communication.

Summary

= The process

of imposing intelligence called modulation. For AM, the RF carrier's amplitude varies in accordance with the envelope of RF that exactly resembles the intelligence. For FM, the RF carrier's frequency is shifted at a rate that is proportional to the intelligence. The formula for calculating the wavelength of a signal is $\lambda = \frac{c}{f}$, where 'L' is the wavelength in meters, 'V' is the velocity of light, and 'f' is the frequency in hertz. The physical size of the magnetron tube cavities determines the tube's frequency of oscillations. Transmission lines carry the RF signal from the transmitter to the antenna.

Flexible

coaxial lines are ideal for transmission of RF

signals below the GHz range.

Waveguides

are used for transmission of microwaves.

Standing

waves on a transmission line are the result of an impedance mismatch at the load. Maximum power transfer to the load occurs when the im-

Pedance

of the

Load

matches that of the transmission line. A half-wave antenna, A Marconi antenna is a Quarter-wave antenna. Maximum radiation of a signal occurs when

the signal to be transmitted is at the resonant frequency of the antenna. Ground

waves are used for low frequency communication. Space waves are usually line-of-sight; UHF frequencies and higher are used for

HF communications are

used for maximum range. They are ideally propagated via sky waves

- The transmitter and receiver are oriented with respect to field Polarization, the direction of the electric field vector.

Antenna directivity is the ratio of the radiation intensity in a given direction to the average radiation intensity.

Antenna directivity and gain are related by the efficiency of the antenna.

Antenna directivity and gain are related by the efficiency of the antenna.

Polarization

Polarization of the electric lines of force of a wave, greater bandwidth than a basic dipole antenna by a factor of 2.

The efficiency of the antenna is the ratio of the power radiated to the power input.

Miscellaneous

Dipole antennas are basic half-wave antennas. Folded dipole antennas are half-wave antennas with a higher impedance. Quarter-wave antennas must be grounded at one end to resonate at the desired frequency. Parasitic arrays are constructed with a driven element and one or more parasitic elements. Parabolic reflector microwave antennas are directional. All illustrations courtesy of the U.S. Navy.

NOTES

For a given antenna, the directivity is the ratio of the radiation intensity in a given direction to the average radiation intensity.

The efficiency of the antenna is the ratio of the power radiated to the power input.

The gain of an antenna is the product of its directivity and its efficiency.

The directivity of an antenna is the ratio of the radiation intensity in a given direction to the average radiation intensity.

where L = wavelength,

wavelength is for

if the lines are ideal for what transmission test is then there is a transmission line with no loss.

where Γ = the transmission line coefficient

where Γ = the transmission line coefficient

it occurs when the total load on the transmission line is infinite.

It is the characteristic impedance of the transmission line.

Is there any radiation of electromagnetic waves? Space waves are for what frequency communications? VLF

HF Line-of-sight UHF Audible frequencies are

8. Electromagnetic field

Orientations:

Polarization is the direction of the electric field vector.

Magnetic Wave?

a. Magnetic lines of force

b. lines of force

9. The Orientation of the transmitter receiver antenna Must match for id
 - a. True
10. Quarter-wave antennas must e a resonance at the desired frequenc shielded In free space oPp

ANSWERS, On page 223

esd by John W. Baldwin, CETsr

Rabe

Rieke

fen here wrreived lexander

G

Belland

Flisha

Gray

coos onconaeen dunng similar 19h century

ee

Talking

machines, as they were called, enabledthe one person tobe transported

to other places where

t

i was not physically located By converting acousical spec sound waves to clectncal signals,copper Wire conductors could be used to extend the range ofa person's audio commu- Micalons. Telephony, as it was later called, let a person'svoice be transmined or transported over great distances. The imper- | sonal telegraph was no longer needed. The personality of a person's speechwas came along with the signalto @distant locaton to be heard by another persoa Refinements allowed both persons to talk atthe same ume in either direction. This is called duplex Today we enjoy portability along with the convenience of i having a private conversation with one ot more persons. Cord- Jess phones and warless phones are replacing the last mile of Connecvity, breaking the tethered connection to the telephone

uy office Vanous standards have nsen to support different trans- Port methods and techniques In this chapter, we will Jook at the basics of telephone systems and how they are allowing us more connecvity than ever before. Wireless data is one of the fastest growing areas to allow access to the Internet any time andanywhere

|| Basic Terms

A few of the basic telephonyterms are defined below. These are necessary to understand the concepts that follow:

Duplex -

Simultaneous transmission in both directions.

Half Duplex -

Transmissionin both directions,but one way al atime.

Simplex -

Transmission in onl

ily one direction, such as a broadcast radio or TV station,

| 28 vars teaching

at the college bevel and

veers

ws

x prodewooret yoann

Capaartieu**Coshege**

Mandate,

M'

at well oat

Riverland**Techedcal**

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in

Farha

en tM my mars forworking pvfewsonals

We bobs o RS degree

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a) hy oni o Indewny »,

ae H see fy

Warten**Saat**

OTPT achmeer

ad

Board

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ng ere om per ma acerca memor

Be FTA Sov ard the SAP ag eaten Mh a oe engtended

Te see, fob wes terecat of GWEC

Wochens

Estuester

of the

Year

award for

year college propre ny,baht

Ynnt

aT nell Deregulation ~ The US government allowing y i Junction ture of devices to an open standard and permitige®

Manu

set wed the ate 1800 and developed tion of these devices to the Bell Telephone netwony

Song

Telephones were first howorked indepen . Divestiture - The 1984 breakup of the AT&T m, 'On local and long distance service into regional el) "Poly companies (RBOCs) Petaling Line - A connection from a local central office to o premise equipment. OM, Trunk - A connection between central office SWitchin, centers, toll centers, regional tandem offices or longdisty carers that canes one conversation in both direction, Central Office - A point of connection for all loca}sub, senbers to connect to the network for service Inter-lata - Telephoneservice connections bet con dip ent area codes fs Intra-lata - Telephoneservice connections betweenyhe same area code. Sometimes called local long distance LATA ~ Local Area TransportAuthority. Boundariesbe. tween area codes are called Jata-lines.

Cordless Telephone

~A telephoneusing a wireless unl. censed radio frequency service for low mobility connection

Wireless Telephone

- Wireless is a name for cellular and personal communications (PCS)service usinglicensedradio frequency spectrum. Mobility
- The degreeof movement allowed in Portable or mobile telephone service and still maintain a connection to th caller. 1G - Analogcellular service known as AMPS,advanced mobile phoneservice, in the US markets and NMT, Nordic mobile telephone, in Europcan markets, 2G - Second generationdigitalcellular i voleneral or PCS wireless

Tm oa) service providing higher wireless ron - quality & et fe 2 wireless service providing Hon of specialcommands needed qn rat to performoperations such as call st ne

ne iI teardown 6 ect. 8 fication known i signaling speci panne! pc analysis

po as a conditioned re- yor? & 5 whal hone ei there's much more wel. a rca The central office has a on 'atts are DC and the subscnber telephone set is sl of

t

pt io ircuit are represented as R1 and wwe Iunes in the mirecoil represents R2. The be ior be as high as 2400 ohms to al-

Fp Tea

ent 1 st A oP is RI SRO

r] ¥: 7 \

R₂

R₃ 26 miles max

$$v_{ik} = -48v / 2400 \text{ ohms} = 20 \text{ mA}$$

$$1 =$$

i tween 16 and 23 mA. laren coil in the phone nas resistance of the lines are

ago ohms» amee, R₁ and R₃, can be 1100 ohms

ae and each same length. ne ways

he cosine

we

$$3, \text{ where } 1100 + 200 + 1100$$

$$= 2400$$

o po+R₃,

tr Rit 26 AWG for tele- 9 AWG to about izes Varyfrom

se lines, This permits total length ye males if the larger 19 gauge al agr i . auge wire, From a wire ; bart each 5 There are 5280 ft
per ome: sigs Pe 10100 ohms / 42.2 ohms per 5 yee sp from the central office. imum maxim Basics or relephone

The DC CURtent 15 "off -hook

Condition

of when the user of subscriber telephone is Placed across the line Two methods are

Sed, called log

When a Current flow js detect B dial tone back to the subscriber

Srvice 15

available Used to detect the

Local Loop ~

ac

Onalysis

All of theVoice or

Speech

"gnals. ay well tones, are AC Stanals sent over the local |

4S possible to send both Dc and AC circuit' I've had SOME electne tans

t The earhest s; Was the 20 . telephone set gnaling frequency

u

Ting Signal ft Was nng of "flag" 1 Sed Over g tel send a low fy, tor, a flag would "dri line

Fequestingsery

Teferred to as "line' erend and drop was to signal was also used t

Cquency 20 Hy 5)

op"

Ie "and "drop" Lin Ward the ce:

The 29 Hz nn

Mng a bell at the

Subscriber 4

and rotary dial 'y Wall mounted and oil with a Song capable f the earliest tclephones Inger inside them, The ringer was always and was not disconnected from the cir- user in any way, Mechanism. The bell Tinger was usuall Consisted of aSingle Electromagnetic

of moving betweentwo bells. Some o did not have a bel} ri connected to the line Cuit by the

sted Amore

We

Lees

could fee! the trckle of ths voltage fs shill used of OnNET' as sen

on the hese oo 2 sto a asfastbusy and one He somet daal tone, busy.

ah yn by

Jing Out-

eens Tani form of "in-band signal!

ae and are cons! Jing uses a separaic careurt oF nc a

ns

cae crrcutt, such aS the 20 Hz

can't hear Yo-handsignals are wha! #¢ can bear. Two-tone frequency com supervisoryalerting signals 360 + 440 H to 620 Ha
5 second

on

nd off" 44g + 620

Hz .

see

on and

ff

+ 480

Hz ,

2 sec on.

£6

Dial tone, Busy.

Fast Busy

Ringback

the number tary dial telephones, ra the DC voltage at 1010

ae

be vecond for each digit dialed Today aor et

pa aad to send information from mos! are

Note that the 1633 Hz 1s

not used yet and is a spare set

of tones for possible expansion. One of the earliest problems to overcome in the telephone system was how to send voice signals in both directions over a single pair of lines, for one subscriber circuit. The easiest way to describe this is to say that the outgoing talk from the hand-set transmitter (mouth) and incoming receiver (ear) signals are out-of-phase with each other. Early systems referred to these as the mouth and ear signals. A hybrid circuit with coils and capacitors in the base unit of the subscriber telephone separates the talk and receive signals from the 4 wire handset, and places both signals on to a 2 wire line. The bandwidth of the line circuit is also filtered for

between 300 and 3400 Hz for voice grade audio

M, Ron, ent "weightings" for filters can change the slope of

ty dig but a common one is called C-message Weighting. This, Fey, reduce the unwanted noise, background noise, or any noise from being added to and transmitted with the voice. The Bell system specifications ensured that a minimum amount of noise was permitted on the network. M

of To enable a user to hear a small amount of their own

No transmit signal in the earpiece of the same handset, some of the audio signal is fed to the earpiece when the phone is connected to a DC source, such as a line from a local central office. This helps to provide the user with the sense that the phone is working and then the person feels that if he can hear himself, others can too! This is called sidetone in the telephone industry. When a subscriber picks up a handset, and blows, breathes into the mouthpiece, it gives the quick response that there is a connection and the system is working.

When the dial tone is also provided as soon as the handset is lifted, the hook-switch connects the hybrid coil to start drawing the current from the line. However, there may be times that the central office is too busy to furnish dial tone back to the subscriber right away, so the user may have to wait for dial tone during times of peak demand. A line sometimes referred to as a twisted pair local-loop connection is also known as a plain old telephone service or POTS line. POTS lines are considered to be analog connections, where digital subscriber loop or DSL is able to

Unanspor

high-speed digital bit rates. The length limitation of 18,000 ft. on DSL lines comes from the fact that it is designed as a line to a single subscriber without line repeaters. Repeater to extend operating distance are available for interoffice high capacity trunk systems, but not individual subscriber lines.

Trunks

A line connects a subscriber to the central office. A trunk connects central offices and other switching facilities together. grade wires.

Normally 4 call at a time In long-haul and trunks, 4 wire trunks are normally used - per pair for the receive signal and 4 for the transmit signal.

id

re 4 four wire trunks between telephone offices is calculated

of users, and not every subscriber at the same time. Trunks are time-shared for channels. Tables based on Erlang's or Poisson

Free resources

learn about trunking efficiencies. One Erlang

second of time lasting for one hour. Traffic tables are

used to

of the probability of a certain percentage of calls being

used to predict levels of congestion in central offices and access

of service. For example, if a subscriber in a city, the probability of a distance is available - company

Y a exchange (PBX) in a company may use a single trunk to serve 150 employees each with 100% of the time. If cables may be used for 100% of the time, it is not effective to use other technology.

HF save on the amounts of copper

a pair of the older telephone poles using knob and tube wiring, each

of the dual inductor, you soon get the idea that we would need shielded wires. Old telephone wires were

a pair of wires to reduce the capacitance and the same concepts allowed bundling of small diameter cable. Twisting pairs reduced magnetic fields to help

a "sound" and the name "twisted pair" is synonymous - can talk. It is one of the multiplexing of conductor cables, multiple

c and time to polieeedeveloped using frequency ne beversatons on trunks. ot co
of

yk bee erin as ppmTrunks

ai iple (FDM) systems used 4 anne up several hundred wot or cack

C k channels. 12 channels are called a He gdwasth to stac Supergroup containing 60 chan- yuo are called a a Mastergroup and
contain 600 | gtd are ca Fe guns are referred to as a Jumbo 'oes

Ad : 600 channels. ich has 3 on" Basics of Telephone i Chapter 15 The

Analog Fy trunk

hierarchy

} voice grade channel =

399 - 4400 Hy

1 Group (12

channels) =

60. 10K kHz :RePETEFOUP

(5

Ciroupa)

312 ~\$52 kHy F home (10

Supergroups)

#36423 On4ky Up (6

Mastergroups)

364.

47.54%

kHz

Digital Tom

Trunks

maintenance in FDM 4. Sistortion and COR of (TDM) systems were Pha ee Te sent avisiom rmutpog system was installed in
New York 4 ay arefirst T.I camer

analog frequencies, the

channels are first used of USING noisy bits using a high enough sampling rate ("On ETE" 10 digital 8 bits are used to describe the intensity of distortion, then 2 bits are used to accurately

Reconstruct

and

Answers

showing

Stipules are

Waveform

a

ity This means that the sampling frequency

audio is 4 kHz, then

ieee

8,000 samples Per second. 110 1101 1100 niente Quantizing shown with Non-linear bands To improve quality of sound with fewer bits per sample, narrower sample bands can be formed for the smaller values of signal voltage near the zero axis of a waveform, as shown

Chapter

15. Separate signaling channels are never sent,

es FON are pean Conference Of Postal and Telexes (the) carrier This reduces quantization signal reconstruction system as channels cannot, because,

are the maximum of the amplitude bits taken from each voice channel; again

i, Up to 16 bits per process TI systems. This process is called companding

North

part of companding of digital transmission - TI carrier systems, and lowers the bit rate but "noise", may be very low level The A-law is used in the 57.6 kb/s (56K) per channel Nallabye 8ny ron systems are widely

US and Neth

AMEDCAM

systems. ie evens and the

a

s to as companding standards In optical fiber trunk systems, the SONET - 's are the standard Telecommunications UNO optical network standards for optical carrier, "hence" by the FTL. the International

and expansion used in the STS (synchronous

a

Sa C is a form of compression data SSS, OC levels of the STS (synchronous transport system) eg 155 Mb/s

s reduce noise in both

= tions. Peas

te sh a Dy ania PS Electrical BitRate -#of sion of all of the bits from Equivalent (Mbps) Voice Chan re Tae 2 010 0011 0010 1101 11001110 iROAtgg consecutive SIMD} baea froen the dhagram shown earlier sTs-1 51.64 672

oc sream can be combinedwith digitalsamme STS-3 155.52 2,016 0C3 iene me and multiplexed ee STS-9 466.56 6,048 0C.9 aesystem In TDM syaeue. pon ais reerred STS-12 62208 -8, 064 0C-12 multiplexedae 'AT-1 resultsin24 tume slots STS-18 933.12 12,096 0C-18

ia ne Naeoned Aine few framing Hel STS-24 1.2Gb/s 16,128 0C-24 each with 64! 000 bits per sect Mul- . 4.19. trangs the bet ate

up

equal ie Pager level multipleXing STS-36 1.9Gb/s -- 24,192 00-36 tnple T-1s can also

be comb STS-48 25Gb/s 32,256 00-48 foe more dita channelcapacity STS-96 46Gb 64,512 00-96 All long distance vor' traffic uses digiual systems i STS-192 9.9 Gb/s 129,024 0C-192 eee ng more than "bx haulers," bch Eanspoe StS-256 133Gb/s 172,032 -0C-256 fom poms pout The specifications for the North Amencan below. digwalhierarchy of TDM systems are shown Higher optical rates are under development, as an OC- 1024 . Tcarrier specificationnow exists, Multiple wavelength sources of Electrical BitRate - #ofChan Designation infrared light can also be used on the same fiber optic cable to Spec Mbps Veloe further increase the capacity of optical systems. Dso

- Gtkbps I

Switching centers

Dsl 1.544 24 T!

DS2 3.152 % T2 Telephonelines were first connected together by manual DS-3 44.736 672 *T3 patchingof a call with the assistance of a switchboard opera.

DS4 274.176 4032 T4M

tor. In 1905 New York City had over 6,000 telephone op- erators in the Bell system. An undertaker named Almon B. Strowger made the first automatic electromechanical switch jn 1892, but it was slow to be adoptedinto the larger Bell system, 73 - Name never existed until later EuropeanCEPT Digital TDM Standards: He also was credited with the invention of the rotary dial tele. CEPT -BitRate -# of Voice Chan oe.

Spee Mbps +

signaling channels There are several types of switching systems, Electrome- chanical designsinclude relay driven step-by-step,also known Fl 2048 30+2 as the Strowger switchingsystem and the crossbar, a matrix 2 cee 12048 (128) form of connecting two lines.

Electronic

switchingis generally i 34.368 wae et ze pde- division and time division, Each line com- ; oat RLM dae gir vitch is assigneda number, and is associated with a i) certain location. i 565.148

7,680 +512 (8192) 124 a

tier and faster than

all have greater switch-

afmes. TheY time is

ow

tyre a

rg! rocesso rcrOP" erm ae tre ded in 1976. | oe a s fae ind pion somberng Plan Association (NAN- lan area (NPA). pe N
yumbering Pp ; w h americat oS disincloding an arca code || aren 7 digits oie cient a oh pctor "en Cente!

ce at® (od) o 3 digits gsi? 3; of: _NXX-XXXX ye NXX 2 theu9

ft where

N

and x7 on

Station

extension 4 digits he central office a code or t of either thea ibers are reserved - ad potesor. mes the operator or for making
The gn 7 cha for cal sucl fi call ivate branch sie ag center, such as @ ratios or 1,000 switching

ps connect 1,000 lines,

li o5x) can «on numbers. A larger switch could «an extension

Mi jon & central office assigned fo one nnect Porto,goor" er switching systems Yeucanid (st 9999.

ge sof HES: lephone 005° ss ofthe NANPA telep! 10" ical analysis© that could be erical 999 numbers tag ulm! 999,

ly asin are 7, igned to one | pla

ea code and 9,999 numbers assig ee ge

a8 i i

ce prefix. 'onof iL an support about 8 mil words, each aan office prefix can support ten wane numbers new device requiring a
a . poxubeer ine for a PC modem, : caas ditional ni such

as

an a pager uses ad nbe -gcess cel phone, i of the rules for i as

i

iw sible jn each me is that the central office prefix an cosine the same 3 digit number.

ea code can?

-nupter 15 Forexample, you Cannot have

507.507 1234 asa telephone

Number

OF overlaying Splitting

Code into smaller

Stvic

area codes to the same ¢ also assigned to cach area.

rs and N11 numbers The

ae toll free 1-800 numb. N11 numbers are:

Unassigned

Non-Emergency

Police and

Other

Goy,

Services

411 Local Directo

Sil

oN emmental ry Assistance Reserved in (: 'anada, Unassignedin the Us.

Wireless

ustomer Care Center of

Deaf

Telecommunicatio

Company

Business

Office

Emergency

800 numbers invol number portability

Provide:

Ve toll free and

Portability

Issues, ty (LNP) is an attempt by federal

ag keepyour Old number even if you move out of your area code.

3) Service

Portability-

You k, Your same features (caller ID, call forwarding, The implementation Of LNP is part of the

Telecommunica-

tions Act of 1996 to allo. W Rew competition into the local tele. Phone markets. New tel or PCS provider,

Call Routing

In routing calls through the PSTN

Switching

centers, a network hierarchy existed before the 1984 divestiture of the Bell System. This included 5 levels of call centers and classes of offices:

Chapter 15 [

opens) Comers

Class!

: a Class 2 Revtrns! Centers po : Perma (enters ai Tol Center (mTbng emer? , ees End cfhoe FO (kee! central office nee | Bell
operat- and cxtahishneot of regrone

a eee UK's} wrth the ard of digital and electronic

a! a non-hicrarchrcal sys-

The Interexchange Camer

(EXC. oF long distance

CONPSNY:

so omg destance networks

ASST

sche dhe ongmnal established ancumDent

an area.

From

the changes

In ct of 1996, 8 LEC can provide long 'sa long distance company theoriginal long connectslocal call: change camer (LE
caltelephone company the Telecommnicans bers if tallow tosell

Jaca!

access

m sts market In the US. distance company was

AT&T MCT

became the second com- pany after st installed microwave links between St. Louis and

Caxcago

and challenged AT&T in court. Sprint ts also one of the largest IXC's.

ALEC

connects to anIXC ata

as point of presence (POP). end offices (EO) after dives- comiolliing theaccess netwi of thea networks. in most arcas. RBOC
After divestiture, there were seven the original LEC is also the RBOCs. These are:

+ Amemiech

BellAtlantic

Bell South

NyNex

Pacifi

Telesis

Southwestern Bell

+ US West Presently, there are 4 RBOCS after recent mergers.

Trunks

connecting

an end office to access tandem or local tandem offices connect calls to either long distance carriers or other local offices within the area code. Also during divest-

Basics of telephone service, service areas called Local Access (LATA'S) were established boundaries.

AN Try called LATA lines and mark areas as customer

area code Local Long distance 14 Intra4 then Long same area code A long distance 2) are Inter-LATA call. 2) are inh, ta" Ody, Switching centers are included in each CO (or FO), access tandem, local tandem't the long calls to the called number

Thee |

xP |

me Circuit switched vs. Packet switching

Chey

Calls can be routed by using either a circuit-switched network. Voice circuit-switched connections, and data over packet-switched networks. Witegy, f

Calls

bay

Chey

Ye best peng ay, form, Circuit-switched networks maintain only open, duration of the call, and then disconnect. Packet-switched networks use many routes and maintain an "average" delay. If a circuit-switched telephone connection

we

\s data call, it is called circuit-switched data (CSD)

eq for: Telephone numbers route calls on a circuit network, using 10 digit numbers such as 612 123 456 789 route data packets over the

Internet

1234 Ip telephone system using a 32-bit domain name system (DNS) of 255.255.255.255. Internet rou

ne span width of connections by Providing multiplexing

it is capable of dynamically assigning a network address (IP) for Internet traffic, it can be "dynamic"

higher bit rates than if channelized for voice transmission

Sued

a }

ic 'ull. ILswitch, ters MAXIMIZEg4h

Signaling

Telephone connections involve both the network of transmission facilities and the central office control. Signaling accomplishes this by providing

for the individual as well as to the system Operator:

the con

ting

of signaling involve signals on the subscriber line and those signals sent between central offices and the network. One common method to connect switching systems to transmission signaling systems is Plessey signaling, or Ear and Mouth Signaling. It is commonly

called F&M for switch-to-switch and switch-to-network connections. Either side may initiate actions. The Ear and Mouth signaling is used to connect trunk equipment to separate signaling channels. Four functions of signaling include: alerting, information, supervision and the transmission

of

transmission. Alerting includes requests for service and USOC. Informing the called party of an incoming call. Addressing is associating the with telephone numbers or Internet Protocol (IP) addresses.

and

common channels and

each is also sent to know whether the trunks

is used for a new call, or if the call can be

established. Signaling centers have been accomplished. He indicated

Channel Signaling

6 wire line Signaling (CCS) Before 1976, transmission over the same trunk that was

in

with 31 signals

any Tones such as 1600 Hz and 2600 Hz

channel signaling is widely deployed. This is a common channel signaling. The channel separates signaling from the data. It is enhanced in the Signaling System 7 (SS7) network in North America and CCITT 17 integrated service digital network (ISDN) features

Like all waiting, caller ID, billing information, etc. Information about special features is sent between offices by the SS7 private network. The 4 network block "star" 2 digit feature codes are national common in the network. Feature codes can be used to provide features such

etc.

of se one

i. ot of bie awe it took 8 as call forwarding, last number

i arate 7 network is derived from its scp. ae a 56k connection using one DS-O } span. In the old days when systems used ich the lo- me for the dialed digits to rea long distance call. Now, using SS7 and

AC tal

ae "he dialed digits are sent on the private SS7 a stching, oe ie quick! ns ppost mined "ee nui scP y to the serving office that the nngback tone ately after entering the last digit of the mber, even on a long distance call. SCP sm STP

\$7

Te vier known pasignal

Transfer

bill etwork has single databases associated with each

n as Signal Control Points (SCP). Redun- Points (STP) are cross-connected for Basics of Telephone

Wireless Telephones

One of the fastest growing areas in telephory has been the use of wireless telephones The freediam that witeiew gives USETS 18 tts greatest asset Two types of wireless connectrvity are the unlicensed cordless phones and the high motniity cet lular phones Cellular and PCS motile phones are approaching 90% market penctratn in the US and some countnes are consulered 100% with more than ONE CONMEction per penen

Cordless Phones

Currently manufactured cordless phones use the unlicensed areas of theUHF frequency spectrum in the 992-928 MH, and

the 2 4 GHz ISM band Older cordless phones also used wplit

frequencies in the | 6 MHz and 49 MHz HF bands Gener-

ally power output ts less than 100 mW for unlicensed devices, ERP rating to go to as vency bands. This ensures that cordless phones are low to medium mobility devices Several types of modulation and access technologies are employed in cordless Phones. Narrowband FM 15 used for

analog phones, and PM jy used for digital phones. Most

digytal phones use QPSK or some form of it to increase the number of bits that can be sent over a Narrow channel bandwidth Like the wired count phones, are duplex in nel frequencies are ni support simultaneot is called frequency Phones also use u channel frequenc: erparts, cordless, as well as cellular operation. That means that 2 RF chan- 'ecded for each direction

Of the call, to us transmission in both directions

This

division duplex (FDD) Some cordless me division duplex (TDD), shanng one RF Yy in both directions, with the base transmit- ting at one time and the cordless phone transmitting at another

Both analog and digital consumer cordless phone

technologies share new frequencies, but the trend is toward digital spread spectrum for Privacy. In business applications, the PACS (Personal Access Communications System) system is used in the US; DECT (Digital European Cordless

Telephone)

ts a strong standard in Europe and the PHS (Personal Handy Phone) is prevalent in Japan. Access technologies describe how wireless phones access base stations. In a cordless phone environment, generally only one-perhaps two cordless phones-share the same base station unit, so there is not as much contention for access in these un-licensed units as compared to cellular phones, where hundreds of users are vying for system access thru the same base station.

and PCS systems now take Come as

Ceteter

andWN (Percent Sacos tet mrbeies avers:

Cetaty

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l ent B Yaect ceded Si boc: ree weete so A Ac and feralserene petaeraen:

we hemng eeveenrd ooo Spsal SNE encremse chemee! capeact) gent

te prcnnde sere

Reames

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Tabane

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C bicbas been sptsO commer sectasms amd has reumtod wore specteas

a Gor harper camer, chee: ofCATE)

EERE

area. Ove 3S bare deviers has hoes spent

to daar for PCS fcemnes mt theUS

AR PCS seme of apt acocssnr The three prevaicer wrccicss axes: technaboges

#e froceency

Greece

madzple access

(FDMA)

time division multiple access

(TDMA)

time division multiple access

(CDMA, TDMA)

and

CDMA

are referred to as 2G and 3G

FDMA

is a frequency

Frequency

channels are separated by guard bands

FDMA**Frequency****Grasce**

multiple access or FDMA puts each user into its own

RF channel

These

are not very spectrally efficient and it is easy to interfere with an FM scanner receiver, so privacy becomes a concern- The first generation (1G) cellular systems around the world use FDMA on different frequencies.

The advanced

mobile phone system

(AMPS)

with the

Amcas

has

channels available from 824 to 894

MHz.

AMPS

uses

a

kHz

channel.

200MHz

most use

we use (RAAS)

Analog

480MHz and the ETACS

(European

Toes?

uses QW Modulation with 25 KHz FTACS also

es

are early analog PMA systems

Whar

makes cellular unique from other rad. allows handoffs

of calls between cells and

ge different market areas.

Thess

is referred to as SM

Petvon

travels to a new area free

a home area net

we must have an agreement with the visited wire? Poameng arabulime path

High

capacity is less

Pomay

In TDMA systems, users share the same

frequency transmitting and receiving

In the diagram

Delon

by a shared one channel.

TDMA

systems also use a to keep transmit (TX) and receive (RX) channels "Dp

Current systems

use 3 to 5 users on one RF channel move spectrally efficient and privacy

is increased,

as

by difficult to intercept calls. Sty Module TX Mobile RX

TOMA

UND

Four

users sharing one RF channel}

Time

of transmit pulses vary from 5

ms to 66

ms

in and are used to send digitally encoded speech data.

Then

called "time slots" and contain 150 to 200 bits of info, depending on the system used. Both the North TDMA TIA-136 system and the European GSM (Global System for mobile communication) system use TDMA technology. Battery life of mobile units is increased as well, the transmitter is not on the air continually, TDMA is only

on analog 30 kHz channels, whereas GSM systems have 20

kHz channel width for higher capacity.

The GPRS (General Packet Radio Service) and EDGE (Enhanced Digital Generation Evolution) technologies are based on GSM and are considered to be 2.7G. With higher data for "always-on" data, services will be delivered simultaneously with voice conversation. You can be on a call and still get your email in the background. These will be an intermediate step to faster 3G services.

Basics of Telephone 3 | Chapter 15 (Owe 12288 Miz RE carer Radin Tramemesscn Textinningy) www.t\ DO oppose for separate channels for voice and data

METV

ICES

are with the EV-DN and

WWRTT

using 108 MHz for RE

Samer

antlNe hav kwand compantie withARTT

CDMA

pattie eT pier Piers Tevtum This is a more cupemnewntunen wnce spectra

oe 8 ' subscriber workta nte waeGSM systems oo milan user in the US use CDMA and GSN texhnology

Korea

wes CDMA, wtnle Euro OPS

me abmoat 100%, on ateswe otecen topes Orgioaty GSM. and migrating tow ands

WCDMA

aM ddvreon came front yl

EE thehue of "OK!

Susanna

froma a eo synchronize transmitters and recenvers using . aan mterface" (TOMA, GSM and spread spectrum technology,

Starting

m

CDMA)

standands chtferent ois a

S around the world, we also have waLed O oe given creuit fordeveloping the RetWurk Protocol stanuants that CommeCT
axe xtahons asd mo nett, grevification. hipaa Senter (MSCx) together In the US. the ANS]

ae nckess (ls 1) standard sets the communtcattoas standards to S fe ee ost sei area es SS? data Retworkfor TOMA and CDMA
wireless

Providers

} p ede arAS esate pain

YY users les In

Europe

and inGSMt Systems,

Providers

that use the mobile channel. 2 S standand uses application protocol (Mf AP) does not forward fearares eran

AU handwodth thathas carmed forward

to

North

Amenean

S4I -

at te systems. ACDMA transmissston appears to ,

Yvems Global

Mamung

boost from VoIP andIP aie | \art wn norse toother receivers. [tisvery secure.

One Networks

forming the backbone for all cont ee coded vouce packet has very low power levels, men eels pel than the overall noise of the channel. Precise Thee vila peess Ss " are many erwarctess standards that we could | go helpsto reduce interference a ye mobiles discuss, such as Bluetooth, WAP and wareless LANS, but thes | pee ipene

Sadnenelgain

(oman Chapter ts on telephone systems, 90 VU end tt here. oot y ust 'hanne! ak sagas

wt | a

References

S. _ oe Noll 4. Michael, Introduction to Telephones and

Telephone

SA, gee Systems, 3nd Edinon, Boston, Artech House Publishers,098 | fone Newton, Harry, Newton \$Telecommamicanons

Dictionary:

16th Eghtton, New York, CommamiWeh, 2000 seunkaneous transausstion pertuts sott hantotts Hioki, Warren, Telecommuntcanons, 4th Editon. New York, F ,

Prentice Hall, 2000

ALA it is also possibleto receive coded

Packets

wa cD than one base station simultaneously. This is fea-

0 OHNE

Tall

of the base stations are

on the same channel . Mobile receivers can decode packets from up to tT ons or multipath signals, This helpsreduce fading i

iggals canceling at the receive point as in other tech- a || se gachas FDMA and TDMA, where multipathcan sieeve. CDMA also makes "soft" handofs possible, 1 ene bre is 90 perceptible changein audio when moving geen cellto cell xewer CDMA systems dominate the US market, with data, syedia messaging, video media and

TV services. 1XxRTT

NOTES

Chapter Quiz

loop

1. On hook DC voltage of a telephone is +24 volts

b. - 8 volts ϕ . ~ 48 volts

4. + 12 volts

2. Dial tone is composed of these two frequencies

a. 350 + 440 Hz

b. 440 + 480 Hz

c. 480 + 620 Hz

a. 440 + 620 Hz

3. The DTMF tones for the number 5 on a telephone keypad is ____."

2. 770 + 1336 Hz

b. 697 + 1209 Hz

c. 940 + 1477 Hz

d. 1209 + 1633 Hz

4. connects to

a. lines

b. two central offices & two trunks

d. two area codes

5. The form of signaling used on telephone systems today is . a. ACS

b. ISDN

c. POP & CCS 6 ACLEC Is. a cable TV company

b. the incumbent telephone company ϕ . a wireless provider

d. either a or c

7. The maximum bit rate available on a T-1 transmission system is

a.

c. 1.544 Mbps

d. 2.4 Gbps

g. A call between area codes is

a. local long distance

b. Inter-LATA

c. Intra-LATA

d. toll free

9. The device that causes the phase change in a telephone network, thereby changing the phase of the audio in a telephone network, is the:

a. hybrid coil

b. delay line

c. hook switch

d. transmitter

10. The 2nd generation digital wire technology that uses code

is less a Pattern of separate

AMPS

TDMA

GSM

CDMA

ANSWERS on page 223

review Questions - RF, Cable connectors use quartz crystals that generate output

one! the effect. effect.

b. Jer effect. avalanche (or Zener) effect. telephone lines have what DC voltage level tip and ring when off-hook?
a. no DC

"48 V DC Depends on length of line velocity factor in a transmission line is:

equal to the termination impedance. by the line length. b. increased by the distributed capacitance and inductance of the line. c. compared to free space velocity. cables of different impedances may be combined for short runs. a. True p, False

The bandwidth of a standard definition analog TV

2 signals is 6 MHz.

p, 16MHz.

o 4 MHz.

4, 88 MHz. part radio receiver section determines the radio's noise level?

4, Carbon microphones are much like variable:

- a. capacitors.
- b, resistors, c. thermistors.
- d. inductors. g and Telecommunications Chapters 13, 14 and 15 | Unit ©

8. In telecommunications work, is replacing twisted pair cabling. microwave wireless fiber optics both b

A length of coaxial cable, used for computers, radio or television signals, is open at both ends. With an ohm meter, the resistance measured from the center conductor to the shield is 50 ohms. What does this indicate? The cable is partially shorted. The cable is completely open. That reading is correct for 50 ohm impedance RF cables. That reading is lower than 72 ohms for video cables. else

10. Waveguides are not utilized at HF or VHF frequencies because:

- a. characteristic impedance is too high. Resistance to high frequencies is too high. c. large waveguide dimensions required at lower frequencies are not practical.
- d. grounding problems.

11. The item shown below is a (no chapter reference): Mini UHF, PL-259, BNC,

N.

ANSWERS on page 223

r Sue ee

Digital Concepts, Gates, Logic

Ul 7

Ci ~ hapter 16, and Computer act o> ~~¥ by Richard B. Kirkland CETsr teacher atthe

Arra

Technical

Trade Center

of

North

nec Techrcce pe 'emer of Las Vegas, Nevada, Richardhay

g

Feecevence

verve

18) He bef fos states {n'tash you

to

do anything Ive din "I #3

Vietnam

sen he Peete PETRI 2 arechen (10 Ye ETD, At Cemfed, CompTIA Profewmanal,

a

Creco

Academy

Instnxtor

(CAI) and

a member are

Sy

4e AY)ra yc etepase

Scheuer

mesa

to eng 87 Tor and

Certification

Administrator

for the FTA bad, thes

oe Signals can be connected to a common bus, or set of pat Digital Pe the signals from various devices have to be stopped
ayy" . | bts(short for binary digit) One at different times, so that they can share the same dpe

Rigenpak

gh or how, orcad loge oe (Nor Way without collisions. ey, ey . the state from 0 to 5 may a 6 gates. "ts of all digital devices are
LOR bcoee cami "meitoa complexmicrocontroller, Digital is on or off with nothing in between hoe the scieaome form of
Og £3tCS Some are

Analog is on or off with everything in between

cuncunts une! to route logic some are designed tochange logy function ond In the case of most digital circuits, zero-volis
would signals, - ter number zero (0), and +5 volts would Tepresen, the lerances. In mber one

(1).This

assignment of vol th variablesand fo és ignment of voltage to number 55 .kat y means enough."

In

= incorporate the voltage measurement to the binary

a The volige doesn't BANE tobe exactly om 7 A few word associations are now in order +

Sv oe the resistor doesn't have

to be eXACLY awe hand ! Deeks a8 follows: logic-I, high.on, yes,*; functron

Drgutal

crcutts

NOS can also

be expressed as follows | des curt to perform ae- For example:an no, or just o 8 Logic-o, low,op pacha Jog cercurt can be
compared to 8 dimmer-switch on the liv-

mg room lamp When the power 8 applied. the oo

hi

= wank of

understanding this chapter will focus on ty, a little. As the dimmer-switch is rotatedthe light gets baghter. tbe of digital
circuit famiies: Transistor Transistor Le The digital circuit, on the other tuand,can be compared tothe

Abele

Complementary

Metal Oxide Semiconductor

standard wall switch the light is either OFF or ON. Another S). Logic levels are standard voltage levels and an example is the variable resistor VS fixed resistor. Under for specific logic families, there

Stand

however, there

is

no perfect device,

yes. there are " In the late 1960's, a positive true logic system began

to some abilities and tolerances

in the digital world too. It emerged as a standard with the development of TTL, Transistor-Transistor logic circuits, which are Transistor-Logic.

This was based on logic 0 =

0 volts and 1 = +5 volts. One of the basic types performing logic

functions of 1 = +5 volts. It gets quite confusing when you look at

these

like AND, OR, NOT, NAND, NOR, XOR and XNOR. levels more closely, and what the minimum and maximum levels are for each type of TTL family. Some of the TTL families are standard, low power (L), and a second area of digital electronics is called sequential. low power (L), low power Schottky, logic. where the basic block is a device called a flip-flop (LS), high speed (HS), and advanced low power Schottky flip-flop. Flip-flops are used to build binary counters and memory (ALS), in the 5400 and 7400 series registers. Many different types of flip-flops have been developed, including RS. Clocked RS, D, JK and Master-Slave JK. Assuming we use +5-volts as our binary digit one (1) and 0-volts as our binary digit zero (0), a TTL device is said to be in a control state called a clock is required to keep all devices in sync (1) when the voltage is between +2-volts and +5-volts, in a system in step with each other. A TTL device is said to

be zero (0) when the voltage

is better. A third area of digital uses devices called tri-state logic, 0-volts and +5-volts. For ease of mind we can say a high exists between 40% and 100% of V_{cc} , and a low exists between 0% and 40% of V_{cc} . Since there are only two logic states, the third type of logic state is a controlled condition where the input and output of a gate can be disabled or un-terminated. This concept is the basis for all digital computers, where 0-volts and +5-volts. (+5 is more than 100% of V_{cc} , but it's easier to

Remember

that way.) So what happens if the voltage is between 40% and 100%? This is a condition known as 'undefined'. See Figure

CMOS

Figure 1

ent which lowered the current required

Logic

et FET devices in logic gates. These are CMOS or

Complementary-Metal-Oxide-Semiconductor

Each CMOS panel and one P-channel MOSFET are used

to form a complementary pair that operates on a voltage of V_{DD} and uses microamps of current rather than the

milliwatts required for standard TTL devices. Many CMOS logic developed into families that are used in building single-chip Newer microprocessor designs use 1.8-volt CMOS for lower current, less heat (power) dissipation.

and faster clock speeds.

to get

the

in TTL (with power supply assumption as for a MOS device is said to be one (1) when the

at

is between +3.5-volts and +5-volts, and zero (0) when 'a' between 0-volts and 1.5-volts. Again, we say a

we have 70% and 100% of V_{DD} , and a low exists between 30% and 40% of V_{DD} . The region between 30% and 70% of V_{DD} is neither ON nor OFF with nothing in between. The region

between 30% and 70% of V_{DD} is neither ON nor OFF with nothing in between. The region between

of logic one causes improper

Operation

of digital logic can be visualized as a top hat shown in Figure 1.

What is a digital signal generated? Three basic circuits to

generate signals are called multivibrators:

1. Monostable - generates a single pulse (one-shot) table - gives a 'clean' high or low 2. Bistable - referred to as a latch

3. Astable - generates a square wave (sometimes referred

to as running or clock) are many different ways to create clocks and multi-bit signals with discrete components or integrated packages, and pulse width characteristics are adjustable in some cases, while other clocks have fixed values designed

into a concepts, Gates, Logic Circuits and Computer Basics

Chapter 16

during manufacturing. An example is a 55510 which can be used to produce an astable multivibrator free running clock or a one-shot pulse generator. Shown below are clock modules for system board applications where precise signal frequency and accuracy are required. Many types of custom clock modules are available and are also known as VCXO, TCXO, OCXO, MCXO, DCTXO which are abbreviations for voltage control crystal (xtal) oscillator (VCXO) or temperature control xtal oscillator (TCXO).

Summary:

1. Analog signals vary gradually, while digital signals are

discrete levels commonly referred to as high and low,

2. Computers, calculators, and many other devices make use of digital circuits

3. Most electronics equipment contains both digital and

analog circuits

4. Logic levels, referred to as high, low, and undefined, vary with different families of semiconductors such as TTL and CMOS.

5. Three basic circuits generate digital signals. They are monostable, bistable, and astable multivibrators. These circuits are sometimes referred to as "one-shots," latches," and 'free running.'

6. Typically, +5-volts is high (red), 0V and 0-volts is low (green), or 0V, realizing that the voltage can be a little less than 5V for high, and a little more than 0V for low. "No indication" on a logic-probe can mean an 'undefined level' or no voltage applied. (If you want voltage, then use a volt-meter,)

Digital Numbering Systems

A technician needs to be familiar with the number systems used by digital circuits and computers. The language of the digital circuit is binary, however, large numbers are usually expressed in the hexadecimal number system, and sometimes there is a need to express a value in the octal number system. All of these numbers may need to be converted to the decimal number system for clarity. Therefore, in addition to the decimal number system, the binary, octal and hexadecimal numbering systems are frequently used.

Aineneemamerswrmammener

anes

ae

Counting in binary the number "10" for difference. For example with two LEDs side by side

both LEDs OFF = binary

left LEDs OFF. net LED ON - binary 01

left LED ON. right LED OFF = binary 10

Neither LED

ON = binary 11

(4) combinations.

Binary 00 = decimal zero 0

decimal one | decimal two 2 This method (LED ON or OFF) gives four See

Figure

3. Both OFF Left ON Right ON Both ON binary states with 2 inputs: 00, 01, 10, 11, where result, with

a Figure 3.4 by Jefferies Using two-character binary code

LED=OFF-0

and LED=ON=1, 4 bits

maximum decimal count of 3. binary states there are several methods binary digit is called a bit To express values greater than 3. (zero) or 1 (one). often used in binary: BIT A BIT can

A 4-bit combination is called a NIBBLE. and an 8-bit combination is called a BYTE. Combinations of more than 8-bits are called WORDS. For example. 16-bit combinations are sometimes called 2-bytes or a 16-bit word. values of 0 (zero) through 7 (seven), 3-bits express decimal which is eight combinations. (Zero is a value too.) values from 0 to 15, or 16 combinations. 4-bits express decimal tens. Now, looking at the binary system:

$2^2 = 4$, $2^3 = 8$, and $2^4 = 16$, the "exponent" represents the number of combinations, and the 'largest value' is the number of combinations less one. An 8-bit binary number (a byte) has 2^8 combinations (256) and a maximum value of 255. A table helps to summarize number of bits with the maximum decimal number expressed. Gates, Logic Circuits and

Comput

here, number 10 and the number 1, binary numbers to express this 1 binary values. decimal three 3

Binary is okay up to a point. Once we

have some strange (hard to remember)

million

one

1024 , $2^{10} = 1024$, $2^{20} = 1,048,576$, etc. A

combinations (65,536) and a max,

to make life easier we change thousands 1024 to 1K, and 65,536 to 64K etc. Millions.

M Value of Using your fin; gers... ten fin; oeny fingers is 2 to the 20";

ae

1,048,576 = mega in bin: 2³⁰th power, or 1 gigibit. If you associate
i your fin, . 32-wires would be 2" or ARE Wires o: Putting this into ive, i
t perspective, if you selecting 8-bit memory nie ais 4-gigabytes (not bits). Remember that 8b Since writing a value of 6:
i

5,53 Si (sixteen r's), Paty into another form: either octal or elon mn for people to handle. You need to be able to

Convert

any of the four (decimal, bi |, binary, octal and hexadecimay "Cen
my ber systems. To convert from binary to octal, group the bjinay

Pass 5}

(2) bit buary 24 to IK, 204g

Stig

We

Ra ome iS 2 to the 19% Pow, s ir Y fingerss a x

tg bus paths % address

SS abit its make o fi of

Ne byte TY Would be iS, binary

id mal to mak, UMbep

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ey groups of three binary characters, Sti

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x cepts, Gates, Logic Circuits and ComputerBasics con pis? ia wl

ol

py

= 17 octal

377 octal 1777 octal yout

yd OFbe ih MN mu.rT iH ouldbe Ful yee! a value between oand

7. Octal num-

ve per frame computers

to hexadecimal, you group the eight binary characters, which are and possible values of 0 through 15, into pairs. For example, the binary number 10110101 is converted to hexadecimal as follows:

Binary Decimal

Example

10110101

To use 'single characters' for each pair, we assign letters to values 10 through 15: 10 is A, 11 is B, 12 is C, 13 is D, 14 is E, and 15 is F. Using the examples above:

10110101 would be B5

10110101 would be B5

10110101 would be B5

10110101 would be B5

10110101

For a value between 0 and 15 (0 and 15): a. each or answer FF, and c. would be 3FF. To convert decimal to hexadecimal you add the place values. (This is a little tricky as all number systems have a commonality of place value.)

Example

In the first column (furthest to the right) has value, 1 (the number-base raised to the zero power, and the value 0) (to the left) has a value of the number-base raised to the first power, etc., you simply multiply the value by its position and add to the other 'place-values.'

Examples

In decimal 2374 would be "Tel eteipe (From right to left)

2374 Or 4 + 70 + 300 + 2000 which is 2374 in decimal In octal, 2374 would be 4556 (from right to left) (Agam, Or 4 + 56 + 192 + 1024, which is 1276 in decimal In hexadecimal, 2374 is 4 x 16⁰ + 74 x 16¹ + 22 x 16² Or. 4 + 112 + 768 + 352, which is 1276 in decimal

Logic Gates

There are seven (7) logic gates used in digital circuits. The first three (3) are called basic logic gates, and the other four (4) are referred to as complex logic gates. The reason for the distinction is we can construct the "complex" gates from the "basic" gates. Basic gates are just that, gates in the simplest form,

Basic Logic Gates

Logic gates have three characteristics. They are the logic symbol, the Boolean expression, and the truth-table. The basic gates consist of the AND, OR, and NOT (NOT is sometimes referred to as INVERTER), as follows: The AND Gate

Symbol

A

B

Output

Boolean

Expression

Chapter

16 E . hing" gate can

Be dcsoribet as the all ornet nigh fiw the our

Another

way os caery bere

Tre ANT 6 Art reteretet

A © "om greOR

Gate

Symbol

o> <

Otput

yrAtB vasil Aor

B, not

Y equals

A plus

B) unced ¥ equals 'any and all" gate. Any be high. Another way to ghequals a high." described

8s

to erter Symbol

Y

Output

The NOT

Gate

orInv

A

Boolean Expression

yA (Pronounced: Y equalsnot

A) -o- - AeA. (Pronounced:

A equals nat

A) Truth-Table Pay | Pi fo | Note: The truth-tablesuse Q and ! to indicate voltage values. 'A'o' indicates a low anda J" indicates a high.

Depending

upon the family (TTL or CMOS), the high and low voltage levels are as discussed earlier. You can assume that a logic probe will be used to indicate both input and output. Also, in digital Concept Ym are some, most circuits that have the in

Putts)

(especially TTL CIRCUITS) wait

needs to connect the on high

test respect to the reactive

Of the troubleshooting tip When you get a bad gate, you may have a disconnected one

It's tricky, Double gate Many people insist that it's "Yer

high gate because it doesn't perform hence The matter of semantics, it depends,

Boolean, 8 On how you

Ignore The NAND Gate Symbol (Pronounced: Y equals not A and B or YHOF

Yeg,

Truth-Table

The NAND gate can be described as a gate. Any input must be low no other way to remember the change; a high." The NOR Gate Symbol

A

Y.

Boolean Expression

as the a for the output expression

i

A

O-, Ynumlee

2D

$$Y = A + B$$

(Pronounced: Y equals not A or B, or Y equals,

as the 'not any' gate Any can be low. Another way to say is 'any high equals a low. rer

Output

$$y = A \oplus B$$

bol between

B) Note: The symbol equals XOR is a plus-sign inside a circle. (Please get the XOR Truth-Table listed as the 'any but not all' gate, it can be used for the output to be XOR

of inputs must be an odd

meuber go remember the character way of 1's equals

1 OR Gate Symbol is

XN

A

Y equals X AND Y, X

Output

Expression poses

EPYAKBE

(Proneamcedy

equally 5 terms B Note The evented between the A and the B is a sum of 4 terms. An odd number of be low
Another Way to remember the character number of 1's equals 0

Combinational

Logic Circuits

term and the MAXterm expression

Both

to the "truth-table" Output column (the Y column) and how it relates to the inputs. The "maxterm" is called "the Product of the sums" "minterm" is called the "sum of the products." Looking at the OR Gate truth table below - output as shown in the truth-table contains three ones in the output column. The zero Output is used to create the MAXterm expression, consisting of one zero and The ones Output are used to create the minterm expression. The maxterm Expression separates the terms (A, and B) with an "OR" symbol, and the minterm expression Separates the terms with an "AND" symbol, Parentheses, in the

Maxterm

Expression are separated by an "AND" symbol, and in the minterm expression are separated by an "OR" symbol.

Qares,

ee bOompur, xs Eco renal Concert i 14 be She mrenveen curmecegn fham the amih table WT

Ne Ye QR) where Yo

Oi "The miner carmecers wrk!

RE peaRy red ATHE TD second i al YaacRieit! artNot finher ihe "ental

Bates

OF the simple OR gate

Nowe

the Now

As the fit That would be an equivalent minterm The three-input AND gate has seven instances of Y0, and one

ve In this Truth-Table we have an expression would be: minterm

of $Y=1$.

The minterm

$$Y = A * B * C$$

A \ possible combination CHCUTT replaces

A

a

y

c Flip-Flops: Sequential logic VAtes Next, we look at sequential logic circuit clock signals, mating Me,

Rog,

hy lean gate me they! UtPUt assum 2

"o change ap. *Mey Bam, mplementeg

by wat A flip-flop is a circuit whose Output either ff to 1)

oF flops (goes from 1 to 0) in response

to input signals, The difference from a Boo! flop includes a latch function, once the 0 state, it latches in that state until it's told we'll see, the latch function is actually 1 Boolean gates and cross-coupling them Three common types of flip-flops are

1, RS and Clocked RS

2. D type

3, JK and Master JK Inputs are labeled with letters depending on the flop and the outputs have the letter Q and NOT. pe off, By definition, a flip flop is SET when the Q output 'Shy' IG The first flip-flop we will look at is the RS tip-top

Hi

R

Q

{0}

Wot-Q)

Inputs on the R-S flip-flop are (as you may have)

Rand \$, The 'R'

stands for "RESET" and tes" ee) 'SET.' " What input state on S will cause Q Output to be

hi

answer is: When $R=0$, when $S=1$, $Q=1$. You can i Ae

yourself by looking back at the NOR gate truth table to

Me that if any input is 1, the output for that NOR is 0,

D wry

pishigh and S is low.

Q islowe.

This

s

ca We condition." fed the eet istow and S is low,

Q tswhat

i sameete

Ramenen

le) aqiert change when one of the inputs moved fr 1620 was already in the required state-the pee cha phe OLD" ied pia bghand Sis igh, Qs Be same wren

weal

to not Q the eace - "PROHIBITED

i casera

aw jean

Algebra)

sequal pot Q; it vio ine clocked R-S flip-flop is shown below.

Ne cucked RS FF's have three inputs The center input is called

Se wclock" input. If the CK (clock)input ts low, the output . oth NANDgates (connected to CK)would be high, and since the two NAND gates are connected the same as the ious RS flip-flop,this puts the output Q into the HOLD ition. THUS, the CK input must be high for the other two ji sto have an effect on the output. Note that the jive, andthe rig diagram) latch w! inner circuit in this fli left-hand ANDs go low when they become ac- ht hand ANDs (whichwere ORs in the RS flip hen one of their inputs go low. Thus, the p-flop is "active low," where the normal tof "active high" is reversed, There is nothing legal or even rare about active-low circuits.

The D

and Sip-fop

i

'S shown

belog.

8? bmn Dee oe So tog Glock stays high, further the Q output. In other Words, the clock is "edge-enable

cropter

1¢ EoD Pierre Concepts, Goresr swwrn Smee =

Nompure, a

a ghe output in step or in Looking at a square-wave, it has four dist

a igi ie crecedan memvawry andl shit tics, they are: met cha Mey, sere with the 6

ve roertet caronits © Low: when the value is 4 steady zero voir, cc aanabably the MEST COMMAN firp-flop in + High: when the value is a steady +5 volt, The BK gen acetrevite Leading-edge: when value ts traversing fron, greBeta, CGN

TATTY

in Traling-edge when value ts traversing from p toy, cienar evn cvocted RS MAN-ACD COP "RM

tg The A fir Bop won Nate the extn links to the We can use these characteristics (o enable a fj, p-

by that hoch date pone inputs. When oth J and K it is "edge-trggered' or 'edge-enabled." All JK 9, Pp, "he, ange NANT wt ANY oweTopete with every clock: edge-enabled. Some are leading-edge, and some

PA arc bagh. the Q and Not pe te These are "preset" and edge. ATE tray, 8 MELO LL Also note the

CATE

ANTS aan

or flop the

i 22 ty (asenchronausly

ip clear" wnpots ony K inpurs which firp of flop the The J-K flip-flop has eight (8) possible conditions eer. clock transitions (synchronously): follows "FQ, a we put it in analyze the circuit, So. instead of asking you

memonze 2 box, and tell you to its characteristics. Note that, in the diagram, state change occurs when the in the symbol, there's a circle clock input goes high. However, 'or "bubble" at the clock input. This indicates that the clock function is inverted; state change occurs when the clock goes low. Many flip-flops are designed this way because experience has shown that it's useful. In this particular case, the clock, preset and clear are all low-enabled. If the CK input is a "line" input, it is "high enabled." If the CK input has a "bubble" input it is "low enabled." If the CK input has an "arrow" (or greater than symbol), it is "leading-edge enabled." (The J-K diagram above is leading-edge enabled.) If the CK input has a "bubble" and an "arrow," it is "trailing-edge enabled." (The J-K symbol above is trailing-edge enabled.)

. Asynchronous Set

. Asynchronous Reset

Asynchronous Hold

Asynchronous Prohibited

Synchronous Set

Synchronous Reset

. Synchronous Hold

. Synchronous Toggle

SI

AARYN-

Asynchronous means not in step with the

Clock

Toggle means change to opposite state. B Bysetting, and K inputs high, the J-K flip-flop is in the toggle erat

each CLK pulse, the output Q will change to opposite =

Q was 0, it will go to 1, and if Q was 1, it will go to 0) This characteristic then leads us to the next topic.

Binary Counters

Counters are digital circuits that count in binary state, using the J-K flip-flop, with the inputs J and K enabled as the output Q will toggle at every instance of the clock enabled. If, for example, we use a trailing-edge enabled flip-flop and set J and K to high, upon each clock enabled the output Q will change to the opposite state. See the circuit below (Figure 1).

Figure 1: J-K flip-flop circuit diagram showing inputs J, K, and clock, and output Q.

Figure 2: Timing diagram for the J-K flip-flop circuit showing the relationship between the clock, J, K, and Q.

So if we will CMOS circuits agree that it takes one complete unit, we see that being the circuit Q to change state (because eventually to cause square-wave) The result in «annonty one baal square-waves, Or, in more technical. weates Soa euene divider-in this case a "divide gate weena divider Guess what? If we add another peep Cet deby four" frequency divider. Look at the

By we gets pK Npelow: al Pe Me Sut ht, add

Fixed, the counter divides by eight, third circuit ppsecie by sixteen, which results in 16

fe JK circuit | tes as summarized in the truth table fol- low counter stal s for

a jenn

$$: Fa = 3$$

wir

a Ble very row, the B- the A-column changes every

om Ear other row, the C-column changes every

en nel D-column changes every 8th i ee, ATO i) into decimal, the value colu -values (binary, these row-val decimal count. i i

ircuit below. f the first flip-flop in the circui

ar rptiiae A). FF-A changes output state (o peat clock pulse.FF-B (the second flip-flop) ree akip Hoe other clock pulse. FF-C

every 4th, and FF-

ou ROGle Circe Sire Cabitinbiticic exami nor"

yn od rd ryt

Crery Rth

Locking at a tryth table few (cmt 090" het doing some Adjustingto LED indicator, [ca TA that "counts"
'S furthest right count from zero in bin

Shift Registers

Registers are Sequential circuits created with flip-flops and other gates. They hold, or store, data as bits. Sometimes, in a microprocessor, a register is referred to as a "flag." A flag is a one-bit register. A register, then, is a storage circuit, or an indicator circuit. Some Register circuits are also considered to

be memory circuits,

Key Concepts: Gates, Logic Circuits and Combinational

and Pulse, By the Figure shows a 4-bit shift register

Arithmetic Circuits

Key Arithmetic circuits are "logic

BIC-C *

Arithmetic functions made up from basic gates. What

is a "half-adder." The first,

is a 4-bit,

Tristate

Input

A 4-bit Sum

Input B

This 41

Carry

One 4-bit flip-flops. This is the shift register is made from four a

in serial-to-parallel, shift-right

Four inputs represent the binary number shift register is called a 4-bit shift register. Below in Figure 10.15 are the four types of shift registers: serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out, and parallel-in/parallel-out.

These

Serial-in/serial-out (SISO) may be called: Serial-in/parallel-out (SIPO) parallel-in/serial-out (PISO)

Port

in-parallel out (PIPO) Note that the Serial data for 8 period of output column is the function of the register

Input A (A=1, B=0, C=1)

as an exclusive OR gate (either A or B is high). To 'load' the above memory the first clock pulse, high, and that the Carry Out column is the same logic as a 2-input AND gate where both A and B inputs are high for

the first clock pulse,

Then

the clock must be "als The Moy, hughfor the fourth wea ap LED's) would change carry is the leftover after you add in Nea
nt

AS areoy ee(the clock being the enable-inpet)A binary adi 'on the next clock pu the data is shafted"(Io this case inary
additiontable is shown below with the clock pulses amve, combinations of os and Is. allPoss, the nght). So... wan "D)," it i i o
pased gh when temenes foo to "D," it is 46 ; a ' ig from "D" io" a dat shifted left when data moves

a o i Registers also come packaged in LC.form, a shown below: i witha cany of

A full-adder istwo half-adders connected to [7a _] following circuit shows: ether, ast,

x

Coyin

a

if

a

nO «the "universal" sbift-register, the 74194.It

This

(above)

is the "universalregister, the can be parallel loaded.It can shift left. {t can shift right,

it can hold its contents. Do you need to know what's inside? No...just memorize the way it works, as follows: I :

m

When

\$o and

\$1=0

it's HOLD

A

Bo

When \$0

=0

and

S1=1 it's

Shift eft

When \$0

$=1$

and

$S1=0$

it's Shift right

When

$S0=1$

and

$S1=I$

it's ParallelLoad

The sigma character (Σ) represents the sum output,
 7" half add. has a Carry Input where the half add-
 ee Let "04 This Carry Input allows the adder to

Carty

yen bs tuple bit columns and account for

next multiple below is a 4 bit adder:

an eye bo

we it can "ADD" two 4-bit numbers. The Output is 5

This

rg the (Co) Carry Output it. inc! pits lers pecot i oder-driver IC to decode mmon is the dec

ne of the a o display on a 7 segment LED or LCD gcpan aceon uses many gates to decode a binary num- as? al er it to. a 7-
 segment display. ber for decoding bits, and driver to IC bas ae oe display. The inputs can be low or comme opel te normal or
 complimented. per abled, and the outputs can high

en ition: a My

h"

ag means the output is high" sera tr means the output is "low' & Output is available in two configurations;

1, common-cathode display 2 common-anode display

Common

TTL types are 7446 and 7447 ICs

Memory

It can be made from shift registers, and is specifically designed for storing data and logic

functions. Many types of semiconductor memory exist with very large capacity and low current requirements, Gates, Logic Circuits and

Computer

Basics

Computers use many

Semiconductor

memory types: RAM

ROM,

Memory

Two types: Static and

Dynamic

Both Bipolar and MOS versions are available,

like ROM Read Only Memory

PROM

Programmable

Read Only

EPROM

Electrically Programmable Read Only Memory

Electrically

Erasable

Read Only

Memory

Flash Memory;

SD

Secure Digital

Mini and Micro size,

Memory**Stick****Compact****Flash**

cs@ailable

Magnetic:**Hard Disk****Tape**

Optical; CD-ROM, co-awyy DVD-RW, DyDsRW,

Tri-State**Logicand****Control Devices****Unidirectional**

devices A one way M-state logic buffer has Oe input, one ourput and a control line. The control line enables or disables the data transfer through the device. When the buffer is disabled, itis said to be ina high-impedance State and disconnects the input from the Output. The input can only be an input and the 'output can only be a Output. Signalscan Pass onl 'y n one direction,and only wher the control line is activated. There are several types of tri-sta buffers:

1. Non inverting,active high control

Input +-- Output**Control**

rc

- 2 Now srverning,active Few contra!

Input**-[> - Output**

Db

2. nverting. active high control

Tnpat

-[> output

Control

4. inverting, active low control top LE one

Control

These devices are used where logic signals only pass one-way. For example, on an address bus output from a CPU, some signal buffering and current drive is provided by these devices so that fan-out (number of gates driven) is improved.

Bi-Directional

low signals to pass in either output and vice versa. If devices as listed Bi-directional tri-state devices allow data in either direction, so that an input can also be

an common I/O pins are typical. The 4 types

of above also apply to the bi-directional tri-state gates. These devices are used on two-way data bus lines to allow a bus to drive multiple devices, while minimizing bus loading. Shown in the next diagram is how simple data transfer is accomplished on a 4-bit data bus. To transfer data between Register A and Register B, a third register is used as temporary storage during the transfer process.

Data in Register A = 1100

Data in Register B = 0011

+ Data in Register C = EMPTY

Digital Concepts, Gates, Logic Circuits and Computers

As;Ss

Tri-state control devices along with 4-bit basic computer model of how data can be transferred from one memory register to another or it is possible that you want to swap the data in register, without losing the original contents. The memory swap sequence

1. Read the data from register A and write

4, Read 1100 from A >>>>

Write 1100

register

TC Register

2. Read the data from register B and write to - Read 0011 from B >>>> Write 0011 to register A

3. Read the data from register C and write

to register C Read 1100 from C >>>> Write 1100 to register A | This is the simplified data transfer of memory between two locations, by using a third memory register - temporary data. This process shows a common data path for microprocessors that in 32-bit or 64-bit computers used today.

Memory is designed to be a storage facility for data instructions. Addressing memory, and retrieving data and instructions can also be accomplished in several different ways.

s,

as 'sequential' for drives, and 'mask' or 'matrix' for rang

Such

'om memories. For example, look at the figure below Point A would be high when Switch A and B are low. the other points (B, C, and D) would be low. Since the 7; State Buffer requires a high to be enabled, only the to would receive a ground or low when Switch A and B m Cte yi, This can be interpreted as addressingthe top circuit eslo just one method of addressing. TMS is

Schmidt-Trigger Devices

Schmidt-Triggers are used to clean up Noisy switch or signals that do not meet TTL specifications,

They

hm noise immunity hystersis characteristic that eliminate ed logic voltage levels.

S UnWant,

r lers coupler/Opto-Coup! t devices for sensing a paper La Ean moving from a mechanical

ly made from an infrared LED and

a sual

Ot Tey

light is blocked from the LED, the eo tor, When the eenstor is cutoff. spo eransts

rs mputer

Basics

co i uter? Depending on whom you ask, you may

wi oat from "the circuit under the hood of your on. some sophisticated "super computer." For this section wndeatwith the common PC (personal computer)

iw order (o qualify as aPC, the unit consists of severa components: a processor, memory, input and output, and power supply. That is a very basic description

In gsstine! we see two basic types of PCs-those built around feM Intel / Microsoft model, and those built around the

a facintoshmodel. Both types use some interchangeable Ap? ents, and each has unique or proprietary components <ompon difference is the language of the microproces- TheF

Se IBM/ Microsoft unit uses an instruction set that is sof oythan that of the Apple/Macintosh unit. This means the aifferer cannot use the same operating system or the same

me without some intermediate communication method,

ee psiation method, Binary code can be interpreted as a eeeand consists of bits, bytes, and words. ter (PC) has a main circuit-board. This "PCB" is Lid a ae to as the motherboard, main board, or the sys- psutlly which connects the microprocessor, ROM, RAM,

interface connectors and ports. r processor speed can be a little confusing, such as

Clock

oF

2.4 GHz. This specification is typically reserved

48 GHz or

= speed, or the speed at which the micropro-

cessor re is the "BUS" speed. Bus speed is typically cally adve tic of memory, or the speed at which information characters! d retrieved, usually at a much lower rate, These issorted

site1066 MHz, 1333 MHz, and 1600 MHz. Hard

tard disk) can also have different characteristics, be (Oe and speed.The PC has an IDE (integrated device guch

aS rahard drive or SATA (serial ATA). Some have a detail computer system interface) hard drive. Both have scsi (s and disadvantages, but are described primarily by

ae anidsuch as | terabyte. This specification is de- ee the number of bytes or 8-bit groups that can be stored piedrive track/ sectors. The other characteristics of the PC are peripheral devices, ychas CD-ROM, DVD, CD-RW, monitor, printer, mouse,

ital Concepts, Gates, Logic Circuits and Computer Basics pig

keyboard, etc The last not typically advertised specification ms

the power supply This box is usually descnbed as 350 Watts

OF up to 150 Watts This is an indicator as to how many periph

eral devices can be added (internal devices such as another hard drive or other device that requires a power source). The larger the number, the more devices the computer (power sup- ply) can handle Computer Literacy is being familiar with the operation of PCs, not necessarily the technical operation of components How to start (boot-up) the computer is part of knowing the basics. A cold boot is a start from power-up A warm boot on PC's ts a restart from the keyboard by pushing Clt-Alt-Del keys together and using the task manager to start over. Computer literacy generally means being able to use a keyboard, with mouse, and use special function keys. such as the Alt and Control characters. For example: the F1 key 's typically used as the Help key in a Windows environment and the continue key from the Start-up sequence (__.press F] to continue). A typical contro! character like Ctrl-C (holding down the Ctrl-Key and typing the letter C) would be 'copy' in a Windows environment. Being literate today also means being able to use the Internet to search, browse, email and use forms to fill in information.

Ports

As a technician, you should be familiar with the interfaces used to connect peripherals. Most connections are made at the back of the computer through some sort of connector. You should be familiar with at least the following: Serial Ports: On most computers you will find a 9-pin male DB9 connector. This is the senal interface where a serial device can be attached, such as a serial mouse. On some older computers you may find a 25-pin DB25 male connector, Both of these connectors (9-pin male and 25-pin male) are called senal ports. A DB9 is a RS232 standard. Senal ports have two other characteristics: The address, and the interrupt, or IRQ.

As a technician you need to be acquainted with these also... as follows:

COM1 Address = 3F8 IRQ=4

COM2 Address =2F8 IRQ =3

COM3 Address =

3E8 IRQ=4

COM4 Address=2E8 IRQ =3

USB universal serial bus Ports have generally replaced COM ports and transfer data much faster, up to 5 Gb/s with version 3.0.

Parallel Port: A PC may have a parallel port (sometimes called the Centronix printer Port) 25-pin female connector. As with serial ports, parallel ports have an address and an IRQ,

as follows (parallel ports also use the designator "LPT" which was used to designate a line printer)

LPT1 Address =378IRQ*7

© LPT2 Address 278IRQ" S S LPT3. Address"

IBCI RQ*7

0378 or 378H as ts

Now Sometimes

the address is shown as use 378 for the serial ports and one other "bit." Most PC's use LPT1, however IBM will typically use SBC as their LPT1 address. In addition this connector is not always a parallel interface. On some other SCSI cards, a 25-pin female connector is used to connect to external

SCSI

devices be either a SCSI or IDE

Other

ports

on

a

PC sound card can support digital high-quality sound port or a MIDI (musical interface) for example. USB, DIN, and PS2 ports often are used for mouse connectivity. used for input keyboard

Network:

ports require connectors such

as RJ-45 for Ethernet cabling for data transmission rates in ranges of 10/10/1000 Mbps. Ports used for monitor connectivity can include VGA, S-video, DVI, and HDMI.

Displays

Historically, in the 1970s, a computer was originally connected to a "dumb terminal" which was nothing more than a low resolution black and white TV monitor with an upper case only keyboard for input. 80 characters wide by 25 lines of text (or 40*25 larger text) were common. Since then, things have changed dramatically. The evolution of display technology has paralleled the television industry for resolution and image quality. The display resolution, and the associated setting such as the dot-pitch is the distance between 'pixels' on a monitor, normally measured in mm {millimeters} (the smaller the number, the closer the pixels) and the resolution, which is a measurement of "dots on the screen" horizontally vs. number of lines vertically. Format Horiz x Vert VGA 640 x 480 [x6] 1024 x 600 1366 x 768 HD etc] don't forget, Gates,

Logic

Circuits

and Computer Basic, Example: 1280 x 1024 means 1280 dots across

lines up and down, Why, To be able to display high resolution images clearly, bandwidth is needed in the circuits that drive the 4-1/2" monitor device. As the number of pixels increase in screen resolution format, each video frame scanned at 60 frames per second

On

(Hz) dots get smaller, therefore more dots are

required to display the image in one full frame

For example, at a horizontal scanning frequency

of

Hz, 320 dots per line, if each line takes 63.5 μ s see 5159

Time

line ($t = 1/f = 1/15,750 \text{ Hz} = 63.5 \mu\text{s}$),

then each dot (pixel) takes $63.5 \mu\text{s} / 320 = 0.2 \mu\text{s}$ of

time This corresponds to a video bandwidth of,

BW (in MHz) $= 1/t = 1/0.2 \mu\text{s} = 5 \text{ MHz}$,

This was typical bandwidth of early NTSC (National,

Television Standards Committee) analog TV with 4 x 3

aspect ratio, (4 wide x 3 high). Today's newer high definition displays double the horizontal

scanning frequency 31,500 Hz with wide screen format so of 16:9

aspect ratio and 1920 dots per line yields x9

Time

per line

$$\text{of } 1/f = 1 / 31,500 = 31.75$$

uS With 1920 pixels per line, 31.75 uS /

$$1920 =$$

or 6 us / doy which translates to a video bandwidth of:

$$BW = F = 1/t = 1 / .016 \text{ uS}$$

$$= 60 \text{ MHz}$$

Digital

signal processing (DSP) creates compression techniques of video and audio signals which will require a higher bandwidth to transmit the same amount of information as analog.

analog systems. In a standard analog NTSC 6 MHz channel,

12 or more standard digital video channels can be broadcast,

or a few less standard digital video channels if HDTV is broadcast. Data rates for HDTV are 15 Mb/s to decode, if properly, slower rates may cause pixelation or dropped and image quality will be poor.

Present

CRT (cathode ray tube) technology is quickly being

replaced with flat panel technology such as plasma, LCD (liquid crystal display), and DLP (digital light processing). Both digital television and computer technology, video formats for displays and monitors can be very snug digital video system. For laptop and notebook PCs are available with double-layer DVD read-write drives, and Blu-ray drives for storage

or playback so that HDTV quality video can be recorded and displayed.

For Concepts, Gates, Logic Circuits and Computer Basics, systems have massive amounts of memory to

quickly process rapidly changing video content. To quickly process a frame at a resolution of 1920 x 1080 takes time for each color displayed and with 256 colors, 307 30,841,600 bits or 66 Mbytes of data for each screen is needed every 1/30 of a second. Superbright LCD high-resolution display panels in car mirrors and cell phones allow video and still pictures to be played under direct sunlight and high illumination,

by

or even brightness comes more battery drain, therefore, light output efficiency becomes important, area

lumens per watt.

is sores

i

A...

aa pigitel more than 1 volt an) Sots 2 vols

s bo more th € mr that 2 Alagic I, CMOS device connected gy supply bs a a more than I volt b more than 1 \$ volts more than 2 volts a more than 3.5 volts a high a bogie low a TTL. logic level a CMOS device for an astable me teivibrator .."

4. Another name a Latch

b. One shot & Free running d Mono in binary:

5. The number 12 in decimal is ____" a 1200 b 1100 ec. 1010 dé 1001

6. The number 25 in octal binary.

a pb. 011001

c. 11011

d. 10010!

7, The three Basic logic gates are

a. AND, OR. NAND b AND. OR. NOT

c. NAND. NOR, Inverter

d. NAND, XOR. NOR

9. The Boolean expression $Y = \text{repr,}$

xg

A . The NAND gate fs sometimeyCalle, all or nothing as any but not all universal inverter acre

a, AND

b. NAND

c. OR N

d. NOR

10. The difference between minterm a, . nd ma, x

a. minterm, $Y \cdot$.

mi $Y = 1$, Least # of g.

maxterm, $Y = 1$, Least # foe SAND

. meal $Y = 0$, Least # of 'SOR

axterm, $Y = 0$, Least # of OR

81S AND

ANSWERS on page 223

works.

A

It is a multipurpose device built on a single 'called a wafer or chip. It can be on a number of information-handling essential; the digital logic functions described in

Cc

Lied

the heart of a PC and Internet data communication

A

of silicon, record

Features all of serves

8S her machines, It is processing use, for controlling other; even hospital patients, and for

Units and calculators. They are classified in terms of the number

of information that can be transferred in parallel and

exists. Intel® developed the first microprocessor in November of 1971. It was the 4004, which; the processor used in a calculator. It contained 2,300 " in a chip, which was impressive in 1971, but a Core i7* is a 64-bit processor and has over

million transistors.

Mk

Instruction

LI

Figure | Address & Data Bus

It contains a simplified representation of a microprocessor. It will be used to describe how a microprocessor contains several sub-parts: Arithmetic Logic Unit (ALU) performs arithmetic operations. Examples of the types of operations are addition, subtraction, logical AND, and logical OR.

Registers are temporary storage for the data used

by: a register decoder and the ALU. Letter identifiers (A, B,

C, etc.) are usually used to designate the registers. The Instruction Decoder determines how to perform the instruction and coordinates its execution.

Microprocessors and

Fundamental Control Systems

Control for SCP

Global

Technologies: He has taught electrical courses at ITT Technical Institute in Bellevue - WA

pce See aeces neers

en cheeses technician operating nuclear power plants on aircraft carriers "e a served 4 The Bus Controller controls the flow of information

Into

and out of the microprocessor

5. The Internal Bus provides a Path for the formation flow within the

Microprocessor =

Microprocessors are

Programmable

using machine language As the microprocessor is a digital logic device, machine language must be in binary

Assembly

language is a more convenient language than the instruction set and information on how it is to be used. (Note: Examples can be found on the Internet at www.intel.com Search for "instruction set") We will use assembly language for the 486 microprocessor. For example, numbers 4 and 3, two different instructions are necessary. The MOV instruction moves data from one location to another. It takes two arguments, destination of the data and the source of the data. The arguments are generally located in the memory locations immediately following the instruction. For example, MOV (register A, Memory). This instruction takes the contents of the memory location specified and places them into register A. To illustrate the operation of the MOV instruction to perform the addition of the assembly language instruction moves data from The ADD instruction adds the contents of register A and register B, then places the sum into register A. Table 1 is a

Representation

of a block of memory. The addresses and data are hexadecimal representations of the binary data. Let's start the step-by-step analysis of adding 4 and 3. The Bus Control has a register that contains the address of the next instruction (in this example the address is 0001).

Step 1 -

The first step is for the Bus Control to retrieve the instruction from that address. In this case the address is 0001 and the instruction is MOV. The instruction is sent to the Instruction Decoder for decoding. The MOV instruction takes two arguments, So the Instruction Decoder directs the Bus

Chapter 17 [

Control to fetch the contents of the next two MeMOry loca- pons. The contents of memory location 0002 (Register

A, the destination of the move) are accessed and sent to the Instruction Decoder Memory location 0003 (0100, the source of the move) is accessed and its data is also sent to the instruction decoder. The Instruction Decoder directs the Bus Control to get the contents of memory location 0100 and put them in Register A. The first instruction is completed. Step 2 - The Bus Control fetches the second instruction, which is located at address 0004. The instruction is MOV. The instruction is sent to the Instruction Decoder for decoding. The Instruction Decoder directs the Bus Control to fetch the contents of the next two memory locations. The contents of memory location 0008 (Register B, the destination of the move) are accessed and sent to the Instruction Decoder Memory location 0006 (0101, the source of the move) is accessed and its data is also sent to the instruction decoder. The Instruction Decoder directs the Bus Control to get the contents of memory location 0101 and put them in Register B. Step 3 - Bus Control then fetches the next instruction at memory location 0007, which is ADD. The Instruction Decoder instructs the ALU to add the contents of registers A (4) and B(3). then put the result back into register A(7). Finally, the instruction at location 0008 is fetched. The instruction is MOV. By using a process similar to the MOV instructions outlined above, the contents of Register A are placed in memory location 0102.

Address Data Address Data

0008 | MOV

Table 1

As seen by the example, it is a complex process just to add two numbers. For most applications, a high level programming language is used and the mechanics of the machine language are hidden. Although there are some applications that still use assembly language programming, they are mostly for smaller microprocessors and for specific applications. microprocessors and

Fundamental

Control Systems

d

For further information on microprocessors, visit the following web site: <http://www.howstuffworks.com/microprocessor.htm>

Microcontroller

A microcontroller is a system that contains, at a minimum, the following four components: Memory, I/O Bus, and the MPU (also called a CPU) is the Microprocessor Unit, the Central Processing Unit discussed in the previous

Section

MPU is the brain of the system where the instructions and data are processed. Memory stores the program(s) and the necessary to accomplish the desired task. I/O connects to

a outside world, through connections to such devices as a j, board, monitor, printer, or modem. The Bus transfers information between all of the devices. Figure 2 shows a simplified microcontroller. A typical device is the 8051 which has self-contained CPU, RAM and ROM. Many common examples of microcontrollers are the PIC, Stamp, and PLC, programmable logic controller. Units consist of a MPU, RAM/ROM, a display, keyboard interface and lots of

of input and output connections. As technology advances, manufacturers can fit more components on a chip and new devices are created, PIC and Stamps fit on small circuit boards, and can control just about anything, from toy robots to home automation. Software programming is accomplished by easy commands and not in native assembly language. Advances in the memory sub-system include RAM and flash memory. Number of new I/O devices is too numerous to list here, For further information see: <http://www.parallax.com>

Basic Controls

A control system is a system of devices that are used to maintain a parameter at a specified level called a setpoint. The parameter can be anything that is measurable, such as

Voltage

current, temperature, or pressure. In any control system there must be at least three components. The transducer is the device that converts the

Parameter

being measured into an electrical signal that can be used by the controller. The controlled element is the device that the controller uses to change the value of the parameter,

1. microprocessors and Fundamental Control Systems Chapter 17 of monitoring temperature, the controlled element - case There is a problem with the heat that will warm the air and thus change in temperature. If the parameter is varying rapidly about the setpoint, it causes the controller to cycle the controlled element on and off excessively, which wastes energy and manipulates the controlled element. Generally reduces the life of the controlled element. To alleviate this problem, hysteresis (also called a deadband) is added to the control system. One of the simplest types, ON/OFF control to the controller. The on and off points are different, with a "deadband" between them. In Figure 5, the setpoint is 80, the ON/OFF control. On setpoint is 78, and the off setpoint is 82. The controller has to change by 4 degrees to cause the controlled element to OFF control operates just like it sounds. The controller changes state only when the controlled element is 100% on or completely off. An example that illustrates this type of control is the typical heating system as shown in Figure 3. The transducer in this system is a thermometer that converts heat into an electrical signal. The controller is the thermostat. The controlled element is the furnace.

Time

Figure 5

The furnace turns

on at points a and b. The furnace turns off at point c. The same temperature curve is used in

Figure 4

On Setpoint Temperature Measurement Air Flow and the number of controlled element cycles is reduced

Figure 3

in Figure 5 by adding the deadband. temperature decreases due to losses to the outside envi-

Inter

ast the signal sent to the controller by the thermometer responds. Once the temperature drops below the set- point, the thermostat sends a signal to the furnace to turn on the heating elements and the fan to turn on. This allows the heat from the heating elements throughout

the room. As the temperature increases above the setpoint due

to room heat, the thermostat sends a signal to the furnace to turn

off

Figure 6

Proportional Control Proportional control is a control system in which the controlled element can be at some partial position. The position of the controlled element is dependent on the difference between the setpoint and the parameter's actual value. The difference is known as the error. When the two values are the same, the controlled element is in a neutral position. The neutral

Position

Time is dependent on the system design. The most common neutral position is at the 50%-of-full position.

Figure 4

shows a graph of temperature over time and the - While temperature remains above the setpoint, the furnace remains off. Once the temperature drops below the setpoint, the furnace turns on. The example used to illustrate this type of control is that of maintaining the water level in a tank as shown in Figure 6. To sense the water level, a float is used. The sensing element

is

the float turns a valve. The controller is part of the valve. As the float's position falls below the setpoint, the valve opens, allowing more water to enter the tank. This cycle continues the entire time the controller is enabled.

, micro outlets at the valve to provide more water

Figure 7

In Figure

7 the flows are equal and the float is lower, which is necessary to maintain the in-flow at the level to keep the system balanced. As you can see, the level is not at the setpoint. This is one of the drawbacks to proportional control. The outlet valve is closed from its normal position. This causes the water in the tank to rise. The inlet valve to the tank reduces the amount of water added to the tank. This results in the water level rising above the setpoint. To correct these drawbacks of the proportional control system, something must be added. The system must be able to compensate for any error that persists which is not corrected by the proportional control. Integral control is used to correct this. It uses the calculus operation of integration. Integration is the operation that takes the area under the curve. The integral

c _ the error over U control signals specified by the propor- For an example of proportional-plus-integral control, let's tank example. Instead of a float connected to a valve, an ultrasonic level sensor is used. The ultrasonic level sensor sends sound waves down toward the water. The sound waves are reflected from the surface of the water back to the sensor. The time it takes for the sound waves to return is proportional to the distance the surface of sensor. Thus it can be converted into an electrical signal that level in the tank and can be understood by the controller. process"

Motor-operated

Inlet Valve

me a secant & MICROPRESSURE Use necessary for inte;

A actuator is a motor-operated valve it is used to open and close the valve. For the process the LD) system, the one in which the level is from its initial position) "the

Operated by the ultrasonic level sensor the error and the proportional control

maintains its neutral position. Since the error still exists, more water enters the tank. Both types of controls will eventually bring the level back to the setpoint.

in more than it was with

valve and together, the level varies

When the level returns to the setpoint

the outflow rate equals the inflow rate to the setpoint - the

level The second case is the one in which the level is from its initial position. This will

maintain the level in which the outlet flow is at its initial position. This will

maintain the level in which the outlet flow is at its initial position. This will

me

Tiered

The level sensor reports this to the, the level control system:

oo

from its initial position. When the level returns to the setpoint

oe

'i. The error level causes the valve to close and the level returns to the setpoint.

Proportional control has an other drawback: the error is changing with time. Proportional control has an other drawback: the error is changing with time. Proportional control has an other drawback: the error is changing with time.

Therefore

the level control system allows it to maintain the level at the setpoint

Teact

to the rate of change of the level control is proportional-plus-derivative

the level control is proportional-plus-derivative. For this, the level control is proportional-plus-derivative.

a

is example, a system similar to the previous example is used, except that the one in the TO] i ivati
e pr portional and derivative clemen controller now | will consider is the one in which the ts. The first case that hich the
We opened' outlet v panaa The result of this is the i
is rapidly
ma

```
=0 aesdecrease
```

quickly enough to me level

Tapidly

enough to prevent
a more control opens the wet
input the tank from emptying. The d

Valve

far eTivative çOn.
poprocesee' vues ~ruumental

Control}

Systems "

asi i Chapter 17

Input

ne

Fa portionalcontrol would have, adding

Machine

Isopope ret team se ye thant ent the tank from being emptied. Figure 9 CONsuMMgtask Sona "intone otero prev ter level and valve position, of the

Microproe:

fe tender ieee el. oe ; ESSOF

It eReCUTES very quickly H, IMe saved in running the

te

eg ost TIME needed to we the progreny wey sifes the extra

ee

Proportions!

only urn cis

Assembly

language,

and is a way of simplifying machine

giving a short usually three-letter mnemonic to represent a language command. It is generally fast-processor and debug assembly-language

Programs

than their language counterparts. In assembly, a mnemonic command and its symbolic operands enter the machine instruction. An "assembler" program translates the "opcodes" (operation codes) and symbols

into binary language. One of the drawbacks to an inflexible control system is that it can be used with only one type of

control. The integral-plus-derivative, also called PID, of

Microprocessor. Therefore

Programmers

who spend

on all three elements described earlier. A time and effort to learn how to program one system changes in the parameter, long term - learn a new set of code work

on an

and compensates for

is controlled by proportional control, and compensated by whether it can be classified by whether they are

Open-Loop control does not have feedback or not. Open-Loop, the water level control can be made into

the level sensor is removed and the controller is programmed to keep the valve open to 75%. which fully opens the outlet valve. Under normal conditions, the water level. If the outlet pipe became a waste tank, the level will start to rise. If no level signal is set to the controller, the controller will correct for this and the tank will overflow. The control is closed on the pump. The example of a control system is in the control uses feedback. The example

of a control system were all closed-loop control. An example of the water level signal was the control:

for proportional control: www.control-engine.com

mp

and four books on programming are numerous books on programming. This section to cover the complex art of programming, but only covers the fundamentals of programming languages,

programs and debugging programs. Writing Programs in C programming

Languages

is a language and is used. A language is the lowest level in the microprocessor. In this language the program-

language is needed. It has a set of assembly language. High-level languages often use as commands that correspond to a list of machine language instructions, print, and open. The commands can be entered from the keyboard or from a program in memory. A translator since the translation is occurring at execution time, the program runs slower than a compiled program. JAVA is an example of an interpreted language.

A Compiler translates the entire program into machine language Prior to execution, so such Programs run as rapidly as though they were written directly in machine language. C and C+ are examples of compiled languages.

Writing Programs

The first step in solving any problem or completing a task is to understand what must be accomplished. Programming is no different. In programming, the first step is to understand what inputs are available, the desired outcome, and the methods available for achieving the desired outcome. For example,

Chapter 17 [

The available inputs are the output of the sum that is 7, and assembly language or any other programming language. The desired methods are machine language. The methods of the high-level languages like C, C++, Java, etc., need to be broken down into state flowcharting, state diagrams, and Venn diagrams. Rehan is breaking down the task and organizing the parts. Figure 10 shows a high-level flow chart for a control system.

c?

rr corres

Fenet

Figure 10

The new step is to determine which methods can

be

and which will be the most efficient (USE). Sometimes there are hardware limitations

on which can

be used. If the task is programming a PLC

Je Logie

Controller

(PLC), the only programming option may be ladder logic programming. If the task is to be performed

by

an embedded micro-processor, the only choice may be assembly like C++, Perl, if the hardware allows higher-level languages like and Visual Basic, a choice as to which language to use must be made. Some languages are better suited to certain tasks than others. For example,

Petl

is very good at pattern matching. For example

o add

and in C+ only requires one line (ant sum

```
= 4 +
```

3). The same operation

in assembly language

is more complicated See

Table

| in this chapter which has the code for adding

and

Now that the mayor decisions have been made, it is time to start working on the specifics.

Take

one of the sections from chart and refine it. Create another flowchart with very detailed steps that must be done to accomplish the task

Keep

in mind the language that should

be used and try

to break the problem into steps that correspond

to the language's instructions. One of the most common mistakes is to not put enough detail into the flowchart. One exercise I do when teaching a programming class is to have the students write detailed instructions on how to make a peanut butter sandwich. In most cases,

do not get to the point of putting peanut butter on the bread due to the students making assumptions. The students often make assumptions about the initial conditions. For example, one assumption may

be that I know how to open the jar

of peanut butter.

Another

assumption may be that I know which end of the knife to use. In programming, assumptions can also have adverse effects on

Microprocessors

and Fundamental Control Systems,

runs the program. For example, assuming that the sensor output is 10 volts when the designer is actually measuring 5 milliamps may cause serious problems,

ly

'Sing4

The to be sorted starts at the same fact the jury Summa

has

one erroneous results,

Tans

again

Ve

Oe

re Once the detailed planning of what's to be

is finished, give the plan to another programmer to review. If that option is not available, find someone as you explain it. Many times in explaining "my" you will find the hidden errors. This review phase is very important as it can catch errors before the program is run on a machine, which could possibly prevent a crash. The review can also detect design errors in the program.

a Prev,

i With the design complete, it is now time to start the program. If the language being used supports the features, use them. If the language does not support

\$e

Of course, you may want to consider using another language. COM languages are helpful when you have to go back later and fix

Comments

make enhancements.

"a program Since the actual writing of the program is large, depend on the language. Any specifics on writing at the end, would not be useful. However, anything that makes the code more readable is a good idea. Using white space to divide the program into functional sections is a common practice to make the code inside a control structure, easy

's : loop, can make the code more readable. In debugging programs finally, after the program is written, it's time to debug the program. This task should be accomplished in a timely fashion. The first check is done by the compiler. The program is written in a compiled language. The compiler

will catch syntax errors.

Be When you are preparing to write a program, you

Shouly

also prepare a test plan. The test plan verifies that the

Operates

correctly under all operating conditions. Then

should include instructions on how to put the system in proper condition to start the test, the test to be performed

a the expected result. This maintains consistency due to testing normal operating conditions, the

's should also provide instructions on how to test the

under error/abnormal conditions. An additional useful strategy is to write a program. The program is then executed and the results are displayed. The drawback with this method is the test code could have an error, resulting in an erroneous result, or the program is incomplete. After the bug has been found and fixed, the tests

the NC tests that

A Mini program to verify that the bug has been

yielded

be run. Far from covering the unaffected portion of the code, this verifies that the fix did not "break" the program

's

wt -oprocessors and Fundamental Conte

Systeme } Chapter 17

Chapter Quiz

1. Which portion of the the addition of two MW microprocessor would perform mbers? a AU

Bh Rus

2. of the can ron directly on the microprocestor?
- a. Assembly language Interpreted language Machine language
4. Compiled language fa microcomputer connects all of 3X Which component o! the sub systems together? a MPU
- b. Memory
 - c. Bus a vO 4& What type of contro! handles the rate of change of the error?
 - a. ON/OFF
 - b. Proportional cc. Integral
 - d. Denvative
5. What type of control handles long term errors?
- a. ON/OFF
 - b. Proportional
 - c. Integral
 - d. Derrative
6. What type of control has the controlled element in either the full open or fall closed position? a ON/OFF
- c. Integral
 - d. Denvauve
7. Which type of program runs the slowest?
- a. Assembly language
 - b. Machine language language
 - d. Compiled language

5 Microprocessors and Fundamental Controt sys; ems

- 8, Atest plan should contain which of the fol
- a. Expected results
 - b. Inital conditions for the test 4. How to perform the test
 - d. above

- 9, What problem does adding hysteresis to an o

Ni

hy control system fix?

- a. Excessive overshoot of the setpoint
- b. Excessive undershoot of the setpoint
- c. Excessive cycling at the setpoint
- d. Excessive setpoint

10, Where are the results of math operations microprocessor? sto ALU

Bus Control

Instruction Decoder

Register

ANSWERS on page 223

req in

Review Questions - Digital Circuits and Mi Chapters 16 and 17 level of a TTL digital signal is called al: or low. © oor high. tor low. i tor high. 65 (decimal) converted to binary would be: «168 110100.

4, 1010010. qne! in binary number 1000000 carries a decimal 3 eis! oft 32.

I

one of the following integrated circuit or i jconductor devices normally requires the most handling to avoid damage by static electricity?

4, 1 MoS p, ECL MOV 4 TH s, What does RAM mean? memory data storage procedure b Random Access Memory

Real Address Mode

Regulated Audio Methodology

jan LCD (liquid crystal display) Is: an inductive coupling device, ba DC source, alow-current digital display, a charge-coupled device. i Hexidecimal 3E is equivalent to a decimal number of: a 62.

b, 46. c 37

4. 48.

8. What is the binary number for number 7

a, 1001 b 010! 1100

d. 010

9, Logic Pulser can be used as a:

a. logic probe

b. digital tracer

c. digital signal generator.

d. filter.

10, In the figure below, if A is low and B is high and C is low and D is high, what is the output level at E? Low

High

Floating

Cannot be determined aoge

oa

11. In the figure below, which of these gates is an OR? aeoPp

13. If SW₂ were to open in the circuit above, the output of |, input NAND has two high inputs and a low Review Questions
- Digital Circuits and Microproces

f° i

Unit 7 | ne Chapters 16 and 17

tn the below, both SW1 and SW2 open, the output will be: high hoe apprenrmately 2 SV. erwre sv OSV

Swi Sw2

the gate circuit would be: a floanng b low.

c. high.

2.5V. input. The output will be: a high

b. low. & not enough informanon to determine. can be neither high nor low.

15. Adding the binary numbers 11011 and 001100 will give which one of the results below? a 0000101

b. c. 1000011

ANSWERS on page 223

—9 chapter 18 by Tom R. Janca CETsr

Pgrdwore-Softwore

Technicians

cians and engineers continue to face a variety of wesand systems that theyare directly responsiblefor essetdrestonng to operation if a failure occurs. Each faces Its own complexities and challenges unique to ict and services.

Compounding

the daily difficulties, ae evolution has fused hardware and software within fs of high-techproblems, The days of ahardware tech-

oe sand asoftware technician being two different specialties g- Modem equipment and systems require that

we nvcin individual be familiar both with its hardware oe and software operation in order to properly diag- wdrestore operation.

ow does one prepare then, for this evolving industry oseoo88 the fact that learningthe common faults with a pseu utoF system is only useful for a few years?To sooth its necessary to developand practice a basic diag- techniquethat will serve you regardless of the technical weetyou are immersed in. jalbegins with an alarm or acustomer complaint. Some- ssn'toperating the way it is expected to, and now the whe ticket or customer is expecting you to correct the situa- poo. Where do you begin? Pee

jn with the customer orsystem. Your most valuable pointis the customer since they, not you, own and the equipment.Secondly,theyknow what the environ- rent ofthe equipmentwas when it failed and what is no lon- ing. This information can be extremely helpful when intermittent problems or preventingthe overlooking of spectfeature failure that does not affect the generalopera- tionof the equipment.

Depending

on your situation, customers canbe questioned directlyor a questionnaire with keyques- tons can be filled out at the time a customer bringsa product infor service.

Diagnostic Techniques

Tor

Jenca

isactrrely working m the communications feld
fi

Power

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te Cached cemaremec store

om the development team for developrng anew

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of experience wits cernsoptical and rakermrtrn arene

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Principles

and techniques i a corperate envimnment and presenily 2 DX

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C aheag

YF compenton

He hes been a member ofthe ETA since 178, winnmng the

Precedents

Award

and serving in various positions in the FTA. Tom was an excellent teacher, writer and speaker on technical subjects, who brings the human element and understanding to the computer and delivery

A few

good questions are: * Please give a description of the problem of your difficulty with the equipment. * What occurred at the time of failure? + Is the problem consistent of intermittent? ¶ If intermittent, does it occur immediately or after a period of time? Effective questions can help identify what mode of operation a piece of equipment was in when it failed. What external situations were taking place that might have affected the equipment, such as a lightning storm or an electrical brownout or surge? Every piece of equipment was designed for an ideal world and every piece of equipment is subject to a world that is less than ideal. Knowing what a piece of equipment has experienced can show you the trailhead to begin your troubleshooting journey. Consider your customer as your first contact in a foreign land and learn as much from him or her as possible before you begin your technical adventure. It could save you a great deal of time. When working with systems, it is important to take advantage of the various alarm and records databases that maintain and store network and system health routinely. System history recording can help you discover what events were taking place when

a given node or segment of the network became operative. Take advantage of these tools to inquire about the status of the system prior to and

at the point of system failure.

If it

is a specific operation that causes failure, it may be a software glitch. Knowing what events took place prior to failure might help you locate or create a software patch to eliminate a software bug. It might even

be a capacity issue that, during certain peak operations, failures occur and these failure records will help you present a convincing argument to management of necessary system upgrades or changes. Using the software interfaces provided to learn as much about the failure as

Possible

before proceeding will reduce the amount of time to locate a system fault and assist in finding an effective solution.

Now What?

All right, Sherlock, you have done your investigation and questioned the witnesses. Now it is time to look at the guilty some body of the crime.

Chapter 18 |

re and gathering the pertinent gunning aha the link has actually has occurred (femance, you need to vent) that when of investigating, you should be on row the devnee fed and be able to rephrase of observe the factshare Your questioning <howikd tell you if it will occur our mmmeharch ae if you need to surest to a bum-in phase first tool or fire UP Teacampatame that before you hit your yews parthen dotlar test equipment you operate the customer campment and when its operation, of nek thereof, through observation Sometimes it may be just an operator issue, and if you have taken out 100 screws and have panels scattered everywhere when you realize this, you might begin to feel a little like a Towe have done a little in the operation, most good SerVICE 'Yew are unfamiliar with a poake the time to read

Manunks

contain

a bef operations thes and become rhe how it is supposed to operate and if there are any cautons you need to

be aware of, ONE the hound hs seen the fon. sony a mater of Me before it iscaught. You need to know: what you are chasing before you beginthe pursuit, Verificanion of the customer complaint

Ah Ha!

Once vou have completed your investigation and verifica- ton of the fault, i is time

to begin

to chase down the cause, Lets start the chase with the oldest set of troubleshooting tools available to you: your own senses. That's nght!Your firs step

ts

to use your eyes, nose, ears and fingers

to collect more clues. Visually check the equipment for signs of damage from a fall, moisture, and anything that might give physical evidence of a problem. Manytimes power amplifiers, diodes and transistors will ex plode and leave a visible sign of failure

in burn marks

on heat- sinks and printed circuit boards. Component failures can also be smelled with the nose. These aren't pleasant smells. Smells can pom! you

m the night direction and sometimes reduce good deal of troubleshooting time. Electrolytic capacitors and semiconductors have the most noticeable smell when they give up the ghost and join the great hereafter. Another useful technique when dealing with equipment is to listen carefully to the sounds present. Televisions with high voltage circuits often whistle loudly when a failure occurs, and you can hear high voltage when it is generated by the static on the froat of a CRT.

Electromechanical

devices often whine 'when

a motor is bound

up

or

an arm is jammed.

A motor that is excessively noisy can indicate bad bearings that cause the motor drive circuit to heat

up and

an over-current fault to occur after a period of time.

Diagnostic Techniques

When troubleshooting logic boards where integrated circuits are present, touch a Prd chip that is operating at a higher temperature,

compensate, thereby indicating the possibility of overheating. Remember that when a short is present, current rises in the

chip always brings a rise in temperature. Some can

be dangerous, however, in interpreting the results of a measurement.

By those are to know the function of the integrated circuit always wear ESD protection. Integrated circuit is designed to power or control power often pulsed. Although a purely logical circuit, and some manufacturers require it to develop an experienced sense of proper temperatures. Once this experience is defined, it is helpful in getting you quickly to the stage of problem

of troubleshooting. The one sense that isn't used is smell unless you're a person who prefers to test by taste. Use your tongue.

TOOL TIME!

What the senses have been unable to find, the brain is covered through logic and discipline using the "tool box."

Be

careful * What common tools are in the box?

Technical Education

Experience

* Service Literature

* Test Equipment

Each of these is necessary to efficiently and accurately test a piece of equipment. Each one is dependent on the strength of the other three. If you have a good amount of experience on a given piece of equipment,

even

but it becomes very important. If you have plenty of experience,

it may require very minimal test equipment or literacy to solve a problem, but it should be common agreement that,

at least four are necessary when encountering new products

or all failures.

Unsy

Service Literature

While service literature will never be found in a class English lecture, it can be exciting and valuable

Teaching

technicians and engineers. The service literature contains following important documents necessary for

Repairing

piece of equipment:

"s

Operating

parameters or specifications

* Basic Operations

Guide

* Interconnection Diagrams

* Block Diagrams

+ Schematic Diagrams

It, it will also contain circuit service manual, along with innovative approaches to systems new

Many theories for each can be useful when choosing a piece not to substitute a signal by informing you that one video. It will prevent you from connecting the wrong source for test. Parameters will technically state what equipment is designed to perform and the test performance that is necessary to accurately evaluate test readings. Don't expect a Porsche when troubleshooting. Both are German and both are cars, but they perform on different in performance,

Is the "on" button? How is it connected and operated if you have never seen it before. However, if you have Honda Airt Force base, probably no manual is so no one knows what it is anyway. Giving a technician, you might see a test question maybe you have seen a familiar puzzle on a video placemat.

Go Which box is "A" connected to? Interconnection diagram can answer this question. It shows the connection point and the cabling assembly that the power, clock, control signal travels through, identifying the connectors at each. If comment similar to J2 to J16, Each connector are

used through PCB silk screening or a label immediately identify to the cable. The identification can also include the name of the connector similar to the following example:

12-3 Ji-14 +5Vdc >> a Diagnostic Techniques

Chapter 18

In this way, a technician can trace the power source from one circuit board to the next

Interconnect

Diagram

In this diagram the individual cable assemblies are labeled at the end. This diagram can also show the primary blocks of sections of the piece of equipment.

Block Diagrams

The above interconnect diagram shows the different blocks of sections that work together to perform the equipment functions for the operator. Each block performs a certain task that is indicated by the section's name. An example is the

Printer

Interface section that has the task of operating and monitoring the printer. If the technician were experiencing a problem with the printer, this section would be a good place to start. A more detailed block diagram might be available for the printer interface section that would give the technician an overview of the different sections involved in forming the printer interface function. The block diagrams allow the technician to divide a circuit quickly into operative and non-operative sections. The technician can then begin to evaluate each section's input and output for proper operation that allows him or her to locate the faulty area. This is what is classically known as 'divide and conquer.' Once the faulty section is identified, a schematic is necessary to follow the signal flow through the individual components that comprise the faulty section. A Single Section Block Diagram See Appendix C-Page 217

Schematics

the electrical schematic diagram

Signal

an example

of a topographical map and a need to map of the material. A detailed signal flow, data voltages and the electrical circuit being able to read

ii

a the component level technician.

a signal trace test and analysis and signal emission. How the technician to given stage of the schematic of components is it.

f 2354) - if A

See Appendix

C-Page 218

the Section, Block, and Component

Once

the tools are in place, the fourth step is to begin

isolating the problem, applying TEST to the problem approach logically. Divide the operating sections from the inoperative sections. If multiple sections are faulty, look for something common to all of them, such as power

of logic control,

as the source. If a single section is bad, break it down into its defective blocks and then begin isolating it to component level.

Verify

proper power levels

Check

for circuit clocks

if present

Always

check IC signals

at the chip body to ensure no other opens are present on PCB

Diagnostic Questions

Every electrical circuit consists of three basic sections

"a source, a conductor and a load. When troubleshooting, it is good practice to first verify the source.

Diagnostic Techniques

+ Is the power applied correctly? + Are the signal levels good? © What is control doing?

Conductor:

* Is the power reaching the load? * Is the signal reaching the load? * Is control able to control the load?

Load:

* Is the load operational? Oftentimes the load is the source of the next

Section

stage, and the troubleshooting repeats itself for the section, block or component.

Methods**Voltage/Resistance Analysis**

When it is necessary and possible to operate the live voltage analysis is used to determine the king

section, block or components. Since Ohm's law

Relates

age, Resistance and Current, the following rules of th

be helpful:

ia

Short -

Resistance decreases.

Parallel

circuity Current increases and voltage decreases, since a short in one branch shorts all branches.

Series

circuit Zero, or a lower voltage is measured across the shorted component affected by the decrease in resistance. Other components show a higher voltage due to an increase in resistance. Open - Resistance increases.

Parallel

branch circuit Total circuit current decreases 'one branch is inoperative.

Series

circuit Entire circuit is inoperative.

Nothing

path open has current, and the open component has total voltage, applied minus the IR drop through the meter.

When

apiece of equipment can't be operated as a result of the failure which is typical with power supply failures, voltage analysis can be useful in locating shorts or open circuits and components. Remember that current seeks the path of least resistance.

therefore use the Ohmmeter to measure and evaluate components;

LS

tion of the circuit,

the equivalent circuit,

Felates

V₀)"

of thumb can be used across the test points in |

Teasesang

UN Past the label

Voltage

SA result of fuses, resistance.

Open fuses

'resistance, data components-

es

el MIE at resistances. To be effective, one must know ae semiconductor biasingto determine good and

LJ

Diagnostic

Techniques

'a

we substitution and Signal Tracing

ao

Es

ss muting good source signal from a calibrated signal he technician removes the equipment source fram

3h equation. An example would be Substitutinga rs source for a television to eliminate the possibility oe cena, CATV or other source. known goodsourse applied, the technician can then Hloscopeor other device to trace the signal from t of the vanous sections, blocks and compo-

sidentify faulty part. Beingable to Properly operate a oe LJ

of signal generators and oscilloscopes will be of great . Also, you must know how the circuit will affect penal Often schematics will provide waveforms at and key test points. If your waveform is missing than what ts expected, You just passed your fault. anne you isolate the first good signal. Now your raidbe between where it is good and where it is bad.

ot

ad Bad ood Problem some introductoryideas to get you started. Each The

i çandexpandon this. It is important that you

wl familiarwith your basic test equipment and ee is presented in the chapteron test equipment.

w test equipment's application will improve your ot and help you diagnosethe variety of problems (oo) es a technician. ror Replacement have isolatedthe fault it is time to replacethe de- one ones) or repairthe wiring or soldering respon- eftthe fault goo cts wyoutrepair will only be as good as your soldering!"

I "ifyou don't observe ESD handlingprecautions you will

z woe rice." cattofetechnician is in how well he or she completes Te repair. Agoodsolderingjobis essential to a lasting

x _ asatisfied customer. Secondly,apply new heat-sink al when removingtransistors and power devices that

use a heat-aink to dissipateheat Fa ling to reapply compound Can cause equipment failure out of the warranty period and leave a bad word in the Customer's mouth regarding your service As a factory technictan at one port, | followed behind another technician by one year The individnal replaced the faulty regulator [C but didn't apply heat-sink compound to the new device, which subsequently failed ina year's me The service literature will also contain a parts list or an exploded diagram to help identify different equipment parts. When possible, use the provided part numbers and OFM parts to properly repair the unit. If

an OEM part is not available, use a replacement cross-reference book to locate a replacement

Verify Operation

Once the parts have been installed, it is critical that you operate the unit for a period of ime and verify that the fault has been corrected. Just replacing a component and sendingit back is a poor troubleshooting technique. You should venfy an equipment's operation for a given duration of time to ensure that all problems have been solved. [t is possible that the m-

mediate problem was the result of some other circuit failure that has not been identified. If the Equipment must be immediately placed back in operation, make it a Point to examine the unit again after it has been in service for at least 24 hours.

Extra Screws, Paperwork and Cleanup...,

Next!

Proper operation has been verified and you are ready to Move on to the next project. Pay close attention to your reassembly of the equipment to avoid little mishaps that leave a bad impression. Verify that all customer controls are accessible and operational with the unit fully assembled. Complete your paperwork in order to be properly credited for the work.

Paperwork should also be done as you work on the equipment to ensure that materials and parts are

Properly

charged to the customer or correct account. This may be the Most tedious part of your job, but it lets others know what work you have done and what materials and parts were used. If you don't do your Paperwork immediately, you might forget to report a part used and this could cause problems with inventory. Worse still, the customer is notified at a later date than Promised or not at all because the unit is sitting on the bench waiting for you to complete the paperwork. With the unit fully assembled, the Paperwork done and the unit returned to the customer, you are now ready to begin your next adventure.

Chapter Quiz

nt?

1. Who knows the most about a piece of equipment?
 - a. The pharmacist
 - b. The operator
 - c. The operator's friend
2. Before beginning to troubleshoot, what should you do?
 - a. the failure
 - b. Call your friend
 - c. Reassemble the equipment
 - d. Divide and Conquer
3. A short allows for an increase of current and a voltage
 - a. resistance
 - b. trouble
4. What is the oldest piece of troubleshooting equipment a technician has?
 - a. Oscilloscope
 - b. Analog Volt Meter
 - c. Galvanometer
 - d. Five Senses
5. Which of the following word(s) is in the acronym:

TEST?

a Time

- b. Ego
 - c. Spectral Efficiency
 - d. Technical Education
6. What is the name of the diagram that shows all the general cable connections of a piece of equipment?
 - a. Schematic diagram
 - b. Block diagram
 - c. Interconnect diagram
 - d. Parts list

- 7. Diagnosing component failures requires the use of a/an
 - a. schematic diagram
 - b. block diagram
 - c. interconnect diagram parts list
- 8. A repair is only as good as
 - a. the test equipment
 - b. the equipment brand
 - c. the part replacement
 - d. the soldering
- 9. Before returning the equipment to the customer, important that you remember
 - a. verify operation
 - b. prepare the bill
 - c. hide extra screws
 - d. call the customer
- 10. What is the most important service you provide?
 - a. Technical expertise
 - b. Equipment repair
 - c. Customer service
 - d. Sales

Intelligence

Customer service

oP

ae

ANSWERS on page 223

With chapter 19 by Tom Schulte, CET, where you are introduced to your career as a service technician, you will not test equipment during a high percentage of your work time, to provide information on a circuit's operating status on of to test the condition of individual circuit components. To be most efficient at servicing, you should have a pre-judging of the proper use of all the test equipment.

How disposal, and you should know just what each instrument

can and cannot

do for you. become a proficient service technician, you need to

know how and when to use each of the different test instruments in your particular service field. Unless you have a good understanding of the full range of information instruments are able to provide, you will often find yourself limited to memorized symptoms and cure analysis, which is only as an initial troubleshooting step or blind parts (never an efficient troubleshooting process). A good pre-judging of the proper operation of test equipment and of the service information each instrument is providing you, will tremendously increase your overall efficiency and will help place you among the ranks of electronic service professionals. Test equipment generally falls into one of three major categories:

1. measurement, circuit stimulus, or component circuit

Measurement Equipment

This category of equipment includes voltmeters, ammeters, vector scopes, logic probes, sound meters, logic gates, color photometers, AC leakage testers, RF level audio, RF or light wattmeters, spectrum analyzers, and aligned circuit analyzers. Their output indicators may be either digital meters, CRT oscillograph displays, or LED status indicators.

These

These instruments are all used to determine the status of a signal within or at the output of energized electrical or electronic circuits. You will use this type of instrument primarily to determine whether a particular circuit is operating within acceptable limits or whether an entire system is producing a desired output result. Circuit measurement equipment is often used during troubleshooting to trace DC supply paths and AC

Test Equipment

John Tom Schulte is an application engineer

at

Sencore, Inc

located in Sioux Falls, SD. He is involved in test instrument design, sales training and support for electronics test equipment users. He has developed numerous training courses and seminars and has presented training seminars throughout the US and Canada for over ten years. Tom previously was an electronics instructor at Mitchell Technical Institute in Mitchell, SD, where he taught consumer electronics servicing, two-way radio servicing, and basic electronics signal paths, to determine at what point the circuit operation becomes faulty. Figure 1: An instrument's display may contain a number of types of output indicators.

Circuit Stimulus Equipment

Often called generators, this type of equipment produces single or multiple outputs that may be very simple or complex and may range from DC to high RF frequencies, or may

be perceptible sound or light. Simple output signals include DC Supply or bias voltages and single frequency AC voltages such as audio tone signals, and unmodulated IF or RF carrier signals. More complex generator signals include swept frequency Voltages, random broadband signals (noise), video signals, logic bit streams, modulated digital data streams, modulated light signals, and modulated IF or RF carrier signals. Generators are often used to provide an unchanging, known-good substitute signal to the input of a complete system, to closely duplicate the signal the system normally processes. This allows you to evaluate the operation of the system while troubleshooting or specification testing, without the confusion of a constantly changing, unknown-quality input signal. A signal generator can help you determine the full range of defect symptoms during an initial performance test of a defective system, and can deductively guide you to defective circuit blocks. You may also use a generator to substitute known-good input signals to individual circuit blocks within a defective system, to help determine which stages are functioning properly and which aren't. A generator needs to have very low output impedance, with AC signal coupling, to effectively substitute signals at the inputs of individual circuit stages without affecting DC circuit voltages. Or, you may use a generator to

Chapter 19 E

errouit or enmplete eystem over its full or. broadband a cwcep poner afew genera) fi evaluate the yn away that wouldn't be pos- Lora simplegenerator exercree an india! ogeratng range (oh as with PRED RORCHRA. OF muloterst \ erent de oyster pT IEMANT atcwath erther the noemal input tena senal

Component Test Equipment

Yoon will ue cxempancnt testers £o test the charactenstics of sample oc cmall groups afcomponents, either out-of-cireutt, or man un-cnerpuzed crrcunt. These testers: include ohmmeters, prancratoc Fosters, sermaconehc Tew CUIVE MTBCETS, impedance fendges.canacttance meters. mductor NNECTS. capacitor'induc- toe anal rors, thyristor testers, IC comparators. CRT testers, caNic testers, and custom componenttest 188 These testers Provideane or more stimulus signalsto the component under trameacare the component output response, and drive

an ape peopnate indicator display: Fig. 2: A transistor tester can test bipolarand field-effect transistors in-circuit for good'bad. or out-of-crrcuit for gain and leakage You will usually use component testers in the final, isola- on phaseof servicing a defective circuit, after you have used functional analyzing to localize aproblemto within one or two specificcircuit blocks (usuallyusing some combination of signalsubstitution and signaltracing).You may also use these testers to confirm the integrity of a replacement partbefore you connect it mto acircuit

or

to performan incomingquality test on components going into a manufactured product.

Commonly Used Test Equipment

Although a wide range of electronic test equipmentis avail- able, much of it is used only infrequently, for specialized test- ing. During most service or calibration, you will typicallyuse only 2coupleof general purpose circuit measurement instru- ments (t.g. a voltmeter and oscilloscope), one or two signal generators appropriate to the systemsyou are servicing,and possiblya coupleof component test instruments.

Test Equipment

AC Isolation Transformer } An isolation transformer is one piece of tey doesn't fit neatlyinto one of the three test "a Tics mentioned above, but should be on even temMe This transformer has isolated primary and se,

te with a 1:1 turns ratio, Whatever AC voltage ay Wn Pnmary is also produced from the secondary

mn hed

MA that the AC voltageproduced from the second, aitage with no fixed reference to earth ground.This le "flay, ment that has intemal AC voltageoffsets from,Had es gat? from causing safetyproblems dunngservicing

= 67 VDC

(approximately) Fig. 3: A non-isolated chassis can ca use damagey to flow from hot chassis groundto grounded - a personnel. PM

Theisolation

transformer should be used to provide

to any equipment you are servicing. While servicing, it is especially important to isolate any equipment that has a full-wave bridge AC rectifier power supply. This is especially true when working on equipment that has a common reference point, such as the earth ground reference used by most test equipment.

A full-wave bridge produces a common reference point, output which is always at a potential of one-half the AC line voltage, compared to the earth ground reference used by most test equipment. Note that a RMS AC voltmeter measures this voltage as even Volts, due to the rectified, non-sinusoidal waveform that equipment of this type has a "hot chassis." If you touch any internal circuit point on non-isolated equipment, and an external earth ground connection, you can receive an AC shock.

Receive

If you connect test equipment to isolated equipment under test, either the test equipment under test will be damaged, or a shock can be created for yourself or others. This is especially true if you are working on equipment that has a common reference point, such as the earth ground reference used by most test equipment.

Test Equipment

Equipment under test should be connected to AC power supply during service.

4. Safety

Equipment under test should be connected to AC power supply during service.

For safety and for the protection of your test equipment, always connect the equipment you are servicing (but not your test equipment) to ground through an AC isolation transformer, rather than directly to ground. For maximum safety and performance, test equipment should be connected to AC power supply during service.

Equipment should always be powered directly from a properly grounded AC line, not through an isolation transformer. Most AC variable transformers are autotransformers and do not provide AC line isolation. If you are going to use a step-down transformer, it should be in combination with an isolation transformer for safety.

There are two general types of DC voltmeters: non-amplified and amplified. The non-amplified DC voltmeter is generally an inexpensive meter that consists of only a D'Arsonval analog meter and a multiplier resistor for each voltage range. The amplified DC voltmeter is often called a Volt-Ohm-Milliammeter (VOM) and is a more sophisticated meter that can measure voltage, resistance, and current.

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The amplified DC voltmeter is often called a Volt-Ohm-Milliammeter (VOM) and is a more sophisticated meter that can measure voltage, resistance, and current.

a Volt-Ohm-Milliammeter (VOM)

Although this type of meter is quite suitable for use in low resistance circuits, it has at least two disadvantages for use in electronic circuits. The meter's internal resistance is relatively low, and its resistance is different for each of the voltage ranges.

meter resistance means that when the voltmeter is connected, the voltage difference between two points in a circuit will draw a small, but in some cases significant, current from the circuit. This additional current drawn by the circuit under test may cause the voltage between the

points to drop significantly when you connect this type of meter.

This is a misleading voltage measurement. This is known as voltmeter loading, and is more pronounced if the internal meter resistance is lower, or when the external

resistances are higher. For an amplified voltmeter, its internal resistance is specified as its input resistance (often called its sensitivity), and is figured by

taking the inverse of the full-scale meter movement current. Thus, a typical meter movement with a full-scale current of microamps has a 20,000 ohm sensitivity ($1/50 \mu\text{A}$). A voltmeter with this sensitivity has 20 kilohms of internal resistance on its 1 Volt range and 2 megohms of internal resistance on its 100 Volt range. If you use this type of voltmeter on its 1 Volt range (20 kilohms of internal resistance) to measure the voltage across a 20 kilohm circuit resistor with a 2 kilohm series resistance, the measured voltage will be one-half what the voltage was before the voltmeter was connected. 20k ohms

$$C^*) \quad 20k =$$

2,000 ohms/volt Fig. 5: If the voltmeter internal resistance isn't significantly higher than the circuit resistance, the voltage measurement will be very inaccurate. To create a simple voltmeter, just add a series multiplier resistor to the basic meter movement. The multiplier resistor for any voltage range will be the resistance value (less the meter movement internal resistance) that limits the meter current

to the meter movement's full-scale current when the maximum voltage for that range is applied. For a 2V range with a 200 μA meter movement, for example, the total meter resistance needs to be 10,000 ohms ($2 \text{ V} / 200 \mu\text{A}$). If the meter resistance is 300 ohms, the multiplier resistor will be 9,700 ohms. $\times$ microamps full scale $\times$ 10^6 $\div$ OSV $\div$ 0.10V $\div$ 20V Fig. 6: The multiplier resistor for each range limits the meter current to its full-scale sensitivity with the full-scale

Range

voltage applied. Amplified DC Voltmeters An amplified (sometimes called electronic) voltmeter uses an amplifier with high input impedance at its input. The input amplifier senses the difference in potential at the voltmeter's

terminals, while drawing very little current from the circuit under test. The input amplifier drives control circuits with

WDE Voltas ranges where then drive either an analog or

digital meter display. The effective input resistance of this type meter is very high (typically 10

or 15 megohms), and is the same for every voltage range. This high input resistance ensures that, even in moderately high resistance circuits, the small amount of current drawn from the circuit under test by the voltmeter is not enough to significantly change the circuit voltage when the voltmeter leads are connected to the circuit. Another advantage of a meter of this type is that, since the meter's input resistance is constant for all ranges, it's quite easy to design a voltage extender probe for the meter. A probe can be designed to extend the voltage measuring range by a factor of ten, for example, by designing a probe that increases a meter's input resistance to be ten times its normal value. For a voltmeter with a normal input resistance of 15 megohms, the extender probe would contain a series resistance of 135 megohms, nine times the meter resistance (Figure 7). This increases the total resistance of the meter plus extender probe to 150 megohms, ten times the resistance of the meter alone. When you connect the meter and extender probe across a circuit voltage potential, one-tenth of

the circuit voltage is dropped across the meter, and the other nine-tenths is dropped across the extender probe. Note that the extender resistor must be capable of withstanding this high voltage. Whatever voltage is measured by the voltmeter, you then simply multiply by ten to determine the actual circuit voltage. Fig. 7: The extender probe is nine times the meter resistance, so nine-tenths of the probe-tip voltage is dropped across the extender probe and one-tenth is measured by the meter. The same method is used to create a 100x high voltage probe, to allow a standard voltmeter to measure the 20 kV to 35 kV high voltage developed in CRT display devices. The extender resistor will be 99 times the meter resistance—990 megohms for a 10 megohmmeter and 1485 megohms for a 15 megohmmeter.

Test

Equipment

Each probe also reduces the voltage loading effect by a factor of 100. This is

important since higher voltage-producing circuits are usually 4).

Resistance and more subject to loading, even with a 10 megohm voltmeter. 100% extend AC Voltmeters
Average Reading, Calibrated to RMS Most AC voltmeters display the voltage of an AC

Signal

way that is precisely correct for sine waves, but is not accurate for

many other AC wave shapes. These voltmeters generally use a precision half-wave rectifier to detect the average value.

For the AC signal, then multiply the average value by 1.11 to derive the displayed RMS value.

This is a generally accepted practice in electronics, so we should understand that the 1.11 conversion factor from average to RMS is correct only for pure sine wave signals (the average value of one-half sine wave cycle is 0.636 of peak value; the RMS value is 0.707 of the peak value). For signals that aren't pure sine waves, the AC voltage

Measurement

by this type meter will be somewhat in error. Some technicians

to use an analog meter to locate intermittent circuit

problem. As long as everybody uses an AC voltmeter of this same design, even if the signal measured is not a pure sine wave, all agree on what AC voltage should be measured at a particular point in a properly functioning circuit.

We have True RMS. There are times, though, that you will need to know the true RMS AC voltage for AC signals which aren't sine waves. In this case, you will need a true RMS AC voltmeter. This is a more expensive voltmeter that includes a more sophisticated true RMS voltage detector. This type detector determines the true "heating effect" equivalence of an AC signal to a DC voltage,

is accurate.

Crest Factor

One of the limitations of a true RMS voltage detector is its inability to accurately measure narrow voltage peaks that are much greater amplitude than the RMS level. The crest factor of a signal is its ratio of peak value to its RMS value. More capable RMS detectors can accurately detect signals with greater crest factors. Typical general-purpose meters

Maintain

is acceptable accuracy with AC voltage crest factors from 1 to 5.

Psa

Test Equipment

Both meters are limited to a particular range of signal to measure. Be sure an AC voltmeter's alternating current does not exceed the highest frequency content.

Pontme

When you are measuring, AC voltmeter bandwidth

AC frequency is to 200 or 300 KHz,

The action of an ohmmeter is to apply a known potential across a component or circuit and measure the resulting current. The meter scale is calibrated in ohms for the voltage.

When you are measuring, AC voltmeter bandwidth

When you are measuring, AC voltmeter bandwidth

Ohms Function

An ohmmeter circuit usually applies enough voltage to turn on forward-biased semiconductor

On forward-biased semiconductor junctions present resistance to the ohmmeter, which can make it difficult to obtain meaningful resistance measurements in-circuit. The objective is to design an ohmmeter circuit that can handle a wide range of voltages to the circuit, to ensure

When

When semiconductor junctions are turned on. This allows you to place a resistor in the circuit, without any effect from junctions in the circuit. One disadvantage

One disadvantage of ohms functions is that they are usually limited to resistance ranges than standard ohms functions. The ohmmeter circuit has a difference of potential across the electrons flowing through the circuit. An ohmmeter "measures" the electron flow (current) through an element. An ohmmeter can be as simple as a meter movement with an appropriate full scale current which is at least slightly more than the current in the circuit to be measured. It will be 60 microamps for 190 ohms

$R_1 = 10$

Ohms with 1mA: An ohmmeter current shunt allows a fixed percentage of input current to bypass the meter movement.

| Chapter 19

For higher current ranges, a precision low-resistance resistor is placed in parallel with the meter movement. The shunt is designed to allow a fixed percentage of the current to bypass the meter movement. The current passing through the meter movement is then always that percentage of the full circuit current. For the simple ohmmeter shown in Figure 8, it can be seen that 950 microamps needs to flow through the shunt resistor when the meter

measuring 1 mA (1000 μ A - 500 μ A). For the shunt current to be 19 times the meter movement current (40 μ A / 50 μ A), the shunt resistor needs to be 1/19th the resistance

of the meter movement internal resistance (190 Ω / 19 = 10

ohms). DMMs Technicians typically use a DMM (Digital Multi

Meter)

for most of their circuit measurement tasks A DMM usually includes, at a minimum, a voltmeter capable of measuring Dc and AC voltages, an ohmmeter, and an ammeter capable of measuring both DC and AC current up to one amp or more. Fig. 9: ADMM contains a number of auto-ranged measure- ment functions in one Convenient instrument A significant advantage with most digital multimeters is that they are able to automatically switch ranges, allowing you to keep your hands on the probe and other tasks, rather than hav- ing to manually switch ranges. In addition, some DMMs may include the following functions:

Diode Test

This

test passes

a constant current through a PN diode junc- tion under test and measures the Voltage drop across the junc- tion. This works well for detecting open or shorted junctions, as well as for testing the front to back ratio of PN junctions.

Continuity Tone

This function places an ohmmeter into its lowest range and sounds a tone when the resistance between the test leads is less than a specified value, such as 50 ohms.

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ctr Threm ten omvement evecafl oantinaity fefor che bingcompem eovrcett pati,

Transistor Test

Transistor test function: may include

transistor test function designed to test the gain of small signal transistors (high-power transistors have very low gain and are not suitable for this function) You can insert a test signal trace with the test lead and get a quick pass/fail result.

Capacitance Meters

Capacitance meters measure the value of two-terminal capacitors. One type determines a capacitor's value indirectly, by measuring its AC impedance at a particular frequency, then calculating the capacitive reactance at that frequency. The other type measures the capacitor's RC charging curve to determine the value of the capacitor impedance. Bridge An impedance bridge applies a single frequency AC voltage

to a capacitor under test, usually at 60 Hz or 1 KHz, across one

leg of a balanced bridge circuit. A variable, calibrated capacitor in the opposite leg of the bridge is then adjusted to balance the voltage across the bridge circuit,

as indicated

by a center zero reading on the balance-indicating meter. The AC impedance of the unknown capacitor is then known to be equal to (or some multiple of) the calibrated value of the variable capacitor, at the test frequency.

Once

the impedance of the capacitor is known, the effective value of the capacitor can be computed according to the capacitive reactance formula

$X_c =$

$$1 / (2\pi f C) \text{, where } C = 10^{-6} \text{ F}$$

A capacitor's effective value usually decreases at higher frequencies, both because of the physical properties of the materials used to form the capacitor (especially the dielectric material) and because of the inductive properties

of parts

of the capacitor (especially in a leaded capacitor). A capacitor's frequency dependence will be greater with some materials than with others. An aluminum electrolytic capacitor's value varies greatly with frequency. The capacitor's effective value is often specified by its manufacturer at a test frequency of 60

Hz. as many electrolytic capacitors are used

to filter low frequencies. If you are using an adjustable frequency impedance bridge, it would be best to test the capacitor as closely as possible to the frequency at which the capacitor will operate. If the meter doesn't provide frequency adjustment, you should at least know what test frequency is used.

Test Equipment

Capacitance RC Charging Curve { The second general type of Capacitor value is the capacitor under test with a constant I, where

$I = \frac{V}{R}$ is the length of time it takes to charge the capacitor from a starting voltage to a higher voltage. The meter then

determines the value of the capacitor. According to

the formula $C = \frac{Q}{V}$ that is, capacitance value equals change in time (charge, divided by change in voltage across the capacitor) multiplied by the charge current.

a Fig. 10: When a capacitor is charged with a constant current it will have a specified V/dt charge slope, determined by capacitance value. the

Capacitor Analyzer

Since capacitors usually fail in ways other than just having wrong capacitance value, a complete capacitor analyzer

Sures

all the normal failure modes of capacitors typically, in electronic circuits, Four capacitor tests cover the failure modes most often encountered in servicing, equivalent resistance (ESR), leakage, dielectric absorption (D/A), and capacitance value.

Capacitor Equivalent Series Resistance

(ESR) All capacitors have equivalent series resistance, which is the resistance in series with the plates of a capacitor due to resistive mechanical connections and lossy dielectric material; Increased ESR limits a capacitor's filtering or signal coupling capability, especially in high current or high frequency circuits. High ESR often occurs as a failure mode in aluminum electrolytic capacitors, due to the drying of the electrolyte solution. It is important to be able to test the ESR of electrolytic capacitors, to isolate defective capacitors in circuits being serviced, Capacitor ESR testers are available as stand-alone

P

o

b and inductor analyzer measures all the components. The AC response for the network function

Leakage

For a fully charged capacitor, extra electrons are exiting the capacitor. They go

to the other plate. However, there is some leakage current

a) a capacitor dielectric (except in the case of

When

a DC test voltage

is applied to a

A capacitor is drawn from the test supply

to charge the capacitor

is charged, the current should be of the type

of capacitors (such

as film),

or

or

formal

in the (such

as ceramic).

in the leakage test at or near the rated voltage

to avoid the voltage stress defects exhibit. For most capacitors, the leakage

CUTEN

Very low, or zero, after the capacitor is charged. The

NSX4 capacitors are designed to continually current, in order to keep the electrolytic H₂ gas from becoming an insulator. The maximum AC leakage for electrolytic capacitors depends on the type (aluminum or tantalum), as well as the rated voltage and value. The leakage tester should apply the capacitor's rated voltage to the capacitor to charge, then check whether the current exceeds acceptable limits for the capacitor being tested. The

of P20

Dielectric Absorption (D/A)

capacitor ion (often called battery effect or soak-erroneous resistance isolation

of portions of a capacitor's value. When a charged capacitor is discharged, the stored capacitance sections aren't completely dissipated. When the discharge is opened, the remaining charge on the isolated sections of the capacitor.

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of nn - _ a: we weg Fg 2A

OREN

AA SY 8, ape VA Cagae. Oth arbres se% fern were OMAK emt

ie

Connphete

capaci analyrer metrxnents ia

Capacitor Value

Inductor Analyzer

Inductors

exhibst only

a couple offatkare modex: a wad. ing opens, leakage develops between wandangs

of a ranie, 'mndmng inductor (transformer). the andor core

ss damaged

oF

a short develops between the terms ofa windse.

Aa open winding

or low resistance between solated wadmnes can

be detected with

an ohmmeter, but the most common defect

m=

inductors, especially power iduciors.

is

a short berweea acia- cent tums

m2 windmg.

Inductor Value

Inductance

value may

be used to identify a component

or

verify that

a proper component is in place in a circuit. A value test may indicate

value change due

a damaged core, but

a

Value

test alone is usually not sufficient to determine whether an inductor is defective.

Chapter 19 [00

' Shorted Turns

A shorted turn in an inductor forms a very low resistance path around the inductor core, absorbing almost all the energy from the inductor magnet field. The shorted turn causes the inductor to overheat and prevents the inductor from producing & normal expanding and contracting magnetic field around the inductor core.

Fig. 13: A specialized test is required to detect a shorted turn in an inductor. This type of defect prevents normal inductor action, but the defect is very difficult to detect without specialized test equipment. An ohmmeter won't detect the small resistance change caused by a single shorted turn, and the inductance change usually isn't significant. The specialized test equipment built to detect inductor shorted turns is called an inductor ringer tester. It applies a resonant meg capacitor in parallel with the inductor under test and pulses the inductor/capacitor tuned circuit. It then counts the number of oscillating pulses produced before the pulses damp below a particular test threshold. The test threshold is set such that ten or more ringing pulses occurring above the threshold indicates that the inductor does not have a shorted winding.

Oscilloscopes

An oscilloscope provides a visual representation of a changing voltage wave shape over time (a line graph). It also measures the amplitude of the voltage changes. A scope waveform shows us much more, though, than just the peak-to-peak amplitude of voltage changes in a circuit. The complete signal information that a scope displays allows you to make informed judgments about the performance of the circuit. This makes an oscilloscope a versatile and effective test instrument in the hands of a capable technician. An oscilloscope's multiple controls and infinitely varied display waveforms can be confusing and intimidating, though, for a technician not fully trained and experienced in its use. An oscilloscope has four main circuit functions; the CRT, horizontal sweep, vertical amplifier, and the trigger circuit. A dual-trace scope has one focus control; alternating horizontal sweeps and dual inputs, position and trigger controls.

a

"7

"Test Equipment

Fig. 14: An oscilloscope can be used to determine at least nine different parameters of an AC signal. The CRT produces a stream (beam) of electrons that are directed, at any one instant, to a single point on the phosphor-coated display screen, producing a point of light. This stream of electrons travels along what is known as the CRT's z-axis. Anything that changes the beam current, such as blanking, is said to be Z-axis modulation. | Controls associated with the CRT circuit are the

Intensity

and focus controls.

Horizontal Sweep Circuit

The horizontal sweep circuit pushes the CRT beam from left to right across the screen (along the X-axis), forming a straight line of light. It then blanks the beam while the deflection is reset to the left edge of the screen to begin another visible trace. Controls associated with the horizontal sweep circuit are the time/division and horizontal position controls.

Vertical Deflection Circuit

The vertical amplifier circuit pushes the CRT beam up and down the screen (along the Y-axis), in step with changes in the input signal voltage. This changes the straight line of light into a "wiggly" line, which duplicates the shape of the input voltage changes. Controls associated with the vertical deflection circuit are the input coupling, volts/division, and vertical position controls. As indicated by the simplified oscilloscope block diagram (Figure 15), any voltage at the signal input is applied, through the switched attenuator stage, to the vertical amplifier stages. The vertical output amplifier applies an amplified replica of

a

Sensitivity

The four main circuit functions of a scope are the horizontal sweep, vertical amplifier, and the trigger. The input impedance of an oscilloscope is actually quite

The Megohm

and approximately 30 pF of capacitance. If you connect the scope vertical input directly to a point (using a direct probe), the low impedance input will tend to load down any DC or AC signal in the case of its frequency (see voltmeter loading). In addition, the 30 pF of capacitance to ground will frequency to ground, increasing the loading at high frequencies.

usually above about 5 kHz, with

characteristic impedance less than 1 MΩ. Because of this, when

to the scope input loading is to use an

extender. This is similar in design to the one we use with a DC

Since

we will use the scope to measure AC as well as DC, we need a capacitive extender element as well

as a resistive one. The typical input impedance of a scope direct probe, a scope

as a direct probe) presents one-tenth the loading resistance and capacitance to a circuit point.

The extender

resistor will be nine megohms; nine times the scope input resistance. The extender capacitor needs a capacitive reactance of nine times the capacitive reactance

of

the scope input capacitance (plus some parallel capacitance of

Test Equipment

" | Chapter 19

the probe itself) That means the extender capacitor needs to be approximately one-ninth the 30 pF input capacitance. In practice, the capacitor is made variable for the slightly different capacitances of different scope models. The x10 extender probe allows only one-tenth the DC or AC probe tip voltage to reach the scope, to be able to compensate

Trigger Circuit

Finally, the trigger circuit locks onto regularly recurring changes in the input signal and resets the horizontal deflection to start at the left edge of the screen at the beginning of each detected waveform trigger point. This causes the occurrence of a recurring waveform to be displayed one after another. This allows us to examine the waveform. Controls associated with the trigger mode, trigger source, trigger level controls. Same portion. Repeatedly, on top of a stationary portion of the trigger circuit are Polarity, and trigger

Oscilloscope Operation

There are a couple of steps to follow when you connect an oscilloscope probe to a circuit test point. This will allow you to obtain a stable, useable waveform on the scope. First, set the trigger circuit for auto trigger, set the trigger source to the channel you will connect to the circuit, set the input coupling switch to AC-coupled, and center the trace on the scope display with the vertical and horizontal

Position

controls. Then, preset the volts/division control to a voltage value between one-fifth and one-tenth of the expected circuit peak-to-peak voltage (divide the expected peak-to-peak voltage by ten, then set the volts/division control to the next largest voltage step above that). Now, when you connect the scope probe to the circuit, you should see some vertical trace deflection. Adjust the volts/division control so that the waveform is at least two divisions tall on the display (don't worry yet if the trace is still a scrambled waveform). Next, adjust the trigger level control until the waveform stabilizes. If necessary, reverse the trigger polarity and readjust the trigger level. You should now have a stable waveform display, although it may be too compressed or stretched out in the horizontal direction. Finally, adjust the time/division control to stretch out or compress the signal to view the desired signal detail. Note: An oscilloscope displays a waveform in time domain view. A spectrum analyzer displays a signal in frequency domain view.

Test Equipment

Quiz

average-reading AC voltmeter displays the average AC voltage value + 0.707 to a peak reading before displaying the voltage. It puts the average value by 1.111 before displaying the voltage. It puts the average value by 0.636 before displaying the voltage. It puts the average value by 2 x 10 scope probe instead of a direct probe: (97 circuit loading. + i the effective input impedance. peak clamping in an oscilloscope is used

It passes the signal to the scope. the above, used as a tool to the Z-axis to blank the vertical retrace soled to the X-axis. It rarely measures a resistance across a transistor's base junction, use a higher ohmmeter range than you normally would. It has a zero ohmmeter range than you normally would. It has a ohms function function

It is used to test AC impedance bridge. It is used to test a shorted turn can be detected reliably using an AC voltmeter. It will measure a non-sine wave signal only if a resistor is used. It is an average reading meter. It is an RMS meter. It is the signal crest factor is very high

- 7. A type of transformer which has a 1:1 ratio
 - a. provides a ten-times voltage step-up ratio
 - b. provides a floating secondary reference point & is an autotransformer . transforms the ground reference at its input to be the same as its output
- 8. Shock hazards exist especially in a power supply which has i
 - a. a full-wave bridge rectifier connected directly to the AC Power source
 - b. an earth ground reference at its voltage output ©. an isolation transformer at its input
 - d. an aluminum electrolytic filter capacitor
- 9. ADC voltmeter multiplier resistor a
 - a. limits the meter movement current to its full scale Value
 - b. multiplies the Voltage across the meter ©. provides an alternate path for current flow
- 4. reduces the meter resistance for less loading effect
- 10. A simple DC voltmeter with a 200 μ A meter movement needs to have how much internal resistance on its 50 volt range? a 4k ohms
 - b. 10k ohms 20k ohms @ 250k ohms

ANSWERS on Page 223

by Dick Glass, CETsr

Soldering

Technicians have always needed skills in replacing electronic Components such as capacitors, resistors, coils, transformers and so forth. In most instances had parts can't be removed and new ones inserted without using a soldering iron to heat and melt the connection for the old defective part. Also solder must be used to make perfect electrical connections for the new component, Soldering is not difficult but there are some definite rules. Rules might be put in categories:

1. Care of soldering irons and equipment

2. Safety considerations

3. Soldering preparation

4. Soldering electronics components

5. Desoldering Care of Soldering Irons It is easy to keep your soldering iron in good shape. Most technicians use the small pencil-type irons which are rated at 25 watts to as much as 50 watts. Higher wattage irons than that might be used to do large jobs such as soldering a UHF fitting on a RG-214 type transmission coax cable, or for securing jacks or brackets to the metal chassis. The small pencil irons have a heating element with a copper or plated tip. The tip may be about 3/16" in diameter and are usually screwed into the larger portion of the heating element. The tip of either type is the problem. After using it for less than an hour, the shiny copper begins to tarnish. Then it builds up a crust of carbon-like material. The plated tip does the same. Either tip works best if it is shiny and has no discoloration buildup of carbon. But the hot copper is "cooking" the air. In all, the irons run at 360°F+ (for 60% tin 40% lead solder).

4.

Vith

copper tips, cleaning can

be done by simply filing the point (which may be chisel, conical or pyramid shaped) together rid of the 'oxide deposits which prevent full transfer of heat to the solder joint. Dick Glass is an author of previous FTA publications as well as more than a dozen books dealing with electronics technology and management. Published by Howard W. Sams Co, TAB Books, and McGraw-Hill. He has also written hundreds of articles in trade journals, magazines and publications. Dick is a former Navy Avionics Technician, Industrial Electronics Computer Technician, Technical Writer, Federal License examination committee Chairman in 1995 and 2008, inductee into the Electronics Hall of Fame and Cup, Sr. | CEO President of FTA.

He has

broad background in all phases of work performed by electronics technicians (after 20. Soldering, Desoldering and PCB Repair, Too)

N. Pitney, Inc. Plated iron tips should be cleaned with an abrasive cloth, never a file, as the file will remove the plating. Whether copper or plated, there are two things that can be done to reduce the buildup of carbon. One is to frequently rub the tip against a wet sponge. This cleans the thin layer of buildup off down to the copper or plating. Secondly, try to keep the

tip covered with a layer of melted solder. Melting a coating of resin flux solder onto the soldering area of the tip reduces oxidation, contaminant buildup. Because the solder coating is in its flowin, or plastic state, a layer of the solder can be wiped off with a wet sponge thus keeping the tip in perfect condition to solder or unsolder a connection.

Lead-Free Solder |

For decades, 60-40 solder, 60% tin and 40% lead, was the standard mixture in solder alloy used in electronic applications. Resin flux core is used-never acid core of pastes. Acid used for cleaning or flux will corrode or arc or cause damage to electrical components. Lead-free solder has universally replaced lead solder for worldwide ISO certification standard. There are other combinations of lead and tin, 63/37 being one that is used in special cases, called 'Eutectic' solder. The object is to get the solder to melt at the lowest temperature, so, solidify immediately upon heat removal and not have any effect on surrounding circuitry or components. 63/37 melts between 355-365°F, whereas 60/40 melts between 361-376°F; Silver solder (2% silver, 59% Tin, 39% Lead) is used for surface mount components (SMT) for stronger bonding. Danger of Lead and Fumes About the turn of the millennium, concern was voiced for the potential health hazards lead solder presents. The Restriction of Hazardous Substance Directive for banning lead type solder took effect July 1, 2006, Lead can enter the blood stream through a cut, or lead can be ingested which is poisonous and toxic to humans. The fumes from melting lead

are

very waite Soldering,

Desoldering

and PCB Repair Tools. of solder are toxic to the lungs from resin flux cores and eye irritation, The entire industrial world has why in solder alloys and has replaced the lead with pure Tin-silver-copper and tin-copper are the choice for rework service jobs. Be aware of the danger - penne eal Work in a ventilated area. Reduce contact with fumes lead-free solder. der

Guns

all service shops used soldering guns. Shaped like a pistol, the "barrel" was a copper or plated brass that was attached to two prongs by set screws (see below). Cleaning requirements are the same as for a pencil. The difference is the size. While few pencil irons rated at over 40 watts, solder guns may be 100 to 250 watts and some even 500 watts. In the early days of radio and TV, components were most often soldered or mounted to terminal lugs or strips. You often resistor

or capacitor lead around the lug stud tightly, you

solder over the mechanical connection to provide a more secure contact.

Today components are mounted on terminal strips. Rather, Few parts of boards contain most of their components and even the PC fiber board - often soldered to both sides and often with the leads poked through a hole. Surface mount technology (SMT) and tiny "chip" placement of PC board traces and components with its aid and precise soldering, thus solder guns are

used for connections to large metal ground lugs or other parts mechanically to the chassis. Soldering guns have two problems which pencil irons do not. Solder in the set-screw attachments one at the gun tip into the two con- ductors will come loose. It is a poor physical connection which is insulating carbon film where the leads

are screwed onto the barrel shafts. It is possible

that the copper under the set-screw is worn away by electrolysis. Whatever the cause, the screws are tightening frequently. If they aren't tightened to make good contact, the resistance at the set screws will result in

excessive heat being concentrated at the set-screws, not in

the solder tip. "Barrels" and tip are actually the secondary of a simple AC transformer, built into the rear bulk of the gun housing. Consider the 115V, 1 AMP power for

a 100 Watt soldering gun: If the

step-down from primary to secondary is 100 to 1, then the step-up in current in the secondary (thru the gun's tip) is now not 1 amp, but

100 amps! 100 amps through the copper tip loop will heat

| Chapter 20

the copper tip to red hot. Crunks use a trigger switch to control the heating of the tip. Electromagnetic damage can occur nearby sensitive components. Soldering of Components It is natural for a person to try soldering, but to not quite "get it right." You will find trained technicians who still leave "cold solder joints." These later can require extremely difficult troubleshooting tasks to ferret them out. It is the technician's job to recognize the reasons for "cold joints" and to assure that none are the result of his or her work. A proper solder joint is one in which the component leads and the PC feed-through or terminal, stud, etc have each been simultaneously heated to where the solder flows over and between them and embeds itself in the metal surfaces of each of the leads, traces or lugs at the joint. A good joint is shiny, not dull and gray. It is natural to cover the joint with solder and assume the connection is OK. It may be that one or more of the leads, traces, lugs or a ground terminal has not been heated sufficiently to embed the solder. Thus you have rosin or an interface between the parts or leads. This creates a gap. In some circuits that gap could even result in an arcing and thus "static generator" within the circuit. A nightmare problem. A joint may "look" secure. Looks are deceiving. Make sure you heat all the connecting leads and any test point or leg connection so that each and every one can melt the solder and so that solder flows onto each component and thus embeds itself in the surface of all leads. You must feel that this has been done. Even when danger to components is a problem, you cannot let that excuse you from "sweating the joint," so to speak.

Tinning

Be sure to have "tinned" your soldering iron tip. This is simply melting solder onto the tip to help prevent oxidation of the copper, and to give the tip a liquid (or plastic) interface with the metal your iron is touching (and attempting to be the melting point of the solder). You will also have wires or component leads which must be soldered better by tinning prior to physically being mated at the lug or joint or PC trace. To try to solder a stranded wire without first tinning the last 1/4" to 1/2" (intended to be soldered to a connection) is to take the chance that one or more strands will not be soldered well. Some of the strands may look soldered, but really they may be "cold." The mechanical security of the wire will be less if all strands aren't secured.

LE

Chapter 20 E

Thinning

the receiver

the wire

the component. It takes a few seconds. It may save a few hours of troubleshooting. Do not damage the equipment. For HV
carnation, always leave a smooth and well-round 4d covering cap solder to reassemble CHIN,

How much

is too much?

'Yumerst

make sure the solder connection onto all of the heads and wires at the connection. Sometimes you will find the heat buildup
in a "stubborn" connection heats

the Component leads to the point where you wonder now about the integrity of the component(s). How do you handle
this? "Sweats" or flows. As a professional technician, you will acquire a "sixth sense" that warns you something is not right - of
that you are about to overheat the connection. Sometimes the plastic insulation on a wire starts melting or snapping back.
Maybe a PCB solder connection has gotten so hot that the connections nearby also appear to be affected by the heat.

Obviously, if a resistor starts "sweating" and discoloring; if a capacitor (Iync) starts "steaming" or the PCB board starts
crumbling and the copper traces are pulling off - you have applied too much soldering iron heat. One reason for this may be
that the parts to be joined have lacquer on them or have oxidized to where solder resists bonding. PCB traces always have a
protective insulating coating. You can heat these impediments enough to blast through to the metal, but in the process you
may destroy nearby components. Transistors and diodes, aside their protective case, have a very delicate and precise junction.
Too much heat in the way of current, as well as too much heat physically, will ruin the junction.

Heat Shunts/Heat Sinks

Heat sinks are merely alligator clips with copper or aluminum welded onto the teeth so that you can use more heat at the
joint, yet not have the heat travel up the transistor leg to overheat the junction. Once you have several heat sinks at your
work station, you will get in the habit of using them regularly with delicate component leads. In today's circuitry, with CMOS
semiconductors that can be destroyed by even small static discharges, you should assure that your soldering iron and your body
is at ground potential. Use a static safe bench and a wrist strap ground and frequently make sure everything in the area is
grounded. This can help reduce soldering-caused troubleshooting problems. In a pinch, you can tighten your long nose pliers on
a lead - and keep the pressure on by using rubber bands on the handles. Another very professional heat sink is built like a doctor's
goldering, Desoldering and PCB Repair Tools scissor-like clamp which tightens up on the lead to a locked position until
you decide to relieve the pressure.

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oy

ure

'y

Desoldering

Some it seems smart to blast all of the solder blobs - but say you are having trouble with a component leads, and you fan : Solder in order to unravel the wires and reunite the component you are replacing. It is tempting "to heat it and before it has time to cool - blast it with a Mireles lected i""ven freon! To do so is to violate a lot of it. A one. For safety and co-worker safety. That is a No. No" end woe (to heat a connection in a small board the winter _ centrifugal force to detach the Solder pot A cut the air blast and the "flick" will preserve the melted solder will not all detach, but it will affect nearby parts or circuits Don't flick

Everanes

' At best desoldering is an imperfect Science. Experienced technician will find some jobs are not possible to do neatly.

Wick

er off

og

in large Need to

ge

Pais

tool universally used by techs is "solder wick." It resembles

be merely the copper braid from coaxial audio and video cables, This was removed in convenient length

When solder was to be removed from a joint, the braid was flattened - dipped in resin flux, then placed against the joint or component lead to be unsoldered, then heated with a soldering iron. Since the braid gets hot first, as the heat moves the joint melts into its plastic state, it is "wicked" up the braid.

Thus the bulk of the solder in a joint or in a feed-through on a

PC board can be wicked away - often leaving such a thin film of solder that leads can easily be wiggled free or unwound.

untangled. Solder wick now can be purchased in rolls -

must for working techs.

sa

Suction

Bench techs have as a major tool of the trade - solder suckers. Several varieties are made, Their action is provided by a small hole in a Teflon tip - then a barrel which contains an air plunger. You "cock" it by pressing the plunger to the top end. A release button lets the spring inside retract the plunger, thus creating a vacuum which sucks air - and melted solder.

Up the plunger tube and away from the heated joint. These work well. They need to have the solder particles (captured inside) removed frequently.

Chip Desolderers

Multi-pin IC chips require patience and delicate handling. Since most modern circuitry contains LSI (large scale integrated)

Or

a

'Soldering, Desoldering

and pcs Repair Tools lace one requires heating up 8 or more pins yet ever one hundred in modern microproces- of
consenough to unsolder one component from a PC An et be done with one which uses dozens of chip

c cua ately, several companies make special shaped

w ta mplatcs which can fit flat against a 14 pin chip melt all legjoints at once allowing the chip to aon . Once extracted, the
chip's 14 leg holes can be ose with wick oF the even nicer air vacuum desolder-

Seance

makes these). "a iustncky to remove all the solder from each leg the PC board, it is not easy to pull a number of the an with a
IC remover tool. It is easy (o rip one or more a from the PC board, They scem to like staying at- oie chip leg and pulling up
from the PC fiber surface.

rs you do a lot of work with (serviceable) PC boards, ont yourself replacing feed-throughsolder rivets, oy portions of copper
circuit traces, jumpering bro- unsandevengold-plating edge connectors that have jan. worl, nee carry tiny drill bits (for
cleaning out feed- ral vets), Scribes, tweezers, kits of replacements PC pve, several sizes of wick, alcohol cleaner, glyptol

st on polycoating and various types of chip desoldering

le

ya foreMount Technology (SMT) gT PC boards are repairable. Fortunately, as PC some

ye becomeSO densely populated and PC traces so wil fiers become necessary, fewer and fewer PC «lost ired - especially in
consumer electronics and wethrow-away products. com ir surface-mount electronics PC boards, chances peera steadystream
of them. You should have the | nor SMT components for the brands of products ied. Wherehistorically a solder connection is
supposed to

si yanically connected, as well as electrically bonded, SM eet| and are held onto the traces by solder alone. benchlights,
tweezers or vacuum component pick wt pasily even a hot air rework system may be a SMT is a specializedsegment of
electronics repair.

y solder paste is frequently used for new and rework. A wie maker may set up a surface-mount station dedicated

r) of thatcompany'sspecific PC boards. Field service xfs ormally would replacethe entire unit, or a major PC san ending teSM
board back to the factory and then to eM station for repair.

H

a Chapter 20

Hand Tools

Most technicians have schooling of expenence with com- Mon hand tools. We won't spend any time on hammers, chisels,
screwdrivers, hacksaws and so forth Power tools, as well as hand tools, are a daily part of technician \ife, past

as soldering equipment is. Safety includes not wearing loose clothes, chains, bracelets or long hair. Safety includes weanng
goggles every time you use a grinder orpower drill. [t mcludes assuring there is adequate ventilation while soldenng or using
toxic cleaning materials. [t includes keeping all tools in perfect condition.

NOTES

·The traditionally used & a metal 2 of rine and copper. Noy consisting

a, Troe

h 3 The most common solder used by electronics consists of: a. aluminum, 50% tin, 50% zinc, 5% copper, 37% bismuth, 4% lead.

3. The proper process for soldering two objects together is:

- a. heat both surfaces at the same time- apply solder.
- b. heat one surface at a time, ensuring full coverage of the heated solder. heat the solder surfaces. clean any resin or flux off the joint until it melts - flow it over the joint

nt = then apply heat.

4 Overheating the component lead will likely damage first. a. transistors

- b. diodes capacitors

4. transformers c. transistors and diodes d. Tinning of, makes the best connections.

- a. soldering iron tip
- b. stranded wire c. component leads
- d. all of the above 6 Movement of any component of the solder connection before the solder has hardened:
 - a. results in a poor electrical connection.
 - b. results in a poor mechanical connection. c. results in a poor mechanical and electrical connection.
 - d. is OK so long as the solder doesn't migrate away from the joint.

7, Which device is capable of causing maximum inductive damage to electronics device or

- a. Pencil soldering iron
- b. High wattage resistance soldering iron Solder suckers & rubber vacuum bulbs
- d. Soldering gun 8 To remove solder from a joint using copper wire, heat:
 - a, heat the joint then quickly dip the melted solder and remove. then
 - b. heat the component leads at first, then the wire, wrap the braid around the soldering iron, hot, wipe off solder from the joint. Then tip and
 - d. touch the joint with the braid - heat it and wick up the braid. Then dip the wire into the flux

9. The best way to remove an IC chip from a PC board is to: solder it

- a. use a desoldering adapter tip which heats simultaneously, means
- b. use wick and remove all solder from the pins. Pry c. heat each pin sequentially while prying it
- d, use a solder-sucker to remove all pins

10, Surface mount components do not require mechanical bonding because:

- a. the components are so small the solder c. holds them adequately,
- b. their sockets provide adequate bonding,
- c. their leads always 'hook' onto the opposite side of the PC Board.
- d. special strength solder is used on SM boards,

ANSWERS on page 223

b obtained on the scope when:

és A waves that are 90 degrees out-of-phase fed to vertical and horizontal inputs, sine waves, in phase, are fed to the vertical inputs, and a sawtooth wave are compared on the same sine waves are fed to the horizontal and inputs.

electronic Is most commonly used for audio work

100 KHz

coupling capacitor used when troubleshooting section? 4. Ign! In the frequency response level present loading of the circuit of impedance match

could be the likely cause of hum in an amplifier, when the CD position is selected? proper capacitor in the power supply. Bleed in the CD power amplifier. 4. Cable from the CD player to the speaker amplifier has severe hum whether CD, AUX, or RADIO are selected. What is the most likely

cause: control shorted; common output transistor grounding. 4. power supply filtering of the below will cause distortion at high audio signal in an amplifier system?

4. 1/2 by tone control. Sent to 4, probing speaker voice coil and transistor. 4. Bussboost control. On. Troubleshooting, Test Equipment and R

@ Chapters 18, 19 and 20 = Unit:

7. The Y axis of an oscilloscope is: brightness, intensity, focus, vertical.

8. A power supply is found to have several 250 uF capacitors connected in parallel. To avoid shock from retained charge after disconnecting the power, the capacitors should be:

- a. ignored.
- b. disconnected, and discharged to chassis ground, with a suitable resistor.
- c. isolated with a non-conductive cover.

9. Dual-trace scope has:

- a. two focus controls.
- b. two sets of position controls, and two vertical inputs,
- c. both b and c.

10. A 10x probe has what effect on circuit testing?

- a. Increases circuit loading.
- b. Increases the signal level by 10 times.
- c. Lowers the effective input impedance.
- d. Decreases circuit loading that a direct probe presents. 11. A voltmeter that converts AC to DC by rectifying the voltage, would be least accurate measuring what types of alternating voltages?
- a. Pulses

b. Sine waves c. Peak voltages UHF signals

12. When connecting an analog voltmeter into a DC circuit, what should be observed?

- a. RMS rating of meter
- b. Loading of meter
- c. Polarity
- d. Peak to Peak voltage of circuit

ANSWERS on page 223

ud Recoru never ce ~ by Robert (Bob)Ing.

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Ny me | Preventive/Scheduled Maintenance Keeping a record for Preventive and scheduled s ih eae enable the technician to ensure that <ainten

os pt in working order, view equipment sin h vine } monitor and order consumable inventory n

Pistongs

ness for scheduled service. CMS In teag

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Service Contracts

Keeping

a record for service contrac the technician to know in advance the Toone cute service a client on a service contract. Resources ir parts, special test equipment or tools, and even in,: technician or technical team will be needed on a pein service call, can all be determined by looking at Witee vice contract record.

Warranty Service Validation

Keeping a record for warranty service validation enable the technician to determine if the unit for "i or out of warranty and is crucial in obtaining a st

or vendor's warranty reimbursement on behalf of the in

I.

Wage/Salary

Verification

record for wage/salary verification purposes hnician to verify the billable hours worked se of invoicing the client, but also to ief

Keeping

a enables the tec! only for the purpo: his or her own remuneration against hours worked.

Profitability

Keeping record for profit ability purposes enables the technician supervisor/manager to ensure that client invoices are properly processed and that parts and other consumable supplies are maintained at acceptable inventory levels.

Business

Planning

Keeping

@record for business planning purposes enables the technician manager to sce how well the department

or business tna etaticties and actual invoiced revenues,

es aderegardin J be made regarding re aco rrdownso8 of others, areas «for keeping a record,

f ri otisnen sales or employer

direct reason for recording what means ensuring that sales tax percentages are accurate and are adequate service call information

Of gas timely invoicing of the client. "4 fs and 'on business revenue can be if) gs aes oe it ae?"

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why a technician might In knowing the work in creating a record even so as - is much easier to get

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is the most common reason - it is no surprise that it tends to overtop Of complement one view listed & listed

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Service/ Repair

Ticket

The service ot Cpairticket is ment the detailsof Who, what,Has was garding the service call TheWhois the h Nit betel item

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Performed

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Invoice

The invoice includes a SUMMARY of the job

Service /

repair ticket with a

Physical Inventory

of Parts charges. Also the invoice will include important work-

Tax

information and payment terms for the client regarding the service performed. As a technician it is Very important that

You

ensure that any information you provide relating

to the invoice is accurate and concise.

Customer Service Record

The customer service record provides details regarding the relationship history between the client and company. It lists all the equipment purchased, the service and repair calls made and other relevant information. Selected information from the service/repair tickets and invoices issued to the client are listed in chronological order with brief notes cross referenced to the original documents.

= record

Keep in

Employee Time Ticket

identifies the billable hours to

the technician. It helps the technician

The employee time ticket and most of all, helps in the client accounts made

by the technician! can keep track of hours billed out and

of the technician's pay check

Personal Field Notebook

requests, to do lists. The best pocket notebook for the job

is the pocket of your service. The most efficient way

to set up the first 2-3 pages at the front of the book for important telephone numbers. At the back of the book should be used for reference information such as formulas, tables, etc. This reference material may

be written in,

or photocopied from technical service manuals using a copier that can reduce the image to a

size that can be pasted in the notebook front

Having

set the notebook, entries are made from a spiral binding. To easily find the next blank or
an elastic band placed around the
pages of the

Book

to be
not after the next blank page. It has the most pages just before!

Today

Technicians also have access
to smartphones
and
as tablets to organize their workday.

They

are small, portable and offer many different 'apps'
to choose from.

Productivity Calculation

"Today is the beginning of a new day. You have been given this day to use
as you will. You can waste it or use it for good. What you do today is important because you are exchanging a day of your life
for it. When tomorrow comes, this day will be gone forever, in its place is something you have left behind... Let it be
something good." All businesses and employees perform work to earn money.

When

most
of the efforts of the business and employee are directly related to generating income, it can be said that the productivity level
is high. However, in the real world obtaining maximum productivity is not always that easy.
I Beha, The trac

Productivity

of a technician is the percentage of time spent on activities that are billable to the
client. Time spent on non-billable activities, also known as overhead activities

Mahe

big
the 3: reduces the productivity of the technician. or rea

The 30-30-30 Rule

Generally, when a service business establishes a labor rule of thumb which states that 30% goes to the technician, 30% goes to overhead, and 30% is set aside as a contingency fund.

For example, if a company has a technician in gross wages, \$30 to cover overhead and administration, etc., \$30 would be set aside for investment in the business and \$10 would be set aside for unforeseen expenses that might occur in the future.

For example, if

this means that in theory up to \$30 of a technician's gross wages, \$30 to cover overhead and administration, etc., \$30 would be set aside for investment in the business and \$10 would be set aside for unforeseen expenses that might occur in the future.

The Productivity Formula

When a technician is billing a client, it is quite clear that the technician's productivity is quite clear.

However, as a technician is scheduled for a specified number of hours net technicians week, not all of their scheduled hours may be billable due to the overhead expense allotment in addition to the technician's wage allotment.

The productivity, or utilization as it is known, of a technician is the ratio between the technician's billable hours to total scheduled hours worked in a given time period (week, month, year). For an employer to know the technician's utilization level of technical staff helps determine if more staff is needed or if efforts should be made to increase the client base.

For a technician, the ability to calculate productivity helps identify time management problems as productivity and can help in performance reviews. Profitable requests or clearly identifying instances when a technician is "maxed out" and needs help comes when calculating productivity, all that is required is:

billable hours

to the client into the total fee

of his or her work for any given time period (week, month, year). The calculation will give productivity as a percentage

The formula is: B/T

$$P = \left(\frac{B}{T} \right) \times 100 =$$

Productivity, %

The total productivity indicated as productivity or utilization rate, is the productivity and age as been scheduled to work

technician led to various clients

as a productivity would be

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are formulae that give us a better idea of productivity. However, the productivity percentage is a rule of thumb discussed earlier in this

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f\crease 7 car. That's the General rule of thumb,

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One to an £7] increase productivity. In some ting.

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we ng the clientele of the business to Fgh iqvol¥é wdfully utilize staff In other instances Belts & Shoes- The Color of yourbett and h oo ak al ume management, training staff to match. If you wear black shoes,

Shoes

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' Grooming

rules is to look and see how your peers dress and Lg y

o, all technicians wore y .

a ee ei wear everything | ane Haig iforms and business suits to casual wear. canon

i clientele and nature of your work will I vest pain One thing is for sure, you and your eee This means if you are working on be polished and kept and soles as Tequired.

Socks -

Socks Should only be black or dark blue, other personal hygieneactivi- r iN. a civilized society. Here are some personal grooming tips: Bath/Shower ~ If you suffer from dryskin, use a liquid

instead of a bar =

Soap for your bath / shower. Deodorant vs. Anti-Perspirant- A. deodorant will teduce body odor, an anui-perspirant will reduce underarm wetness and odor, however your body will increase Perspiration else- where to compensate.

Fingemails ~

Technicians should keep their fingernails short and clean,

Hair -

Long hair should be tied or put up so as not to get caught in machinery or contaminate an area or equipment you are working in and on. Remember, a static discharge can come from the end of a single human hair and that a loose human

Chapter 21

a micro-electromechanical device that has found its way into many areas. Over a period of time, the device has the potential to do data

Personal Communication

How you communicate with others, both verbally and through body language, sets the type of relationship you will have with them. Even if you know that you may never see the person again, it is always best to communicate with them as if you will. How you communicate will have an impact on the level of respect, trust, intelligence and sincerity people will associate with you. Here are some tips to make you a better communicator: **Introduce Yourself** - The first words that you should say when you see a client on a service call are your name, your company and the reason for the call.

Be Personal -

Always try to use the name of the person you are talking to at least twice in the conversation. **Face-To-Face Contact** - When talking to someone, always look at the area of their face. Some books suggest eye contact but this may make some people feel uncomfortable, particularly if you are talking with someone of the opposite sex in a closed area. **Avoid Slang or Lazy Words and Phrases** - While we could fill an entire book with these, here are some common ones and what you should really say:

* Yeah - Yes

+ You Guys - You

* O.K. or Okey Dokey - All Right + Fix - Repair or Service + [don't know - I will have to look into that]

Cheap - Inexpensive

How can I help

What

can I do for you

Crapped

Out - Stopped Working Hiding Your Hands - Research has shown that people who talk to others with their hands in their pockets or behind their backs give the impression that they are not being totally honest or are withholding information. As a technician this is not the message you want to send to clients, so keep your hands out of your pockets and where your client can see them during a conversation. **Chewing Gum & Mints** - Chewing gum or having a mint in your mouth while in the presence of a client is a definite way to look unprofessional. Don't do it. - **It is never acceptable to smoke in front of a client or on client premises.** As a technician you are there to **Record Keeping, Productivity Calculation & Personal Behavior**, perform service and a client may take exception if you were not completely focused on the

NOTES

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cial requests Jp the technician keep track of spe ue invoice or receipt is me anon the spot a by other technicians nt provided by the company disa

Chapter, 21

8. The formula for im Productivity

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b = = py,

© P!T(B) = po,

d + The 30-30-39 Tule refers to Cc. wages, scheduled hours, OVerh cach Wages, overhead te, a ate tack ba safety pin tte bar

4. none of the above

ANSWERS on Page 223

Chapter 22 by David E. Caldwell, CET Sr. Career Mailer Practitioner

TECHNICAL WRITING

For his chosen field it was fit To match his learning and his wit

DEFINITION

Writing is a means of making known the

essence of the writer to a reader who is interested in it. Technical writing involves the communication of knowledge and ideas, precisely and in such a manner that a specific audience will find it understandable and essential in all fields of engineering, technology, and the like. The technical writer must have proficiency in the technical field in which he works. He or she must write skillfully and precisely and describe things unfamiliar to the reader. Uncommunicative writing or heretofore to the reader will be useless. Technical material will be pointless -

if it is presented in an accepted format.

To describe some standard techniques and arrange

them for you in this chapter. However, the proper use of punctuation and spelling will still play a key role

in

technical writing success. Applied science and the information

LEVEL

Information

can

be technical and yet

be written

on various levels. If you follow all of the guidelines for proper editing, you might still encounter problems.

As

a writer, it is perfectly clear

to you may seem confusing

the reader cannot use out-of-the-way terms and take for granted that the reader understands everything you write.

Since

there is no official technical dictionary for both you and the reader. Use it is important that your writing

be properly directed to the particular level of the reader. You must take into account the amount

of training and experience the reader has.

Readers

will not read material that

is written too high above their vocabulary level. Very few people were from the 16,
 benefited electromechanical technicians in avionics and communications. He retired AF here

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He then took on the job, NAech pope before developing and teaching the current two-year electromechanical technician course for the
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He has written several articles for

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Hany

ie Will do additional research Just to increase their com, ston of awniten article Your writing effort should be e a express
 ideas rather than impress the reader with your y lary skills. You must also avoid trying to oversimplify i by using words
 beneath the reader's

Vocabularylev} The

ideal level should provide the reader with mostly new fact, at least stimulate the reader into thinking about the core You
 might consider dividing your read- Ts into one of three general levels. The first level can be described as elemen- tary. This
 level consists of amateurs or beginners. Anyone who has received only minimal training might qualify for this category. A
 newspaper or magazine article should be written at this level The second level can be designated as the intermediate level.
 This level is comprised of technicians and or students. Any person who has Specialized training and experience would qualify
 for this level. Amemo about changes

ly equipment or maintenance procedures should be written at this level. The third level would be characterized as the
 advanced level. This level consists of scientists, senior engineers and technicians. Anyone holding a degree and having
 extensive experience in the subject field could be considered for this category. A report on research concerning a new
 theory may be authored at this level. Since these various reader levels do exist, the writer

Must

adjust this style to the specific purpose at hand. When writing at lower levels, remember to keep the language simple. Use more illustrations, present less complicated theory, give

More

definitions, keep instructions short and offer more Step-by-step explanations.

STYLE

When writing about technical topics, keep your COMPOSITION simple, logical and objective. Technical writing should be accurate rather than convincing, specific rather than theoretical, reasonable rather than impulsive. Technical writing should be crystal-clear, to the point and precise rather than poetic and imaginary. A good technical writer will be sincere, direct and utilize clear composition. A clever technical composition will be critical and complete, leaving no unanswered questions.

technical Writing

use discipline and tact, omitting

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| Chapter 22

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Teport:

FORMAT OF THE FORMAL

REPORT

Cover and Title Page Letter of Transmittal

Acknowledgements

Distribution List*

Purpose of the Report Summary of Conclusions and Recommendation: . Table of Contents . List of Illustrations
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Chapter 22 [: "A

Introductions

Text of Ready of Report

Canhistons

Recommendations

Appendices

Glossary

Bibliography or Reference

Index

BOL

Zr ren Opnonal. depending on specifications of report,

WRITING THE REPORT

A. THINK AND PLAN

Choose

the purpose

of your report and then ask

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yourself some of the following questions Who am I writing the report for?" "What dol hse

uw to e achreve "What do | want to gct across?" "How do | present just enough imformation. ba- much" "How long should the report be x is the level of the reader?" "How and where wt i get the data needed?" "Do I have @ deadline to meet?" "Have I completed a ume schedule to estimate the dates when } will finish each step oF section of my work?" "Have ! allowed enough time for the approvals needed from my supervisor?

B. GATHER YOUR INFORMATION will need for the report As- You must decide what you Peerage eaialh, neld semble your notes, records, pul 4 soneresults, surveys. IIE IE. AO ebooks, c ind yourself researc q grams. etc. You may also fi formation. If you do lopedias or even the Internet for in uyalanorg be sure you take notes sO you may aca quotations or mention your sources of information, The use o 3" x 5" cards for note-taking wall allow you to easily arrange your data. Wnte down the following for each source you sult: ttle, author, publisher and facts of publicauon.

wi need this information for your bibliography or hist of refer- ences at the end of the report. Read your data over and reflect upon it for a while before completing the next step.

C. MAKE AN OUTLINE Choose the order in which you will present your data. When your information is arranged in the best manner, It will be easier for you to wnite and for the reader to read. Proper arrangement will help you to put just enough emphasis on each topic. Your outline should show what items are to be included and each item's relative importance in the overall report. You may use any of the types of outlines

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D. WRITE THE ROUGH

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Yo notes to yourself inthe Margi, since probauly Pi thts draft but you Mark where Your iMust AbiY no On, Follow your outline and try to rough dung Will, sequence Remember, you are goin the tite 4 not the full report in its final order finished before you can write your pu summary of conclusions and recomniee

ot should contain the following parts ation

a You WItte the teext

OU text wij pt! he 5 Your, Introduction - a brief, Separate section of the report in which you tell the reader exactly what subject is covered in the re. port. It is important that the introduction be as interesting as possible because the read, will decide at this point whether fo cont teading your report.

ne ell Body - describes all of the mai inferesting fashion, begin to tell wee Your te Use headings and subheadings in the bole th sectlong passages and to keep the teader jn es Tey is being discussed next.

Headings

of mayor dy TMed as to in capital letters and may begin a new ike Ff'SIONS she subheadings, the first letter of all significant eet Wish capitalized. Both may be placed on the left Psi 'S us, or in the middle of the page, as you choose _ TEIN. Inde, with the pattern you use. JUSt be con IC\$I

Conclusion -

explain the significance of wha Just described in the text and, for the first time own opinions. State your conclusions in order of Conclusions must be logical, reasoned and sou ear by evidence presented in the text. Summanze esly su ing paragraph(s), end all discussion and Bive th ae con, of completeness. Emphasize and focus on yea

e reader a, information so that the reader is left with the dolor

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SIMPLICITY ~

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style suitable for the report? PREPARE THE FINAL Copy Your final draft, after typed in standard manuscript style Use 1/2" x 1" Paper with ample MARGINS on sides. 1 After completing the review the fine touches to your report. For forma

1) your work. If your report is informal, you may omit some of them Footnotes are used with direct Quotations or with

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If listing the book or magazine title statement appears, Use a reference in Passage and the same number in footnotes consecutively. A bibliography is an alphabetical list of things you considered in writing your report. [of books and articles You want to list as other information or reference books. The books and writings are indexed much the same as in footnotes, except that the author's last name appears first and the number of pages in the book is listed. A list of references is used when you do not footnote, In the text, at the end of the sentence you want to acknowledge, Place the number in Parentheses. The list of references will follow the report, listing each reference by consecutive number in the order cited, not alphabetically. 'References' can also consist

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graphy tables, mathematics! calculations, etc. An endnote is a supplemental detail that is appropriate, detailed or distracting to method in the text Pages should be numbered with Arabic numerals in the margin of right-hand pages or at the bottom center of the page Take into consideration any full-page illustration or table

Numbering

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A=}, A-2, etc. For

those pages before the text (table of contents, abstracts, etc.) use small Roman numerals at the bottom center

A table of contents can be made once the pages are numbered. It lists in sequence all sections down to the level of major subheads, exactly as in the text, and page number of each. If you have formulated a good, organized outline, you may have this job almost finished. The running header, or cover letter, can be placed within the report or sent separately to the designated reader of the report. This better identifies the report and formally presents it to the designated reader. It should contain the title and number of the report, a brief description of the report and its findings, and the signature of the person responsible for the report.

An index

should be included if the report is detailed and long so that readers can find information about a specific topic quickly. The best way to compile one

is to write out a 3" x

5" card for each topic, making sure to cross-index everything. When you have cards covering every possible topic, arrange them alphabetically. A glossary is an alphabetically arranged short dictionary of special terms in the report.

Commonly

used or known terms should not be included in this list. Illustrations should be listed by number, title and page number and placed

on

a separate page unless the list is a short one that can be included

at the end of the table of contents.

A distribution list lets the reader know who else has read the report,

in case the reader wants or needs to discuss the report with other readers.

Include

each recipient's name, title, company and the copy number of the report. An acknowledgment page contains a sentence or paragraph on this page to briefly offer credit where credit

is due to persons-

Sources

or companies for information used in the creation of the report. Any permission granted for reprinting material is also acknowledged here.

Technical Writing

The cover and title page give information about subject and purpose. The title should not be too long. The cover should be made of protective and a good quality paper. It should contain the title, author's name and the date the report was issued, the number, the contract number, and, if appropriate, the security classification. The control number. The title page, just inside the cover, should have, plus perhaps the authorization for the release of all

of the report or a routing schedule. This page has no number. The

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ENDING COMMENTS

Technical writers put much research to work, and sometimes that work needs or changes

all persons who have no right to read or use such information is considered restricted information.

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I wish you the best of luck,

NOTES

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Or x " Technical Writing written in the most detailed, id be wow yel would be directed oem an advanced Feve we" ot on neal students · ta aud include mee at page numbering the same ecu i ll be rity classification wii : ment 5 cover i ef i tified ly would be iden ya A frst text Pl a ry. H Wis ution list Lid le pABE Juation et test and eval ie rformance te VD pe :

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8. After Printing your first draft of a Publication, which of these should occur next?
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- e . One of these should be a Separate l
 - a. Subjects
 - b. Illustrations

Chronography

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10. Which Of these wou
 - a. Outline Teport
 - b. Compile list Of illustrations ©. Consider and Note objectives

d. Make table of contents

ANSWERS on Page 223

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Appendix A

xq t Associate CET Practice Exam #1 hee we tan practice tests of 100 each These are typical of hat a tee! Aa Geld of a wel cotinined IKK KOK would find hey wok the FTA NO se meters ty the full wavelength in air for what RF ies: fa ngth in air 7 Which Wire size is physically largest?

a. #8

a 72 MHz b. #12

c. #16 © d. #24

16.66 MHz

8. Adding the binary numbers 101011 and 00 with Rhy 2 Decimal 164 would appear as in binary? you what result? & 10010101 a. 111001

b. 10100100 © c. 001010 0110100 d.

3. A resistor has color stri ; ver. It ie ipes yellow, violet, violet, silver. 9. Acomponent that reduces In re a is a negative temperature coeffici th

a 46 Meg0hm +/- 10%. b ee

470 si 5 : : ee coefficient thermistor

©. 37 Meg0hm +/- 10%. d. isa thermocouple

4. 48 kOhm 5%.

10. An autotransformer:

4. What is the total capacitance of three capacitors valued at 100pf, 100pf and 50pf when connected in a series? a. has the secondary

b. has a single winding. ©. is used only in automobiles. a 25uf

b. d. uses a buck-boost coil. ce 25 pf

d. 250 pf 11. A dual trace oscilloscope differs from a single that only the dual trace scope: ace in

5. What is the ripple frequency of a full-wave single phase

bridge rectifier operated at 60Hz? a. has two cathode ray guns,

- b. has a Z input.
- a. 180 Hz c. alternates sweeps.
- b. 90Hz d. has intensified traces.
- c. 360 Hz
- d. 120Hz 12. If an RF signal voltage is reduced by 6 dB, it is then:

6. When heated or cooled, a emits a voltage. a, 25% less,

- b. half the original size.
- a. thermocouple c. 75% less.
- b. thermometer d. not changed in voltage level,
- c. thermostat
- d. thermocline

realize it the Pend and negative

to read digital VOM over an analog preference when using a portable operator

Petes

access intermittent. the video amplifier In a radar or TV receiver the response of the antenna frequency is a function of the antenna mode that utilizes the basic noise components of the side bands: see graph

new technology:

fiber optics to transfer a signal, the light in fiber wave

telescope. perturbation noted.

¥, generated 30 GHz.

Heisenberg is stored In a coil is measured In: energy levels Joule

a tele: A telephone communication term 'Cellular' refers

to telephone radio channels. Mobile Line service. The areas covered by each base station. ~ practice Exams on,

4 Appendix A

20. Integrated Circuit failures Often open or short Conditions,

- b. solder leakage. © audible noise radiation hazards.

21. Which of these we LED to light?

a 3mVAC

- b. 3mVDC ©. Dc

22. A capacitor 3/8" x 1/8" ceramic or mica) has a 47pF/-10%,

- b. 4.7 uF +/- 50%, ©. 47 20%,
- d. 47 mF +/- 10%,

23. A Digital Storage

- a. uses micro processing and Memory to allow close
- b. is Specially designed for analyzing

converts analog Input signals to a d

- d. alternatives between digital and

24. Carbon microphones are much like various Capacitors, Resistors, Thermistors, Inductors.

25. An antenna that is longer than another:

- a. will have a higher Resonant frequency.
- b. will have a lower Resonant frequency, etc. has a higher Q.
- d. delivers higher Power,

26. An 8-bit binary code allows logical combinations,

a

- b. 64
- d. 512

Appendix A |

22. An interface resets AC current flow appears as AC current reduces power dissipation as a parallel short to AC current 2K
A printer directly connected to a personal computer would use connection, a VGA

Ethernet 100 Base-T

USB of parallel Centronics and DRIS ing characters by using

20. The digital method of representation is called: combinations of zeroes and ones

- a. decimal and duplex. and binary.

30. FM where AM utilize signal level variations. frequency variations power variations sideband modulations period

31. What decimal number is represented by the binary number 01101010 ? a 64

- b, 53
- c. 106
- d. 152

32. Decimal 50 would appear as ____ in Hex. a 32

- b. 62 101000 and 110010

33. Of these choices, the display of a DMM would probably be: a LCD.

- b. Pointer.
- c. CRT.

d. Floating ball.

Practice Exams

44, A disadvantage of LCD (liquid crystal display) is high current demands environmental temperature sensitivity slow reactance to input vibration sensitivity limitations, edge

48, A series DC motor, with no load: tends to stop or not start may be damaged by excessive speed reverses direction becomes an AC generator.

436, Which motor maintains a constant speed changing loads?

- a, Series
- b. Shunt

c. Servo

- d. Buck-boost

37. A small power drill or electric mixer would properly be powered by a/an motor, namely

- a, three-Phase
- b, induction series
- d. parallel

38, The rotor in an induction motor is not connected to AC or DC power. Is it?

- a, True
- b. False
- c. It depends on speed
- d, Induction motors have no rotor.

39, 0 dBm, when referring to audio signals means;

- a, decibels referenced to 1 milliwatt.
- b. decibels referenced to 1 millivolt pk-pk, or, decibels referenced to 1 volt pk-pk.
- d. zero signal output.

40, If a class A differential amplifier is using only two possible inputs, what will the output signal shape be?

- a, Zero
- b, Exactly the same as the one input
- c, Equal in amplitude and frequency to the input
- d. Smaller than if two in-phase inputs were applied

oat

1. : difference in electron flow due to resistance is a term describing many op-amp circuits. The gain of an op-amp is inversely related to bandwidth frequency increases.

Ho

Aways

Ne 1 can be changed without affecting the other pine rect bandwidth but higher gain affects SO little that it is effectively insignificant jo detector circuit in an AM radio most · we Is a circuit. tial amplifier with two identical in-phase ad on the inputs will have an output igo a of ter the input, but same shape 4 cannot be determined with this information one the input signal ranning oscillator is also called a monostable

Af THe

False

use which has its fusible wire wrapped around an insulator Is: 3 Slo-Blo type fuse. te an Inductive fuse. · citive fuse. ; | ntended for high voltage use.

Appendix *

48. The power transformer in power & always steps voltages up b always steps voltages down © might step voltages up of down always steps current up

49. Which of the below would he considered 'havie' components of a switching power supply? Zener, regulator, transformer, SCR Regulator, diodes, AC power, load Transformer. bndge. capacitor, load Filter, rectifier, transformer acge

50. The average of a full sine wave rectified voltage is: $0.637 \times \text{Peak}$. $0.707 \times \text{Peak}$ $0.707 \times \text{Peak to Peak}$.

1.414 x Average aeges SI. The ripple frequency for a half wave rectifier

operating from 60 Hz electrical power ts:

- a. b. 60 Hz.
- c. d. 240 Hz.

52. The effective voltage of a 100V Peak-to-Peak sine wave

- a, 35.35 VAC.
- b. 70.70 VAC,
- c. 141.4 VAC.
- d. 31,8 VAC.

53. Api filter is an inductor with: a capacitor at each end to common. a capacitor at each end in series, a resistor at each end to common. a resistor at each end in senes.

54. An inductor opposes any change in: voltage. current. frequency. reactance.

gx any change current frequency

PORTA

CN A will tn cae RC theme? to what percent of full charge

60. Moving one plate of a capacitor further away from the other will: a. decrease capacitance. b. increase capacitance. c. decrease voltage rating. d. increase mutual transconductance. && The fact that an inductor opposes change in current is true of a. AC signals only.

b. DC signals only. c. DC signals. d. signals more than UHF.

61. If you increase the value of a capacitor, the capacitive reactance will: a. increase.

b. decrease. c. depend on connected components. d. not be affected by the value. . In checking a capacitor to see if it is shorted, open or you would use the _ ohm meter range. a. lowest

b. highest You can't check a capacitor this way. @ You will ruin the ohmmeter unless the capacitor is charged. |. The capacitor in a power supply capacitive filter:

a. is in series with the load.

b. is in parallel with the load. c. a series choke. d. must be in parallel with a choke coil. e. Patines con

62. The time required to discharge a capacitor is proportional to the ____ Its given by

a. the total capacitance times the total resistance

c. the total inductance

d. the combination of LRC

63. With a given filter type, pure filter in a low pass filter: a. low frequencies b. high frequencies c. mid frequencies. d. any frequency

64. Using a resistor instead of an inductor in a pi filter: a. is more expensive

a. is less expensive.

b. is less efficient. c. does not filter the AC as well

d. all of the above.

65. A power supply whose output voltage varies whether the load is light or heavy percent of regulation? RE does p, Would have

a. 100

b. 50

d. 0

66. A power supply which has no load and 40 volts under a full load regulation using the % of regulation formula is

a. 100

b. 50

d. 0

67. Correctly connecting a properly rated Zener diode across a power supply load will: a. protect the power transformer.

b. increase the output DC. c. limit the output DC, eliminate ripple.

68. The diode with the largest diameter? a. should NOT be of concern b. conductors? c. wire gauge | one + We 2 sectional area in a circuit d. voltage for each cell is approximately 1.1V

a

It is made of type solder for electronics work is frequently used for electronics circuit-board connections (approximately): it provides 60 watts.

As it goes into the soldering iron, the foreign material on the metal to be soldered, the temperature of the solder, the conductor-to-solder-to-conductor chemical junctions, etc.

• | Appendix

75. Laminations used for the core of a transformer must physically make metal to metal contact. The adjacent laminations must be insulated from each other to increase eddy currents.

d. reduce counter emf in the

76. $I = 1$ amp, $R = 10,000$ ohms, What is the value of the voltage?

a.

b. 10 volts c. 100 volts

d. 10,000 volts

77. If $R = 200$ ohms and $I = 4$ amps. What is the value of P (power)? a OR watts

b. 8 watts c. 4 watts d. 8 watts

78. Which frequency will have a higher loss when using coaxial cable transmission line?

550 kHz

27 MHz

470 MHz

1 GHz

po SP

79. A is a reference antenna that all other antennas with gain are compared to.

Yagi

omni mast dipole

80. A high frequency antenna has a low frequency one.

a. is thicker

b. is shorter

c. has more gain

d. is longer

81. Standing waves on a transmission cable:

a. indicate maximum power out.

b. indicate less than optimum signal transmission, should cause no concern. c. can be caused by poor termination.

d. can be located with a voltmeter.

er

Appendix A

ma are when a radio frequency

a A transmitted fadto a & radiated into space. power level KX Doudting the power in # transmitter output signal fs

___ dB gain.

od

84, 0 dBm\` is wbat level of received TY signal? a Zero b 0001 Volt & 1000 merovolts

da && is: a 4bits. db. S bits

c. 16 bits. a 32 bits.

86. Ina AND gate: a 1 input must be high in order to have a high output.

b. 2 mputs must be high for a high output. 3 put must be high for a high output. all three inputs must be high to have a low output.

87. An LCD operates by polarizing the light through filters. The fluid rotates the light degrees. a 30

b. 45

ad 180

88. To test an SCR with an ohmmeter:

a. this cannot be done.

b. check resistance the same as transistors.

c. with + to anode and - to cathode, short gate and anode together. Should go from near 'open' to near 'short.'

d. with- to anode and + to cathode, short gate and cathode together. Should go from near 'open' to near 'short'

Practice Exams

89. Using the $R \times 100$ range and » source and drain of an N channel MOSFET

- a. readings should be equal in both
- b. with positive on anode, reading that of the opposite polarity. 4, with negative on anode, reading that of the opposite polarity, should be V_g
- d. this check provides no useful information

90. With + voltage to drain or source and the gate on an N channel JFET, the JFET will be: At least 4

- a. low. high
- c. high to drain, low to source.
- d. low to source, high to drain,

91. Which Is LEAST likely to cause ESD MOSFET? P damage to

- a. Flux
- b. Solder sucker
- c. Human finger
- d. Wick

92. Probe:

- a. injects signals into logic circuits,
- b. indicates high and low voltage Conditions.

c. measures dB levels.

- d. displays voltage levels on an analog scale.

93. In a typical electrical generator, a small AC Voltage applied to the field windings is able to control output level of the generator. Is this true?

- a. True
- b. False

94. When working with and testing lead-acid batteries:

- a. wear eye protection goggles if battery can
- b. short between terminals only if in a well ventilated
- c. when charging, keep room closed off.
- d. always wear eye protection, never Cause a charge in a ventilated room. Spark and

me ref ji

Poke Practice Exams :

a such as the output from DVD players. Appendi=

; vers and preamplifiers, usually are of ϕ for input to a T.Y. modulator or other get? < peak-to-peak | · peak-to-peak | 4 wae volt peak-to-peak 2 \ ·ot ϕ signal parameters docs @ spectrum als

et end phase Fat capacitance i ee signal separation vier" and frequency ag yolatil src oie jatile, non-removable data storage yter

Weal

p^v

Ji

ob peused first to clean a corroded plated ou! wine jroo tip? emery cloth while cold 73 coat with acid core solder after heating b

ae t that a thermal shunt be in place. f solder. | with a smooth and rounded high connections: ie imum o}

b of solder. connection with rough solder tips.

by eration of an audio amplifier means that | or flows through the active device (tube, | FET, IC) for ___ · degrees of the input waves |) 430 | h tal 180 but less than 360 | P ethan 180 | um on page 223

eee

ge P i Practica CET Practice Exam #2 ee ere rue pene io guess ·These are Lot ta ever we re the maternal contamed in ees Cy took the ETA Assoc hom,

i, on a computer removabte disk is: 7 me, Coaxial cable line losses at RF frequencies a ane record, : duet, into sectors. 7 wire resistance, capacitive and inductive Teac as data is written. 4 wire resistance, dielectric fading and tran @ cme path. capacitive and inductive reactance and braid le; ee

d. magnetic effects of nearby materials Akag J hy 2 APC Ae mem 8. You are troubleshooting audio reproduction equ; (ow kdtol nd E ded (abo a and measure the signal to be input into meg) and Exten ve What approximate RMS level would you expec megabyte. What other names for internal memory tobe? Dect that types do PCs have? Rha) a Cache a. Audio signals vary oo much to be measured

b. 1 microvolt a va c. 1 millivolt Resistor array d. 1,000 mi aE millivolts . 9. The noise level in a rad veri

4. Which video system has the highest resolution? the: jewel in arad 4 ren a. IF stages. A b. Local Oscillator. c SV GA c. Audio section. a@ XVGA RF section.

4. You would want to use # dual trace oscilloscope to io. The fast c measure: should not exceed, % of its AHr, or ampere-hour rating.

2. signal differences between two points. ; ~ i ircuit. a

b. suspected resistance problems one

c. one faulty circuit in voltage evel while another b. 100 js faulty im a current level, Fi done fault video and the other 10 audio.

5, Common Intermediate Frequencies (IF's) for AM and FM commercial broadcast receivers are:
operation:

- a. Serial port board
- a. 455 and 10.7
- b, 268 kHz and c. CGA controller

455 KHz and 10.7 MHz. d. IDE controller

4. and 10.7 MHz. ED: all compression: 12, 6 radio receiver sign a, only emits visible light when forward bias reverse D/A converters. b. can be used as a fiber optic signal source. c. does not need high current

b, reduces the high frequency receiver may send or receive light signals.

may have connected as an inverting by the external 50k ohm op-amp determined the condition would be significant for 100 miles to be received at a distance of 500 miles? if signal NAND has two low inputs and one high output is:

the power of Low. nor low. you would be considered a component of a 100V supply? the power circuit with rectifier and the DOK short circuit protection is & 4 al is defined as: 10 sign 1" cd RE lig al including sound information. ji 4 including visual information.

RF

4. towbar

Practice Exams

20, 21. 22, 23. 24,

Appendix A

ee a power supply and have nected power, It has sev eral 250 oF connected in paraltel. To avoid shock, tapecito

a. should be disconnected may be ignored should be discharged to cha ground, through suitable resistor : oo should be isolated with a avoided. used in electronic servicing:

a. shows the size and shape of all the components mm the unit.

b. is sometimes called a block diagram. shows the location and arrangement of components in the unit,

d. is much like a road map for following signals. High voltage should be discharged before servicing a product. After disabling AC input power, discharge an HV PS by:

a. shorting output to the plastic cabinet.

b. disconnecting the HV lead with rubber gloves. ¢, draining through a large value resistor to chassis ground.

d. shorting output of the collector heat sinks. Which of these would be an important electrical consideration in the use of a transformer? The operating frequency . Space requirements Environmental temperatures Number of laminations aese If a 12v product is dissipating 50w, how much load resistance does the device represent? .24 ohms 24 ohms

4.16 ohms

2.9 ohms

Beep

An 'Audio signal' is often defined as:

a. frequencies in the 20 Hz to 15,000 Hz range.

b. baseband including visual information.

c. frequencies above ultrasonic.

d. broadband RF.

Appendix A

seam a generator rather than AC.

PS

a

h key

a

27. You suspect the hasan internal short. To prevent further damage when you apply power, you should: a short around the main fuse OF watching for heat or arcing bh replace the unit wathout aftermpn! y repaired safely Time oF Cost breaker and proceed, ng repairs. as 'or within reasons! a om resistance high (100 resistor sn With the power

28. Aripple counter:

- a. that indicates the power SUPPLY ripple frequency.
- b. isa network of several XNAND gates that regulates the computer oscillator. isa synchronous counter that stores the counts of other counters in a computer. dis two of more connected sequennally with the input pulses toggling the first one. tage is applied to a capacitor: 2 current is maximum at first, then reduces 8S the capacitor becomes charged. is reached.
- b. current is constant until full charge ! cc. the voltage difference between the plates reduces as full charge is reached. there is no difference between the plates after begins.

30. An electrolytic cap transformer and the rectii

- a. protects the diodes from power surges.
- b. eliminates the need for filter capacitors.
- c. is part ofa voltage doubler circuit.
- d. provides that earth ground be isolated from DC ground.

29. When a vol acitor in series between the power fiers in a power supply:

Practice Exams

uM. Information travels through optical fibers via light.

he protons flow, means of, photons. electrons, are also, You would most likely prefer

for: an analog video,

tr, intermittent fluctuations, accuracy. convenience. safety. aa, Which of these circuits would diode? ure gy a A receiver I F section

- b. A television tuner A power regulator
- d. A bridge rectifier

44. A block in a digital diagram is called a No true. R, NY high i

- a. also be called a NAND.
- b. It is a NOT OR or Inverted OR. cc. It will pass any single negative b » But not than one.
- d. It must be reset after any positive level input,

More

ich of these statements is most true regarding, ating €, lo

35. Whi matic diagrams of electronic devices? Ss? sche!

- a. b. They components.
- c. They represent signal paths.
- d. They provide user operational procedures in Personal Computer technology, ROM indicates; ly manufacturing specifications of IS Of

36. I

- a. memory that may not be addressed randomly!
- b. memory that may only be read from.
- c. real address mode memory.
- d. resident operating system memory.

37. A light emitting diode will light when:

- a. the gate is turned on. avalanche.
- c, it is forward biased.
- d. it is reverse biased. They provide pictorial representations of components,

Exe, te to limit the voltage and the gate voltage in order to 2H 6 vV if activated by light FS volts met net € is rated 6.3 V. This tetty you: ry RIT on

= tection.

er SE On and of ss ly turn equip for ontically knowledgeable about 'ESD'; mus jan nie soldering device ry ject the proper ple (o sel sensitive devices come in numerous o voltage | use & the proper test equipment for i he will hav that in them stat pecan ents.

a discharge can destroy many ic dis ter-emf voltage induced in the or, th nie source current. adds decrease in uency, thus longer. g much mation the antenna length is not Vis row while CB is vertical, cB! jarize pol c is at _ ' ing Power supplies: chin rters. wi DC to DC converters from remote locations. tro! m . or sormall con 117v AC power to a higher incoming

b. not convert ly. , nly. 4 erate off AC sources o ust o gm WA Power g, Dasa en

be Compared to What cou 0) =e

system? . suc!

& A valve

A pump cA filter aA Pipeline

Exams

Appendix AL indication of the quacity of an 86. A current-carryin: iz co nductoy 'l ga, Which of these fa Feed field vi

analog meter: move Wits Mines arte = .__direeti f

lines of flux, as determing the force Neng voltage Scale Of mat tat The highest ee: boa me mone . a perpendicular MRS number of resistance seales b. a parallel 'ang &. an opposite eq, The vertical ants of an oseiltoscoPe display fs also a the same the: 57. The antenna used to com, ome antennas is called a Pare the pai time hase. _ _

= a. dipole b. rhombic

&. conical disk cone

52. Ifyou increase the value of an inductor, the inductive S reactance will: 58. Th ae effectively bi turned on, Is turned off by a

b. decrease. © anode to cathode is rt 'on connected components. us 'mn

a. silicon controlled rectifi, *

b. Schottky diode.

53. Which would least likely be a high-voltage (above

1,000V) capacitor? d. field effect transistor.

59. Which of these terms would yo

b. Ceramic when referring to types of resi Most expe : istors? St tO seg we a. Electrolytic Paper b. FET

54. The rechargeable cells currently used in many oral

- d. Henry ducts would be: consumer portable pro cid. 6o. 'Microwave,' as used in radio e}
- b. Carbon-zinc. Sctronics, refe
- c. Lithium. a kitchen appliance. Tsto 1 GHz - 100 GHz radio frequencies a style of television. an oscilloscope display interpretatio mn.
- d. Lithium-Ion. peop

55. A solid-state regulator:

61. A differential amplifier who: Se two i equal, but 180 degrees out of phase. output signal. , Put signals ay

- a. is made up of SCRs. will have ala
- b. varies conduction to maintain a
- c. is simply a Zener diode. level output voltage. ut voltage level and vary
- d._ uses Diacs to sense the outp the input voltage. a. maximum b. zero
- c. undeterminable
- d. one half

Practice Exam

to control the output level used

be

capture winding

a arm: the field winding at the A connects to the coil at the other end. Includes: inductance.

new and Inductively, active and passive

Final RF line is RF transmission line in a circuit: resist!

or hen «ween the center conductor line jens" rial bel sic inated.

14. terminal

ne n be used in circuits

a Omax 250V, 65 ly voltage of: parked se: supP ope! > Seq ith higher voltage. With heat. output erature. temp ance with ai deer
ges resist heat or c chang e'ected by

a! sum fers to; what most oe the secondary? ansforme ut voltage net ns the eter Nals are ims to secondary tu Yea /an jo of Prim
al ratio ve materi The wer rating load © he secondary

Ke

a b3 Merease

6 Meteage

\s Merease TL. Which of Semtcong, Fequires th, TEMES nor. damay a TTL ECL Mov

4. CMOS,

72. Which of these 12 output of a TON a ANiCad ba

b. A sens Ve diode ang An altemato

a. 73. Three Series Inductors have Va and 40 my Fespectively, Total indu a 90 mH.

b. 13.33 mH. © 075 my a On mu.

Appendix A i

2,

re vat light electrical signal path

a bh regres 1 rps in fe eat rede sehen BETTY a cant tO THY measurements. block diagram. will 7a. AC power outlet strips
with surge suppressors include at least one: a MOV.

b. MOS. Circuit breaker.

4. Zener diode. of: 05 nF, 208

77. Connecting 3 capacitors series results in a total capacitance of a 60.05 pF

- b. 0.498 pF.
- c. 19.95 pF. a 65 pF.

78. You would use the formula ($L_t =$ series circuit. nce. a ina capacitive series circu!

- b. to find total inductance in & for seres-parallel mj dina resistive paralle! circuit.

79. The fifth color band on a 5% resistor is yellow. This tells you:

- a. ithas an extremely high degree of accuracy.
- b. the manufacturer.
- c. tolerance.
- d. the recommended frequency range.

80. VHF or UHF antennas don't use waveguides as feed lines because: a they create grounding problems.

- b. their resistance to high frequency is too high.
- c. the charactenstic impedance is too high. . the large dimensions at these wavelengths makes other transmissions lines more practical. F and 40

Practice Exams x

81. To trace a logic path, you Use. a Logic probe a DMM a Logic Pulser a Pulse Tracer,
82. Ordinary alpha/numeric information computer is converted to binary Using: a Decoder. a Crystal a Multiplexor. an Encoder.
83. In electronic devices, what term refers magnetism? (Select one), if
- a, Hysteresis
 - b. Flux
 - c, Induction
 - d. Kinetic energy
84. Some variable capacitors are made interleave. The capacitance varies as (separate) and close. As they open: SO that
- a. the capacitance increases,
 - b. the capacitance decreases.
 - c. the voltage rating increases
 - d. the dielectric becomes more dense
85. In VOM's, voltage multiplier resistors are important, Ant;
- a. in the voltage scales.
 - b. in the current scales.
 - c. in the resistance scales.
 - d. in protecting the movement.
86. In a series-tuned circuit, to increase the resonance frequency, you need to: want
- a. decrease the inductance.
 - b. add resistance.
 - c. increase the inductance.
 - d. increase the input signal level.
87. AMOSFET:
- a. may be called a Zener.
 - b. typically has a low input impedance. It retains data even when input power is lost.
 - d. amplifies signal voltage variations.

ec

Chapter 9

en

in transformers for:

ed all the flux, magnetic. or down. external influences, if it contains a voltage loop circuit, then The output of the + of the transformer

I

ti j cont dl [] abl signal

Ie

ms depends or aps riable.

a

es yey a ible. on adjustal ae" being the + gel on back. for feed! le i r} as Cer) ty Rul s below the trigger point. drop' scR tor. on.
the collec! | ne gcR seen on ce and reactance which tan resis! nt is called: rnating curre of alte ate flow mm: | nce: i ese | et clot
t likely be le would most likely mocoup her! fal

at

a: | wi at. eis: n an oscilloscop is or

Lass

tne

o,

Practice Exams

voltages are maintained, the ϕ a cathode

98. A emission) t_g Ta, Mt my . rong hus Shetg

a. PER sing lig length der Wick hie Ore of normathy be

Paper

b Femite

© Rosin

d

97. The i, binary Number 19009 a 32 a

b. 10000 © 10 a 16.

98. You would Choose anLep ifyoq Nee aoa Sou,

b. alow Current digital display © an induct, Soupling devicg, doa devicg

99. Germanium diodes: are also Called Tunnel Diodes,

b. normally drop al V with forward Cu © may be used as Zen, diodes,

4. more commonly are Called Varact

100. RE signal Propagation consid 'line-of-sight' Would be;

a. HF.

b. MF,

c. UHF.

d. VLE

ANSWERS on page 223

Certification Administrators

(CAs) ETA Certification Administrator Liss;

h

The foietng: red cermanan adiminteruinn atther heer artheCPT ondPOU testing program

Withoan

them there would

he

no profewsional certific ee ee te Comeensal

License

exams wrk!require traveling hing distances and being there

on specific date,es

We ote tehee Ae

Whee

eee aenee remdh feamen either tepe aferam cruvtac the clearet

CA

Understand

that some CAs are electronics instructors in schools whe dere

Stet

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yr Aah the

CA when ftmight

he comeniont for he/she and the instihition to monitor vee cone fey GC A Aenetch ceevom

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mn the military eer toe. your exhcation office and the

DANTIS TCO

will obtain exam and ceenn

Please inform you to take the exam code CV Tac FURear

in most of the education offices are usually aware of the details

of the CET and ICC programs to get the details you should be aware

of the rules and options of the program and help to make sure you are taking the APCH sth, either on the POL and OFT program the correct exam and Siding by the MIHUIR for such things

Weer

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When free of time to write for that exam

A money order, or your charge card numbers are the preferred method of exam payment

Some

CAs may charge MT! Vow con

Retp

with CA and the FTA

Testing

Sifhy

making sure you include the examination number

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on exc N answer sheet on the answer paper) Be sure every word you write

on the information form is legible

If your school teaches electronics and

not

an established CA site have the choice of ETA by phone (RAD) DAK. TRI4

OF e-enactment

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at

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 pappter 7: 1-b, 2-a, 3-b, 4-d, 5-b, 6-c, 7-a, 8-a, 9-c, 10-c wait & 7 1-a, 2-d, 3-b, 4-a, 5-c, 6-a, 7-b, 8-a, 9-b, unl 10-b, 12-a, 13-c, 14-
 c, 15-a, 16-¢, 17-a, 18-a, 19-c, 20-b, 21-a, 22-b, 23-d 2-8, 3-b, 4-¢, 5-d, 6-c, 7-d, 8-a, 9-b, 10-d coe ter 9: 1-8, 2-b, 3-a, 4-c, 5-b, 6-
 a, 7-b, 8-b, 9-c, 10-a 10: 1-c, 2-d, 3-b, 4-b, 5-a, 6-d, 7-a, 8-c, 9-c, 10-¢ 'ai 4 (8, 9 10): Be 2-b, 3-b, 4-a, 5-a, 6-d, 7-b, 8-a, 9-b, o-
 b, 11-a 2-b, 3-b, 4-a, 5-d, 6-b, 7-b, 8-a, 9-b, 10-c, ter 11: I-e, chap H-a, 12-b chapter 12: I-c, sot (1 & 12: 1d, 2-d, 3-d, 4-b, 5-a, 6-c,
 7-b, 8-b, 9-d, 10-d 2-d, 3-(a-4, b-3, c-5, d-6, e-1, f-2), 4-b, 5-b, ter 13: T-a, 8-C, 9-a, 10-d

Chapter 14: 1-b, 2-a, 3-d, 4-b, 5-a, 6-b, 7-c, 8-b, 9-a, 10-d

chapter 15: 1-2-8, 3-8, 4, 5-d, 6-d, 7-o, 8-b, 9-2, 10-d unit 6 (13, 14 & 15): 1-b, 2c, 3-d, 4-b, 5-a, 6-a, 7-b, 8-d, 9-a, 10-c, 11-c

Answers to Quizzes and =

4 i?

Chapter 16: 2-d, 3-6, 4-c, 5-6, &@, 16.

(a4

Chapter 17; |-a, 2-c, 3-c, 4-d, 8, Rd. 94,

Unit 7 (16 & 37): 2-d, 3-c, 4-4, 5. Be, VAL

10-b, 12-9. 13-b, Mea,

Chapter 18: |-c, 2-a, 3-c, 4-d, 6-c, 7-4, 8-4, 9-4, [hd

Chapter 19: |-cç, 2-d, 3-b, 4-d, 5-b, 6, 7-b, Ra, 9-2, |\0-4

Chapter 20: 1-b, 2-d, 3-a, 4-ç, 6-a, 7-4, 8-4, 9-a, 10-2

Unit 8 (18, 19 & 20): 1-a, 2 -d, 3-b, 4-c. 5-4, 6b, 7-4, Re

9-d, 10-d, 12-c

Chapter 21: I-d, 2-a, 3-b, 4-a,b orc, S-a, 6-c, T-a, -b, 9-4,

IO-C

Chapter 22: 1-d, 2-a, 3-a, 4c, 5-b, 6-c, T-a, 8-d, 9-b, 10- <

Unit 9 (21 & 22): 1-a, 2-a, 3-a

2-c, 3-b, 4-b-d-g, 5-c, 6-b, 7-e, 8-b, 9-d, 10-a

Practice Exams

Practice Exam 1:

I-c, 2-b, 3-b, 4-c, 5-d, 6-a, 7-a, 8-a, 9-a, 10-b, 11-c, 12-5 13-a, 14-a, 15-c, 16-a, 17-d, 18-b, 19-d, 20-a, 21-d. 22-4. 23-a, b, or c, 24-b, 25-b, 26-c, 27-a, 28-c, 29-d, 30-a, + 32-a, 33-a, 34-b, 35-b, 36-b, 37-c, 38-a, 39-a, 40-b, 41a 42-a, 43-a, 44-d, 45-a, 46-b, 47-a, 48-c, 49-4, 50-a, 51a 52-a, 53-a, 54-b, 55-a, 56-c, 57-a, 58-c, 59-b, 60-b, 61-4, 62-d, 63-b, 64-d, 65-d, 66-c, 67-c, 68-4, 69-a, 70-¢. 71, 72-c, 73-b, 74-b, 75-b, 77-a, 78-d, 79-d, 81.0 82-a, 83-b, 84-c, 85-b, 86-c, 87-c, 88-c, 89-a, 90-b, 92-b, 93-b, 94-d, 95-d, 96-d, 97-b, 98-d, 99-¢, 100.

Practice Exam 2:

1-b, 2-a, 3-d, 4-a, 5-c, 6-b, 7-a, 8-d, 9 =

13-d, 14-d, 15-b, 16-d, 17-b, 18-a, 19-a, 20-a, 21-a, 22-a, 23-a, 24-d, 25-a, 26-d, 27-a, 28-d, 29-d, 30-d, 31-d, 32-d, 33-c, 34-b, 35-c, 36-b, 37-c, 38-a, 39-d, 40-d, 41-d, 42-d, 43-a, 44-b, 45-c, 46-b, 47-c, 48-a, 49-b, 50-c, 51-d, 52-d, 53-b, 54-a, 55-a, 56-a, 57-a, 58-a, 59-c, 60-b, 61-b, 62-h, 63-c, 64-c, 65-a, 66-c, 67-a, 68-c, 69-a, 70-b, 71-a, 72-a, 73-a, 74-d, 75-d, 76-a, 77-b, 78-b, 79-d, 80-d, 81-d, 82-a, 83-a, 84-a, 85-d, 86-a, 87-d, 88-a, 89-c, 90-b, 91-c, 92-a, 93-b, 94-d, 95-c, 96-b, 97-a, 98-b, 99-b, 100-d

Formulas

Factors

$$3.14 \times 6.28$$

$$= 0.497$$

$$\text{meter} = 3.28 \text{ feet}$$

$$1 \text{ inch} = 2.54 \text{ centimeters}$$

$$1 \text{ radian} = 57.3^\circ$$

Resonant frequency formulas,

Where f is in kHz, L is in microhenries, C is in microfarads

$$f = 189.2 \text{ NIC}$$

2 Vic

Frequency & Wavelength formulas = frequency, λ = wavelength

$$180^\circ = \text{halfwave} \quad 0.25\lambda = 90^\circ = \text{quarter wave}$$

$$f = \frac{v}{\lambda} \text{ or } f = \frac{v}{\lambda} = 984 + \frac{v}{\lambda}$$

$$\lambda = \frac{v}{f} \text{ or } \lambda = \frac{v}{f} = 984 + \frac{v}{f}$$

Sine wave conversion (RMS = root mean square).

$$\text{Effective value (RMS)} = 0.707 \times \text{Peak Value} = 1.11 \times \text{Average Value}$$

$$\text{Peak Value} = 1.414 \times \text{Effective Value (RMS)} = 1.57 \times \text{Average Value}$$

$$\text{Average Value} = 0.637 \times \text{Peak Value} = 0.9 \times \text{Effective Value (RMS)}$$

Identify: Waveform, Peak (amplitude), RMS, | cycle over time period (frequency). Peak to peak, and practical average.

v

Periodic time T (Frequency = $1/T$)

Voltage gain in decibels

Gain dB = $20 \log (V_{out} / V_{in})$

Ratio of 2 power levels in decibels

Gain dB = $10 \log (P_1 + P_2)$

Resistors in series

$R = R_1 + R_2 + \dots$

Resistors in parallel

IR Inductors connected in series

$L = L_1 + L_2 + \dots$

Inductors connected in parallel

$1/L = (1/L_1) + (1/L_2) + \dots$

Reactance of inductors where X_L is reactance, f is frequency, and L is inductance Sine wave figure © Eric Coates MA BSc (Hons) Used with permission, 2013.

Common Formula,

Ohm's Law

$E = \text{Voltage}$

$P = \text{Power}$ $R = \text{Resistance}$

$P = I^2 E$, the power being dissipated by the resistor is a

product of the current and the applied voltage. Time constants

T (Greek Tau), R (ohms), C (microfarads), L (microhenries)

RL circuit: $1 T (\text{sec}) = L (\mu H) / R (\Omega)$

RC circuit: $1 T (\text{sec}) = R (\Omega) \times C (\mu f)$

How to Compute Charge or Quantity of Electricity where Q is the charge (in coulombs), C is the capacitance

(in farads), and V is the potential difference (in volts).

$$Q = C \times V$$

Energy Storage in a Capacitor where W is the energy (in Joules), C is the capacitance

farads), and V is the potential difference (in volts).

Capacitors connected in parallel

$$C = C_1 + C_2 + C_3 + \dots$$

Capacitors connected in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

(in Reactance of capacitors

$$X_C = \frac{1}{2\pi f C}$$

Impedance Formulas for a Series Circuit

$$Z = \sqrt{R^2 + X^2}$$
 where Z is impedance

Impedance Formulas for R and X in Parallel

RX Battery internal resistance

$$V_{out} = EMF - I r_{internal}$$

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